Symbols and abbreviations

➔	cross reference
🖱	website
❗	warning
⚠	warning
▶▶	don't dawdle
💣	controversial topic
⚠	rare but fascinating
1°	primary
2°	secondary
~	approximately
=	equal to
>	greater than
<	less than
≥	equal to or greater than
+/-	plus/minus
%	percent
α	alpha
β	beta
δ	delta
γ	gamma
κ	kappa
♀	female
♂	male
°C	degree Celsius
®	registered trademark
™	trademark
AAT	α-1-antitrypsin
ABPI	ankle–brachial pressure index
ACE	angiotensin-converting enzyme
ACR	American College of Rheumatology
ACTH	adrenocorticotrophic hormone
AD	autosomal dominant

ADHD	attention-deficit/hyperactivity disorder
AGEP	acute generalized exanthematous pustulosis
AIDS	acquired immune deficiency syndrome
AJCC	American Joint Committee on Cancer
AMP	antimicrobial peptide
ANA	antinuclear antibody
ANCA	anti-neutrophil cytoplasmic antibody
APECED	autoimmune polyendocrinopathy–candidiasis–ectodermal dystrophy
AR	autosomal recessive
AST	aspartate aminotransferase
ATLL	adult T-cell leukaemia/lymphoma
AV	arteriovenous
BAD	British Association of Dermatologists
BCC	basal cell carcinoma
BCG	bacille Calmette–Guérin
bd	*bis die* (twice daily)
BMI	body mass index
BMZ	basement membrane zone
BP	bullous pemphigoid
BSA	body surface area
CAPS	cryopyrin-associated periodic syndromes
CCLE	chronic cutaneous lupus erythematosus
CCP	cyclic citrullinated peptide
CFTR	cystic fibrosis transmembrane conductance regulator
CH	congenital haemangioma
CINCA	chronic infantile neurological cutaneous and articular syndrome
CLASI	Cutaneous Lupus Disease Activity and Severity Index
CM	capillary malformation
CMV	cytomegalovirus
CNS	central nervous system
CRP	C-reactive protein
CT	computed tomography

CTCL	cutaneous T-cell lymphoma
CTLA-4	cytotoxic T-lymphocyte-associated protein 4
CXR	chest X-ray
DH	dermatitis herpetiformis
DIC	disseminated intravascular coagulation
DIP	distal interphalangeal
DLE	discoid lupus erythematosus
DLI	donor lymphocyte infusion
DLQI	Dermatology Life Quality Index
DMARD	disease-modifying antirheumatic drug
DMSO	dimethyl sulfoxide
DNA	deoxyribonucleic acid
DRESS	Drug rash, eosinophilia, and systemic symptoms
ds	double-stranded
DSSI	Dermatomyositis Skin Severity Index
DVT	deep venous thrombosis
EASI	Eczema Area and Severity Index
EB	epidermolysis bullosa
EBA	epidermolysis bullosa acquisita
EBV	Epstein–Barr virus
ECG	electrocardiogram
ECM	extracellular matrix
ECP	extracorporeal photopheresis
ED	ectodermal dysplasia
EDS	Ehlers–Danlos syndrome
eGFR	estimated glomerular filtration rate
EGFR	epidermal growth factor receptor
EGPA	eosinophilic granulomatosis with polyangiitis
EKV	erythrokeratoderma variabilis
EKVP	erythrokeratodermia variabilis et progressiva
EM	erythema multiforme
EMPD	extramammary Paget disease
EN	erythema nodosum
ENA	extractable nuclear antigen
ENT	ear, nose, and throat

EPP	erythropoietic protoporphyria
ESR	erythrocyte sedimentation rate
ET	epidermolytic toxins
FAE	fumaric acid ester
FBC	full blood count
FCAS	familial cold auto-inflammatory syndrome
FGFR2	fibroblast growth factor receptor 2
FMF	familial Mediterranean fever
FSH	follicle-stimulating hormone
FTU	fingertip unit
g	gram
G-CSF	granulocyte colony-stimulating factor
GFR	glomerular filtration rate
GI	gastrointestinal
GIT	gastrointestinal tract
GLUT1	glucose transporter 1
GP	general practitioner
GPA	granulomatosis with polyangiitis
G6PD	glucose-6-phosphate dehydrogenase
GPP	generalized pustular psoriasis
GTN	glyceryl trinitrate
GVHD	graft-versus-host disease
HAART	highly active antiretroviral therapy
HBV	hepatitis B virus
HCV	hepatitis C virus
HDL	high-density lipoprotein
HED	hypohidrotic ectodermal dysplasia
HHV	human herpesvirus
HIAA	hydroxyindoleacetic acid
HIDS	hyperimmunoglobulin D syndrome
HIV	human immunodeficiency virus
HLA	human leucocyte antigen
HPV	human papillomavirus
HSCT	haemopoietic stem cell transplant
HSV	herpes simplex virus
HTLV	human T-lymphotropic virus

Hz	hertz
IBD	inflammatory bowel disease
ICU	intensive care unit
IFN	interferon
Ig	immunoglobulin
IL	interleukin
ILVEN	inflammatory linear verrucous epidermal naevus
IM	intramuscular
IMF	immunofluorescence
IP	incontinentia pigmenti
IQ	intelligence quotient
IRIS	immune reconstitution inflammatory syndrome
IU	international unit
IV	intravenous
JAK	janus kinase
kcal	kilocalorie
kDa	kilodalton
KOH	potassium hydroxide
KS	Kaposi sarcoma
L	litre
LDH	lactate dehydrogenase
LDL	low-density lipoprotein
LE	lupus erythematosus
LEKTI	lymphoepithelial Kazal-type 5 serine protease inhibitor
LFT	liver function test
LH	luteinizing hormone
LP	lichen planus
LPC	liquor picis carbonis
LPP	lichen planopilaris
m	metre
M	molar
MALT	mucosa-associated lymphoid tissue
MAPK	mitogen-activated protein kinase
MC1R	melanocortin 1 receptor

MCP	metacarpophalangeal
MCTD	mixed connective tissue disease
MEK	mitogen-activated protein kinase
MEN	multiple endocrine neoplasia
MF	mycosis fungoides
mg	milligram
MHC	major histocompatibility complex
MHRA	Medicines and Healthcare Products Regulatory Agency
mL	millilitre
mm	millimetre
mmol	millimole
6-MP	6-mercaptopurine
MRA	magnetic resonance angiography
MRI	magnetic resonance imaging
MRP6	multidrug resistance-associated protein 6
MRSA	meticillin-resistant *Staphylococcus aureus*
MSSA	meticillin-sensitive *Staphylococcus aureus*
mTOR	mammalian target of rapamycin
NAPSI	Nail Psoriasis Severity Index
NF1	neurofibromatosis type 1
NF2	neurofibromatosis type 2
NF-κB	nuclear factor kappa B
ng	nanogram
NHS	National Health Service
n	nanometre
NTT	non-treponemal test
NOMID	neonatal-onset multisystem inflammatory disease
NSAID	non-steroidal anti-inflammatory drug
NSF	nephrogenic systemic fibrosis
PAN	polyarteritis nodosa
PAPA	pyogenic arthritis, pyoderma gangrenosum, and acne syndrome
PAS	periodic acid–Schiff
PASH	pyoderma gangrenosum, acne, and hidradenitis suppurativa

PASI	Psoriasis Area and Severity Index
PCOS	polycystic ovary syndrome
PCR	polymerase chain reaction
PCT	porphyria cutanea tarda
PDE	phosphodiesterase
PEST	Psoriasis Epidemiology Screening Tool
PET	positron emission tomography
PG	pyoderma gangrenosum
PHACE	Posterior fossa malformations, Haemangioma, Arterial abnormalities, Coarctation of the aorta, and Eye abnormalities
PIIINP	type III procollagen peptide
PIP	proximal interphalangeal
PLE	polymorphic light eruption
POEM	patient-orientated eczema measure
PPK	palmoplantar keratoderma
PRN	*pro re nata* (as required)
PSEK	progressive symmetric erythrokeratoderma
PTH	parathyroid hormone
PUVA	psoralen with ultraviolet A
PVL	Panton–Valentine leukocidin
PWS	port wine stain
PXE	pseudoxanthoma elasticum
qds	*quater die sumendum* (four times daily)
RA	rheumatoid arthritis
RBC	red blood cell
RCLASI	revised Cutaneous Lupus Disease Activity and Severity Index
RDD	Rosai–Dorfman disease
RF	rheumatoid factor
RTR	renal transplant recipient
SCC	squamous cell carcinoma
SCLE	subacute cutaneous lupus erythematosus
SCORAD	SCORing Atopic Dermatitis
SDRIFE	symmetrical drug-related intertriginous and flexural exanthema

SF-MPQ	short-form McGill Pain Questionnaire
SJS	Stevens–Johnson syndrome
SLE	systemic lupus erythematosus
SLICC	Systemic Lupus International Collaborating Clinics
SSM	superficial spreading melanoma
SSRI	selective serotonin reuptake inhibitor
SSSS	staphylococcal scalded skin syndrome
STD	sexually transmitted disease
TB	tuberculosis
tds	*ter die sumendum* (three times daily)
TEN	toxic epidermal necrolysis
TGF	transforming growth factor
TGM1	transglutaminase-1
Th	T-helper
TNF	tumour necrosis factor
TORCH	toxoplasmosis, other, rubella, cytomegalovirus, herpes simplex
TPMT	thiopurine methyltransferase
TRAPS	tumour necrosis factor receptor superfamily 1A-associated periodic fever syndrome
Treg	T-regulatory
TSS	toxic shock syndrome
TT	treponemal test
UAS	urticaria activity score
UK	United Kingdom
USA	United States of America
UV	ultraviolet light
UVA	ultraviolet A
UVA1	ultraviolet A1
UVB	ultraviolet B
UVR	ultraviolet radiation
VAS	visual analogue scale
VLDL	very low-density lipoprotein
VZV	varicella-zoster virus
WCC	white cell count

WHO	World Health Organization
WSP	white soft paraffin
XP	xeroderma pigmentosum

Definitions

Isomorphic response (Koebner phenomenon): appearance of new lesions of a pre-existing disorder at a site of injury.

Isotopic response: occurrence of a new skin disorder exactly at the site of another unrelated and already healed skin disorder.

Pathergy test: hyper-reactivity of the skin to needle-prick.

Nikolsky sign: applying firm sliding pressure to the skin causes the epidermis to separate from the dermis, producing an erosion.

Chapter 1

Structure and function of the skin

Contents

Introduction

The skin is the largest organ in the body and the heaviest—an adult's skin weighs 4–5kg. Skin is not an inert barrier but plays an active part in defending against insults (microbial, physical, and chemical) from the external environment.

In this chapter, we aim to give you an insight into what your skin is doing for you, in the hope that this will encourage you to care for the skin of your patients (and your own skin) with the respect it deserves. An understanding of cutaneous physiology will also help you to analyse what has gone wrong or what might happen in patients with skin problems, including skin failure.

We discuss the structure of the skin and consider how this finely tuned organ functions as a barrier ('keeping the inside in and the outside out') and reduces water loss, is part of the immune system, is a metabolic organ synthesizing vitamin D and cytokines, regulates body temperature, and senses noxious stimuli (skin on the tips of your fingers is particularly sensitive; see Box 1.1).

Most of us will, at some stage, worry about the appearance of our skin, hair, and/or nails. 'Looking good' gives us confidence, and we may try to enhance the appearance (or smell) of our skin if we want to display ourselves to make a positive impression or attract a sexual partner. It is no surprise that fortunes are spent on lotions, potions, and procedures designed to conceal blemishes (actual or imagined) and to restore the appearance of youth. The psychological aspects of skin problems are discussed in more detail in ➔ Chapter 28, pp. 570–1.

Box 1.1 Dermatoglyphics (fingerprints)

- Fingertips, palms, soles, and toes are covered with a pattern of epidermal ridges called dermatoglyphics.
- The three basic patterns (loops, arches, and whorls) are unique to each individual.
- Characteristic dermatoglyphic patterns accompany many chromosomal abnormalities.
- The ridges amplify vibrations when your finger brushes across a rough surface.
- The ridges enhance grip.
- Autosomal dominant adermatoglyphia (*SMARCAD1* mutation)—absence of epidermal ridges, also known as 'immigration delay disease'. Also absent in some rare genodermatoses and in cases of palmoplantar dysaesthesias secondary to chemotherapy drugs.

Summary of functions of the skin

Prevention of water loss

- Stratum corneum—overlapping cells and intercellular lipid.

Immune defence

- Structural integrity of stratum corneum.
- Keratinocytes produce antimicrobial peptides (AMPs) including defensins, cathelicidins, and members of the granin family, e.g. catestatin.
- Langerhans cells trap antigens and migrate to lymph nodes where antigens are presented to T-cells (see ➔ p. 14).
- Cytokines secreted by lymphocytes, macrophages, and keratinocytes regulate inflammatory and immune responses.
- Acid pH of sweat and stratum corneum.
- Fungistatic activity of sebaceous secretions.
- Dermcidin (AMP) produced by eccrine sweat glands activates keratinocytes to produce cytokines/chemokines important in skin immunity.

Protection against ultraviolet damage

- Melanin synthesized by melanocytes protects keratinocyte nuclei from harmful effects of ultraviolet (UV) radiation by absorbing and scattering rays and by scavenging free radicals (see ➔ Chapter 16, pp. 320–1).
- Enzymes repair UV-damaged deoxyribonucleic acid (DNA) (see ➔ Chapter 16, pp. 320–1).

Temperature regulation

- Vasoconstriction and vasodilation control blood flow and transfer of heat to the body surface.
- Evaporation of sweat cools the body.

Synthesis of vitamin D

- Skin is the 1° source of vitamin D. 7-dehydrocholesterol is photoactivated to cholecalciferol (vitamin D_3), which is metabolized in the liver to 25-(OH)D_3 and in the kidney to the active form of vitamin D calcitriol (1,25-$(OH)_2D_3$). Only small amounts of vitamin D are obtained from the diet. Vitamin D is required for calcium absorption and has essential roles in bone metabolism, neuromuscular function, and immune function. Deficiency can lead to rickets (in children), osteomalacia, and osteoporosis (see ➔ Chapter 16, pp. 319–39).

Sensation

- Free nerve endings detect potentially harmful stimuli (heat, pain).
- Specialized end-organs detect pressure, vibration, and touch (see Box 1.1).
- Autonomic nerves supply blood vessels, sweat glands, and arrector pili muscles.

Aesthetic

- The skin has an important role in social interaction.

Epidermis

The epidermis originates from embryonic neuroectodermal cells. The epidermis is a stratified squamous epithelium composed of layers of keratinocytes that differentiate as they move towards the skin surface (see Box 1.2). The epidermis is attached to an underlying collagenous dermis, which contains the blood vessels that nourish the epidermis. Downward projections of epidermal rete pegs interlock with upward projecting dermal papillae, stabilizing the structure and making it difficult to shear the epidermis from the dermis (see Fig. 1.1).

Structure of the epidermis

(See Fig. 1.2.)

- A single layer of columnar keratinocytes in the deepest layer of the epidermis (basal layer) is attached to a basement membrane that is an interface between the epidermis and the underlying dermis. The basal cells are anchored to the basement membrane by adhesion junctions called hemidesmosomes (see ➔ p. 262).
- Regeneration of the epidermis (and hair follicles; see ➔ p. 10) depends on populations of epidermal stem cells.
- The middle layers of the epidermis (spinous or prickle cell layers) have a spiky appearance under a light microscope, because of the intercellular junctions (desmosomes) that are the main adhesive force between adjacent keratinocytes (see ➔ p. 262).
- The outermost horny layer or stratum corneum is composed of layers of flattened keratinocytes (corneocytes) locked together by modified desmosomes (corneodesmosomes) in a lipid matrix.
- All keratinocytes contain keratin intermediate filaments, but the structure of the keratins changes as the cells differentiate (see Box 1.2).
- Melanocytes are dendritic cells that are interspersed amongst the basal keratinocytes (approximate ratio 1:6) (see ➔ pp. 12–3).
- Epidermal Langerhans cells (dendritic antigen-presenting cells) are found throughout the epidermis (see ➔ p. 14).
- Migratory leucocytes are present in small numbers in the epidermis.

Proliferation and shedding

- The thickness of normal epidermis (0.05–0.1mm) is regulated by the balance between proliferation of basal keratinocytes and shedding of cells at the surface (desquamation).
- It takes ~40 days for a keratinocyte to move upwards from the basal layer to the horny layer where, after proteolytic breakdown of adhesion junctions, cells flake off (desquamate) (see Box 1.2).
- The activity of epidermal proteases and their inhibitors regulate cornification and desquamation.

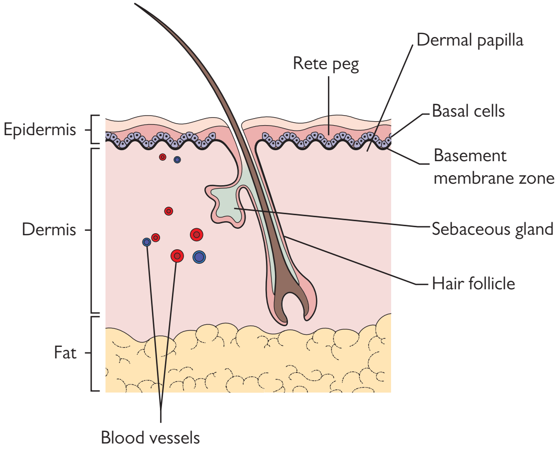

Fig. 1.1 Diagram of skin: epidermis, dermis, and fat.

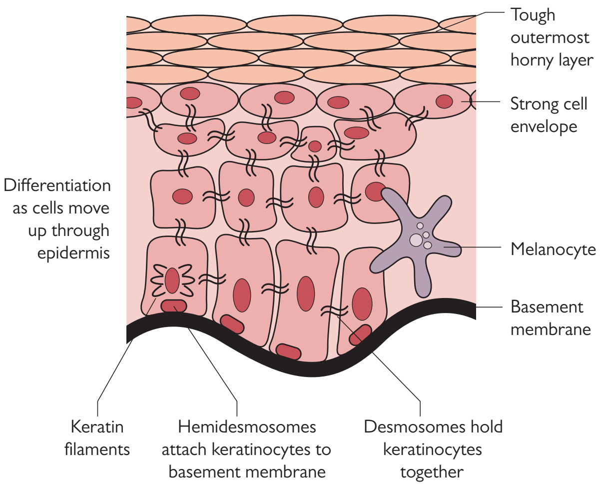

Fig. 1.2 Structure of epidermis.

Differentiation and the skin barrier

Box 1.2 Keeping the outside out

- Each keratinocyte has a cytoskeleton of microfilaments containing actin, microtubules containing tubulin, and intermediate keratin filaments composed of type I and type II keratins (from the Greek 'keras' meaning horn).
- Keratinocytes are held together by desmosomes (see Fig. 1.2). A glycoprotein intercellular substance aids cell cohesion. In acute eczema, hyaluronan secreted by keratinocytes into the intercellular space takes up water and may dilute noxious chemicals (see ➲ pp. 214–16).
- Keratinocytes change (differentiate), as they move from the basal layer towards the skin surface:
  - The types of keratins in the cells and the composition of the desmosomes change.
  - A protein called filaggrin aggregates and cross-links the keratin filaments, giving the keratinocytes internal strength. Loss-of-function mutations in filaggrin are associated with ichthyosis vulgaris, atopic eczema, and food allergies in older children.
  - The cell envelope is strengthened by Ca^{2+}-dependent cross-linking of proteins, such as involucrin and loricrin, and is catalysed by transglutaminase-1 (TGM1). Mutations in *TGM1* cause ichthyosis (abnormalities in the cell envelope and desquamation).
  - Lipid is synthesized by keratinocytes and accumulates in the intercellular space.
  - By the time the keratinocytes reach the outermost horny layer, they have lost their nuclei and subcellular organelles and are densely packed with keratin filaments.
- The outermost horny layer is composed of ~20 layers of flattened horny cells (corneocytes) with tough insoluble cell membranes (cornified envelopes) locked together by corneodesmosomes and a thick intercellular layer of lipids (like mortar). This impermeable layer enables the skin to withstand chemical and mechanical injury and restricts loss of water and electrolytes. Darkly pigmented skin seems to provide a better epidermal barrier than lightly pigmented skin—the stratum corneum is less permeable and more cohesive.
- Permeability of the skin depends primarily on the balance between intercellular cohesion and desquamation in the stratum corneum. Serine proteases, such as kallikreins, degrade corneodesmosomes, so cells can desquamate. Inhibitors, such as lymphoepithelial Kazal-type 5 serine protease inhibitor (LEKTI), control protease activity. In Netherton syndrome, mutations in the *SPINK5* gene encoding LEKTI lead to defective LEKTI expression, unregulated proteolytic activity, increased desquamation, and highly permeable skin (see ➲ Box 31.11, p. 603).

Further reading

Cork MJ *et al. J Invest Dermatol* 2009;**129**:1892–908.

Dermis and glands

The dermis is a layer of connective tissue beneath the epidermis. A layer of subcutaneous fat separates the dermis from the underlying fascia and muscle. The dermis has a rich supply of blood vessels, lymphatics, nerves, and sensory receptors. The thickness of the dermis varies with body site and may measure as much as 5mm on the back.

Structural components

- Collagen, mainly type I but some type III, gives the dermis tensile strength. Ageing skin is characterized by reduced collagen synthesis and increased collagen breakdown by matrix metalloproteinases.
- Elastic fibres, containing a core of elastin, supply elasticity and resilience, but elastin functions poorly in aged skin.
- Ground substance of proteoglycans and glycoproteins that binds water and hydrates the dermis.
- Skin appendages: hair follicles, sebaceous glands, eccrine sweat glands, and apocrine sweat glands (see ➔ p. 9, and p. 10).
- Blood vessels: superficial and deep vascular plexuses. Vasodilatation and vasoconstriction help to regulate heat loss.
- Lymphatics: afferent capillaries in dermal papillae pass via a superficial plexus to deeper horizontal plexuses and collecting lymphatics (see ➔ Skin immune system, p. 14).
- Nerve fibres: most sensory nerves end in the dermis, but a few penetrate the epidermis. Free nerve endings detect heat and pain; Pacinian corpuscles detect pressure and vibration, and Meissner corpuscles detect pressure and touch. Autonomic innervation is cholinergic to the eccrine sweat glands, and adrenergic to eccrine and apocrine glands, arterioles, and arrector pili muscle.

Cellular components

- Fibroblasts synthesize proteins, such as collagen and elastin, as well as glycosaminoglycans in the dermal ground substance.
- Dendritic cells (dermal dendritic cells) are involved in antigen presentation (see ➔ p. 14).
- Macrophages scavenge cell debris and foreign material.
- Mast cells: a few near blood vessels. Granules contain inflammatory mediators, such as histamine, prostaglandins, leukotrienes, and other chemokines, that may be involved in inflammatory responses.

Langer lines

Langer lines (cleavage lines) correspond to the alignment of collagen fibres within the dermis. They define the direction along which the skin has least flexibility. Surgical incisions carried out parallel to favourable skin tension lines heal with less scarring.

Eccrine glands

- Eccrine glands cover most of the body surface but are most numerous on the palms and soles.
- A secretory coil deep in the dermis is attached to a duct that conveys sweat to the surface of the skin.
- Glands are innervated by the sympathetic nervous system.
- Eccrine glands secrete water, electrolytes, lactate, urea, and ammonia.
- Sweating helps to regulate body temperature.
- Sweat may have antimicrobial properties; eccrine glands produce dermcidin peptides (antimicrobial peptides)—DCD-1 and DCD-1L.

Apocrine glands

- Apocrine glands are found mainly in the axillae, anogenital region, female breast, eyelids, and external auditory canal.
- A secretory coil in the deep dermis leads to a duct that opens into the upper portion of a hair follicle.
- They produce an oily secretion of protein, carbohydrate, ammonia, and lipid.
- Apocrine glands become active at puberty.
- Sympathetic nerve fibres control secretion.
- The action of bacteria on secretions produces body odour.

Sebaceous glands

- Sebaceous glands are associated with the upper portion of each hair follicle. The hair follicle and associated glands are known as a pilosebaceous unit (see Fig. 1.3).
- The size of glands varies at different body sites. The largest glands are on the face and upper trunk (sites prone to acne).
- Sebocytes secrete lipid-rich sebum with emollient properties.
- Sebocytes produce cathelicidin, which has antimicrobial properties, in response to vitamin D and infectious agents.
- Sebum in the sebaceous duct enters the upper portion of the hair follicle and protects the surface of the skin.

Further reading

Peng Y *et al. Ageing Res Rev* 2015;**19**:8–21.

Hair and nails

Hair

- Hair is found all over the body, except for the palms and soles, but the density and size of hair follicles vary at different sites.
- The hair follicle opens onto the surface of the epidermis and is a potential portal of entry for pathogenic microbes.
- The size of the follicle determines the size of the hair shaft (the product of the follicle). Short, fine, soft, non-pigmented vellus hair covers most of the body (you can just feel vellus hair on your forehead), and coarse, pigmented terminal hair grows at sites such as the scalp, eyebrows, eyelashes, limbs, genitalia, and axillae. Androgens influence the hair type—some vellus hair changes to terminal hair in hirsutism (see ➔ p. 474).
- The cylindrical follicular root sheath moulds the shape of the hair shaft.
- A bulge region in the mid portion of the root sheath contains epithelial hair follicle stem cells from which the follicle arises (see Fig. 1.3). These stem cells regulate the hair cycle (see Box 1.3). Damage to stem cells may contribute to irreversible hair loss (scarring alopecia) in inflammatory conditions such as cutaneous lupus erythematosus.
- The hair shaft arises from dividing cells in the bulb at the bottom of the follicle. The shaft gradually keratinizes, producing a highly cohesive structure, as it moves up through the follicle root sheath. Blood vessels and sensory nerves in the associated dermal papilla supply the cells of the hair bulb (see Fig. 1.3).
- Melanocyte stem cells are found in the bulge region from where they migrate to the bulb.
- The arrector pili muscles attach to the mid portion of follicles. These smooth muscles, supplied by adrenergic nerves, make hair 'stand on end' in the cold or during emotional stress (goose pimples).
- The sebaceous and apocrine ducts penetrate the follicular root sheath at the start of the upper third of the follicle.

Nails

- A nail is a plate of keratinized cells (onychocytes) with a free edge.
- The nail plate is formed by specialized keratinocytes in the nail matrix, which lies under the proximal nail fold and extends distally to the lunula but proximally to the insertion of the extensor tendon.
- The nail plate grows forward over a nail bed that is tightly connected to the nail plate.
- Lateral and proximal nail folds cover the sides and base of the nail plate. The cuticle forms a seal between the nail plate and the proximal nail fold.
- Fingernails grow by ~1mm per week (toenails more slowly). If growth in the matrix is slowed by serious illness or drugs, horizontal lines or dents appear at the same place in all nail plates (Beau lines). The lines will only become apparent 8–12 weeks after the insult when the nail plate emerges from under the cuticle. For a diagram of a normal nail, see ➔ Fig. 3.3, p. 47.

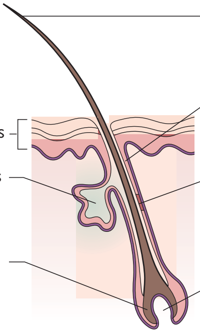

Fig. 1.3 Structure of a hair follicle. The arrector pili muscle (not shown) attaches to the mid portion of the follicle.

Box 1.3 The hair follicle cycle

- Hair grows by 0.3–0.4mm/day. About 85–90% of the 100 000 scalp hairs are growing. Each hair follicle cycles through phases of:
  - Growth (anagen): active phase of hair production (2–6 years on the scalp). The length of hair is determined by the duration of anagen. The duration of anagen is determined genetically.
  - Involution (catagen): conversion from active growth to resting phase (2–4 weeks).
  - Rest (telogen): 15% of scalp hairs are in telogen (2–4 months).
- At the end of telogen, hair is shed, and a new cycle starts. It is normal to lose 100–150 scalp hairs a day.
- Each hair follicle cycle is independent of neighbouring follicles, so humans do not moult, unlike many other mammals.

Further reading

Harries MJ *et al. Autoimmun Rev* 2009;**8**:478–83.

Melanocytes and colour

Melanin, synthesized by melanocytes, provides some defence against ultraviolet radiation (UVR) as well as giving colour to skin and hair. Skin colour and ease of tanning are important determinants of the risk of skin cancer (see ➔ pp. 320–1 and p. 346, and Chapters 16 and 17).

- Melanocytes are dendritic cells that migrate from the neural crest to the epidermis and hair follicles in the third month of fetal development.
- Melanocytes are interspersed amongst the basal keratinocytes in the epidermis. Melanocyte stem cells are found in the bulge region of hair follicles from where they migrate to the hair bulb (see Fig. 1.3).
- Melanocytes synthesize both brown/black eumelanin and red/yellow phaeomelanin from tyrosine.
- Melanin is passed in packages (melanosomes) along dendritic processes into keratinocytes (see Fig. 1.2) where melanosomes sit in a cap or 'parasol' over the upper sun-exposed side of the keratinocyte nucleus. Each melanocyte links to a number of keratinocytes, forming an epidermal melanin unit. Melanin provides UV protection by absorbing visible light and UVR (see ➔ pp. 320–1).
- UVR induces tanning by stimulating oxidation of pre-existing melanin, by triggering the synthesis of new melanin, and by changing the distribution of melanosomes (see ➔ pp. 320–1).
- Melanin is the main determinant of the colour of skin and hair (see Box 1.4). The colour depends on the number, size, and distribution of melanosomes within keratinocytes and the type of melanin, rather than the number of melanocytes. Darkly pigmented skin has similar numbers of melanocytes to lightly pigmented skin, but more and slightly differently packaged melanin (melanosomes) within keratinocytes.
- Skin colour is not uniform (see Box 1.5).
- Darkly pigmented skin has better epidermal barrier function than lightly pigmented skin (see Box 1.2).
- Genetic variation in the amino acid sequence of the melanocortin 1 receptor (MC1R) is a major determinant of skin and hair colour. Some variants, e.g. redheads, do not tan in response to sun. Polymorphisms in MC1R variants confer susceptibility to skin cancers (see ➔ p. 347).
- The first step in the synthesis of melanin is hydroxylation of tyrosine to dopaquinone. Oculocutaneous albinism is caused by abnormalities in the function of tyrosinase or other enzymes in the synthetic pathway of melanin.
- Defective migration of melanocytes from the neural crest or abnormalities in the maturation or trafficking of melanosomes may present as a pigmentary disorder.
- Depletion of melanocyte stem cells in hair follicles may contribute to age-related greying of hair.

Box 1.4 Skin colour

- The colour of normal skin comes from a mixture of pigments, but melanin is the main contributor to skin colour.
- Oxyhaemoglobin in blood gives untanned Caucasian skin a pink colour.
- Carotene in subcutaneous fat and the horny layer of the epidermis adds a yellow hue to normal skin.
- Abnormalities in skin colour may result from an imbalance of pigments (e.g. in cyanosis, chloasma, and carotenaemia) or the presence of abnormal pigments (e.g. haemosiderin).
- Genetic abnormalities in the synthetic pathway of melanin present with abnormal skin colour.
- Post-inflammatory hypo- or hyperpigmentation is common, particularly in darker skin.
- Damage to the basal layer of the epidermis (interface or lichenoid reaction) is associated with the release of melanin into the dermis and hyperpigmentation.

Box 1.5 Pigmentary demarcation lines (Voigt–Futcher lines)

Dorsal skin surfaces are more pigmented than ventral surfaces. Lines of demarcation between darker dorsal and paler ventral surfaces are apparent in about 20% of people with dark skin. The lines, which are symmetrical and bilateral, are present from infancy and have no clinical significance.

- Type A: vertical line on the lateral aspect of the upper arm that may extend into the pectoral region (commonest).
- Type B: curved line on back of the thigh (posteromedial) that extends from the perineum to the popliteal fossa and occasionally the ankle.
- Type C: vertical or curved hypopigmented band on the mid chest (contains two parallel lines).
- Type D: vertical line on the posteromedial area of the spine.
- Type E: bilateral hypopigmented streaks, bands, or patches on the upper chest in the zone between the mid third of the clavicle and the periareolar skin.

Facial patterns, which appear around puberty, have been described in the Indian subpopulation:

- Type F: 'V'-shaped hyperpigmented lines between the malar prominence and the temple.
- Type G: 'W'-shaped hyperpigmented lines between the malar prominence and the temple.
- Type H: linear bands of hyperpigmentation from the angle of the mouth to the lateral aspects of the chin.

Further reading

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Somani VK *et al. Indian J Dermatol Venereol Leprol* 2004;**70**:336–41.

Skin immune system

The skin immune system (both innate and adaptive) provides vital defence against external cutaneous pathogens or physical injury.

Keratinocytes

Keratinocytes respond to cutaneous injury or infection by releasing antimicrobial peptides (AMPs), such as β-defensins and cathelicidins, that have broad-spectrum antimicrobial activity against bacteria, fungi, and viruses. AMPs may also reduce the likelihood of microbes entering the skin through follicular openings. AMPs are part of the innate immune system (see Box 1.6). Keratinocytes can also produce a wide range of cytokines: pro- or anti-inflammatory, immunomodulatory, and immunosuppressive. Alarmin cytokines, such as interleukin (IL)-33, can be released in response to cell damage. Cathelicidins induce the production of pro-inflammatory cytokines that enhance cell migration and promote wound healing. Vitamin D_3 seems to be involved in the regulation of cathelicidin expression in the epidermis. Keratinocytes are also able to present antigens to T-cells.

Dendritic cells

Dendritic antigen-presenting cells, such as epidermal Langerhans cells and dermal dendritic cells, express major histocompatibility complex (MHC) class II antigens and CD1 on their surface and 'police' the skin. They are part of the 1st line of defence if the skin is breached. The cells trap, ingest, process, and present antigens, in association with MHC (peptide antigens) or CD1 (lipid antigens), to the receptors of T-cells.

Lymphatics

Langerhans cells and dermal dendritic cells that have trapped antigens migrate via afferent lymphatics to draining lymph nodes where the processed antigen is presented to naïve T-cells that have the receptor that will bind that antigen. This triggers the generation of activated and memory T-cells imprinted with specific combinations of adhesion molecules and chemokine receptors that target their migration to the skin.

T-cells

Memory T-cells can be resident in the skin or can patrol the body for foreign antigens, recirculating through lymph nodes, lymphatics, blood vessels, and skin. Endothelial adhesion molecules aid trafficking by tethering flowing T-cells, so they can migrate through endothelial junctions. Chemokine gradients direct migration in the skin.

The receptor of each memory T-lymphocyte will bind to a specific peptide antigen presented on the surface of target cells in association with MHC class I (for $CD8^+$ T-cells) or class II (for $CD4^+$ T-cells). Some T-cells can recognize lipid antigens presented by the CD1 family of molecules. The antigen–MHC complex interacts with the T-cell receptor, which, along with co-stimulatory molecules, triggers the release of cytokines that attract more lymphocytes into the skin to augment the inflammatory response.

Box 1.6 What is the innate immune system?

The innate immune system is the first line of defence against environmental insults. The response is rapid and efficient but is less specific than adaptive immunity and does not improve with repeated exposure.

Receptors (transmembrane and intracellular), such as Toll-like receptors, recognize highly conserved molecular patterns that are common to many classes of pathogen, e.g. components of the wall of microbes.

Activated receptors stimulate the production of cytokines, chemokines, and AMPs. Immune cells, such as innate lymphoid cells (e.g. natural killer cells), and neutrophils are activated and recruited to the site of injury or infection. This innate response kills pathogens, promotes angiogenesis, and initiates repair after injury.

The major components of the skin innate immune system are:

- Intact physical barrier: stratum corneum and intercellular junctions.
- Antigen-presenting cells, keratinocytes, mast cells, innate lymphoid cells and neutrophils.
- AMPs, cytokines, and chemokines.

Further reading

Salimi M and Ogg G. *BMC Dermatol* 2014;**14**:18.
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Chapter 2

The history in dermatology

Contents

The dermatological history: 1

Setting the scene

Introduce yourself; make the patient comfortable; ensure privacy (difficult on a ward), and allow sufficient time to establish rapport and for the patient to explain what has happened. Remember that skin problems can be embarrassing. Your history will be guided by the nature of the problem—a rash, a leg ulcer, or a skin tumour—but 'Please tell me about your skin problem' is a good opening question.

Finding out what the patient means

Clarify what the patient means by words such as blemish, wheal, hive, scar, or blister. Patients, as well as many doctors, find it difficult to describe rashes. You may think that you know what is meant, but make sure you use these words to describe the same thing as the patient. For example, some people call any raised lesion a 'blister' but do not mean a bump filled with clear fluid, and people may use the word 'scar' to describe post-inflammatory pigmentation without any dermal damage. Even 'sunburn' means different things to different people. Some people use this term to describe getting a suntan.

Sometimes the rash will have changed or even disappeared, and you will have nothing but the patient's description to help you make a diagnosis, so the description should be as accurate as possible (see Box 2.1). The patient may bring photographs to clinic.

Getting on the same wavelength

Clarify the patient's preconceptions about the cause of the problem, as well as their expectations of the consultation and your role. It is helpful to know more about the patient's perception of the current severity of the condition, particularly in those with fluctuating inflammatory rashes. This may differ markedly from your own assessment (see Box 2.2). You need to understand what matters to the patient, so ask 'What is most important to you?'. This will help you and the patient to set goals when planning investigation and treatment (see ➔ p. 32).

Symptoms, causation, and impact

'How does your rash/skin condition bother you?' is another good open question to start off the discussion (see Box 2.3).

Document the relationship of the condition to work, hobbies, and drugs, including herbal medicines and health foods. Might the rash be related to the menstrual cycle (a catamenial dermatosis)?

Skin conditions can be both disfiguring and embarrassing, and treatments can be messy, unpleasant, and time-consuming, so find out how the skin condition is affecting the life of the patient, as well as that of the family and/or carers.

More information about questions to ask in specific circumstances is given later in this chapter (also see ➔ DLQI under Quality of life, p. 32).

Box 2.1 Questions to ask about a rash

- Where the rash started, appearance at onset, and how it evolved.
- Is the surface smooth or scaly, or is the rash flat or raised?
- The colour of the rash: if red, did the patient notice what happened if he/she applied light pressure, i.e. did it blanch, suggesting an erythema as opposed to purpura?
- Is the rash persistent, or do lesions come and go? Is it cyclical?
- If transient—how long do individual lesions last, and, when they fade, what does the skin look like? Are there any changes in skin colour or any marking (scarring)?
- Are hair, nails, or mucosal surfaces affected?

Box 2.2 Grading the skin condition: the patient's perception

It helps to have some insight into the patient's perception of the severity of his/her skin condition as it is now. Ask the patient to grade the condition:

'You have explained that your rash/skin condition can vary a bit. I would like to get a feel for how bad it is for you today by asking you to grade your skin condition. Let us say that a grade of 10 means it is the worst it has ever been for you and zero means that you have no skin problem. What grade would you give your skin today?'

Box 2.3 Questions to ask when exploring symptoms and impact of a skin condition

- What does the rash feel like? Is the skin tender, itchy, or painful?
- What brings on the itch/pain? What does the patient do about it?
- Itch—try to quantify the impact. Does it affect sleep or concentration?
- Is anyone else itchy? Ask about family and all close contacts, e.g. sexual partner (could it be scabies?).
- Pain: what is the character of the pain? Sharp, dull, throbbing, paraesthesiae, continuous, or intermittent?
- Does the skin ooze clear fluid, pus, or blood?
- Is the skin fragile? Does it bruise easily, or does the skin blister?
- Has the patient noticed a smell from the skin (ulcer)? Patients may welcome a chance to discuss this awkward and embarrassing symptom if you introduce the subject in a sensitive manner.
- What makes the rash better or worse? Is there any relationship to recent travel, work, or hobbies? Does the problem improve or worsen at the weekends or in holidays? Does sunlight make a difference? Any relationship to cosmetics being used or to drugs/health foods?
- How does the skin condition affect activities of everyday living—school, work, or social?
- How long does applying treatment take, and what is the treatment costing?
- Is the skin condition affecting mood?
- Are carers coping, e.g. the parents of a child with eczema?

The dermatological history: 2

Drug history

- What has been prescribed, purchased, begged, or borrowed? Patients may be using a variety of topical preparations, some of which they will not regard as medicaments at all—and they will probably not remember all the names.
- Is the product an ointment (greasy, petroleum jelly-like, sits on the surface of the skin), a cream (white, 'disappears into the skin'), a gel, a foam, or a lotion (liquid)?
- What does the patient do with the treatment? How much is used—how long does the tube or tub last? Where, and how often, is it applied? If the patient has more than one topical preparation, how and where are they used—at the same time or at different times?
- Is any moisturizer being used, and how often is it applied?
- What soaps, cosmetics, or other toiletries are used?
- Is anything added to the bath, e.g. bath salts?
- Is the patient taking other drugs, including recreational drugs, and how does the use of these relate to the onset of the rash?

Past medical history

- Enquire about previous skin problems, in particular has the patient had a similar skin condition before, and, if so, how long did it last, and what was done?
- Is there a history of atopy such as infantile eczema or 'dry skin', asthma, or hay fever?
- Document other medical conditions.

Allergies

- Ensure you know what the patient means if he/she uses the term 'allergy'.
- Has the patient noticed 'reactions' when anything is in contact with the skin, e.g. topical medicaments, jewellery (usually nickel allergy), cosmetics, or perfumes? What sort of reaction and how long did it last? Has allergy been investigated by patch or prick testing?
- Have there been 'reactions' to oral medications? What happened, and what was done?

Social and personal history

- Explore lifestyle, exposure to nicotine and alcohol, sexual history, travel, hobbies, and occupation.

Family history

- Does anyone at home have a similar skin problem? Scabies and skin infections are contagious. Other skin conditions, such as atopic eczema or psoriasis, may have a strong genetic predisposition.
- Draw a family tree if you think that you are dealing with an inherited condition (see ➔ p. 622).

Functional enquiry

Complete the history with a review of all systems (functional enquiry). Itching, ulcers, or purpuric rashes can be manifestations of a systemic disease. A careful history may provide clues to a diagnosis, such as underlying malignancy or connective tissue disease, and guide the examination as well as investigation.

Patients do not always volunteer such additional information without prompting, because they do not relate the skin problem to their other symptoms.

History in patients with allergy

Type I hypersensitivity

Urticaria (hives, wheals, welts) is an immediate type I hypersensitivity reaction that may occur on its own or in association with angio-oedema and/or anaphylaxis. Urticaria is commoner in atopics.

Acute urticaria may be triggered by drugs, e.g. penicillin, or foods, e.g. peanuts, fish. The patient can often identify the allergen. In contrast, it is difficult to pinpoint the trigger in most patients with chronic urticaria. Physical urticarias can be triggered by firm pressure (dermographism), heat, cold, sun, or exercise.

The wheals of urticaria are 2° to dermal oedema. They are raised, smooth, erythematous, and usually itchy. Angio-oedema implies oedema of the deep dermis and subcutaneous tissues, as well as mucosal and submucosal oedema. Patients describe the sudden onset of large tender swellings involving the skin, mucosae, and submucosal tissues (see ➔ Chapter 5, pp. 104–5, and Chapter 11, p. 228 and pp. 230–1).

The history should be directed at finding out exactly what happened to confirm the diagnosis of urticaria and/or angio-oedema, as the rash may not be present when you see the patient (see Boxes 2.4 and 2.5). Does the patient have one or both problems? ⚠ Rarely, angio-oedema can be caused by a deficiency of C1 esterase inhibitor (inherited or acquired), but these patients do ***not*** have urticaria (see ➔ p. 104).

Prick testing may be helpful to investigate the cause in some cases of acute urticaria, particularly if there is a history suggesting food allergy, but such testing is unlikely to be helpful in chronic urticaria.

Irritant contact dermatitis or type IV hypersensitivity

Dermatitis (eczema) caused by an external agent is known as contact dermatitis. Contact dermatitis may be irritant or, less often, allergic.

Irritant contact dermatitis is caused by chronic exposure to irritants (wet work, detergents, chemicals) and is commonest in individuals with atopic eczema who already have an inadequate skin barrier.

Allergic contact dermatitis is a type IV hypersensitivity. The patient has to be sensitized by prior exposure to the allergen, sometimes for months or years, before the onset of allergy. The eczematous response is maximal 5–7 days after exposure to the allergen. Allergic contact dermatitis is less common in children than in adults. Irritant contact dermatitis is commoner than allergy.

If you suspect an allergic contact dermatitis, direct your history towards elucidating the cause (see Box 2.6) by exploring the relationship of the rash to activities (work, hobbies). Patch testing is the investigation of choice (see ➔ Chapter 10, pp. 214–16).

A 2° allergic contact dermatitis may complicate the picture in an individual with long-standing dermatitis of another type, particularly when many different medicaments or over-the-counter preparations have been used over a prolonged period. Consider this possibility in individuals with chronic problems such as leg ulcers, pruritus ani, hand dermatitis, facial dermatitis, or otitis externa.

Box 2.4 The history in suspected urticaria

- Triggers?
- Confirm the diagnosis by asking about the appearance and behaviour of the rash:
  - Urticarial wheals are raised pink papules or plaques.
  - The wheals may be itchy.
  - The surface of the skin is smooth.
  - The wheals are erythematous, i.e. blanch with light pressure.
  - Individual wheals generally last no more than 24 h (48 h at the very most).
  - Wheals come and go in different places.
  - Wheals fade to leave normal skin.
- What happens if the skin is rubbed—does it wheal? = dermographism

Box 2.5 The history in suspected angio-oedema

- Triggers? Dental procedures or vaginal pressure during sexual intercourse may precipitate attacks of angio-oedema in people with C1 esterase inhibitor deficiency. These patients do not develop urticaria, only angio-oedema.
- The patient may describe one or more of these problems:
  - Sudden onset of deep tender swellings of hands, feet, and genitals.
  - Swelling of lips, tongue, and/or larynx.
  - Puffiness around eyes and lips.
  - Difficulty swallowing or difficulty breathing.
  - Stridor or wheeze.
- Gastrointestinal symptoms, such as vomiting or abdominal pain, occur in patients with C1 esterase inhibitor deficiency.

Box 2.6 The history in suspected contact dermatitis

- Remember that the allergen may be a product that the patient has used for some time.
- Document symptoms (itch?) and appearance (red, scaly, blisters?).
- Explore patients' occupations and hobbies in detail. Find out exactly what they do and with what they are in contact. Do they protect the skin, e.g. with gloves?
- Does the rash improve at weekends or during holidays?
- Is the skin exposed to any prescribed topical medicaments or over-the-counter remedies, and how long have they been used?
- How does the patient care for his/her skin—what is used to wash the skin or to moisturize? What cosmetics are being applied?
- Is the patient atopic? Irritant contact dermatitis is commoner in atopics (dry skin with poor barrier function is easily irritated).

History in patients with hair loss

Hair loss is a cause of great distress and may be difficult or impossible to reverse. Androgenetic alopecia is patterned in men (male-pattern baldness), with loss being most marked on the vertex of the scalp and/or the temples. Women with androgenetic alopecia describe diffuse thinning.

The commonest cause of sudden diffuse shedding is telogen effluvium (see Box 2.7)—a great diagnosis to make, because you can reassure your patient that he/she will not lose all their hair and that the hair will grow back. Drugs may also cause reversible hair loss (see Box 2.8).

What should I ask the patient?

- When did the problem start? Has this happened before?
- Establish if hair was shed suddenly or gradually.
- Is the loss diffuse or in patches?
- Is the loss complete, leaving bald patches, or partial?
- Is hair being lost at other sites, e.g. eyebrows, eyelashes, axillae? Alopecia areata can be widespread—alopecia totalis (all scalp hair) or universalis (all body hair).
- Are teeth or nails abnormal (indication of an ectodermal problem)?
- Is the patient still shedding excess hair? Remember it is normal to shed 100–150 hairs a day, and this may appear to be a lot to an anxious patient, particularly if the hair is long.
- Is the loss associated with other symptoms such as itching or scaling? Eczema and psoriasis are not usually associated with hair loss, but fungal infections certainly are. Some fungal infections cause no inflammation and minimal scale.
- How does the patient care for the hair—hot combs, perms, hair straighteners, colouring?
- Is the hair ever tied back tightly and for how long each day?
- Does the patient pull or pluck the hair?
- Did the patient start any new medications prior to the hair loss? (See Box 2.8.)
- Any exposure to radiation?
- Is there a family history of hair loss?
- Do close contacts have similar problems?
- Was there an event 8–12 weeks prior to the onset of loss, e.g. serious illness that might have triggered telogen effluvium? (See Box 2.7.)
- Has the patient just had a baby? The growing phase (anagen) is prolonged in pregnancy, but hair is shed post-partum.
- Has the patient any other medical problems? Iron deficiency and thyroid disease (hypo- and hyperthyroidism) may be associated with diffuse hair loss.

Box 2.7 The hair cycle and telogen effluvium

Hair grows by 0.3–0.4mm a day, but growth is asynchronous. About 90% of the 100 000 scalp hairs are growing. Each hair follicle cycles through phases of growth (anagen), involution (catagen), and rest (telogen) (see ➔ p. 10). At the end of the telogen phase, hair is shed, and a new cycle is initiated. It is normal to shed 100–150 hairs per day.

Telogen effluvium, 2–3 months after an insult such as severe illness or major surgery, is a common cause of acute diffuse hair loss. The hair follicle switches from a growing phase into a resting phase, and then the hair is shed when the new hair starts to form.

Box 2.8 Drugs causing reversible hair loss

- Anticoagulants.
- Antineoplastic agents.
- Antiretroviral drugs.
- Oral contraceptives.
- Progestin-releasing implants.
- Gonadotropin-releasing hormone agonist.
- Anti-oestrogens and aromatase inhibitors.
- Esterified oestrogens–methyltestosterone replacement therapy.
- Extracorporeal membrane oxygenation.
- Immunosuppressive drugs.
- Interferons.
- Minoxidil.
- Psychotropic drugs:
  - Antidepressants.
  - Anxiolytics.
  - Dopaminergic therapy.
  - Mood stabilizers.
- Retinol (vitamin A)/retinoids.

History in patients with skin tumours

Skin cancers are the commonest type of cancer. The non-melanocytic skin cancers, basal cell carcinoma (BCC; the commonest skin cancer) and squamous cell carcinoma (SCC), are commoner than the melanocytic cancer malignant melanoma, but all are rising in incidence.

Most elderly Caucasian patients have sun-damaged skin and are at risk of skin cancer. In any patient with a cutaneous tumour or an isolated scaly lesion, the history should include specific questions that will help you to decide if this patient is likely to have a skin cancer.

Sun damage and skin cancers are discussed in more detail in ➔ Chapter 16, pp. 320–1 and pp. 336–7, and Chapter 17, pp. 341–63.

The history

Find out:

- How long the patient has had the lesion(s).
- If it is changing (many skin cancers grow slowly, and the patient may not have noticed any recent change).
- If the patient is aware of any reason for the 'bump' to have changed, e.g. a history of trauma.
- If it is symptomatic—some non-melanocytic skin cancers may itch or 'tickle'; some SCCs can be tender or painful.
- If there was a pre-existing mark, such as a brown 'spot' or 'mole', but most malignant melanomas arise *de novo*, not in a pre-existing melanocytic lesion.

Risk factors

Document:

- Skin phototype—what happens to the patient's skin in the sun? (See Table 2.1.)
- History of sunburn, particularly in childhood.
- Chronic sun exposure—explore occupation, hobbies, holiday destinations, time spent in hot climates, e.g. military service or living abroad in childhood. How was the skin protected?
- Previous skin cancers.
- Chronic immunosuppression or chronic photosensitivity, e.g. drug-induced. Both increase the risk of skin cancer.
- Exposure to carcinogens, such as industrial tars or oils, pipe smoking, especially the clay pipes still smoked in India (SCC of the lower lip), X-irradiation, or arsenic (a constituent of some old-fashioned 'tonics' such as Fowler's solution; a contaminant in drinking water drawn from wells in Bangladesh and parts of India and elsewhere, e.g. Taiwan).
- Family history of skin cancers.

Skin phototype

Skin phototype affects the likelihood of burning in the sun, as well as the risk of skin cancer.

Table 2.1 Fitzpatrick classification

Skin type	Response of skin to sun	Appearance
I	Always burns, never tans	White or pale skin, many freckles, blond or red hair, blue or green eyes
II	Burns easily, tans poorly	Pale skin, possibly some freckles, blond hair, blue or green eyes
III	Sometimes burns, tans lightly	Darker white skin, dark hair, brown eyes
IV	Burns minimally, tans easily	Olive skin, brown or black hair, brown eyes. Mediterranean
V	Rarely burns, always tans	Naturally black-brown skin. Often has dark brown eyes and hair. Asian, Latin American, Middle Eastern
VI	Never burns, tans darkly	Naturally black-brown skin. Usually has black-brown eyes and hair. Black African

History in photosensitivity

Photosensitivity is defined as an abnormal cutaneous response involving the interaction between photosensitizing substances and sunlight or filtered or artificial light. People who are photosensitive are easily sunburnt. The skin is uncomfortable or develops a rash when exposed to sunlight or filtered or artificial light. You may suspect photosensitivity from the history or if you find a rash predominantly on sun-exposed surfaces (see ➔ Box 4.6, p. 73). Photosensitivity is most often an acquired problem (see Boxes 2.9 and 2.10).

Sometimes conditions, such as atopic eczema or psoriasis, are made worse by sunlight (photoaggravated).

What should I ask?

Specific questions will help you to pinpoint the type and cause of the photosensitivity.

- At what age did the problem start?
- Were any drugs or topical agents started prior to the onset of the problem? Most cases will present within 6 weeks of starting the drug, but drugs started in late autumn may not cause a rash until the following spring or summer.
- What are the symptoms—itch, pain, burning?
- Is there a rash? What is the distribution, and what does it look like—redness, swelling, blisters, pigmentation?
- How quickly does the problem occur after exposure to sunlight?
- How long does the problem persist—minutes, hours, or days?
- Does the rash leave any pigmentation when it fades?
- What time of year does the problem present—is there any seasonal variation? Symptoms usually commence in the spring.
- What sort of exposure is required to trigger the problem?
    - How much exposure is required to cause the rash?
    - Does sunlight in the United Kingdom (UK) produce the eruption, or does it only happen when the patient is abroad?
    - Does it happen if the light comes through glass, e.g. when sitting next to a window or when driving? Ultraviolet B (UVB), unlike ultraviolet A (UVA) or visible light, cannot penetrate glass (see ➔ p. 320).
    - Does lying on a sunbed (UVA) trigger the problem?
- Is there any evidence of 'hardening', i.e. does the skin become tolerant after repeated exposures to sunlight or a sunbed? Paradoxically, some problems improve, as summer progresses, and the rash may spare chronically exposed skin on the face.
- Are antihistamines, sunscreen creams, or clothing protective?
- What impact does the problem have on the quality of life?
- Is there a history of skin cancer, and did skin cancers occur at an earlier age than one would expect, given the skin type and history of sun exposure? Some photosensitive patients are more prone to skin cancers (see ➔ pp. 336–7).
- What hobbies or occupation does the patient have?
- Is there a family history of photosensitivity?

Box 2.9 Types of photosensitivity

1° photosensitivity (uncommon or rare)

- Idiopathic photodermatoses.
- Cutaneous porphyrias.
- Genophotodermatoses: very rare.

Acquired photosensitivity (phototoxicity or photoallergy)

- Drug-induced (commonest cause of photosensitivity).
- Contact photosensitivity:
  - Soaps, tars, perfumes.
  - Phytophotodermatitis (plants containing psoralen).

Box 2.10 Drugs that cause photosensitivity

Phototoxicity

(common, erythematous like an exaggerated sunburn)

- Phenothiazines, amiodarone, thiazides, non-steroidal anti-inflammatory drugs (NSAIDs), quinine, tetracyclines, sulfonamides, retinoids, psoralens, vemurafenib, voriconazole.

Photoallergy

(eczematous scaly rash, may not occur on 1st exposure)

- Sulfonamides, phenothiazines, thiazides.

Lupus erythematosus

- Hydralazine, procainamide, thiazides.

Pseudoporphyria

- Furosemide, nalidixic acid, amiodarone, ciprofloxacin, bumetanide, NSAIDs.

Lichen planus

(may be photoaggravated)

- Thiazides, quinine.

Pellagra

- Isoniazid.

For more information, see ➔ Box 4.6, p. 73, and Chapter 16, pp. 319–39.

Why the skin condition has not responded to treatment

Patients often say that the treatment ‘has not worked’. It is important to explore exactly what is meant by this statement. It is not pleasant using messy ointments, and some people, particularly older people and those living alone, may find it difficult to apply the treatment. It is hardly surprising if patients do not adhere to treatment plans.

What should I ask the patient?

- What does the patient mean by the phrase ‘has not worked’—are the patient’s expectations realistic? Most skin conditions can be controlled, but cure is the exception, rather than the rule.
- Have you established what matters to the patient by asking ‘What is most important to you?’. The answer will help you and the patient to agree treatment goals (see Box 2.11).
- Has the patient given the treatment enough time to ‘work’? It may take 3 or 4 months before some treatments have a maximal effect.
- If the patient has stopped the treatment, what was the reason? Were there any problems applying the treatment or reactions to the treatment? Was the patient under the impression that it was a ‘course’ of treatment that would be stopped, rather than an ongoing treatment?
- Did the patient collect the prescribed medicaments or more medicaments from the general practitioner (GP) after the initial prescription?
- What was prescribed? It is not sufficient to know the name. You need more information before agreeing with the patient that ‘it does not work’ (see Box 2.12). It will help if the patient can show you the preparations being used—they may not be what you think were prescribed!
- When, where, and how is the preparation applied? Ask open questions to find out exactly what the patient is doing: ‘Tell me more about how you use your treatment’.
- How much is being applied, and how often? Underuse is common.
  - Steroid phobia is a common cause of treatment failure. How long does the 30g or 100g tube of steroid ointment last? Some parents may be very reluctant to apply topical steroids to their child’s eczematous skin (see ➔ pp. 608–9).
  - Emollients should be used liberally and frequently. Was the patient given a 500g tub?
- It can be particularly confusing for the patient if several topical preparations were prescribed. Are they to be used at the same time or sequentially; all over the skin or just on localized areas?
- Has anyone shown the patient how to use the treatment? There is a knack to applying topical medicaments. Ideally, all patients with chronic skin problems would see a dermatology nurse. The nurse can answer questions, explain (again) what the treatment is likely to do, and show the patient (or carer) how to use the treatments. If you do not have access to a dermatology nurse, you should be prepared to demonstrate the treatments yourself.

Box 2.11 Shared decision-making

Decisions about care should be patient-centred and collaborative. Your role, as a clinician, is to help the patient to understand the options and to reach a decision after considering his values, beliefs, and goals. Patient-centred care, utilizing shared decision-making, is associated with improved patient satisfaction, improved adherence to medication, and better outcomes.

Shared decision-making involves three steps:

1. Explaining/discussing the need to consider alternatives (team talk).
2. Describing the alternatives in more detail (option talk). At this stage, it may be helpful to introduce an option grid: a one-page evidence-based summary of options and frequently asked questions (see Option Grids website, available at www.optiongrid.org).
3. Helping the patient to explore their personal preferences and to reach a decision (decision talk). But give the patient time to consider all the information—it may be entirely reasonable to defer reaching a decision until another day.

Box 2.12 Questions to ask when exploring what was prescribed

- Is the product a cream, an ointment, or a lotion?
- Ointments have fewer preservatives and are less likely to irritate the skin than creams. They are also more effective than creams in dry, scaly conditions. But some people stop treatment, because they dislike the feel of the greasy film left on the surface of the skin or the mess left on clothes. They may also have the erroneous perception that the ointment does not 'get into the skin'. The answer may be to use an ointment in the evenings and a cream in the daytime.
- Some people find the skin is itchier under an occlusive ointment than under a cream.
- Preservatives in creams can sting inflamed skin or cause transient erythema. People who misinterpret these problems as a sign of 'allergy' will stop the treatment.
- Was the patient prescribed enough of the preparation? Hopefully, a 500g tub of moisturizer (emollient), rather than a small tube? Most medicaments come in 30g or 100g tubes.

Further reading

Elwyn G *et al*. *J Gen Intern Med* 2012;**27**:1361–7 (shared decision-making).
Elwyn G *et al*. *Patient Educ Couns* 2013;**90**:207–12 (Option Grids).
Elwyn G *et al*. *Ann Fam Med* 2014;**12**:270–5 (shared decision-making).

Assessing the impact of skin conditions

Impact of skin diseases

Our appearance affects the way we feel about ourselves and how well we function. The skin is always on show, particularly on the face and hands. If we think that we 'look good', we are more likely to 'feel good' and have confidence.

The psychosocial and physical impact of highly visible inflammatory skin diseases, such as psoriasis, acne, or atopic eczema, may be profound. Skin diseases disrupt school, work, and social life. Low self-esteem, common in teenagers with acne, may affect personal relationships, as well as prospects of employment. Inherited skin diseases, such as epidermolysis bullosa (EB), may be even more devastating. Patients, their families, and friends may find it difficult to cope.

Patients with chronic scaly skin conditions may also have a deep-seated, and probably quite irrational, fear of being contagious. 'I feel like a leper'. This may be reinforced by the reaction of family and/or friends who have an equally deep-seated fear of 'catching something'. The patient withdraws from physical and social contact.

Skin conditions in patients with other medical problems may be the 'last straw', e.g. the intractable itching in chronic liver or renal failure which can be more difficult to control than pain; photosensitivity in diseases, such as systemic lupus erythematosus (SLE), which may have a major impact on normal daily activities; hair loss or pigmentary changes which may be disabling; and highly visible adverse effects of drugs (see ➔ pp. 372–91) or the result of conditions such as chronic graft-versus-host disease (GVHD) (see ➔ pp. 534–6).

Quality of life

Consider measuring quality of life objectively, using a validated questionnaire such as the Dermatology Life Quality Index (DLQI) (www.cardiff.ac.uk/dermatology/quality-of-life/dermatology-quality-of-life-index-dlqi/) (see Boxes 2.13 and 2.14). This tool will give you insight into patients' perceptions of the impact of their skin conditions on their lives and also help you to assess the effectiveness of interventions such as education, counselling, and, of course, treatment. The reaction of patients is not predictable and sometimes may seem out of proportion to the severity of the skin disease. The DLQI provides insight into that reaction.

The DLQI is self-explanatory and is designed for patients over the age of 16 years. It can be handed to the patient before the consultation and takes 1 or 2min to complete. A similar questionnaire has been validated for use in children. Skindex is another dermatology-specific health-related quality of life instrument that can be self-administered. Tools have also been developed for specific conditions, e.g. hand eczema.

Itch severity scale

Itch is the commonest symptom in dermatology, and severe itch may be extremely debilitating. A questionnaire (completed by the patient) quantifies how much of the day itch is a problem, the quality of itch, the intensity of itch, and the impact of itch (including impact on sleep) (see ➔ Further reading).

Box 2.13 Dermatology Life Quality Index© (DLQI)

DERMATOLOGY LIFE QUALITY INDEX

DLQI

Hospital No: Date:

Name: Score:

Address: Diagnosis:

The aim of this questionnaire is to measure how much your skin problem has affected your life OVER THE LAST WEEK. Please tick one for each question.

1.	Over the last week, how **itchy**, **sore**, **painful**, or **stinging** has your skin been?	Very much A lot A little Not at all	❒ ❒ ❒ ❒	
2.	Over the last week, how **embarrassed** or **self-conscious** have you been because of your skin?	Very much A lot A little Not at all	❒ ❒ ❒ ❒	
3.	**Over the last week, how much has your skin interfered with your going shopping or looking after your home or garden?**	Very much A lot A little Not at all	❒ ❒ ❒ ❒	Not relevant ❒
4.	Over the last week, how much has your skin influenced the **clothes** you wear?	Very much A lot A little Not at all	❒ ❒ ❒ ❒	Not relevant ❒
5.	Over the last week, how much has your skin affected any **social** or **leisure** activities?	Very much A lot A little Not at all	❒ ❒ ❒ ❒	Not relevant ❒
6.	**Over the last week, how much has your skin made it difficult for you to do any sport?**	Very much A lot A little Not at all	❒ ❒ ❒ ❒	Not relevant ❒
7.	**Over the last week, has your skin prevented you from working or studying?**	Yes No	❒ ❒	Not relevant ❒
8.	Over the last week, how much has your skin created problems with your **partner** or any of your **close friends** or **relatives**?	A lot A little Not at all	❒ ❒ ❒	
9.	**Over the last week, how much has your skin caused any sexual difficulties?**	Very much A lot A little Not at all	❒ ❒ ❒ ❒	Not relevant ❒
10.	Over the last week, how much of a problem has the **treatment** for your skin been, for example by making your home messy or by taking up time?	Very much A lot A little Not at all	❒ ❒ ❒ ❒	Not relevant ❒

Please check you have answered EVERY question. Thank you.

Box 2.14 Calculating and interpreting the DLQI score

- Very much, scored 3.
- A lot, scored 2.
- A little, scored 1.
- Not at all, scored 0.
- Not relevant, scored 0.

Sum the score of the answers:

- 0–1 = no effect on the patient's life.
- 2–5 = small effect on the patient's life.
- 6–10 = moderate effect on the patient's life.
- 11–20 = very large effect on the patient's life.
- 21–30 = extremely large effect on the patient's life.

Further reading

Chren MM *et al. J Invest Dermatol* 1996;**107**:707–13.
Finlay AY and Khan GT. *Clin Exp Dermatol* 1994;**19**:210–16.
Hongbo Y *et al. J Invest Dermatol* 2005;**125**:659–64.
Majeski CJ *et al. Br J Dermatol* 2007;**156**:667–73 (itch severity scale).

Skin and the psyche

Impact of skin disease on mood

People with skin disease may feel shame or embarrassment and have poor self-image or low self-esteem. Ten percent of people with psoriasis have had suicidal thoughts. Some patients become increasingly introspective, isolated, and depressed. Treatments prescribed for the skin condition, such as systemic corticosteroids, can also affect mood.

But anxiety or depression may be quite unrelated to skin disease, and the potential link between the mood change and the skin problem should be explored in the history.

Dysmorphophobia and dermatological non-disease

Each of us has a clear idea of how we look, but some individuals have a distorted body image.

Dysmorphophobia (body dysmorphic disorder) is a psychiatric condition in which patients have an unshakeable belief that they have a major flaw in their appearance. If present, the lesion is trivial, but some patients have no visible pathology (dermatological non-disease). Most patients lack insight into their behaviour but want a dermatological answer to the intractable problem.

Before diagnosing dysmorphophobia or dermatological non-disease, it is important to be clear what the patient is describing and what they are doing to their skin, e.g. picking may scar the skin, hair plucking induces folliculitis. Many inflammatory skin conditions do fluctuate in severity, and some, such as urticaria, may disappear completely.

Ask the patient to grade the skin problem as it is today, with 10 being the worst possible and zero being normal skin. If the patient says it is dreadful and gives the problem a grade of 10, yet you can see no gross pathology, it is likely that your patient is delusional (see ➔ p. 570).

Delusions of parasitosis

The patient believes that he/she is infested and quite often will have convinced other members of the family that this is the case. He/she may describe a crawling or biting sensation, lumps on the skin, and digging out the 'parasites' to provide relief. You may be given a bag or box containing specimens of these 'parasites' (fragments of skin scale, crust, or hair). For more information, see ➔ Delusions, pp. 570–1.

Also see ➔ Neuropathic itch, pp. 548–9.

Chapter 3

Examination of the skin

Contents

Preparation

Get the basics right, and examination of the skin will be much easier. You should inspect all the skin and aim to describe the signs just as accurately as you would signs in the respiratory or cardiovascular systems. You should be able to interpret what you see, and, although you may not reach a diagnosis, try to pull together a reasonable differential.

Explain what you would like to do, and obtain the patient's consent to examine all the skin, not just the areas that seem to be affected. Ensure the patient is warm and comfortable. You will need to expose the skin, but you can do this in stages, so have a sheet or blanket to hand so that you can cover the patient. Ask if the skin is tender before you begin.

Remember to examine the nails, scalp, hair, and mucous membranes, as well as the skin. Look under dressings, and invite patients to remove make-up, dentures, or wigs. You may need a chaperone, particularly if you are going to examine sensitive areas such as the genitalia. You may also need a nurse to help you with dressings.

Can you see?

Lighting in most clinical areas is inadequate for examining the skin. Ideally, you should be working in bright ambient lighting that is equivalent to daylight, and you should have access to mobile supplementary illumination for some tasks, e.g. inspecting inside the mouth or illuminating awkward skinfolds.

Magnification

Some dermatologists use a hand lens or magnifying loupes for close inspection of skin lesions. Magnification can also help to assess the small blood vessels in the nail folds—you can use an ophthalmoscope with a drop of immersion oil, but a dermatoscope is even better. A dermatoscope can also confirm the diagnosis of scabies by revealing mites in burrows. A dermatoscope is used for assessing pigmented lesions, but it takes expertise to interpret the signs (see ➔ p. 640).

Wood light (black light)

A Wood light emitting UVR in the long-wave UVA region may be used in the assessment of hypo- or hyperpigmentation or some cutaneous infections (see ➔ p. 640).

Recording the signs

Draw pictures, or use a body map, to show the distribution of any rash or tumours (see Fig. 3.1). It is good practice to record the maximum dimensions of any tumour. It may be helpful to record the appearance of a rash or tumour photographically, but do obtain consent before you take clinical photographs.

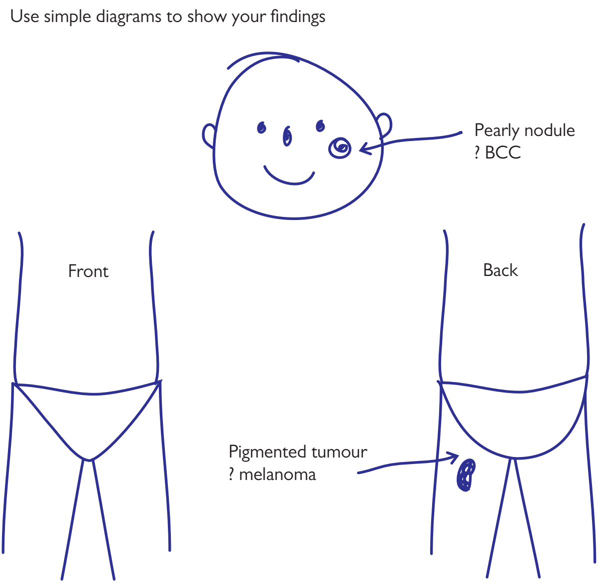

Fig. 3.1 Recording signs. Include the maximum dimensions of any tumour.

Rashes

Inspection

- Start by inspecting the patient from the end of the examination couch or bed to assess general health. Is this a life-threatening problem? Is the patient unwell (septic, malnourished) or looking healthy?
- Decide if the rash is:
  - Localized or generalized.
  - Predominantly symmetrical, suggesting a systemic (endogenous) cause, or asymmetrical suggesting an external cause such as infection, trauma, or contact dermatitis.
- Does the rash have a predilection for certain areas, e.g. extensor surfaces, flexural surfaces, light-exposed skin (see ➔ Box 4.6, p. 73), or the extremities—fingers, toes, ears, and nose (an acral distribution)?
- Does the rash demonstrate the Koebner phenomenon (show a predilection for scars or areas of trauma)?
- Are lesions grouped, scattered, generalized, linear, annular (in a ring), reticulate (net-like), or serpiginous (snake-like) (see Fig. 3.2)?

Palpation and interpreting the signs

Now look more closely at the individual lesions, and be prepared to feel the skin, but do ask the patient if the rash is tender before you begin. Try to describe a 1° lesion as precisely as possible, and decide whether the problem is predominantly epidermal or dermal. Ask yourself the following questions:

- How does the rash start, and what shape are the 1° lesions?
- Is the border well demarcated or indistinct?
- Is the surface scaly, indicating an epidermal pathology, or smooth suggesting a predominantly dermal pathology? Do not be afraid to gently scratch the surface of the skin.
- Are the lesions raised or flat? Palpate to answer this question.
- What colour is the rash? Does redness blanch with gentle pressure, indicating erythema (increased blood in small vessels), or persist indicating purpura (leakage of blood from the vessels into the dermis)?
- Is the involved skin hotter or cooler than normal?
- If there are blisters, what size are they? Do they rupture easily, suggesting an intra-epidermal blister, or are they tense with a thick roof suggesting a subepidermal blister, with the whole of the epidermis forming the roof? (see ➔ Fig. 13.1, p. 261.) Any milia? These tiny (<3mm), white intradermal keratinous cysts are sometimes found in subepidermal blistering diseases.
- Are there excoriations or erosions 2° to scratching?
- Is the skin thickened (lichenified), suggesting chronic rubbing?
- What happens as lesions evolve? Ask the patient to show you lesions at different stages. Do the lesions leave any colour change or scars?
- Is the problem malodorous, suggesting an anaerobic infection?

For terminology, see ➔ inside front cover.

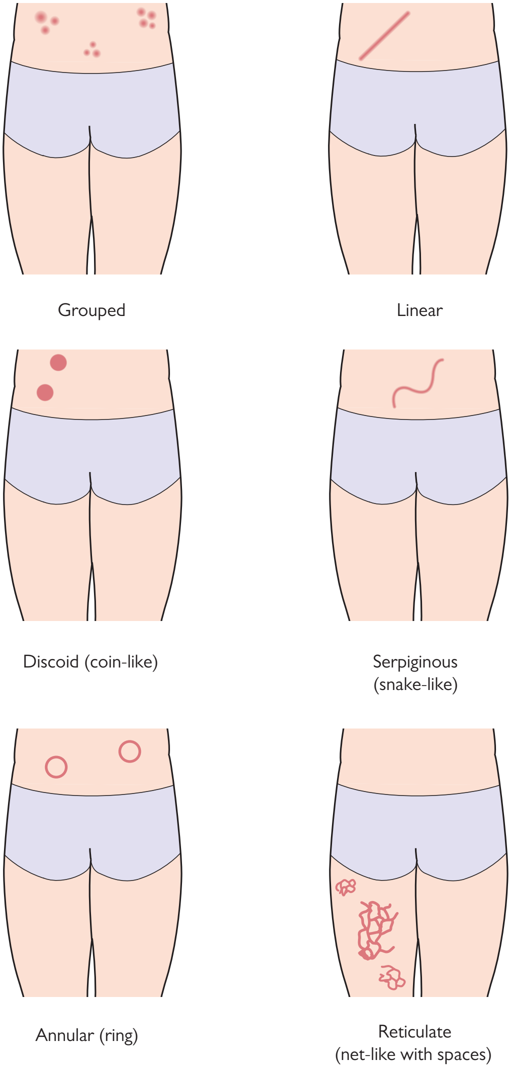

Fig. 3.2 Diagram of signs in the skin and patterns.

Skin tumours

Take an interest in the tumours that you see when you are examining patients, so that you become familiar with the common tumours and can distinguish the many benign 'bumps' that can be safely ignored from the malignant tumour that needs treatment. Skin tumours are common, particularly in older people, and skin cancer is the commonest of all malignancies (see ➔ Chapter 17, pp. 341–64). Seek a second opinion, if you are not sure about what you find. Do not be ashamed to ask—skin cancers diagnosed early are curable. You are in an excellent position to make that diagnosis when you are clerking patients.

Record your clinical findings as accurately as possible.

The examination

(Also see ➔ Box 4.8, p. 83.)

- Check all the skin, taking a particular interest in your patient's back—he/she cannot see it.
- Record the skin colouring (skin phototype) (see ➔ Table 2.1, p. 27).
- Are there signs of chronic sun damage (see Box 3.1)?
- Does one pigmented tumour stand out from the others? Take a careful look—this 'ugly duckling' may be malignant.
- For any tumour, ask yourself:
  - Where has this tumour arisen—epidermis, dermis, or deeper?
  - What cells/tissues are probably involved, e.g. keratinocytes, melanocytes, vascular, fibroblasts, neural, lipocytes, or other?
  - Is it likely to be benign or malignant (might it be a metastasis)?
- Take a systematic approach to the examination of the tumour—for pigmented tumours, use the ABCDE criteria on ➔ p. 354.
  - What colour is it, e.g. brown, purple, erythematous, or a mix?
  - What are the maximal dimensions?
  - Is the outline regular or irregular?
  - Is the surface smooth, shiny (pearly), crusted, keratotic, or ulcerated?
  - Is there a punctum? Seen in epidermoid cysts and other tumours centred on hair follicles such as sebaceous hyperplasia (see Box 3.1).
  - Are there telangiectasia running over the surface (suggests BCC)?
- Stretch the skin over the tumour between your thumb and index finger to accentuate the pearly appearance and borders of a nodular BCC, as well as the thready margin of a superficial BCC.
- Is the edge rolled, undermined, or indurated?
- Could it be vascular—can you compress the tumour?
- Might it be cystic—does it transilluminate?
- Palpate to check the consistency (soft, firm, hard) and thickness (depth).
- Pinch the skin gently on each side—does it 'sink down' into the dermis?—the buttonhole sign in dermatofibromas (see ➔ p. 342).
- Is it mobile or deeply fixed?
- Check the regional lymph nodes if you suspect malignancy.

Chronic sun damage (photoageing)

(See Box 3.1.)

Patients with fair skin, blue eyes, and blond or red hair are most at risk of sun damage and skin cancer. UVR is responsible for many of the signs we associate with old age, including coarse wrinkles and blotchy abnormal pigmentation.

Sun-damaged skin looks old—perhaps the most effective message to give to those addicted to 'cooking' themselves on beaches!

Chronic photosensitivity, e.g. caused by some drugs, is also linked to photoageing and skin cancer.

Box 3.1 **Signs of chronic sun damage**

- Solar lentigines: persistent flat, brown spots (liver spots) on exposed skin, e.g. backs of hands, forearms, face.
- Easy bruising (senile purpura) which resolves, leaving white stellate pseudoscars—most obvious on forearms and backs of hands. Elastic tissue damaged by chronic sun exposure provides the small cutaneous blood vessels with inadequate support.
- Waxy coarse, yellowish skin (solar elastosis) on the forehead, temples, cheeks, and back of the neck, associated with deep furrows.
- Sebaceous hyperplasia: scattered small, yellowish papules on the face. Often misdiagnosed as BCC but lacks marked telangiectasia (check with magnification). Look for a central punctum.
- Favre–Racouchot syndrome: thick, yellowish plaques (solar elastosis) studded with comedones (dilated follicles, blackheads) on the back of the neck and face, particularly the skin around the eyes.
- Telangiectasia on the sides of the neck and cheeks with reddish brown pigmentation. Known as poikiloderma of Civatte and seen most often in women with fair skin.
- Hypopigmented macules, alternating with areas of increased pigment—most obvious on the forearms and bald scalp.
- Dry skin.
- Solar keratoses: erythematous or pigmented rough, hyperkeratotic areas on chronically sun-exposed skin such as bald scalp, face, dorsum of hands.
- Bowen disease (SCC *in situ*) on lower legs: scaly, red patches. Usually asymptomatic. Differentiate from eczema or tinea.
- Skin cancers (BCC, SCC, malignant melanoma).

Scalp and hair

Examining the scalp is part of the general dermatological examination but assumes particular importance in any patient who is complaining of hair thinning, excessive shedding, or balding (complete loss).

Hair loss (alopecia) may be caused by a variety of insults, including fungal infections, drugs, or systemic disease. The loss may be diffuse or localized. Loss manifest by increased shedding may be apparent to the patient but not the physician. Normal hair density may be reduced by as much as 50%, before thinning becomes obvious. The loss may affect hair elsewhere, e.g. eyelashes, eyebrows, beard hair, or body hair (see ➔ p. 94).

Your findings should help you to decide if the pathology is likely to be a non-scarring alopecia (you will still be able to see the hair follicles in the affected area) or scarring alopecia (the hair follicles cannot be seen easily).

What to look for: the scalp

- Is the hair loss complete, and/or is the hair density reduced? Is the loss diffuse or patchy? Is there any sign of regrowth?
- Does the hair loss affect particular parts of the scalp, i.e. is it patterned? Androgenetic alopecia in men predominantly affects the vertex of the scalp and/or the temporal areas. Check the frontal hairline (see ➔ p. 480).
- Does the loss affect white hairs as well as pigmented hairs? White hairs are spared in alopecia areata (see ➔ p. 488).
- Does the scalp appear normal, or is it erythematous or scaly? Are there follicular pustules, perifollicular erythema, or follicular plugging?
- Are there broken hairs? Is the hair shaft smooth or rough?
- Can you see exclamation-mark hairs (short and tapering to a point) at the edge of patches of complete loss—pathognomonic of alopecia areata, but not found very often.
- Do you think there is scarring? (See Box 3.2.)

Hair pull and hair pluck

Telogen effluvium is a common cause of acute diffuse hair loss (➔ Box 2.7, p. 25). The hair becomes less dense, but patients do not lose all their hair, and the scalp looks entirely normal. In contrast, radiotherapy or drugs, such as chemotherapeutic agents, cause growing anagen hairs to fall out. Consider performing a hair pull or hair pluck (see Boxes 3.3 and 3.4).

What to look for: the hair shaft

You have examined the scalp, but you may also need to look at the morphology of individual hairs.

The hairs in some inherited ectodermal conditions are abnormal. Cut off a sample of a few hairs, and examine the hair shafts under a light microscope (or send them to an expert). If the hair shafts are abnormal, does the patient have problems with other ectodermal tissues, e.g. nails, teeth, or sweat glands? (see ➔ p. 634–5.)

Box 3.2 Diagnosis of scarring alopecia

The diagnostic sign is loss of follicular ostia. Use a hand lens or dermatoscope to examine the involved areas of the scalp.

Other signs will depend on the cause but might include:

- Epidermal atrophy—shiny, thin skin that wrinkles excessively if pinched gently.
- Perifollicular hyperkeratosis and perifollicular erythema.
- Boggy swelling and inflammation.
- Follicular plugging or prominent follicular openings without hair.
- Follicular pustules.
- Keloidal scarring.

To confirm the diagnosis, take two deep 4mm punch biopsies, orientated parallel to the hair shafts, from hair-bearing skin where the disease is active clinically. Ask your pathologist to section one specimen vertically and the other horizontally.

Box 3.3 Hair pull

This test involves gently pulling a group of 25–50 hairs, while running your fingers from the base to the terminal ends. Normally, only one or two hair shafts are dislodged.

In active telogen effluvium, a gentle hair pull will yield at least ten hairs with each pull. Light microscopy will demonstrate that these are in the telogen phase (see Box 3.4).

A gentle hair pull at the edge of a patch of alopecia areata will also remove hairs if the disease is active.

Box 3.4 Hair pluck or trichogram

A hair pluck involves removing a tuft of 10–20 hairs forcefully (use artery forceps).

The hairs are placed on a glass slide, covered with a glass coverslip, and the coverslip held in place by adhesive tape. Long hairs may be folded in a figure of eight to fit them under the coverslip. The hair shafts can be examined under the light microscope. Telogen hairs have club-shaped root 'bulbs' without much pigment. Growing anagen hairs have much smaller root bulbs. These bulbs tend to be soft and easily distorted, and they may have tissue adhering to them (root sheaths).

The normal anagen/telogen ratio is 4:1. In telogen effluvium, the ratio is decreased or reversed. More than 25% of the hairs may be in telogen.

Nails

Inspection of fingernails and toenails is an important part of the physical examination. Inspect the nail fold, nail plate, and nail bed (see Fig. 3.3).

Acquired changes may be a manifestation of drugs or systemic disease. Congenital problems may indicate a more widespread ectodermal problem that might also affect hair, teeth, and sweating. Nails may be partially absent, misshapen, or totally lost (see ➔ p. 634).

Equipment

Side-lighting makes it easier to pick up changes in the surface of the nail plate. Consider magnification with an ophthalmoscope or a dermatoscope to look at the blood vessels in the nail fold.

What to look for

Inspect:

- Nail folds, both lateral and proximal. Is there erythema or swelling?
- Blood vessels in the proximal nail fold. You will need magnification to assess blood vessels, e.g. dermoscopy (see ➔ p. 38).
- Cuticle: is this present, intact, and firmly attached to the nail plate?
- Shape of the nail plates.
- Colour of the nail plates: punctate white areas are common in normal individuals; linear splinter haemorrhages near the distal free edge of the nail plate in one or two nails may be a sign of trauma. Is any colour change uniform, or does it involve part of the nail?
- Surface of the nail plates: longitudinal grooves are common in the elderly, when they are of no significance.
- Thickness of the nail plates.
- Attachment of the nail plate to the underlying nail bed. Separation (onycholysis) may be caused by problems such as psoriasis.
- Nail beds (look beneath the nail plate). Is there hyperkeratosis or a tumour such as a subungual wart?

Interpretation

Which nails are involved?

The distribution of the nail problem, just as in the skin, will provide some guide to the likely cause. If the abnormality is widespread and symmetrical, it is more likely to have an endogenous than an exogenous explanation.

If just a few nails appear abnormal, particularly if these are toenails, consider whether the changes might be due to trauma or a fungal infection. What does the patient do to his/her nails—biting, manicuring, trauma at work? If infection is a possibility, you may wish to take nail clippings for mycological examination and culture (see ➔ p. 646).

For more information about the causes of a nail dystrophy, see ➔ p. 96.

For drug-induced nail problems, see ➔ pp. 380–1.

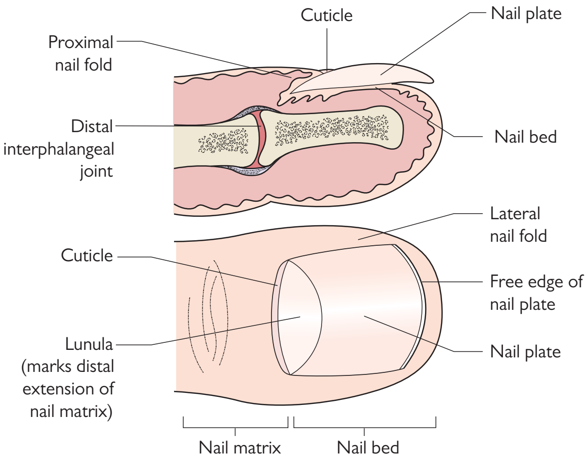

Fig. 3.3 Normal nail. Adapted with permission from De Berker D, Hair and nails. In Warrell, Firth, and Cox, *Oxford Textbook of Medicine* 5th edn, 2010. Oxford: Oxford University Press.

Mucosal surfaces

Mucosal inflammation or ulceration may occur in skin diseases, systemic diseases with cutaneous manifestations, such as graft-versus-host disease (GVHD), and in some drug eruptions.

The mouth

Equipment and preparation

You will need:

- Gloves.
- Two wooden tongue spatulas or oral mirrors.
- A gauze swab and a bright light.

Ask the patient to remove dentures, before you start the examination.

What to look for

- Assess the lips, including the outline of the vermillion border, which may appear rather 'ragged' in sun damage or discoid lupus erythematosus (DLE).
- Palpate tumours or ulcers to assess infiltration.
- Are the lips swollen, scaly, or cracked?
- Lichen planus (LP) gives the lips a lacy, white appearance.
- Look at the hard and soft palate.
- Check the attached gingiva for bright erythema (desquamative gingivitis), bleeding, or ulceration—seen in LP or pemphigus.
- To examine the buccal mucosa, ask the patient to move his/her tongue across to one side of the mouth, so that you can see the opposite side, and repeat this for the other side. Use spatulas to help you to visualize the buccal mucosa. Are asymmetrical areas of erythema or ulcers adjacent to broken teeth? Are pigmented areas near teeth that are filled with amalgam?
- Grasp the tongue gently with a gauze swab, and examine the dorsal and ventral surfaces. Move the tongue to the right and left to examine the retromolar region ('the coffin trap' and a common site of oral cancer).
- Check the dentures, if the signs suggest trauma.

Genitalia

In patients with oral mucosal lesions, you should examine the genitalia where you may find similar mucosal signs. Sometimes these are not symptomatic or the patient may be too embarrassed to complain.

What to look for

- Erythema, swelling, rashes, erosions, ulcers, or pigment change.
  - Assess the labia, the folds between them, and the clitoris.
  - Inspect the scrotum and shaft of the penis.
- Distortion of the normal anatomy or scarring.
  - Does the foreskin retract normally? In uncircumcised patients, pull back the foreskin gently to check for tightening and to inspect the glans.
  - Are the labia majora and minora visible or obliterated by scar tissue?

- Palpate to assess for tenderness and infiltration.
- Is there any discharge from the vagina or penis?
- Inspect the perianal skin and rectal mucosa.

The eye

The eye may be involved in a number of systemic diseases with cutaneous manifestations, including GVHD and connective tissue diseases, but pay particular attention to the eyes in any patient with a history of a red gritty eye, blisters, or mucosal ulcers.

What to look for

- Assess the lids, e.g. the violaceous discoloration of dermatomyositis.
- Are the eyelashes normal? These may be lost in alopecia areata.
- Is there blepharitis, or are there chalazions, e.g. in rosacea?
- Is there ectropion—seen in chronic erythroderma?
- If a patient with ectropion closes the eye, is the cornea covered? If not, refer to an ophthalmologist.
- Is the eye red? For causes of a red eye, see ➔ p. 86.
- Is the eye sticky?
- Is the patient photophobic?
- Is there subconjunctival blistering or scarring? Gently pull down the lower lid to see if the sulcus between bulbar and lid conjunctival surfaces is retained (loss of the sulcus suggests scarring) or surfaces are adhering together because of scars (synechiae). Synechiae may be more obvious if you ask the patient to look laterally and then medially.

Assessing the severity of a rash

Clinicians should consider using an objective tool to assess and monitor the extent and severity of skin involvement in inflammatory conditions such as psoriasis, atopic eczema, lupus erythematosus (LE), or dermatomyositis, particularly when new therapies are being introduced. Objective scoring is equally important in patients with life-threatening problems such as toxic epidermal necrolysis (TEN).

The percentage of body surface area (BSA) affected can be calculated (see Box 3.5), but this provides no other insight into the severity. A number of instruments have been shown to be reliable in specific skin conditions. These clinical scoring systems supplement information provided by a tool like the DLQI that assesses the patient's perception of the impact of the condition on quality of life (see ➔ p. 32). A few systems are mentioned here.

Psoriasis Area and Severity Index (PASI)

One of the best known disease-specific tools for assessing an inflammatory skin disease (see ➔ Chapter 9, pp. 200–1).

Atopic eczema score (SCORAD or EASI)

See ➔ Chapter 31, pp. 606–7.

Cutaneous Lupus Disease Activity and Severity Index (CLASI) and Dermatomyositis Skin Severity Index (DSSI)

Indices validated by both dermatologists and rheumatologists. Changes in scores correlate well with changes in global assessments, as well as with symptoms such as pain or itch (see ➔ p. 404 and p. 410).

Scorten

Severity-of-illness score developed to predict mortality in toxic epidermolysis/Stevens–Johnson syndrome (SJS).

See ➔ p. 119.

Box 3.5 Calculating the body surface area involved

The rule of nines was devised for determining the percentage of the body surface area (BSA) involved in adults with thermal burns (see Fig. 3.4).

It is calculated as follows:

- Head and neck = 9%.
- Anterior chest = 9%.
- Posterior chest = 9%.
- Anterior abdomen = 9%.
- Posterior back and buttocks = 9%.
- Upper limbs = 9% each.
- Lower limbs = 18% each.
- Perineum = 1%.

Alternatively, you can use the patient's palm to measure the BSA. Each palm without fingers = 0.5% of total BSA. (Palm plus fingers and thumb = 1% of total BSA, but this is less reliable than using the palm alone).

Scarisbrick JJ and Morris S. *Br J Dermatol* 2013;**169**:260–5.

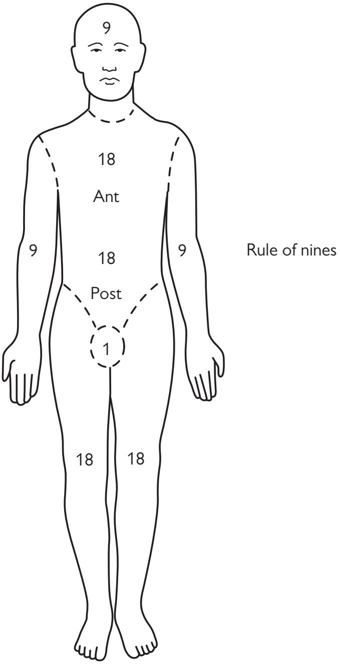

Fig. 3.4 Wallace rule of nines. Reproduced with permission from Giele H and Cassell O. *Plastic and Reconstructive Surgery*, 2008. Oxford: Oxford University Press.

Chapter 4

What is the diagnosis?

Contents

Epidemiology of skin disease

The prevalence of the skin problems that you see will be influenced by the setting in which you are working—hospital or community, rural or urban, tropical or temperate, specialty or general—but always remember that common diseases are commonest! Atypical presentations of common skin diseases are much more frequent than typical presentations of rare skin diseases.

In the community, you are more likely to come across skin infections, viral exanthems, skin tumours, and inflammatory dermatoses, such as acne, eczema, psoriasis, or urticaria, than manifestations of systemic disease. Chronic leg ulcers are common in 1° and 2° care, as are drug eruptions. In 2° care, you may see manifestations of systemic disease and acute problems, including conditions that cause skin failure.

But even in the hospital setting, please bear in mind what is common. Do not forget scabies as a cause of itching when you are caring for patients with chronic renal failure, many of whom are going to be itchy; remember fungal infections when you are looking at asymmetrical scaly rashes; exclude infection or trauma when you see blisters; and think of other causes of photosensitivity, e.g. a photosensitizing drug, before diagnosing LE.

What is common?

- Skin infections:
  - Bacterial cellulitis, impetigo.
  - Viral: herpes simplex, varicella-zoster, warts, molluscum.
  - Fungal: dermatophyte, yeast.
- Skin infestations:
  - Scabies, fleas, lice.
- Inflammatory diseases:
  - Acne.
  - Eczema (dermatitis) of any type.
  - Psoriasis.
  - Urticaria.
- Skin tumours:
  - Benign.
  - Premalignant (solar keratoses, Bowen disease).
  - Malignant (BCC, SCC, malignant melanoma).
- Hair loss (alopecia) or gain (hirsutes, hypertrichosis).
- Chronic leg ulcers.
- Pigment change (gain or loss).

Prevalence of skin disease

In 1975, the Lambeth Study of 2180 adults in the UK found that 55.5% of them had a skin condition. The prevalence of individual skin conditions in this community is shown on a bar chart in Fig. 4.1.

In developing countries, skin diseases are a major burden, particularly in children. These are predominantly bacterial infections, fungal infections, and infestations.

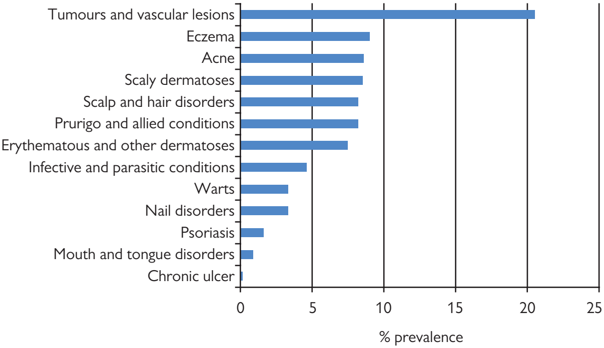

Fig. 4.1 Community prevalence of skin conditions. Data from Lambeth Study (Rea JN *et al. Br J Prev Soc Med* 1976;**30**:107–14).

Further reading

Schofield JK *et al. Br J Dermatol* 2011;**165**:1044–50.
Williams HC. Epidemiology of skin disease. In: Burns T *et al.* (eds) *Rook's Textbook of Dermatology*, 8th edn, 2010. Oxford: Blackwell Scientific Publications.

Reaching the diagnosis

The principles

- The introductory chapters on history and examination (see Chapter 2, pp. 17–36, and Chapter 3, pp. 37–51) will help you to take a logical approach.
- The history is just as important as the physical examination, but experts often take short cuts, using the signs to guide history taking. If you are a novice, you should take a full dermatological history.
- Consider the whole patient, not just the skin—after all, dermatologists are physicians of the skin and its contents. Systemic problems, such as immunosuppression, malnourishment, or rheumatological disease, may have specific cutaneous manifestations.
- A rash in a patient who is unwell should alert you to a different differential diagnosis than a rash in a patient who is well (see p. 58).
- Check the mucous membranes, hair, and nails, as well as the skin.
- Remove dressings, so you can see what is beneath them.
- Integrate the information you gather, and formulate a differential diagnosis, using analytical rules as well as pattern recognition (see Box 4.1).

Box 4.1 Analytical processing and pattern recognition

- Reaching a diagnosis depends partly on ordering and structuring information (analytical processing) and partly on pattern recognition.
- Experts use their knowledge and previous experience to integrate and make sense of clinical information.
- Pattern recognition ('spot diagnosis') is said to be an almost unconscious process that involves recognizing similarities between cases, based on the overall form of the cases. Expertise is equated with rapid, efficient, and accurate pattern recognition.
- Your skills in pattern recognition will improve when you have been exposed to many clinical examples of a condition and thought about their similarities. Concentrate on the variations in the presentation of common skin problems in the first instance. Why is this likely to be seborrhoeic dermatitis, and not asteatotic eczema? What makes this venous stasis, rather than a fungal infection? Read around the conditions that you see, and reflect on the variations you observe with your peers.
- To develop your analytical skills, adopt an organized approach to each case, so that you gather relevant information (experts know what questions to ask), detect important clinical signs (experts know what to look for), integrate the information, and formulate a differential diagnosis. The introductory chapters on history and examination will help you.
- Pattern recognition may have a powerful influence on the eventual diagnosis, but even experts use analytical rules as a safeguard against error.

A framework for analysing cutaneous signs

- Distribution: symmetrical (suggests an endogenous process) or asymmetrical (consider an external cause).
- Is there a predilection for certain sites, e.g. sun-exposed, acral (cold), flexures, extensors?
- What is the pattern, e.g. unilateral, generalized, grouped, linear (see ➔ pp. 62–3), annular (ring-like), reticulate (may be vascular), serpiginous? (see ➔ Fig. 3.2, p. 41.)
- Surface: scaly (epidermal problem—can scratch off scale), smooth (predominantly dermal), hyperkeratotic (thick horny layer, does not scratch off easily), or crusted (dried exudate—look underneath).
- Colour: red—erythema (blanches) or purpura (non-blanching).
- Hypo- or hyperpigmentation: this may be post-inflammatory.
- Identify a 1° lesion, e.g.:
  - Erythematous macules and papules: 2–10mm in diameter, with a tendency to confluence—morbilliform rash.
  - Erythematous scaly papules (papulosquamous rash) or scaly plaques.
  - Smooth papules or nodules (dermal involvement, e.g. granulomas).
  - Lichenoid papules: these have a purplish colour and a relatively smooth surface.
  - Urticarial wheal: smooth, raised erythematous papules or plaques that last no more than 48h and fade to leave normal skin.
  - Urticated erythema: smooth, raised erythematous papules or plaques that persist for more than 48h.
  - Purpuric macules or papules.
  - Blister: is it thick-walled (subepidermal) or thin-walled and easy to rupture (intra-epidermal)?
  - Pustule.
  - Thick plaque that feels as if the process extends below the dermis into the subcutaneous tissue.
- What does the involved skin feel like—soft, firm, hard (like bone)?
- Identify 2° lesions:
  - Excoriations: sign of scratching.
  - Lichenification: sign of chronic rubbing.
  - Prurigo: nodular reaction to chronic rubbing, typically spares the middle of the back which cannot be reached easily.
  - Erosion, ulcer, or scar.

Generalized rash in a sick patient

Has the patient been travelling? Is the patient immunosuppressed or taking any medication? Has the patient a fever?

Generalized erythematous rash in a sick patient

- Drug rash, eosinophilia, and systemic symptoms (DRESS) (see ➔ pp. 122–3).
- Infection (see Boxes 4.2 and 4.3).
- Erythroderma (red and scaly) 2° to drugs, eczema, psoriasis, or T-cell lymphoma but may be idiopathic (see ➔ p. 108).
- Acute GVHD (see ➔ pp. 532–3)—facial involvement suggests GVHD, rather than a drug reaction.
- Early TEN (see ➔ p. 116).
- Toxic shock syndrome (TSS): may be rapidly fatal (see ➔ p. 101).

Anaphylactic reaction with urticaria

- Reaction to drugs, venoms, or foods (see ➔ pp. 104–5).

Purpura in a sick patient

- Extensive irregularly outlined ecchymoses, bullae, and gangrene: disseminated intravascular coagulation (DIC) (purpura fulminans)—does the patient have meningococcal septicaemia? (see ➔ pp. 102–3.)
- Other infection (see Box 4.4).
- Small-vessel cutaneous vasculitis 2° to infection, drugs, or a systemic vasculitis (see ➔ p. 440).
- Drug reaction, e.g. DRESS (see ➔ pp. 122–3).

Widespread blisters and/or erosions in a sick patient

- Examine the mucosal surfaces (ocular, oral, genital) for blisters or erosions.
- Blisters that rupture easily, leaving erosions: consider staphylococcal scalded skin syndrome (SSSS), SJS, TEN, DRESS, acute GVHD, and pemphigus (may be 2° to drugs, including angiotensin-converting enzyme (ACE) inhibitors) (see ➔ p. 112 and p. 114).
- Generalized herpes simplex infection (eczema herpeticum) or herpes zoster. Vesicles evolve into pustules and become crusted (see ➔ Box 5.11, p. 113).
- Tense blisters: bullous pemphigoid (BP), but patients are usually systemically well, unless there is 2° infection (see ➔ pp. 270–1).

Generalized pustules in a sick patient

- Herpes simplex infection in atopic patient (eczema herpeticum): vesicles evolve into vesicopustules (see ➔ p. 112).
- Varicella-zoster infection (chickenpox): vesicles evolve into vesicopustules (see ➔ pp. 156–7).
- Bacterial infection.
- Sterile pustules: pustular psoriasis, acute generalized exanthematous pustulosis (AGEP) usually 2° to drugs (see ➔ pp. 124–5).

For dermatological emergencies, see ➔ Chapter 5, pp. 99–127.

Box 4.2 Some infections that may present with a morbilliform (measles-like) rash and fever

(see ➔ Chapter 6, pp. 129–48, and Chapter 7, pp. 149–69.)

- Measles (rubeola): also cough, coryza, and Koplik spots on the buccal mucosa.
- Rubella (German measles).
- Erythema infectiosum (parvovirus B19): 'slapped cheek' appearance on the face of children.
- Roseola infantum (human herpesvirus 6, HHV-6) = exanthem subitum: affects children.
- Enterovirus: may have pharyngitis.
- Infectious mononucleosis (Epstein–Barr virus, EBV): malaise, pharyngitis. A rash develops in 90% of patients with EBV who are treated with antibiotics such as amoxicillin.
- Leptospirosis.
- Acute retroviral syndrome (HIV).
- 2° syphilis: weeks to months after initial chancre.
- Typhoid fever: vomiting, diarrhoea, pink papules on trunk = rose spots (culture of rose spots may yield *Salmonella typhi*).
- Streptococcal scarlet fever, staphylococcal toxic shock syndrome.

Box 4.3 Infections that may cause a morbilliform (measles-like) rash and fever in travellers

- Chikungunya fever: Asia, Africa, Indian Ocean.
- Dengue fever: South East Asia, Central America, South America, Caribbean, South East United States of America (USA).
- West Nile virus: Asia, Africa Europe, south-eastern USA.
- O'nyong-nyong fever: sub-Saharan Africa.
- Mayaro virus: South America.
- Sindbis virus: Europe, Africa, Asia, Australia.
- Ross River disease: Australia, Papua New Guinea, Fiji, Samoa.

Box 4.4 Infections that may cause a purpuric rash and fever

- Meningococcal septicaemia.
- Rocky Mountain spotted fever: North America, Central America, and South America.
- Dengue fever: South East Asia, Central America, South America, Caribbean, south-eastern USA.
- Viral haemorrhagic fever: arbovirus/arenavirus infection.
- Yellow fever: sub-Saharan Africa, Amazon basin of South America.
- Epidemic louse-borne typhus.
- Leptospirosis.
- Atypical measles.

Itch

Generalized itch

The history is crucial. You need to decide, for example, if the itch (pruritus, not pruritis!) is a manifestation of a skin problem or an infestation, reflects an underlying systemic disease, or might be neuropathic or psychogenic in origin (see ➔ Box 10.5, p. 223, and Box 28.1, p. 567). What is the character of the itch? Ask about symptoms, such as burning, tingling, hypoaesthesia, or hyperalgesia that might suggest an underlying sensory neuropathy. How much impact is the itch having on the patient? Is it disturbing sleep? Is anyone else itchy? Ask about drugs, including substance abuse.

When you are examining the patient, decide what signs are 2° to scratching or rubbing (excoriations, lichenification, nodules, ulcers, scars) or if you can find evidence of a 1° dermatosis. Patients cannot easily reach the middle of the upper back, and usually this is spared if signs are just 2° to scratching. Look carefully for evidence of scabies (and look a 2nd time if the patient returns still itching).

1° skin condition—the differential includes

- Scabies (are any close contacts itchy?) (see ➔ p. 172).
- Eczema (see ➔ pp. 211–24).
- Psoriasis (see ➔ pp. 189–207).
- Urticaria (the rash may have gone by the time you see the patient). Ask about the rash and how it behaved. Stroke the skin to test for dermographism (a physical urticaria) (see ➔ p. 228, and Fig. 11.2, p. 231).
- Insect bites.
- LP.
- Prurigo nodularis (see ➔ Fig. 10.6, p. 223).
- The prodrome of BP: look for urticated erythema and blisters (see ➔ pp. 270–1).

Itch may be 2° to a systemic disease or drugs

Sensory neuropathy may cause itch or contribute to itch in some of these conditions (see ➔ pp. 548–9). The differential includes:

- Pregnancy-related (see ➔ pp. 584–5).
- Iron deficiency anaemia.
- Hypo- or hyperthyroidism.
- Chronic renal failure.
- Biliary obstruction, including primary biliary cirrhosis.
- Polycythaemia.
- Internal malignancy, including Hodgkin disease.
- Neuropathic: centrally driven neuropathic itch in association with brain tumours, strokes, spinal tumours, and multiple sclerosis. Human immunodeficiency virus (HIV) and diabetes may be associated with neuropathic itch (see ➔ p. 548).

- Drugs may cause pruritus (see ➔ p. 375), including:
  - Statins.
  - ACE inhibitors.
  - Opiates, barbiturates, and recreational drugs.
  - Antidepressants.
  - Oral retinoids.
  - Hydroxyethyl starch products (colloid volume expanders). Attacks of itch may last for years. Withdrawn in the UK in 2013, because of risk of renal damage.

Localized itch

Skin that is persistently rubbed becomes thickened with increased markings (lichenified) or may become hyperpigmented, nodular, or even ulcerated.

Consider:

- Contact dermatitis (see ➔ pp. 214–16).
- Lichen simplex (see ➔ p. 222).
- Picker's nodule (localized prurigo nodularis) (see ➔ p. 222).
- Neuropathic itch such as notalgia paraesthetica affecting the lower border of the scapula or brachioradial pruritus affecting the dorsolateral aspect of the arms (see ➔ pp. 548–9, and Boxes 27.1 and 27.2, p. 548 and p. 549).
- Rarely, localized itch is the presenting sign of an intracranial tumour, e.g. pruritus of the nostrils caused by brain tumours extending to the base of the fourth ventricle, intramedullary tumours causing unilateral itch of the upper limb, brainstem or spinal cord tumours causing itch of the head/neck/upper extremity.
- Trigeminal trophic syndrome (see ➔ Box 27.3, p. 549).

Linear patterns and sharp demarcations

If a rash is linear or displays a sharp cut-off, consider an external cause such as an excoriation, pressure of a waistband, or brushing against a plant (see ➲ Box 16.4, p. 327). External agents usually produce strikingly asymmetrical rashes. Some inflammatory dermatoses, such as LP, psoriasis, or vitiligo, demonstrate the Koebner phenomenon, and the rash develops at sites of trauma, e.g. where the skin has been scratched, and some cutaneous infections may be auto-inoculated into a scratch, e.g. viral warts or molluscum contagiosum. Ask about trauma such as scratching.

Once you have excluded an outside cause, seek an internal explanation, remembering the comment of that distinguished skin pathologist Ian Whimster: '*Some invisible intersegmental boundaries, whose existence we have been taught to expect by comparative anatomy and embryology, are only revealed by disease*'.

Venous or lymphatic drainage

Does the problem follow a vein or lymphatic? Stand the patient up to look for varicose veins. Conditions, such as stasis dermatitis or vitiligo, may follow varicose veins. Thrombophlebitis presents with tender nodules along a vein. The extending red line of lymphangitis is a typical finding in cellulitis, but other infections, including sporotrichosis, may spread along lymphatics, as may a malignant infiltrate.

Dermatome

A dermatome is the cutaneous area supplied by one spinal dorsal nerve root. Almost all areas of the skin are innervated by two or more spinal roots, so that adjacent dermatomes overlap to a large and variable extent, excluding those separated by the ventral axial line (see next section). Herpes zoster involves the skin in a dermatomal pattern (see ➲ p. 156–7 and Fig. 4.2, pp. 64–5).

Embryonic ventral axial line

The embryonic ventral axial line marks the border between the C5/C6 and C8/T1/T2 dermatomes. There is no crossover of sensory nerves or sensory function along this line that separates the cranial and caudal dermatomes. Contiguous dermatomes on either side of the ventral axial line are supplied from discontinuous spinal cord segments: C5 and C6 dorsal; and C8, T1, and T2 ventral (see Fig. 4.2, pp. 64–5).

Some morbilliform drug eruptions show a sharp 'drug line' that corresponds to this axial line, with well-demarcated involvement of the T1, T2, T3 spinal nerve dermatomes, but sparing of the adjacent C5 nerve dermatome.

Pigmentary demaration lines (Voigt–Futcher lines)

(see ➲ Box 1.5, p. 13.)

The demarcation between darker dorsal and paler ventral surfaces is apparent in about 20% of people with dark skin. Pigmentary demarcation lines in the upper limb that correspond to the ventral axial line (see previous section) were first described by Voigt and Futcher. The lines are normal variants in skin colour that may appear pathological.

Blaschko lines

These lines seem to correspond to the pathways followed by keratinocytes migrating from the neural crest during embryogenesis. Dermatoses following Blaschko lines suggest keratinocyte mosaicism (see ➔ pp. 620–1, Fig. 32.1, p. 621, and inside back cover). The lines follow a V shape over the spine, an S shape on the front and sides of the trunk that is unlike dermatomes, and a linear pattern on the limbs.

A number of inflammatory conditions can follow Blaschko lines, including lichen striatus (self-limiting inflammatory rash of childhood), linear LP, inflammatory linear verrucous epidermal naevus (ILVEN), and blashkitis (Blaschko dermatitis)—a rare, self-limiting linear or whorled itching erythematous papulovesicular eruption on the trunk and limbs that may be the adult equivalent of lichen striatus. Rare inherited conditions with cutaneous features following Blaschko lines include incontinentia pigmenti and Goltz syndrome (see ➔ pp. 636–7).

Embryonic cleft

Congenital anomalies in the midline may be associated with the lines of fusion between embryonic tissues.

Wallace line

This demarcation line marks the anatomical boundary on the lateral aspect of the palms and soles where the glabrous (i.e. skin devoid of hair) plantar or palmar skin meets hair-bearing dorsal skin. Some inflammatory problems involving the palms or soles, such as LP, pompholyx (vesicular eczema), or the erythematous rash of Kawasaki disease display a sharp cut-off at the Wallace line.

Langer lines of cleavage

see ➔ p. 8.

Further reading

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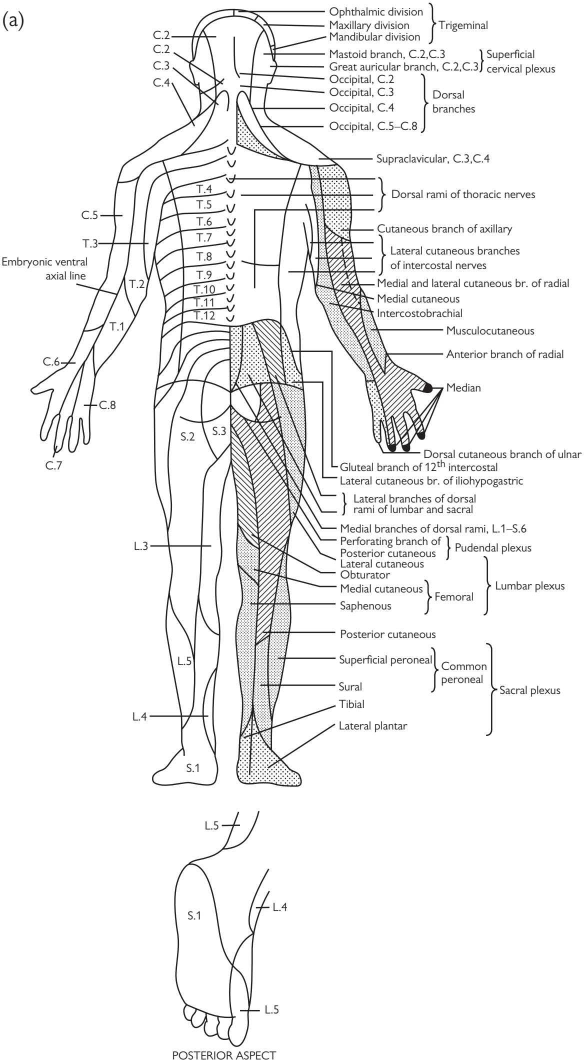

Fig. 4.2 Dermatomes. Reproduced with permission from Longmore, Wilkinson, and Rajagopalan *Oxford Handbook of Clinical Medicine*, 6th edn, 2004, Oxford: Oxford University Press.

(b)

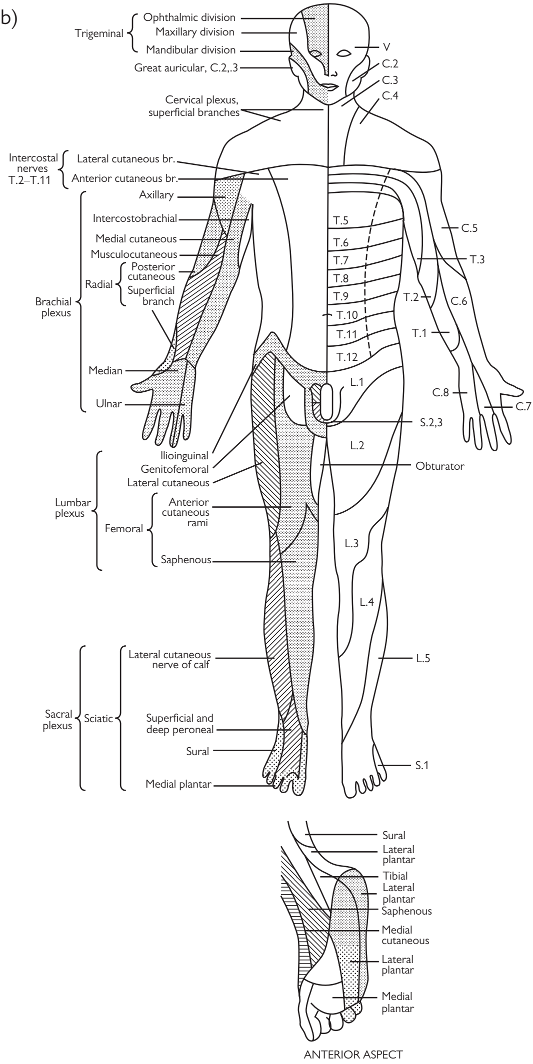

Fig. 4.2 (*Contd.*)

Scaly or hyperkeratotic red rashes

Erythematous scaly papules or plaques

Scale suggests the pathology involves the epidermis. Scratch the skin with your fingernail to reveal scaling. Consider:

Relatively localized

- Lichen simplex: localized itchy patch. The skin is thickened (see ➔ p. 222).
- Varicose (gravitational, venous stasis) eczema: usually not terribly itchy. A single patch may develop over a varicosity—stand the patient up to see the dilated vein (see ➔ p. 220).
- Fungal infection: usually asymmetrical and often annular. Scale may be lost if topical corticosteroids have been applied. Not terribly itchy (see ➔ p. 164).
- Herald patch of pityriasis rosea (see ➔ p. 151).
- Irritant or allergic contact dermatitis (see ➔ pp. 214–16).
- Bowen disease: not itchy (see ➔ pp. 348–9).
- Red, oozing nipple: atopic eczema or Paget disease (see ➔ p. 360).
- Asymmetric periflexural exanthem of childhood: papular scaly rash, more obvious on one side of the body.

Relatively widespread

(see ➔ Box 4.5.)

- Scabies: itchy papules, nodules, and excoriations (see ➔ p. 172).
- Eczema: endogenous (symmetrical) (see ➔ Chapter 10, pp. 211–24):
  - Atopic: very itchy.
  - Seborrhoeic: scaly, greasy-looking erythema with relatively little itch. Signs may overlap with those of psoriasis. Look for diffuse scale 'dandruff' in the scalp, flaky eyebrows, central facial erythema (nasolabial folds), flexural erythema, and scaly papules in the centre of the chest and back. May have an annular pattern.
  - Discoid (nummular): generally very itchy, and often crusted and weeping.
  - Asteatotic (dry skin): itch is variable. Often limited to the shins.
- Psoriasis: the amount of itch varies (see ➔ pp. 189–208).
- Pityriasis rosea (see ➔ p. 151) or pityriasis versicolor (often pigmented or hypopigmented) (see ➔ pp. 168–9).
- Drug reaction (see ➔ p. 374).
- 2° syphilis (see ➔ p. 146).
- Photodermatitis: uncommon (see ➔ p. 324).
- Cutaneous T-cell lymphoma (CTCL): uncommon (see ➔ pp. 358–9).

Scaly annular erythematous lesions

Look for scale, either following the erythema or at the expanding edge of the ring. Consider:

- Dermatophyte infection (tinea, ringworm) (see ➔ p. 164). Usually localized. (Scale may be lost if topical corticosteroids have been applied.)
- Pityriasis rosea (see ➔ p. 151).

- Seborrhoeic dermatitis (central trunk, flexures, face, scalp) (see ➔ p. 219).
- Psoriasis (see ➔ pp. 189–208) or rarely CTCL (see ➔ pp. 358–9).
- Subacute cutaneous LE: remember drugs as a cause. The patient is photosensitive, but paradoxically the face is often spared (see ➔ p. 402).
- Annular erythema: uncommon—a few lesions (see ➔ pp. 238–9).

Hyperkeratotic papules and plaques

Rough hyperkeratosis, a sign of a thick horny layer, is not removed as easily as scale by scratching. Is the rash follicular? Consider:

Relatively localized

- Tumours: viral wart, corns (pressure sites), seborrhoeic wart (may be numerous), actinic (solar) keratosis (sun-exposed skin), keratoacanthoma, SCC (sun-exposed skin) (see ➔ Chapter 17, pp. 341–63).
- Keratosis pilaris: perifollicular rash most often on the upper lateral arms and thighs, giving the skin a dry rough texture (see ➔ Box 31.11, p. 603).
- Lichen simplex: itchy nodules or plaques. On the legs may resemble LP—see next bullet point (see ➔ p. 222).
- Hypertrophic LP: itchy purplish nodules or plaques that may become very thickened (hypertrophic) on the leg. Resembles lichen simplex. Look for LP at other sites, including buccal mucosa (see ➔ p. 224).
- DLE: look for scars, follicular plugging, loss of hair (see ➔ p. 404). May only be one or two plaques.

More widespread

- Chronic psoriasis: occasionally very hyperkeratotic, particularly on the palms and soles (see ➔ pp. 189–208). Reiter syndrome, which is pustular and hyperkeratotic, is probably a variant of psoriasis (see ➔ p. 208). Usually very symmetrical.
- Follicular eczema.
- Follicular hyperkeratosis: uncommon. May be caused by some inflammatory diseases that target the hair follicle, including LP, hereditary conditions, and nutritional deficiency (phrynoderma) (see ➔ p. 512).

Box 4.5 Erythroderma

⚠ If the patient is red and scaly all over, and this has developed suddenly, the patient may be very unwell. Skin failure is a dermatological emergency (see ➔ p. 108).

Consider:

- Eczema, particularly atopic or seborrhoeic.
- Psoriasis.
- Drug eruption.
- Cutaneous T-cell lymphoma.

Smooth erythematous rashes

Smooth erythematous patches, papules, or plaques

Lack of scale suggests the pathology is predominantly dermal, but the signs may have been modified by treatment. Consider:

Relatively localized

(Also see ➔ Red faces, p. 72.)

- Insect bites or papular urticaria (an urticarial reaction to bites).
- Cellulitis: asymmetrical distribution. Hot, tender, erythematous, and swollen. May blister (see ➔ pp. 138–9 and pp. 234–5).
- Acute contact dermatitis: may be smooth and oedematous (sometimes blisters), rather than scaly. Skin may be tender and itchy (see ➔ pp. 214–16).
- Lipodermatosclerosis: tender, indurated erythema on the leg (see ➔ p. 308; for eosinophilic fasciitis, see ➔ p. 421).
- Necrotizing fasciitis: rapidly spreading, poorly demarcated purplish erythema in an ill patient with high temperature, tachycardia, and low blood pressure (see ➔ pp. 126–7).
- Treated or flexural psoriasis: well-demarcated shiny erythema.
- Treated eczema also loses scale.
- Erythema migrans (not always annular) (see ➔ p. 148).
- Carcinoma erysipeloides (spreading erythema on the breast) (see ➔ pp. 360–1).
- Very rarely, an erythematous patch or plaque may overlie a tumour of the bone (plasmacytoma, sarcoma).

More generalized

- Urticaria: crops of itchy erythematous papules or plaques (wheals) that may evolve into rings with normal central skin. Wheals last no more than 48h. The skin is normal when wheals fade. May be associated with angio-oedema (deep swelling). No blisters. Stroke the skin with a spatula to test for dermographism (a physical urticaria) (see ➔ pp. 230–1 and p. 372).
- Morbilliform or anaphylactoid drug eruption: symmetrical (see ➔ p. 372).
- Viral exanthem: symmetrical.
- DRESS (see ➔ p. 122–3): papules coalesce into plaques. May have angio-oedema. Toxic shock syndrome: may be mild or rapidly fatal (see ➔ Box 5.2, p. 101).
- Erythema multiforme (EM): papules that last at least 7 days. May spread into rings with a dusky or blistered centre (target lesions). May have mucosal ulcers (see ➔ p. 111).
- Prodrome of BP: itchy, erythematous, urticated papules. Look for blisters (see ➔ pp. 270–1).
- Scleredema (rare): indurated erythematous skin (see ➔ Box 22.2, p. 467). Eosinophilic cellulitis (Wells syndrome): rare. Painful itchy, erythematous oedematous plaques and nodules. Resembles cellulitis. Tissue and usually circulating eosinophilia.

Smooth annular erythematous lesions

Without scale. Consider:

- Urticaria: transient and fades to leave normal skin. Common.
- BCC (central ulceration): common (see ➔ p. 350).
- Granuloma annulare: fairly common (see ➔ Box 22.3, p. 469).
- LP (little scale): often penis or axillary (see ➔ p. 224).
- Sarcoid (may have some scale). Diascopy reveals yellow-brown 'apple jelly' nodules (see ➔ pp. 524–5). Uncommon.
- Erythema migrans (see ➔ Lyme disease, p. 148): uncommon.

Livedo reticularis: a net-like erythema often on the legs

- Physiological reticulated erythema is continuous and disappears when the skin is warmed. Differentiate from the fixed brownish net-like discoloration that develops after prolonged exposure to external heat—erythema ab igne (see ➔ Fig. 20.5, p. 454).
- Discontinuous reticulate erythema that does not disappear on warming. May be purpuric (retiform purpura). Consider:
  - Occlusive vasculopathy caused by cholesterol emboli or thrombi, e.g. in antiphospholipid syndrome(see ➔ Box 20.10, p. 453, p. 312, p. 314). Look for scars with a telangiectatic margin.
  - Vasculitis affecting medium and/or large vessels (see ➔ pp. 450–1). Look for purpuric nodules or ulcers with a purpuric rim.

Purpura and telangiectasia

Purpura

Exclude trauma, the commonest cause. Is the patient taking aspirin, another platelet inhibitor, or an anticoagulant that might potentiate bleeding into the skin? For discussion of purpura in sick patients, see ➲ pp. 102–3.

Flat (macular) purpura

Leakage of red blood cells (RBCs) without inflammation of vessel walls. Consider:

- Legs: non-specific, associated with inflammatory rashes, e.g. stasis eczema, psoriasis.
- Forearms: minor trauma to sun-damaged atrophic skin with poorly supported vessels, particularly in elderly patients. Also common in patients with atrophic skin 2° to prolonged treatment with oral or very potent topical corticosteroids.
- Scurvy: perifollicular purpura, corkscrew hairs, poor wound healing (see ➲ p. 513).
- Thrombocytopenia or hyperglobulinaemic purpura (see ➲ p. 539).
- Palmoplantar petechiae or purpuric macules in dermatitis herpetiformis (DH) (see ➲ p. 274).

Palpable purpura

- Purpuric papules: small-vessel cutaneous vasculitis, with or without systemic disease (see ➲ p. 440).
- Urticarial vasculitis: erythematous wheals become purpuric and fade to leave bruising (see ➲ p. 443).

Purpura, livedo reticularis, and nodules

- Cholesterol emboli: asymmetrical acral petechiae, subcutaneous nodules, and livedo reticularis (see ➲ p. 452).
- Antiphospholipid syndrome (see ➲ Box 20.10, p. 453).
- Necrotizing vasculitis affecting deeper cutaneous vessels (see ➲ pp. 450–1).
- Extensive irregularly outlined ecchymoses, bullae, and gangrene: DIC with purpura fulminans (see ➲ pp. 102–3).
- Erythematous purpuric plaques with livedo and ulceration at the site of application of ice packs (used to relieve chronic pain). Uncommon.

Telangiectasia (small dilated cutaneous vessels)

- Sun damage (face), rosacea (face) (see Fig. 4.3).
- Oestrogen-related: liver disease, pregnancy, exogenous oestrogens.
- Prolonged use of potent topical corticosteroids.
- Unilateral naevoid telangiectasia: congenital or acquired. Usually in trigeminal or upper cervical dermatomes.
- General essential telangiectasia: usually legs of women.
- Cutaneous collagenous vasculopathy: symmetrical, starts on legs.
- Hereditary haemorrhagic telangiectasia (see ➲ Other rare gastrointestinal conditions, p. 522).

- Mucocutaneous and gastrointestinal (GI) telangiectasia: visible on mucous membranes in adolescence.
- Diffuse or limited cutaneous systemic sclerosis: matt telangiectasia on the face, hands, and/or lips (see ➔ p. 414).
- Poikiloderma, e.g. in chronic dermatomyositis, chronic GVHD, mycosis fungoides (MF).
- Telangiectasia macularis eruptiva perstans (mastocytosis) (see ➔ Box 31.5, p. 593).
- Carcinoid syndrome: rare (see ➔ pp. 520–1).
- Genodermatoses (rare), e.g. Klippel–Trenaunay syndrome, ataxia telangiectasia, xeroderma pigmentosum (XP), Goltz syndrome.

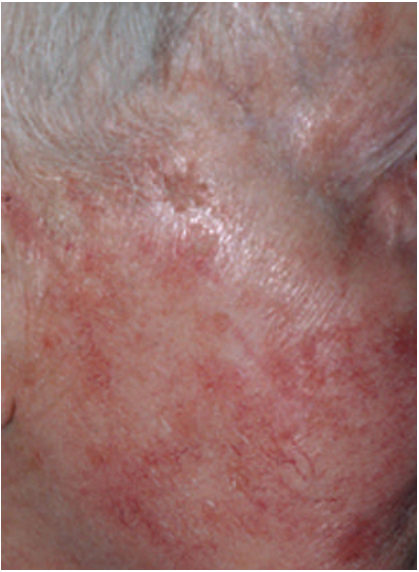

Fig. 4.3 Telangiectasia and solar keratoses in sun-damaged skin.

Red faces

A red face may indicate photosensitivity, but consider other causes:

- *Seborrhoeic dermatitis*: scaly, greasy-looking erythema with little itch. Tends to involve the nasolabial folds, centre of the forehead, and skin behind the ears, as well as other sites (see ➔ p. 219).
- *Cellulitis or erysipelas*: is there fever or malaise? Look for swelling and tenderness, as well as erythema. The rash is often asymmetrical.
- *Contact dermatitis*: is the skin itchy or scaly? Is the face swollen? Any vesicles? Might this be an acute contact dermatitis (irritant or allergic) to a cosmetic? Remember that the product may have been used for some time before the patient is sensitized (see ➔ pp. 214–16).
- *Acne*: pustules, papules, comedones, nodules, scars (see ➔ pp. 242–7).
- *Rosacea*: pustules, as well as telangiectasia, erythema, and swelling, but some patients have much more of one component than another. Comedones are not a feature of rosacea (see ➔ pp. 254–5, and Box 12.3, p. 247). Morbihan disease (late-stage rosacea)—look for persistent erythema and solid oedema of upper 2/3 of the face.
- *Herpes simplex or herpes zoster*: any prodrome of tingling or discomfort? Look for vesicles progressing to small pustules. Herpes zoster presents in a dermatomal distribution.
- *Corticosteroid-induced acneiform rash*: prolonged application of potent topical corticosteroids to the face can produce a telangiectatic rosacea-like eruption with pustules, but no comedones.
- *Angio-oedema*: usually transient. Swelling more marked than erythema. Mucous membranes may be affected (see ➔ p. 104). Subcutaneous emphysema (rare complication of dental treatment) may cause periorbital swelling, mimicking angio-oedema. Check for crepitation.
- *Photosensitivity*: what is the distribution of the redness? Check the eyelids, under the hair, behind the ears, and under the chin. Compare exposed and covered sites (see Box 4.6 and ➔ p. 28 and p. 324).
- *Flushing*: if repeated, may become fixed. Investigation is only required if flushing is of sudden onset and associated with systemic symptoms (see ➔ Box 24.7, p. 521). Ask about triggers. Consider:
  - Blushing: flushing triggered by emotions.
  - Physiological flushing after exercise, heat, hot drinks, alcohol.
  - Menopausal hot flushes (flashes).
  - Rosacea: labile flushing progresses to fixed oedematous erythema, telangiectasia, and pustules (see ➔ pp. 254–5).
  - Carcinoid syndrome (very rare): look for rosacea-like vascular changes, oedema and induration of the face, wheezing, severe diarrhoea, hypotension and tachycardia, features of pellagra (scaly sun-exposed skin, glossitis, angular stomatitis) (see ➔ pp. 520–1).
- *Superior vena cava obstruction*: the face is swollen and red. Look for dilated veins on the neck, chest, and arms. Symptoms include dyspnoea, cough, and headache. Associated with malignancy (bronchial carcinoma, lymphoma) or, less often, thrombosis, e.g. after pacemaker insertion.
- *Other*: cutaneous LE, dermatomyositis.

Box 4.6 Is the patient photosensitive?

What should I look for?

The signs will depend on the cause of photosensitivity but may include erythema, urticaria, papules, blisters, and/or pigmentation. Look for a rash that is maximal on exposed sites and spares covered skin. The skin may be normal between episodes. Chronic sun exposure may induce tolerance, so-called 'hardening', and so, paradoxically, the face may be spared. Drugs are a common cause of photosensitivity (see ➔ Box 16.3, p. 327).

Photosensitive rashes may involve one or more of the following sites:

- The prominences of the forehead, nose, cheeks, particularly over the cheekbones (malar), and chin.
- The back of the neck, if not covered by hair.
- The helix of the ear, if not covered by hair.
- The V on the upper chest where the shirt sits open.
- The extensor surfaces of forearms, the back of the hands.
- The nails: tender onycholysis with a rim of pigmentation (photo-onycholysis) (associated with some drug-induced photosensitivity).
- The dorsum of the feet (if exposed).

Are there signs of chronic sun damage such as freckling on exposed skin (photoageing) or skin cancer (see ➔ p. 43)? Are these signs appropriate for the age and lifestyle of the patient or greater than you would expect?

Is there sparing of covered skin, with a sharp cut-off between involved and uninvolved skin? Check:

- Around the orbit (ask the patient to close his/her eyes to see the eyelids).
- Under the frame of spectacles.
- The upper lip.
- Under the nose.
- Deep wrinkles on the face.
- Below the chin.
- Behind the ears.
- Under the hair—on the forehead or at the back of the neck. Is a spared area exposed, because the hair has been cut recently?
- Under watch straps.
- The distal phalanges, because the hands often sit with fingers curled.
- Between the fingers.
- Under the shoe.

Red legs and leg ulcers

Red leg(s)

Erythematous

- Cellulitis (unlikely to be bilateral): hot, oedematous, and tender.
- Lipodermatosclerosis: often bilateral. Painful, indurated erythema that may simulate cellulitis in acute stages.
- Gravitational eczema: associated with venous stasis and oedema. Stand the patient up to look for varicose veins. Often bilateral.
- Pseudo-Kaposi sarcoma (acroangiodermatitis): purplish patches, papules, and plaques in association with chronic venous stasis (also seen in arteriovenous (AV) malformations or AV fistulae).
- Acute allergic or irritant contact dermatitis (may be bilateral): oedematous, erythematous, and may blister (see ➔ Fig. 10.1, p. 216). Itchy, but also uncomfortable if swollen.
- Deep venous thrombosis (DVT): unilateral swelling with erythema and calf tenderness.
- Lymphoedema: initially erythematous and pitting oedema. Later indurated with thick hyperkeratotic skin. Predisposes to cellulitis (see ➔ pp. 316–17).
- Erythema nodosum (EN): tender erythematous nodules on the shins (see ➔ pp. 458–9).
- Necrobiosis lipoidica: well-defined reddish yellow plaques with telangiectasia, an atrophic centre (that may ulcerate), and a raised erythematous rim (see ➔ p. 468). Associated with insulin-dependent diabetes.
- Livedo reticularis: net-like mottled purplish pattern (see ➔ p. 452, and Fig. 3.2, p. 41). Patients may have leg ulcers and nodular vasculitis or an obstructive vasculopathy (see ➔ p. 452, p. 312, and p. 314). May be purpuric in places (retiform purpura) (see ➔ p. 452). Differentiate from erythema ab igne—fixed brownish reticulate marking where the skin has been repeatedly exposed to heat (see ➔ p. 452).
- Erythermalgia (erythromelalgia erythralgia): painful burning erythema. May be idiopathic, associated with thrombocythaemia, or familial. Involves acral sites (see ➔ p. 551).

Purpuric

- Some flat purpura is not uncommon in inflammatory rashes on legs, e.g. psoriasis, eczema, simply because of leakage of RBCs.
- Capillaritis (pigmented purpura) produces flat purpuric spots (likened to cayenne pepper) and brownish yellow (haemosiderin) pigmentation but does not signify systemic disease. Capillaritis is commonest on the legs. The cause is unknown (rarely, drugs cause capillaritis). Lymphocytic inflammation around capillaries is associated with extravasation of RBCs into the skin.
- Palpable purpura. If purpura is palpable, consider small-vessel cutaneous vasculitis (usually bilateral), and exclude systemic disease (see ➔ p. 440).

- Retiform purpura (purpura in a livedo pattern): seen with occlusive vasculopathies, including cholesterol emboli, antiphospholipid syndrome, deficiency of protein C or protein S, cryoglobulinaemia (see ➔ p. 452).

Chronic leg ulcers

The history and examination should help you to pinpoint the cause of the chronic ulcer. More details are given on ➔ pp. 294–5. Consider:

Common

- Venous ulcers: risk factors include DVT, trauma, operations to hips or knees, obesity, and immobility.
- Arterial disease: look for intermittent claudication, rest pain, night pain, pain worse when the leg is elevated. Occurs in tobacco smokers.
- Mixed AV: symptoms of both venous and arterial disease.
- Neuropathic: develop at sites of pressure—painless ulcers. Commonest on the feet of diabetics.
- Malignancy: do not accept the label 'leg ulcer' without looking under the dressing. BCCs are often misdiagnosed as leg ulcers.

Less common

- Pyoderma gangrenosum (PG): rapidly growing, painful ulcer with an undermined bluish margin. May be associated with rheumatoid arthritis (RA), inflammatory bowel disease (IBD), or myeloproliferative disorders, but 50% of patients have no underlying disease (see ➔ p. 310). Nicorandil-induced ulcers may simulate PG.
- Vasculitis: is the rim purpuric? Associated with connective tissue diseases, particularly RA (see ➔ Box 19.1, p. 395).
- Occlusive vasculopathy (see ➔ pp. 314–15 and p. 452): suggested by the presence of livedo reticularis. Causes include inherited disorders of coagulation—mutations in protein C, protein S, antithrombin III, fibrinogen, or factor V genes.
- Necrobiosis lipoidica: associated with insulin-dependent diabetes (see ➔ p. 468).
- Infection, e.g. deep fungal, treponemal (syphilitic gumma).

Hands, feet, and other extremities

The skin is cool at acral sites such as the hands, feet, ears, and tip of the nose. These sites are also exposed to sun and trauma. Conditions that tend to favour acral sites include:

- Scabies: crusting between fingers and burrows on palms. Blisters on the palms and soles in children (see ➜ p. 172).
- Some infections (see Box 4.7).
- Contact dermatitis (irritant or allergic) is common on hands.
- Chilblains (perniosis): warm erythematous nodules that may become purplish. Itch or cause burning pain. Occur in cold weather in damp temperate climates. Seen most often in the young, especially children, and elderly, but less frequent now homes are centrally heated. Self-limiting. May be associated with acrocyanosis—see later in list.
- Sarcoidosis: may involve the tip of the nose—dusky purplish discoloration and swelling (lupus pernio) (see ➜ p. 524).
- Chronic cutaneous LE: hyperkeratotic papules on fingers (chilblain lupus) or photoaggravated disease on the nose (see ➜ p. 404).
- Raynaud phenomenon (sudden pallor, followed by cyanosis, and finally erythema with swelling and tingling): may affect fingers, toes, nose, and/or earlobes (see ➜ Box 19.18, p. 414, p. 444).
- Acrocyanosis: persistent dusky, mottled discoloration of hands and feet. May be bright red when very cold. Associated with chilblains (see ➜ pp. 444–5).
- Erythromelalgia (erythermalgia, erythralgia): attacks of painful burning erythema affecting feet, legs, and, less often, hands (see ➜ p. 551). May be idiopathic, associated with thrombocythaemia, or familial. Sometimes also associated with acrocyanosis.
- Chemotherapy-induced hand–foot skin reaction (see ➜ p. 382).
- EM is associated with erythematous papules and target lesions on palms and soles. Check mucosae (see ➜ p. 111).
- Acute GVHD: macular blotchy erythema of palms, soles, and face; a common early sign (see ➜ p. 532).
- In systemic sclerosis, fingers are tight and puffy (sclerodactyly)—look for calcinosis and vascular changes in nail folds (see ➜ p. 414).
- Cutaneous vasculitis and occlusive vasculopathy favour cool sites where blood flow is slow—check toes (see ➜ p. 440, p. 452).
- Neutrophilic dermatosis of dorsal hands (see ➜ pp. 518–9).
- Cryoglobulinaemia or cryofibrinogenaemia: mottling, retiform purpura, ulceration, or blotchy cyanosis at acral sites exposed to cold—helices of the ears, as well as fingers and toes. May be associated with Raynaud phenomenon and livedo reticularis (see ➜ pp. 444–5).
- Cholesterol embolus: blue toe or peripheral gangrene associated with livedo reticularis. May have retiform purpura. Seen when arterial catheterization disrupts an atheromatous plaque or after prolonged anticoagulation when a clot becomes friable (see ➜ p. 452).
- Palmoplantar pustulosis: sterile pustules on palms and/or soles (see ➜ p. 198); also see ➜ Zinc deficiency, p. 513).
- In porphyria cutanea tarda (PCT), blisters and erosions on the dorsum of hands (trauma and the skin is fragile) (see ➜ p. 332).

Box 4.7 Infections that may produce acral signs

- Herpes simplex causing a whitlow on a finger (see ➔ p. 152).
- Orf or milker's nodule on a finger (see ➔ pp. 158–9).
- Hand–foot–mouth disease associated with infections by Coxsackie viruses A10 and A16 and with enterovirus 71. Vesicopustules on hands and feet, as well as mucosal ulcers.
- Gianotti–Crosti syndrome (papular acrodermatitis of childhood) (see ➔ p. 150): flat-topped, brownish pink papules or papulovesicles on hands and feet. Most often associated with hepatitis B and Epstein–Barr viral infections in children.
- Purpuric rash in a gloves-and-socks distribution caused by human parvovirus B19 (see ➔ p. 150).
- 2° syphilis: copper-coloured macules on palms (see ➔ p. 146).
- Atypical mycobacterial infection with nodule on finger (see ➔ p. 144).
- Mycobacterial infections favour the nose and ears: leprosy, cutaneous tuberculosis (TB) (lupus vulgaris) (see ➔ p. 144).

Flexural rashes

Intertrigo is another name for a flexural rash but is not a diagnosis. Consider these causes of intertrigo.

Common

- *Candidiasis*: look for a bright red erythema and satellite pustules with a collarette of scale. The rash is often asymmetrical, and the skin is sore, rather than itchy. Candidiasis is commoner in obese patients, those with diabetes, and the immunocompromised. Take swabs to exclude a streptococcal infection. May complicate an irritant dermatitis (see ➔ p. 166).
- *Dermatophyte infection*: the rash gradually spreads out from the flexure. Look for scale at the edge of the erythematous patch. Tinea cruris (groin) is common. Infection is often asymmetrical. Take a scraping from the scaly margin to confirm the diagnosis (see ➔ p. 164).
- *Seborrhoeic dermatitis*: look for 'dandruff' in the scalp (ill-defined diffuse scaling) and behind the ears, or erythematous, greasy, scaly plaques in the centre of the face, centre of the chest, or centre of the back. The features overlap with those of flexural psoriasis (sebopsoriasis). Look for signs of psoriasis in the nails, e.g. pitting, onycholysis, or subungual hyperkeratosis (see ➔ p. 219).
- *Flexural psoriasis*: presents as a well-demarcated, shiny erythema. Skin is moist, rather than scaly. Is there plaque psoriasis on extensors? Check for pitting of the nails or onycholysis, and look in the scalp for scaling plaques. Any family history of psoriasis? (see ➔ p. 194.).
- *Irritant contact dermatitis*: in obese, sweaty, or incontinent patients (see ➔ p. 214).
- *Erythrasma*: look for symmetrical orange-brown, slightly scaly plaques in the flexures. The skin fluoresces coral pink under Wood light (UVA). Caused by infection with *Corynebacterium minutissimum*. Commoner in warm humid climates, diabetes, obesity, and old age (see ➔ p. 132).
- *Streptococcal infection*: bright red, painful perianal rash.

Less common

- *Allergic contact dermatitis*, e.g. to deodorants, clothing dyes: eczema involves the edge of axillae and spares the axillary vault (see ➔ p. 214).
- *Pseudomonas infection*: brownish, scaly rash with painful fissuring. Potentiated by washes with antibacterial lotions (see ➔ p. 132).
- *Hidradenitis suppurativa*: patients present with a history of recurrent 'boils'. Look for comedones, inflamed nodules, pustules, sinuses, and scars (see ➔ p. 250).
- *Acanthosis nigricans*: velvety, hyperpigmented thickening of flexural skin, associated with skin tags. Seen most often in obese Asian patients with insulin resistance ('pseudoacanthosis nigricans') (see ➔ p. 466–7). Presents rarely as a manifestation of an underlying malignancy, usually adenocarcinoma of the stomach (see ➔ p. 546).

Pustules

Pustules are small dome-shaped bumps filled with milky fluid containing neutrophils or sometimes eosinophils. Pustules are often caused by bacterial infection of a hair follicle (folliculitis) but may be sterile. Decide if pustules are arising in association with hair follicles (perifollicular pustules). Vesicles (small blisters that contain clear fluid) may evolve into pustules (vesicopustules) or may simulate pustules, when on palms and soles and covered by a thick horny layer. Puncturing the vesicle to let out some fluid may help, if you are not sure if it is clear or milky.

Common

- Bacterial folliculitis (usually staphylococcal): may be a 2° bacterial folliculitis in conditions such as atopic eczema (see pp. 134–5).
- Bullous impetigo: superficial vesicles and pustules rapidly rupture, leaving golden crusting (see p. 136).
- *Candida*: satellite pustules at the margin of the erythematous flexural rash (see p. 166).
- Irritant or occlusive folliculitis, e.g. with thick applications of greasy emollients (see Box 6.3, p. 135).
- Acne: follicular lesions with comedones, inflammatory papules, pustules, nodules, cysts, and scars. Involves the face and may involve the trunk (see p. 242–3).
- Herpes simplex virus, including eczema herpeticum; herpes zoster virus infection (dermatomal): vesicles (the 1° lesion) evolve into pustules or vesicopustules (see pp. 152–7).
- Periorifical 'dermatitis': pustulovesicular facial rash, most often caused by potent topical corticosteroids (see pp. 256–7).
- Rosacea: look for erythema, telangiectasia, papules, and oedema (see p. 254). No comedones. Involves the face but spares the trunk.
- Steroid-induced acne: look for sterile pustules without comedones.

Less common

- Hidradenitis suppurativa: pustules in flexures with grouped comedones, nodules, scars, and sinuses (see p. 250).
- Chronic palmoplantar pustulosis (sterile non-follicular pustules) (see p. 198).
- PG may be preceded by a tender pustule (see p. 310).
- Demodicosis in immunocompromised (see p. 178).

Rare

- Bowel-associated dermatosis–arthritis syndrome or Sweet syndrome (see p. 519).
- AGEP: sterile non-follicular pustules (see p. 124).
- DRESS: morbilliform rash with 'juicy' papules that may coalesce into plaques. Sometimes sterile pustules or vesicles (see pp. 122–3).
- Pustular psoriasis: waves of non-follicular pustules appear at the edge of erythematous tender plaques or patches. Sick patient with fever and neutrophilia (see pp. 124–5 and p. 198).
- Subcorneal pustular dermatosis (see Box 13.4, p. 269).
- Behçet disease (see pp. 284–5).

Blisters

A blister is a bump, which may be large (bulla) or small (vesicle), filled with clear fluid (serum). Blisters may form just below the stratum corneum, within the epidermis, or within or below the dermo-epidermal junction (see ➔ Fig. 13.1, p. 261). Causes of blisters include:

Fairly localized blisters

Common

- Physical causes:
  - Friction, pressure, ischaemia, scalds or cold injury, sunlight, radiation.
- Cutaneous infections:
  - Viral: herpes simplex or varicella-zoster (shingles); Coxsackie virus A16 (hand–foot–mouth disease).
  - Bacterial: bullous impetigo caused by *Staphylococcus aureus*; blisters in association with oedema 2° to streptococcal infection (cellulitis, erysipelas, and necrotizing fasciitis).
  - Fungal: dermatophyte.
- Insect bites and stings.
- Scabies (vesicles or vesicopustules on the palms or soles in infants).
- Contact dermatitis: irritant or allergic, e.g. nickel sensitivity (see ➔ Fig.10.1, p. 216).
- Pompholyx (itchy vesicles on hands and feet) (see ➔ p. 221).
- Photosensitivity: drug-induced and phytophotodermatitis (see ➔ p. 326, and Box 16.4, p. 327).
- Oedema blisters: sudden peripheral oedema in elderly patients with atrophic skin may cause blisters on the legs.

Uncommon

- Diabetic blister: on legs (see ➔ pp. 466–7).
- Damage to basal cells (lichenoid reaction):
  - EM (see ➔ Chapter 5, p. 111).
  - Fixed drug eruption (see ➔ Chapter 18, p. 378).
- PCT or pseudoporphyria on back of hands (see ➔ pp. 332–4).
- Mastocytoma: a localized increase in mast cells in the skin—seen in infants (see ➔ Box 31.5, p. 593).
- Coma blisters: originally described with barbiturates (see ➔ p. 379).

Rare

- Genetic:
  - EB simplex (see ➔ p. 624–5).
  - Incontinentia pigmenti (see ➔ pp. 636–7).
  - Acrodermatitis enteropathica (see ➔ p. 513).

More widespread blisters

Common

- Eczema:
  - Contact eczema may generalize after starting in a localized pattern.
  - Atopic eczema (exclude infection if you see blistering in atopic eczema).
- Cutaneous infections:
  - Viral: herpes simplex (may generalize in conditions such as atopic eczema) or varicella-zoster (chickenpox); poxvirus (vaccinia).
  - Bacterial: widespread infection in atopic eczema.
- Miliaria (caused by blockage of eccrine ducts and sweating in a hot, humid environment; the itchy vesicles may be superficial or deep).

Uncommon

- Cutaneous infections:
  - Bacterial: SSSS caused by a circulating exotoxin produced by *S. aureus* (see ➲ Fig. 6.2, p. 131 and pp. 136–7).
- Damage to basal cells (lichenoid reaction):
  - SJS (see ➲ p. 116).
  - TEN with full-thickness necrosis of the epidermis (see ➲ p. 116).
  - LP (see ➲ p. 224).
  - Vesiculobullous cutaneous LE (see ➲ Box 19.9, p. 405).
- Autoimmune blistering diseases:
  - BP (commonest of the autoimmune blistering diseases in Western Europe) (see ➲ pp. 270–1).
  - Pemphigus foliaceus and vulgaris (see ➲ pp. 268–9).
  - DH (see ➲ pp. 274–5).
- Bullous cutaneous vasculitis (bullae will be haemorrhagic) (see ➲ p. 440).
- Urticaria pigmentosa (increased mast cells in the skin) (see ➲ Box 31.5, p. 593).
- PCT (sporadic commoner than inherited):
  - Associated with iron overload (haemochromatosis), alcohol, oestrogens, and hepatitis C (see ➲ p. 332–3).

Rare

- Autoimmune blistering diseases, including:
  - Paraneoplastic pemphigus (see ➲ Box 13.4, p. 269).
  - Mucous membrane (cicatricial) pemphigoid (see ➲ p. 272).
  - Pemphigoid gestationis (see ➲ p. 584).
  - Epidermolysis bullosa acquisita (EBA) (see ➲ p. 272).
  - Linear IgA dermatosis (see ➲ p. 272).
- Genetic:
  - Hailey–Hailey disease (see ➲ Box 32.4, p. 629).
  - EB group of diseases (see ➲ p. 624–5).
  - Epidermolytic ichthyosis (see ➲ Box 32.2, p. 627)
  - Cutaneous porphyrias associated with subepidermal blistering:
    —Variegate porphyria.
    —Hereditary coproporphyria.
    —Congenital erythropoietic porphyria.

Nodules and solitary cutaneous ulcers

Nodules suggest involvement of the dermis and sometimes subcutaneous tissues. They may be tender or painless and sometimes ulcerate.

Nodules

- Cysts: epidermoid ('sebaceous' with a punctum—face, trunk, scalp), pilar (no punctum—scalp), dermoid (congenital and usually in midline on the face), acne (inflammatory on the face).
- Thrombophlebitis: tender nodules along a vein.
- Prurigo nodularis: excoriated dome-shaped nodules (see ➔ p. 222).
- Granulomatous, e.g. foreign body with granulomatous reaction, sarcoidosis, rheumatoid nodules (see ➔ pp. 394–5).
- Gouty tophi: hands, elbows, ears (see ➔ Box 19.2, p. 395).
- Calcinosis cutis: hard, white nodule that extrudes chalk-like material. Usually hands or pressure sites. May occur in systemic sclerosis (see ➔ pp. 416–17 and Fig. 19.11, p. 419).
- Tumour (see ➔ pp. 42–3, Chapter 17, pp. 341–63, and Box 4.8), including:
  - BCC, SCC, melanoma.
  - Lipoma, muscle tumour, neural tumour.
  - Vascular tumour.
  - Appendageal tumours.
  - Lymphoma (T-cell or B-cell).
  - Metastatic carcinoma (breast, renal, colon, lung, ovary, gastric).
  - Kaposi sarcoma (KS).
- Infection, including:
  - Furuncle ('boil'): hot, tender, erythematous.
  - Sporotrichosis or atypical mycobacterial infection: nodules spread proximally along a lymphatic.
  - Leprosy, deep fungal infection.
- Xanthoma: elbows, knees, ankles, hands. Yellowish tinge (see ➔ p. 470).
- Panniculitis, e.g. erythema nodosum, traumatic panniculitis, nodular vasculitis (see ➔ Chapter 21, pp. 455–63).
- Perforating disease: rare (see ➔ p. 494 and Box 19.28, p. 431). Nodules with a central keratin plug.
- Multicentric reticulohistiocytosis (rare): symmetrical nodules on fingers, papules around nail folds, destructive arthritis (see ➔ p. 426).

Solitary cutaneous ulcer

For leg ulcers, see ➔ p. 75 and pp. 291–318.

- Trauma: including insect/arthropod bites, intravenous drug abuse, pressure—is the ulcer neuropathic?
- Infection: ecthyma—full-thickness infection of the skin, usually by *S. aureus* or *Streptococcus pyogenes*, with ulceration and crusting (see ➔ p. 137); herpes simplex virus (HSV) infection (see ➔ p. 152).
- Malignant tumour.
- Temporal arteritis: ulcer on the forehead or scalp (see ➔ pp. 450–1).
- Pyoderma gangrenosum. Drug related—nicorandil, hydroxyurea.
- Dermatitis artefacta: deliberate self-harm (see ➔ pp. 566–7).

Box 4.8 Skin checks and skin cancer

Skin cancers are the commonest malignancy. You should carry out a full skin examination when assessing patients. But cutaneous tumours are common, and you need to 'get your eye in', so you know when to refer. For more information, see ➔ Chapter 17, pp. 341–63. These pointers may be helpful:

- Patients with fair skin who burn in the sun are at greatest risk of skin cancers. Chronic immunosuppression with drugs, such as azathioprine, or chronic photosensitivity greatly increase the chances of developing skin cancer (see ➔ p. 346).
- Most skin cancers grow slowly, an important exception being nodular malignant melanoma.
- BCCs and malignant melanomas are usually painless, but some SCCs may be tender, particularly on the ear.
- Most malignant melanomas arise *de novo*, and not from a pre-existing melanocytic naevus.
- Carry out a skin check in all patients. Examine sun-exposed skin on the face, bald scalp, ears, forearms, and back of the hands. Look at the skin on the back and legs, and check behind the ears.
- Look for signs of chronic sun damage, e.g. mottled pigmentation, telangiectasia, yellowing, wrinkling, and keratoses (see ➔ Box 3.1, p. 43). The more sun damage, the greater the likelihood of developing a skin cancer—eventually.
- Remember deeply tanned skin is generally unhappy skin!
- Solar keratoses are common and do not need treatment, unless the keratosis is thickened (indurated) when you should suspect an SCC. But solar keratoses do indicate too much sun exposure, and you should advise your patient accordingly.
- Put the skin on the stretch to see the pearly border of a BCC (the commonest skin cancer).
- Think again before you diagnose melanocytic naevus in an elderly patient. Benign melanocytic naevi, common in youth, gradually regress with age. A pigmented tumour in an elderly patient is much more likely to be a seborrhoeic wart (raised, well-defined margin, 'stuck-on' appearance, rough with a pitted, crinkled surface) or a solar lentigo (flat, smooth, even pigmentation), but it just might be a malignant melanoma.
- Use the ABCDE criteria (see ➔ p. 354) to assess the likelihood of any pigmented lesion being a malignant melanoma, and remember to look twice at moles that stand out from their neighbours—the 'ugly ducklings'.
- Record your findings, including the presence of sun damage and the fact that you carried out a skin check.
- Advise patients how to protect their skin from sun damage to minimize the risk of skin cancer. But temper your advice with a little common sense—we all need some sun to lift our spirits and maintain our vitamin D levels! (See ➔ pp. 346–7.)

Induration

Induration (woody hardness) and thickening suggest deep involvement with fibrosis.

- Scar or changes after radiotherapy (see ➲ pp. 530–1).
- Lipodermatosclerosis (leg in association with venous stasis) (see ➲ p. 308).
- Morphoea, systemic sclerosis, or drug-induced scleroderma (see ➲ p. 414–5 and p. 420).
- Chronic GVHD: sclerodermoid form (see ➲ pp. 534–6).
- Scleredema: associated with infection, diabetes, monoclonal gammopathy. Firm non-pitting oedema of the face, neck, upper back (see ➲ Box 22.2, p. 467).
- Scleromyxoedema: associated with paraproteinaemia, myeloma, lymphoma, and leukaemia (see ➲ Box 26.10, p. 541). Waxy, flesh-coloured papules on the face, trunk, and extremities.
- Nephrogenic systemic fibrosis (see ➲ p. 498). Seen in patients with renal failure exposed to contrast medium containing gadolinium. Features similar to scleromyxoedema (see ➲ Box 26.10, p. 541). Now rare.
- PCT: waxy thickening of sun-exposed skin (see ➲ p. 332).

Eyes, mouth, and genitalia

Causes of a red eye

- Conjunctivitis: redness involves the entire surface of the eye. Examine the conjunctiva, both bulbar and palpebral (lid).
  - If the eye is also sticky, this is likely to be a bacterial or viral conjunctival infection.
- Dry eye. Is the conjunctiva irritated, because the eye is dry? Dry eyes burn and feel gritty. Vision may be blurred. Confirm the diagnosis by performing a Schirmer strip test.
- Allergic conjunctivitis: common in atopic eczema.
- Iritis (anterior uveitis): the eye is painful and photophobic. The eye is not sticky. Redness is most marked around the cornea. Small pupil.
  - Iritis occurs in systemic diseases that may also have cutaneous manifestations including sarcoidosis, IBD, Behçet disease, herpes virus infection, and Lyme disease.

Causes of oral ulcers

⚠ Biopsy any long-standing unexplained ulcer to exclude malignancy.

Common

- Trauma: check teeth and/or dentures.
- Inflammatory:
  - Aphthous ulcers: minor, major, or herpetiform (see ➔ pp. 282–3).
  - Erosive LP (see ➔ p. 286).
- Infections:
  - Herpes simplex, varicella-zoster virus (VZV), hand–foot–mouth disease, *Candida*, syphilis, HIV.
- Drugs:
  - Cytotoxic agents, co-trimoxazole, antithyroid drugs, nicorandil, beta-blockers, clopidogrel, alendronate, protease inhibitors, non-steroidal anti-inflammatory drugs (NSAIDs), anticholinergic bronchodilators, and antihypertensives (captopril, enalapril).
  - Cocaine (ask about recreational drugs).
  - Ulcers heal when the drug is withdrawn.
- Radiation to the head and neck.
- Oral SCC (solitary ulcer, may be asymptomatic).

Uncommon

- Inflammatory:
  - EM, SJS, and TEN (see ➔ p. 111 and pp. 116–21).
  - GVHD (see ➔ pp. 534–6).
  - Eosinophilic ulcer: uncommon benign self-limiting ulcer, probably 2° to trauma. Usually a large ulcer on the tongue.
  - LE (see ➔ pp. 396–400).
  - Behçet disease: rare (see ➔ pp. 284–5).
- Infection in immunosuppressed patients:
  - TB (prevalence increasing as a complication of HIV infection), fungus (*Cryptococcus*, histoplasmosis, *Aspergillus*), leishmaniasis.

Rare

- Inflammatory:
  - Angina bullosa haemorrhagica (blood-filled blisters that rupture).
  - Immunobullous diseases, such as pemphigus vulgaris, which may affect the oral mucosa before the skin (see ➔ p. 268).
  - EB (see ➔ pp. 624–5).

Causes of genital ulcers

⚠ Biopsy any long-standing unexplained ulcer to exclude malignancy.

- Trauma: including sexual abuse.
- Infections:
  - HSV: may be extensive and chronic in immunosuppressed patients (see ➔ pp. 280–1).
  - EBV (see ➔ Box 14.2, p. 281).
  - Ecthyma gangrenosum caused by *Pseudomonas aeruginosa* in immunosuppressed patients (see ➔ p. 137).
  - Syphilis: *Treponema pallidum*.
  - Chancroid: *Haemophilus ducreyi*.
  - Lymphogranuloma venereum: *Chlamydia trachomatis*.
  - Granuloma inguinale (donovanosis): *Klebsiella granulomatis*.
- Inflammatory:
  - Aphthous ulcers: possibly triggered by local injury or infection, including HIV. Most patients also have oral aphthous ulcers (see ➔ p. 282–3). Genital aphthous ulcers usually measure 1–3cm in diameter and may be quite deep and either round or irregular in outline. Patients usually only have a few genital aphthous ulcers (often just one). Large ulcers may heal with scarring.
  - Erosive LP: check the mouth (see ➔ pp. 286–7).
  - EM, SJS, and TEN (see ➔ p. 111 and pp. 116–7).
  - Hidradenitis suppurativa: sinus tracts develop into chronic ulcers (see ➔ p. 250).
  - Crohn disease (see ➔ pp. 508–9).
  - PG (see ➔ p. 310).
  - Drugs: including nicorandil.
  - Behçet disease: rare (see ➔ pp. 284–5).
  - Immunobullous disease such as pemphigoid or pemphigus. The skin is fragile, and genital vesicles rupture rapidly to form painful superficial erosions or ulcers (rare) (see ➔ p. 268–9 and pp. 270–1).
- Malignancy:
  - Carcinoma: BCC or SCC.
  - Leukaemia or lymphoma.
  - Extramammary Paget disease.

Change in skin colour

The commonest cause of change in colour is previous inflammation. Ask when the change appeared (was it present at birth?) and if anything preceded it. A Wood light may determine if the pigment is epidermal or dermal and will accentuate the epidermal change (see ➔ p. 640).

Acquired hyperpigmentation

Strictly speaking, hyperpigmentation means brown skin, but this list encompasses colours ranging from brown to blue or grey.

Common

- Normal racial variation or suntan.
- Stasis dermatitis: the pigment is a mix of melanin and haemosiderin.
- Melasma (usually facial): brown colour caused by melanin (see ➔ Box 30.1, p. 583).
- Post-inflammatory hyperpigmentation, particularly in dark skin. Common after acne or lichenoid eruptions such as LP or cutaneous LE. After lichenoid eruptions, the skin has a greyish tinge, as the pigment is deep in the dermis (see ➔ p. 224).
- Ten percent of normal people have one or two café au lait spots (see ➔ Box 27.6, p. 553).
- Erythema ab igne: repeated local heating of the skin from a hot water bottle or fire causes localized fixed reticulate pigmentation.
- Phytophotodermatitis: linear streaks of brown pigmentation are preceded by erythema and sometimes blisters (see ➔ Fig. 31.11, p. 615).
- Dermatitis neglecta: occasionally, patients avoid touching a patch of skin. The unwashed skin builds up brown scale.
- Drugs, including minocycline, antimalarials, amiodarone, and heavy metals, may give the skin a bluish grey tinge (see ➔ pp. 376–7).

Less common

- Neuropathic itch or chronic rubbing (see ➔ Box 27.2, p. 549).
- Malabsorption, pellagra.
- Cutaneous systemic sclerosis and sclerodermoid chronic GVHD. The skin is thickened, often with perifollicular hypopigmentation (see ➔ p. 414 and Box 26.4, p. 535).
- Pseudo-ochronosis 2° to hydroquinone in skin-lightening creams (may also cause confetti-like loss of pigment).
- Primary biliary cirrhosis, haemochromatosis.

Rare

- Café au lait spots: neurofibromatosis, McCune–Albright syndrome, multiple mucosal neuromas syndrome (see ➔ Box 27.6, p. 553).
- Widespread freckling in children may be associated with XP, multiple lentigines syndrome, Carney complex, and Peutz–Jeghers syndrome (see ➔ p. 613).
- Generalized pigmentation: adrenal insufficiency, Nelson syndrome, ectopic adrenocorticotrophic hormone (ACTH)-producing tumours, POEMS syndrome in plasma cell disorders (see ➔ Box 26.8, p. 540).
- Alkaptonuria: blue black pigment of helices of the ear and sclerae.

Hypopigmentation and depigmentation

Hypopigmented skin: loss of pigment is partial. The tone of the skin is creamy, rather than absolutely white. Depigmented skin is white and fluoresces bright white under Wood light, e.g. vitiligo.

Common

- Pityriasis alba: hypopigmented cheeks in children. Subtle scale.
- Pityriasis versicolor: scaly in the active phase, but macular post-inflammatory hypopigmentation may persist for months, until melanocytes are stimulated by sun exposure.
- Idiopathic guttate hypomelanosis: pale macules on sun-damaged forearms of adults.
- Progressive macular hypomelanosis: common in young Afro-Caribbean adults. Progressive symmetrical hypopigmentation in midline of the trunk.
- Post-inflammatory hypopigmentation: most often in dark skin. Causes, e.g. psoriasis, sarcoidosis, leprosy, pinta, and kwashiorkor.
- Vitiligo: smooth depigmented macules or patches (see ➔ pp. 486–7).
- Halo naevus: children or young adults. A ring of white skin appears around a melanocytic naevus. The brown 'mole' gradually turns pink and eventually disappears, leaving a depigmented macule.
- Scars (may also be hyperpigmented).
- Atrophie blanche: pale scar with a rim of telangiectasia. Leg in venous disease.

Less common

- Contact leukoderma after exposure to chemicals, e.g. aromatic or aliphatic derivatives of phenols or catechols, hydroquinone in skin-lightening creams, betel leaves, fentanyl patches.
- 2° syphilis: hypopigmented macules superimposed on hyperpigmented, reticulate patches (syphilitic leucomelanoderma). Neck, chest, and back. Six months after 1° disease.
- Tuberous sclerosis: oval or confetti-like hypopigmentation (see ➔ pp. 556–7).
- Cutaneous LE: hypo- and hyperpigmentation (see ➔ pp. 396–407).
- Morphoea or cutaneous systemic sclerosis: perifollicular hypopigmentation in thickened skin. May also be hyperpigmentation (see ➔ p. 414 and pp. 420–1).
- Chronic GVHD (see ➔ Box 26.4, p. 535) (also hyperpigmentation).
- Antiphospholipid syndrome: porcelain white scars with telangiectatic rim (like atrophie blanche) (see ➔ Box 20.10, p. 453). Associated with livedo reticularis. Differentiate from Degos disease (see ➔ p. 90).
- Chronic arsenic ingestion: 'raindrop' hypopigmentation.
- Extragenital lichen sclerosus (see ➔ p. 288): crinkly, shiny white papules with follicular plugging. Look for genital disease.
- Naevus depigmentosus: localized hypopigmented skin with discrete, regular, or serrated margins. Stable appearance (also see ➔ p. 641).
- Naevus anaemicus: jagged outline, caused by vasoconstriction (see ➔ p. 562 and p. 641).
- Cutaneous T-cell lymphoma (see ➔ pp. 358–9).

Rare

- Malignant atrophic papulosis (Degos disease). Erythematous papules evolve into porcelain white scars with a rim of telangiectasia. Linked to fatal vascular occlusion in the gastrointestinal tract (GIT) or central nervous system (CNS). Differentiate from antiphospholipid syndrome.
- Pigmentary mosaicism: swirling hypopigmented patches (see ➔ Box 32.7, p. 637).
- Focal dermal hypoplasia of Goltz (see ➔ Box 32.8, p. 637).
- Albinism: total body depigmentation, light blue iris, nystagmus.
- Waardenburg syndrome (a form of piebaldism). Autosomal dominant (AD) inheritance. Symmetrical patches of hypopigmentation on the face, scalp, back, and proximal extremities, with a stripe of normal-coloured skin down the centre of the back. Also white forelock, neurosensory deafness, widening of the bridge of the nose, and heterochromia of the iris.

Further reading

Vachiramon V *et al*. *Clin Exp Dermatol* 2011;**37**:97–103.

Skin of colour

Signs in dark skin may be difficult to assess, and some conditions are much commoner in certain ethnic groups, while others may present differently (see Box 4.9). Pigmentary disorders may be particularly distressing and disfiguring.

- Traditional medicines, cultural practices, or cosmetics may affect the skin, so ask what the patient is using on the skin.
- Pigmentary demarcation lines may simulate pathology (see ➔ Box 1.5, p. 13).
- Benign melanocytic naevi are common in the nail bed of black patients and produce hyperpigmented linear bands of varying width in the nail plate. These may be single or multiple, and their number tends to increase with age. Pigment does not extend onto the skin of the surrounding nail fold, although some pigment may be visible beneath the translucent cuticle. If pigment is detected in the nail fold (Hutchinson sign), arrange a biopsy to exclude malignant melanoma.
- Some black patients have harmless diffuse nail pigmentation.
- Palms and soles are paler than the rest of the skin in dark-skinned patients.
- Palmar creases tend to be hyperpigmented and, in black patients, may contain punctate conical pits (keratosis punctata).
- Asymptomatic hyperpigmented macules varying in shape and size are common on plantar surfaces in black patients, particularly the ball of the foot and the heel.
- Brown pigmentation of the oral mucosa, including the tongue, buccal mucosa, and palate, is a normal finding in many black patients.
- Gingival tattooing (blue-grey) is common in parts of Africa.
- Periorbital hyperpigmentation is common in the Indo-Asian population. Colour change (may be familial) starts below the eyes around puberty.
- Dermatosis papulosa nigra (1–3mm pigmented, warty papules like small seborrhoeic warts) is common on the cheeks of Africans, African Americans, and dark-skinned South East Asians.
- Redness (erythema) in dark skin may be difficult to assess, and the skin may merely appear slightly darker than normal. Ask the patient to show you what is normal or abnormal. Even erythroderma (generalized erythema) may not be obvious. Look for signs of skin failure such as shivering, thirst, and widespread scaling (see ➔ p. 106).
- Long-lasting post-inflammatory hypopigmentation and/or hyperpigmentation are common after conditions, such as acne, psoriasis, or eczema, as well as after lichenoid inflammatory conditions such as LP or LE. Ask if anything preceded the colour change.
- Prolonged hypopigmentation may be caused by intralesional corticosteroids or very potent topical corticosteroids, but it is much commoner for the colour change to be 2° to the inflammatory skin condition for which the corticosteroids were prescribed.
- Even individuals with dark-coloured skin may be photosensitive, so do not assume anything!

Box 4.9 Presentations commoner in skin of colour

Hair follicles and follicular rashes

- Alopecia related to hair straightening (with chemicals or hot combs) or to tight braiding (traction alopecia) in black patients.
- Pomade acne on the forehead caused by greasy hair oils.
- Acne keloidalis nuchae: firm papules on the nape of the neck or scalp, common in black patients with curly hair.
- Pseudofolliculitis barbae: hyperpigmented papules/pustules on the neck/cheek where skin shaved. Triggered by ingrowing hairs.
- Atopic eczema in a follicular pattern is common in black children, as is follicular accentuation in other inflammatory conditions, including lichen planus and pityriasis versicolor.
- Disseminated and recurrent infundibulofolliculitis: itchy follicular papules on the chest, back, and buttocks of black patients.

Darker colour (often post-inflammatory) (Also see ➔ p. 88.)

- Scaly plaques of psoriasis look blue-black, rather than red.
- LP presents with deeply coloured purple papules in dark skin. LP pigmentosa is commoner in Asian patients and presents with ashy grey macules.
- Pseudo-acanthosis nigricans: hyperpigmentation, velvety thickening, and skin tags in the flexures of obese Asian patients. Associated with diabetes and insulin resistance (see ➔ pp. 466–7).
- Macular amyloidosis: itchy rash with rippled grey-brown pigmentation, often on the upper back. Commoner in Asian patients (see ➔ Box 23.3, p. 497).
- Acquired ochronosis: hyperpigmentation caused by skin-lightening creams containing hydroquinone.
- Suction cups used in the traditional practice of cupping leave circular hyperpigmented macules that may simulate bruising.

Paler colour (often post-inflammatory) (Also see ➔ pp. 89–90.)

- Pityriasis alba: pale oval patches and fine scale. Most often on the face but may involve the trunk. Seen in prepubertal children (see ➔ p. 89).
- Trichrome vitiligo: zones of hypopigmentation, as well as depigmentation and normal pigmentation.
- Leprosy produces anaesthetic macules (see ➔ Box 6.8, p. 145).
- Dyspigmentation with scarring in discoid LE (see ➔ p. 404).
- Confetti-like hypopigmentation on the face caused by skin-lightening creams containing hydroquinone.

Wounds and scars

- Sickle-cell leg ulcers (see ➔ Box 15.2, p. 295).
- Keloids (smooth nodular scars that extend beyond original wound) common on the upper chest, shoulders, and upper back. Avoid taking skin biopsies from these sites in patients with black skin.

Neonates (see ➔ p. 590)

- Mongolian spots (dermal melanocytes). Light blue to slate grey macules, most often lumbosacral. May be multiple. Fade slowly (see ➔ Fig. 31.10, p. 615). Naevus of Ota affects periorbital skin and sclera.
- Transient neonatal pustular melanosis (see ➔ Box 31.4, p. 593).
- Infantile acropustulosis (see ➔ Box 31.4, p. 593). Differentiate from scabies.

Hair: too much or too little

The history and examination in hair problems are described on ➲ p. 24 and p. 44. Excess hair may grow anywhere there are hair follicles. Hair loss may be diffuse, localized, complete (bald patches), or partial (thinning). Scalp may be normal or scaly. Decide if alopecia is non-scarring and potentially reversible, or scarring when loss is irreversible (see ➲ Box 3.2, p. 45).

Hypertrichosis

Definition: excessive hair on any part of the body.

- *Acquired*: generalized, most often drugs (see ➲ p. 380). Rarely indicates systemic disease, including malnutrition (anorexia nervosa; see ➲ pp. 512–13 and p. 572), advanced HIV infection, PCT (see ➲ p. 332), POEMS syndrome (see ➲ Box 26.8, p. 540), and disorders of the CNS. Localized hypertrichosis occurs after chronic cutaneous irritation or inflammation and may develop in association with some tumours (melanocytic naevi, Becker naevi, plexiform neurofibromas, smooth muscle hamartomas).
- *Congenital*: localized hypertrichosis in the midline over the scalp or spine may indicate an underlying neural tube defect, e.g. a hair collar in the scalp or a lumbosacral faun tail (see ➲ p. 594). Hypertrichosis is associated with some rare hereditary diseases, including porphyrias, mucopolysaccharidoses, fetal alcohol syndrome, and Cornelia de Lange syndrome.

Hirsutes

- Increased growth of terminal hairs in ♀ in an androgen-dependent pattern (normal ♂ sexual pattern)—face, lips, chest, arms, thighs. Caused by androgen overproduction, androgenic drugs, or increased sensitivity of the hair follicle to androgens (see ➲ p. 474).

Localized non-scarring alopecia

- Alopecia areata: well-defined areas of loss. The scalp is smooth, not scaly. You may find exclamation-mark hairs. At first, regrowing hair is not pigmented (see ➲ p. 488 and p. 44).
- Tinea capitis (dermatophyte infection, ringworm): scalp is itchy and scaly, and may be inflamed.
- Hair pulling (trichotillomania): short, stubby hairs (<2cm). Longer hairs can be pulled out. Common in children.
- Trauma: hair may be pulled out when thick scale is removed in psoriasis or seborrhoeic dermatitis.
- Traction alopecia: affects the margins of the scalp where hair has been pulled back tightly. No scale or exclamation-mark hairs.

Diffuse non-scarring alopecia

- Telogen effluvium (illness in previous 2–3 months?), post-partum.
- Androgenetic alopecia (see ➔ p. 480).
- Thyroid disease or iron deficiency anaemia.
- Drugs (see ➔ p. 380).
- Rare congenital causes of hypotrichosis include ectodermal dysplasias and disorders in which the hair shaft is abnormal and brittle (see ➔ pp. 634–5).

Scarring alopecia

Follicles may be replaced by a tumour. In inflammatory diseases, such as DLE, damage to follicular stem cells (see ➔ p. 10) probably contributes to permanent hair loss.

Commoner

- Trauma: including radiotherapy.
- Tumours:
  - Benign: naevus sebaceous—present at birth (see ➔ Fig. 31.2, p. 595).
  - Malignant: BCC, SCC, metastatic deposits.
- DLE (see ➔ p. 404).
- Acne keloidalis: most often in black men. Firm follicular papules (keloids) at the nape of the neck and/or occipital scalp. May also have pustules.

Uncommon

- Aplasia cutis: present at birth (see ➔ Box 31.6, p. 595).
- Linear morphoea.
- Lichen planopilaris (variant of LP that affects hair follicles).
- Folliculitis decalvans: erythematous perifollicular pustules and scarring. Cause unclear but may involve *S. aureus*.
- Dissecting cellulitis of scalp: occurs most often in young black-skinned men. Firm, deep nodules become fluctuant, discharge malodorous pus, and develop interconnecting sinuses. May be associated with hidradenitis suppurativa and acne conglobata (see ➔ p. 250).

Funny nails

As with the skin, the distribution of changes points towards the cause. If many nails are affected (symmetry), the problem is likely to be endogenous; if only one nail is affected (asymmetry), the problem is likely to be exogenous. Inflammation of periungual skin in conditions, such as eczema, leads to 2° changes in the nail plate.

Common or important signs

- *Changes in nail colour* (see Box 4.10).
- *Transverse depressions*: temporary loss of mitotic activity in the proximal nail matrix. Think of trauma if only one nail is affected. Some patients damage the nail plate by repeatedly pushing down the cuticle. Beau lines affect all the nails and indicate slowing of nail growth in the previous 1–2 months, e.g. during a serious illness or chemotherapy (see ➔ pp. 380–1). Arrest of all nail matrix activity causes nail shedding (onychomadesis).
- *Nail pitting*: foci of abnormal keratinization of nail matrix. Psoriasis is the commonest cause of coarse pits. Many fine pits occur in alopecia areata. May see pits in eczema.
- *Thinning, longitudinal ridging, and fissuring* (onychorrhexis): diffuse damage to nail matrix. Mild disease is common in older people. Also seen with impaired vascular supply, LP, chronic GVHD, and amyloidosis. May cause permanent loss of the nail plate with dorsal pterygium (adhesion between nail fold and nail bed).
- *Onycholysis*: the nail plate separates from an abnormal nail bed. If distal and only a few nails, consider fungus, trauma, or subungual tumour. How does the patient manicure the nails? Is something being pushed underneath the nail plate to remove accumulated debris? This will damage the nail bed and worsen onycholysis. Psoriasis may cause proximal oil spots or salmon patches, as well as distal onycholysis. Also seen with hyperthyroidism. Drugs, particularly tetracyclines and taxanes, cause painful photo-onycholysis (see ➔ Box 16.3, p. 327).
- *Subungual hyperkeratosis and discoloured thickened nail plate*: fungus, psoriasis. Likely to be a fungal infection if only a few nails. Crumbling starts at the free edge of the nail and moves proximally. Exclude psoriasis by looking for pitting and/or onycholysis in finger nails (see ➔ p. 196).
- *Splinter haemorrhages*: indicate damage to nail bed capillaries. Trauma is likely if only one nail is affected and the haemorrhages are distal, rather than proximal. Splinter haemorrhages may develop when the nail plate is abnormal, e.g. in psoriasis. Numerous proximal splinter haemorrhages are more likely to indicate systemic disease such as infective endocarditis.
- *Nail clubbing*: associated with systemic problems, including chronic lung disease, cyanotic heart disease, IBD, and thyroid disease, but may be idiopathic or familial.
- *Spoon-shaped nails* (koilonychia): the soft nail plate in iron deficiency becomes concave. Children have physiological koilonychia.
- *Abnormalities in nail fold* (see Box 4.11).

Box 4.10 Changes in nail colour

- Dark discoloration: diffuse darkening or longitudinal bands of pigment are common in individuals with dark skin and are caused by melanocytic naevi in the nail matrix.
- Addison disease and ectopic ACTH production may lead to dark nails or bands of melanonychia.
- Does pigment extend onto the skin of the proximal or lateral nail fold? Discoloration caused by subungual haematoma (e.g. after repetitive trauma such as running) or fungal infection never spreads onto surrounding skin, but pigmentation associated with subungual malignant melanoma may affect adjacent skin (Hutchinson sign).
- Onycholytic nails become dark green when the space beneath the nail plate is colonized by *Pseudomonas aeruginosa*.
- Drugs, e.g. antimalarials, can darken nails (see ➔ pp. 380–1).
- Acquired leuconychia: fungal infection, trauma (punctate leuconychia). Hypoalbuminaemia, e.g. chronic liver disease, and some drugs cause apparent leuconychia in all the nails, because of changes in the nail bed, but discoloration fades with pressure.
- Congenital leuconychia: rare, affects all nails.
- Transverse white lines may occur after trauma, systemic illness, poisoning with arsenic (Mee lines), or thallium.
- Half and half nails (white proximally and reddish brown distally) are a rare sign of chronic renal failure.
- Yellow nails. Nails grow slowly and become thick and greenish yellow in the yellow nail syndrome, which is associated with lymphatic hypoplasia, peripheral oedema, and pleural effusions (rare).
- Red and white horizontal bands in HIV infection.

Box 4.11 Abnormalities in nail fold

- Boggy, swollen erythematous nail fold: most likely to indicate chronic paronychia. The cuticle is missing; low-grade *Candida* infection beneath the nail fold leads to ridging and discoloration of the nail. *S. aureus* may cause episodes of tender acute paronychia, when beads of pus exude from the swollen nail fold.
- Dilated nail fold capillaries and periungual erythema are valuable signs of a connective tissue disease (see ➔ Fig. 19.9, p. 410).
- Digital mucous cyst: the smooth swelling in the nail fold produces a vertical furrow in the nail plate.
- Papules in nail fold:
  - Viral warts; periungual fibroma in tuberous sclerosis; beaded papules in multicentric reticulohistiocytosis (see ➔ p. 426).
- Pustules around the nail and beneath the nail plate: seen in reactive arthritis (see ➔ p. 208), as well as in acropustulosis and parakeratosis pustulosa (both rare and possible forms of psoriasis) (see ➔ Box 9.9, p. 199).

Chapter 5

Skin failure and emergency dermatology

Contents

Rash in a sick adult with a fever

Ill patients with rash and fever pose an urgent diagnostic problem. Life-threatening causes include infections, severe drug reactions, and acute GVHD (see Boxes 5.1 and 5.2). Taking a systematic approach to history and examination will help to establish the cause.

What should I ask?

- Explore symptoms, e.g. rigors, sweats, headache, photophobia, arthralgia, myalgia, nausea, or diarrhoea. Any mucosal problems, e.g. conjunctivitis, sore throat, oral or genital ulcers?
- Is fever sustained, remittent (elevated each day and returns to baseline, but not normal), or intermittent (intermittently elevated but returns to normal)?
- When did the rash develop, and how has it evolved?
- Any history of contact with an infectious disease?
- Document a travel history (were any preventative measures taken, e.g. vaccines, avoiding mosquito bites?) and sexual history.
- Explore the drug history (including use of recreational drugs).
- Any past history of drug reaction, infectious disease, or skin disease?
- Might the patient be immunocompromised?

What should I look for?

- Cutaneous signs:
  - Does the rash blanch with light pressure (erythematous), or is it purpuric (non-blanching)? Purpura in a febrile patient raises the possibility of meningococcal septicaemia (see ➔ pp. 102–3).
  - Is the rash morbilliform (measles-like erythematous macules and papules)? Is there any desquamation?
  - Are there erosions (denuded areas where the epidermis has been lost), blisters, or pustules? (See Box 5.1.)
  - Is the skin scaly, or is there superficial peeling (desquamation)?
  - Are the palms and soles involved? Is there oedema?
- Examine the conjunctiva, buccal mucosa, throat, and tongue.
- Check for lymphadenopathy.
- Is the neck stiff, or are the muscles tender?
- Examine the joints for warmth, erythema, swelling, or tenderness.
- Examine the cardiovascular, respiratory, GI, genitourinary, and neurological systems for localizing signs.

What should I do?

Investigations will be guided by your findings, but consider:

- Full blood count (FBC) and erythrocyte sedimentation rate (ESR); C-reactive protein (CRP), clotting screen.
- Liver function and renal function; urinalysis.
- Blood cultures, wound swabs, lumbar puncture, skin biopsy.

Further reading

Low DE. *Crit Care Clin* 2013;**29**:651–75.
The RegiSCAR Project. Available at: www.regiscar.org/ (severe cutaneous drug reactions).
Young AE and Thornton KL. *Arch Dis Child Educ Pract Ed* 2007;**92**:ep97–100.

Box 5.1 Causes of rash and fever in sick adult patient

- Infections, e.g.:
  - Severe bacteraemia, e.g. meningococcal, pneumococcal.
  - TSS (see Box 5.2).
  - Rickettsiosis, e.g. Rocky Mountain spotted fever.
  - Viral haemorrhagic fevers and dengue.
- Severe cutaneous adverse drug reactions, e.g.:
  - TEN.
  - DRESS.
- Acute GVHD.
- Rheumatological disease, e.g.:
  - SLE, systemic vasculitis.

Box 5.2 Toxic shock syndrome

! TSS is a multisystem disease, usually caused by an exotoxin produced by *Staphylococcus aureus* which acts as a superantigen, resulting in immune cell expansion and a cytokine storm. Predisposing factors include surgical packing, contraceptive sponges, post-partum infections, and deep abscesses. TSS is also a potentially fatal complication of small burns in young children.

TSS can be produced by other bacterial exotoxins, including group A *Streptococcus, Pseudomonas*, and *Klebsiella* strains. TSS may be mild or rapidly fatal. Mortality for streptococcal TSS is around 50%.

What to look for in TSS

- Sudden onset of high fever >39°C, with myalgia, vomiting, diarrhoea, and headache.
- Hypotension. Disorientation or altered consciousness.
- Diffuse macular erythematous (not purpuric) rash that starts on the trunk and spreads outwards. If you find vesicles or bullae, think of SSSS (see ➔ p. 112) or TEN (see ➔ p. 116). If purpuric, consider meningococcaemia (see ➔ pp. 102–3).
- Erythema and oedema of palms and soles.
- Erythema of mucous membranes with strawberry tongue and conjunctival hyperaemia. No ulceration, unlike TEN.
- Organ dysfunction: renal failure, abnormal liver function.
- Lymphopenia is common. DIC (if severe)—deranged clotting and low platelets.
- Blood cultures frequently negative, and diagnosis is clinical.

►► *Management of TSS*

- Resuscitate with intravenous (IV) fluids and vasopressor agents, if required.
- Treat with IV antibiotics (suppress toxin production).
- Remove wound dressings; clean wounds, and remove any potential nidus of infection such as a foreign body.
- IV immunoglobulin has been used in severe TSS.
- Skin desquamates, particularly on the palms and soles, after 1–2 weeks.

Vasculitis and purpura fulminans

Palpable purpura is the hallmark of a small-vessel cutaneous vasculitis, such as IgA vasculitis (Henoch–Schönlein purpura), whereas flat (macular) purpura is associated with non-inflammatory pathology (see ➔ pp. 436–9). In purpura fulminans, fibrin thrombi occlude dermal vessels, as well as vessels in the subcutis, leading to haemorrhage, ischaemia, and infarction. Purpura fulminans most commonly occurs in meningococcal septicaemia (see Box 5.3).

Meningococcal septicaemia (*Neisseria meningitidis*)

❗ Meningococcal infection may cause devastating septicaemia (acute meningococcaemia) and death within hours of first symptoms. The non-specific flu-like presentation, with fever, vomiting, headache, abdominal pain, and muscle aches, may be misleading.

Although 80–90% of patients develop purpura within 12–36h of onset of infection, the rash may be subtle and lesions few in number. You should examine the skin in a good light.

What should I look for?

- Early: a transient erythematous, maculopapular rash (may simulate TSS (see ➔ Box 5.2, p. 101).
- After 12–36h: small purpuric lesions (petechiae) with an irregular outline and (usually) raised centres. You will find petechiae most often on the limbs and trunk, but examine the head, palms, soles, and mucous membranes. Also look for purpuric lesions in pressure areas, e.g. under the waistband, or at sites of friction.

►► *What should I do?*

- Consider the diagnosis in any febrile patient with a petechial rash. Frequently, meningitis is not clinically apparent.
- Commence immediate treatment with intravenous (IV) antibiotics, e.g. ceftriaxone 2g, without waiting to confirm your suspicions.
- ❗ Call the critical care team urgently.
- If not in hospital, treat with IV or intramuscular (IM) benzylpenicillin 1.2g, and call an ambulance.

Complications

- DIC and purpura fulminans (see Box 5.3). Extensive irregular haemorrhagic areas (ecchymoses) become bullous and may progress to well-demarcated blue-black gangrene of digits or whole limbs.
- Limb compartment syndrome 2° to thrombosis, venous congestion, and oedema. Muscle infarction and rhabdomyolysis.
- Shock, hypotension (late sign in young), confusion, tachypnoea.
- Waterhouse–Friderichsen syndrome (caused by haemorrhage into the adrenal cortex).
- Coma and multi-organ failure.

Box 5.3 What is purpura fulminans?

The term purpura fulminans describes a devastating illness that is characterized by rapidly progressive and extensive purpura with dermal vascular thrombosis and haemorrhagic infarction, but without vasculitis (see Fig. 5.1). In severe cases, subcutaneous vessels or major veins are thrombosed. Patients have a consumptive coagulopathy. The mortality is high. Survivors may be left with extensive scarring, missing digits, or amputated limbs.

Causes include

- Acute bacterial infection (commonest cause), e.g. *Neisseria meningitidis* or, much less often, β-haemolytic *Streptococcus, Streptococcus pneumoniae, Staphylococcus aureus*, and other bacteria.
- After infection (usually children 1–3 weeks after varicella or streptococcal infection).
- Congenital deficiency of protein C or protein S (presents in neonates).
- Acquired deficiency of protein C or protein S associated with drugs (warfarin) (see ➔ p. 376) or disease (renal dialysis, nephrotic syndrome, cholestasis, bone marrow transplantation).
- Antiphospholipid syndrome (see ➔ Box 20.10, p. 453).
- Heparin-induced skin necrosis (antibody-mediated platelet aggregation usually seen at the injection site of subcutaneous heparin) (see ➔ p. 376).
- Toxins or poisons (spider bites, snake bites).

Fig. 5.1 Extensive purpura fulminans.

Further reading

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Anaphylaxis, angio-oedema, and urticaria

What is anaphylaxis?

❗ Anaphylaxis is a severe, generalized or systemic hypersensitivity reaction associated with release of histamine and capillary leak. Characterized by rapidly developing, life-threatening airway and/or breathing and/or circulatory collapse, usually associated with urticaria and mucosal swelling (angio-oedema).

Triggers include foods, drugs (see Box 5.4), and venoms (wasp, bee). Reaction develops within minutes of exposure to trigger, but often no cause identified (these are non-IgE-mediated) (see Box 5.5).

What should I look for?

- Sudden onset and rapid progression of symptoms.
- Life-threatening airway ± breathing ± circulatory collapse that may include stridor, hoarse voice, difficulty swallowing, wheeze, cyanosis, tachycardia, hypotension, reduced consciousness.
- Skin and/or mucosal changes (flushing, urticaria, angio-oedema)—signs may be subtle or absent in up to 20% of reactions.

What is angio-oedema?

Angio-oedema implies oedema of the deep dermis, subcutaneous tissues, mucosa, and/or submucosa. Laryngeal oedema, which may cause difficulty swallowing and respiratory obstruction, is life-threatening.

Angio-oedema occurs in anaphylactic reactions. ⚠ Rarely, angio-oedema is caused by C1 esterase inhibitor deficiency (inherited—AD inheritance or acquired). These patients do ***not*** have urticaria. Angio-oedema is usually triggered by mucosal trauma, e.g. dental procedures or, in ♀, sexual intercourse. Some patients complain of vomiting/abdominal pain. Prodromal features reported more commonly in women include fatigue, malaise, and short temper.

α-1-antitrypsin (AAT) deficiency is a rare cause (see ➔ p. 462).

Facial swelling in acute contact dermatitis (see ➔ pp. 214–16) or acute dermatomyositis (see ➔ pp. 408–9) may mimic angio-oedema (see ➔ Fig. 19.8, p. 409).

What is urticaria?

Urticaria (hives, wheals, welts) is a hypersensitivity reaction that may occur in isolation or associated with angio-oedema and/or anaphylaxis. Wheals are 2° to dermal oedema. They are raised, smooth, erythematous, and itchy. They come and go, leaving normal skin, but generally do not last >24h. Some patients provide a clear history of precipitants, such as drugs, latex, or foods (fruit, fish, peanuts), but many, especially with chronic urticaria, cannot identify a trigger. Many idiopathic (spontaneous) cases are not mediated by IgE. Allergy testing in chronic urticaria is unrewarding.

Urticaria may be acute or chronic, and often confused with EM (see Box 5.10). Most patients with urticaria do not develop anaphylaxis. Urticaria is a nuisance but, unless associated with airway problems (ask about throat swelling or wheeze), is not life-threatening (see ➲ p. 228). Non-sedating antihistamines are the mainstay of treatment (see ➲ pp. 232–3).

Box 5.4 Drugs linked to anaphylaxis

Virtually any class of drug can be implicated:
- Antibiotics, including penicillin and cephalosporin.
- Aspirin.
- NSAIDs.
- ACE inhibitors.
- Anaesthetic drugs, including suxamethonium, vecuronium, and atracurium.
- Contrast media, particularly iodinated media.

►► Box 5.5 Management of anaphylaxis in adults

Use an ABCDE approach to assess and treat anaphylactic reactions:
- Place the patient in a comfortable position. Conscious patients with breathing difficulty may prefer to sit up.
- Remove the trigger, if possible.
- Adrenaline (epinephrine) 0.5mg IM (0.5mL of 1:1000)—repeat after 5min, if no better.
- Oxygen: 100% high flow as soon as possible.
- Rapid IV fluid challenge to restore tissue perfusion, e.g. 1L of warmed 0.9% saline infused in 5–10min. Monitor response, and give more fluids, if necessary, aiming to restore the patient's normal blood pressure.
- If wheeze persists, give bronchodilators.

After initial resuscitation:
- Chlorphenamine 10mg IM or IV slowly.
- Hydrocortisone 200mg IM or IV slowly.
- Ensure the patient is monitored.

❗ *Note that patients with angio-oedema 2° to C1 esterase inhibitor deficiency do not have urticaria and will not respond to adrenalin. They should be treated with C1 esterase inhibitor.*

Patients who have had an anaphylactic reaction should be referred to an allergy clinic both for further investigation and education, as these patients have a substantial risk of further reactions.

Adverse drug reactions should be reported to the Medicines and Healthcare products Regulatory Agency using the yellow card scheme.

Skin failure

What is skin failure?

Skin failure occurs following sudden onset of a widespread severe inflammatory dermatosis that impairs the normal physiological functions of the skin. The stratum corneum may be disrupted or the epidermis may be lost completely, leaving a denuded dermis.

Skin failure has a number of causes but is most often either 2° to deterioration of a pre-existing dermatosis or manifestation of a drug reaction (see Box 5.6).

What should I look for?

❗ Patients with skin failure have an inflammatory skin condition involving at least 90% of the BSA. Skin failure occurs in:

- Generalized scaly erythema (erythroderma, exfoliative dermatitis).
- Generalized erythema with pustules.
- Widespread blistering, with or without loss of epidermis.

The other features of skin failure may include:

- Cutaneous pain or widespread itching.
- Impaired temperature regulation: hypo- or hyperthermia. Heat is lost due to increased dermal blood flow. Patients feel cold and shiver to raise the temperature.
- Infection (the skin barrier is disrupted).
- Loss of fluid (4 or 5L of fluid are lost each day if 50% of the BSA is involved and the stratum corneum is destroyed).
- Loss of electrolytes and protein.
- Increased energy requirements.
- Oedema (vasodilated cutaneous blood vessels are more permeable).
- High-output cardiac failure (particularly in the elderly).
- Mucosal ulceration in some blistering disorders makes eating or drinking painful.
- Hair and nails may be lost acutely or 2–3 months later.

Causes of skin failure should be explored, and treatment directed towards treating the underlying disease and relieving discomfort (see Boxes 5.6 and 5.7). Skin failure may be life-threatening—patients may develop infection, hypothermia, or cardiac failure. Older patients, patients with extensive skin involvement, and those already taking corticosteroids/immunosuppressants are at risk of life-threatening complications.

Box 5.6 Causes of skin failure

- Erythroderma (generalized erythema and scale) 2° to eczema, psoriasis, or drugs.
- Cutaneous T-cell lymphoma (CTCL) is a rare cause.
- SSSS: causes very superficial blistering.
- Generalized pustular psoriasis.
- SJS/TEN caused by drugs.
- Widespread blistering:
  - Skin loss caused by thermal burns or phototoxicity.
  - Autoimmune bullous disorders, e.g. BP, pemphigus.
  - Inherited group of mechanobullous diseases, i.e. EB (rare).

Box 5.7 General approach to managing acute skin failure

- Search for the cause, and stop all unnecessary drugs.
- Record the evolution of the skin disease as objectively as possible—photographs may be helpful.
- Nurse patients with blistering disorders or denuded areas on a low-pressure (air) mattress.
- Minimize the risk of infection with reverse barrier nursing.
- Monitor fluid and electrolyte losses.
- Keep the patient (and room) warm, and monitor the core body temperature; the skin may look red and feel hot—this can be misleading, and the core temperature may be low. Hyperthermia is less common.
- Take skin swabs every 3–4 days.
- Take blood cultures; monitor for systemic infection, and limit indwelling lines to reduce the risk of infection.
- Ensure nutrition is adequate; protein and calorie requirements will increase, requiring high calorific food and protein supplements.
- Be alert for cardiac failure in the elderly with erythroderma.
- Apply bland emollients to soothe red scaly skin, e.g. 50% white soft paraffin (WSP) in liquid paraffin 3–4 times/day (you will need 500g tubs), or use Emollin® spray.
- Use an emollient as soap substitute.
- Ensure adequate pain relief.
- Prescribe sedating antihistamines, such as hydroxyzine 25–50mg four times a day (qds), to relieve intractable irritation.
- Consider low-dose heparin to reduce the risk of venous thrombosis—patients may be dehydrated and immobile.
- Contact a dermatologist urgently to discuss the diagnosis, investigation, and management.
- The disfigurement caused by widespread skin disease can be very distressing, and, as they recover, patients may need psychological support.

Erythroderma

What is erythroderma (also known as exfoliative dermatitis)?

Erythroderma is defined as erythema, with variable amounts of scale, affecting 90% or more of the body surface.

What causes erythroderma?

Search for the cause when you are taking the history, but remember the commonest cause is deterioration of a pre-existing skin disorder or drugs. Erythroderma may develop suddenly but, when associated with skin problems such as psoriasis, may evolve gradually. Causes include:

- Dermatitis of any type, e.g. atopic, seborrhoeic, contact, stasis, asteatotic (15–40% of cases).
- Psoriasis (8–25% of cases).
- Drugs (10–28% of cases)—numerous drugs have been implicated. Also see Red man syndrome, ➔ p. 372.
- Cutaneous T-cell lymphoma or leukaemia (15% of cases).
- Rare skin diseases, e.g. pityriasis rubra pilaris, pemphigus foliaceus, ichthyosiform erythroderma (<1% of cases).

In up to 30% of patients, erythroderma is classified as 'idiopathic', i.e. no cause found. Consider skin biopsy before prescribing treatments, such as topical or oral corticosteroids, which will reduce inflammation and mask signs. Skin biopsies may not reveal the underlying cause.

What should I look for?

- Itchy, erythematous scaly skin (see Fig. 5.2).
- Cutaneous oedema.
- Oozing of serous fluid.
- Pustules (suggesting infection or pustular psoriasis).
- Superficial blisters (suggesting acute dermatitis or, much less likely, pemphigus foliaceus).
- Keratoderma (thick skin on palms or soles).
- Ectropion if erythroderma affects the face.
- Hair loss.
- Thickened or ridged nails or loss of nails.
- Lymphadenopathy (dermatopathic lymphadenopathy is common in erythroderma and is not caused by lymphoma, but consider a biopsy if the nodes are large and rubbery).
- Evidence of a 1° skin disease, e.g. nail pitting suggesting psoriasis, psoriatic plaques on extensor surfaces or in flexures, stasis eczema around a leg ulcer.
- Non-specific symptoms and signs such as fever and malaise.

What should I do?

See Box 5.8 for immediate management.

Box 5.8 Immediate management of erythroderma

- Stop all non-essential systemic drugs.
- Take swabs for culture from oozing skin.
- Apply emollients such as Cetraben® cream or 50% WSP in liquid paraffin (prescribe 500g tubs) 3–4 times/day or Emollin® spray.
- Bath or shower daily, using a soap substitute such as Epaderm® ointment or Dermol® 500. A bath oil, such as Oilatum®, can be added to the bath.
- Prescribe a sedating antihistamine, e.g. hydroxyzine 25–50mg qds for severe itching.
- Treat 2° infection with oral antibiotics.
- Check fluid balance and body temperature, and monitor for other complications of skin failure (see Box 5.7).

❗ Moderately potent topical corticosteroids may be helpful, but do not prescribe very potent topical corticosteroids or oral corticosteroids, until the investigation and management have been discussed with a dermatologist. Corticosteroids are helpful in eczema but can make psoriasis much more unstable and difficult to manage.

Fig. 5.2 Generalized erythema and scale in this patient with drug-induced erythroderma.

Localized blisters

What is a blister?

- A blister is a bump filled with clear fluid (serum).
- Small blisters, <1cm in diameter, are called vesicles; blisters >1cm in diameter are known as bullae.
- Blisters worry both patients and their doctors, but localized blisters are unlikely to be life-threatening. Always find out precisely what the patient or referring doctor means if they use the term blister.
- see ➔ Chapter 13, pp. 259–75 for details of blistering diseases.

Localized blistering that may present as an emergency

Acute contact dermatitis

(see ➔ Chapter 10, pp. 214–16.) The history may suggest the cause. The rash is itchy, rather than painful (as it is in cellulitis), and the skin is erythematous and oedematous. Scale takes time to develop and may not be apparent at presentation if the eruption is acute. A potent topical corticosteroid cream will act promptly. Patch testing is the investigation of choice if you suspect allergy, rather than irritancy.

Erythema multiforme

(see ➔ p. 111 and Fig. 5.3.)

Cutaneous infections

(see ➔ Chapter 6, pp. 129–48 and Chapter 7, pp. 149–69.)

- 1° herpes simplex infections may be oedematous and painful.
- Herpes zoster.
- Bullous impetigo.
- Blisters in association with cellulitis or erysipelas.
- Blisters associated with dermatophyte infection, usually on the feet.

Physical causes of blisters

- Insect bites: exclude 2° infections, and treat with a potent topical corticosteroid cream applied twice daily (bd).
- Sunburn and photosensitivity, including phytophotodermatitis (plant-induced photosensitivity) and drug-induced photosensitivity. Treat with a potent topical corticosteroid cream applied bd.
- Cold injury.
- Thermal burns: emergency care—remove burnt or hot wet clothes (unless stuck to skin), jewellery, etc. Cool the wound with running cold water for 20min—do not use ice, which causes vasoconstriction and more damage. Clinical findings indicate the depth of the burn:
  - Epidermal: painful, red, may be shiny and wet.
  - Superficial dermal: painful, pink, may have small blisters.
  - Mid-dermal: may lose sensation, slow capillary refill, dark pink, large blisters.
  - Deep dermal: absent sensation, absent capillary refill, blotchy red, may have blisters.
  - Full thickness: absent capillary refill and sensation, no blisters, thick white leathery appearance. Will heal with scarring.

Refer patients with burns to a plastic surgeon for definitive treatment.

Erythema multiforme

EM is an acute, self-limited illness triggered by infections, most often HSV. Other causes include orf, histoplasmosis, VZV, and cytomegalovirus (CMV). Drugs cause <10% of cases. EM resolves within 4 weeks, but some patients have recurrent episodes, usually triggered by herpes simplex infections. Persistent EM (rare) associated with atypical inflammatory lesions is caused by viruses (EBV, CMV, and HSV), IBD, and malignancy.

What should I look for?

(See Box 5.9.)

- Symmetrical, well-defined, round erythematous macules on knees, elbows ± palms; evolve into papules and target lesions, which may blister over days. Infrequently, the rash is widespread (see Fig. 5.3).
- Lesions are at different stages of development (multiform).
- Patients may have few ulcers on one mucosal surface—usually oral.
- EM can be often confused with urticaria (see Box 5.10).

What should I do?

- Take a skin biopsy to confirm the diagnosis.
- Take swabs from oral lesions for viral culture.
- Mouth ulcers: prescribe a mouthwash for pain relief, if required (see ➔ Box 14.4, p. 283).
- Cutaneous lesions: a potent topical corticosteroid cream applied bd may relieve discomfort but will probably not speed resolution.
- Recurrent disease: consider prophylaxis with aciclovir 400mg bd.

Box 5.9 Clinical criteria for diagnosing erythema multiforme

- Acute, self-limited illness with duration of <4 weeks.
- Symmetrical, discrete, round erythematous macules/papules that persist at the same site for at least 7 days.
- Some papules evolve into target lesions comprising two or three concentric zones of colour change. A central dusky purple zone is surrounded by an outer red zone and sometimes a white middle zone. The centre may blister or crust after several days.
- No mucosal involvement/one mucosal surface involved.

Box 5.10 Differences between erythema multiforme and urticaria

- Urticarial lesions resolve within 48h (usually much quicker) and disappear to leave normal skin; EM papules persist for ≥7 days.
- New crops of urticarial lesions continue to appear daily for weeks. All lesions of EM appear within the first 3 days.
- Urticarial papules may evolve into rings, but the central skin is normal. The central zone in the target lesions of EM is dusky, bullous, or crusted.
- Urticaria, unlike EM, may be associated with angio-oedema (deep swelling) of subcutaneous tissues or mucosa.
- EM, unlike urticaria, may be associated with mucosal ulcers.

Generalized blisters

! Generalized herpes simplex infection (eczema herpeticum)

A generalized infection commonly seen in children with atopic eczema. Eczema herpeticum is life-threatening, particularly in children, when it is generally a 1° infection (see ➔ pp. 152–5). Twenty percent of children will have localized recurrent infections. Generalized herpes simplex infection may also complicate burns and some rare skin diseases (pemphigus, Darier disease, ichthyosis).

What should I look for?
See Box 5.11.

What should I do?
See Box 5.12.

! Staphylococcal scalded skin syndrome

(see ➔ pp. 136–7.) SSSS occurs most often in infancy or children, and less often in the elderly or immunosuppressed. Patients have localized focus of staphylococcal infection, often in the nasopharynx. Staphylococci release epidermolytic toxins (ET) that cleave the skin high in the epidermis.

What should I look for?

- The skin is erythematous and tender. The erythema is accentuated in flexural and periorificial skin (see ➔ Fig. 6.2, p. 131).
- The thin-walled blisters rupture rapidly but are extremely superficial and, unlike the blisters of TEN, do not involve all the epidermis.
- Mucosal surfaces are never involved in SSSS. This and the superficial blistering help to distinguish SSSS from TEN.
- The diffusely erythematous rash in SSSS may resemble TSS in early stages, before blisters appear, but patients with TSS are more ill with high fever and hypotension (see ➔ Box 5.2, p. 101).

►► *What should I do?*

- Culture swabs from possible foci of infection, e.g. nasopharynx. Culture of skin swabs from the blisters is negative in SSSS, because the condition is mediated by a toxin.
- If it is difficult to differentiate SSSS from TEN and if you want a rapid answer, gently peel the roof off a fresh blister; roll the fragment of tissue into a scroll, and take the fresh tissue to the pathology laboratory. Ask the pathologist to examine a frozen section. In SSSS, you will see the horny layer and very little else, whereas in TEN you will see full-thickness necrotic epidermis.
- Alternatively, you can make a Tzank preparation from the base of a fresh blister (see ➔ p. 647). In SSSS, there will be some keratinocytes in the smear, because the split is high within the epidermis. In TEN, the Tzank preparation from the base of a fresh blister has no keratinocytes, because the entire epidermis forms the blister roof.
- Treat with flucloxacillin—the prognosis is generally excellent.

Box 5.11 Signs of eczema herpeticum

(see Fig. 7.3, p. 155.)

- Sudden deterioration of atopic eczema.
- Pain as well as intense itching.
- Malaise, vomiting, anorexia, diarrhoea.
- Fever in variable number of cases.
- Vesicles (1° lesions), papules, and pustules (older lesions).
- Small punched-out ulcers, sometimes clustered, but these may coalesce into ulcers with irregular outlines.
- Oozing, sometimes haemorrhagic, and crusting.
- Lymphadenopathy.
- Signs may be difficult to differentiate from VZV (see pp. 156–7) or bacterial infection. 2° bacterial infection may complicate the picture.

Box 5.12 Management of eczema herpeticum

- Take swabs ideally from vesicles for both viral studies (polymerase chain reaction, PCR) and bacterial culture (see p. 644).
- A Tzank smear preparation can be used to confirm the diagnosis (see p. 647).
- Treat with an oral antiviral, e.g. aciclovir, as well as an antibiotic such as flucloxacillin. Severe cases will need IV aciclovir.
- Provide antipyretics, parenteral fluids, and pain relief.
- Eczema: in severe cases, all topical corticosteroids should be avoided until infection is controlled. In less severe cases, it may be reasonable to treat eczema with a mild topical corticosteroid ointment, once antivirals have been started.
- For less common causes of generalized blisters, see Box 5.13.

Box 5.13 Less common causes of generalized blisters that may present as an emergency

- Damage to basal cells (lichenoid reaction):
  - SJS (see p. 116).
  - TEN (full-thickness necrosis of epidermis) (see p. 116).
  - Acute GVHD (see pp. 532–3).
- Autoimmune blistering diseases (see pp. 264–5):
  - BP—the commonest.
  - Pemphigus foliaceus, pemphigus vulgaris.
  - Other rare autoimmune blistering diseases.
- Bullous cutaneous vasculitis (bullae will be haemorrhagic) (see p. 440).

Severe cutaneous adverse reactions

Cutaneous adverse drug reactions are common (2–3% of hospitalized patients). Most drug rashes are morbilliform and settle quickly when the drug is stopped (see ➔ p. 372), but rarely a morbilliform rash is the first sign of a much more serious problem such as TEN (see ➔ p. 370).

Life-threatening reactions that are discussed in this book include:

- Anaphylaxis, angio-oedema, and serum sickness (see ➔ pp. 104–5 and p. 373).
- Erythroderma (see ➔ p. 108).
- SJS (see ➔ Fig. 5.4, pp. 116–20).
- TEN (see ➔ Fig. 5.5, pp. 116–20).
- DRESS (see ➔ pp. 122–3).
- Anticoagulant-induced purpura fulminans and skin necrosis (see ➔ pp. 102–3).

What should I look for to help me to decide if this adverse drug reaction might be life-threatening?

see ➔ Box 5.14 and Chapter 18, p. 370 for information about adverse drug reactions.

! Box 5.14 Indications of severe cutaneous adverse reactions

- General:
  - Fever >40°C, hypotension, lymphadenopathy.
  - Arthralgia, arthritis.
  - Dyspnoea, wheeze.
- Mucocutaneous:
  - Erythroderma (generalized scaly erythema).
  - Swollen face, swelling of tongue, urticaria (anaphylaxis).
  - Skin pain or burning (TEN).
  - Erosions; shearing stress detaches epidermis from dermis; bullae or mucosal erosions (TEN or SJS) (see Fig. 5.5).
  - Vasculitis (palpable purpura) or extensive flat purpura.
- Laboratory results:
  - Eosinophilia >1000mm^3
  - Lymphocytosis with abnormal lymphocytes
  - Abnormal liver function tests (LFTs).

Fig. 5.3 Well-defined target lesions on the palms in erythema multiforme.

Stevens–Johnson syndrome and toxic epidermal necrolysis

These severe reactions with overlapping features (see Box 5.15) are commonly caused by drugs (see Box 5.16). Onset 2–3 weeks after commencing the drug. Patients often have underlying diseases, e.g. acquired immune deficiency syndrome (AIDS).

❗ If you see any patient with a painful erythematous 'drug rash' and mucosal signs, alarm bells should ring.

Stevens–Johnson syndrome

What should I look for?

- Vague upper respiratory tract symptoms (fever, cough, headache, sore throat, rhinorrhoea, and malaise) for 1–3 days prior to onset.
- Symmetrical, poorly defined painful erythematous macules evolving rapidly to papules and target lesions, with 2/3 concentric zones of colour change and dark purpuric centres.
- Large bullae rupture, leaving denuded skin, but typical SJS affects <10% of BSA (see Fig 5.4).
- The rash is usually maximal in 4 days.
- ❗ Severe mucosal ulceration with involvement of at least two mucosal surfaces (lip, oral cavity, conjunctiva, nasal, urethra, vagina, GI tract, respiratory tract). The lips become crusted and haemorrhagic. Painful stomatitis interferes with eating and drinking. Purulent conjunctivitis is associated with photophobia.
- Late ocular complications include dry eye, scarring, and loss of vision (see Box 5.20).

Toxic epidermal necrolysis

What should I look for?

- Flu-like symptoms may precede TEN. Fever is higher than in SJS.
- ❗ Complaints of painful skin or a burning discomfort should alert you to the possibility of TEN. These symptoms are not a feature of morbilliform drug reactions and are rare in other conditions.
- Tender, dusky erythema resembling SJS progressing to widespread subepidermal blistering (the roof of blisters is formed by dead epidermis). The flaccid, thin-walled blisters rupture easily (see Fig. 5.5). For differential diagnosis, see Box 5.17.
- Large sheets of necrotic epidermis that detach, leaving a painful denuded oozing dermis involving >30% of BSA.
- Gentle pressure extends the blisters, and the erythematous epidermis is detached from the underlying dermis by lateral pressure (Nikolsky sign).
- ❗ Inadvertent shearing pressure when handling the patient may cause further detachment.
- Examine the mouth, eyes, and genitalia. Mucosae usually involved (unlike SSSS). Ulceration causes dysphagia, photophobia, or painful micturition.
- Epithelia of the respiratory tract and GI tract may be involved.
- Sequelae may include scarring, irregular pigmentation, dry eye, corneal scarring, photophobia, visual impairment, phimosis, and vaginal synechiae (see Box 5.20).

Box 5.15 Classification of Stevens–Johnson syndrome and toxic epidermal necrolysis

The features in severe SJS overlap with those of TEN:

- SJS: <10% of BSA involved.
- Overlap SJS–TEN: 10–30% of BSA involved.
- TEN: >30% of BSA involved.

Box 5.16 Causes of Stevens–Johnson syndrome and toxic epidermal necrolysis

- Drugs (85%): sulfonamides, NSAIDs, antibiotics—penicillin-related and cephalosporin, anti-epileptics, barbiturates, allopurinol, tetracyclines.
- Viral infections: AIDS, HSV, EBV, influenza.
- Bacterial infections: *Mycoplasma pneumoniae*, typhoid, group A streptococci.
- Fungal infections: dermatophyte, histoplasmosis.
- Protozoal infections.
- Malignancy: Hodgkin lymphoma, leukaemia.
- GVHD.

Box 5.17 Toxic epidermal necrolysis: differential diagnosis

No mucosal signs

- TSS (no blisters, but may simulate early TEN) (see ➔ Box 5.2, p. 101).
- SSSS (also has blisters) (see ➔ p. 112, and Fig. 6.2, p. 131).
- Morbilliform drug rash (no blisters, but may simulate early TEN).
- DRESS (see ➔ pp. 122–3).
- Generalized herpes simplex infection: widespread vesicles (see ➔ p. 112).
- Varicella-zoster infection: widespread vesicles (see ➔ p. 156).

Mucosal involvement

- Acute GVHD.
- Pemphigus vulgaris (may be drug-induced).

Management

see ➔ Management in Stevens–Johnson syndrome or toxic epidermal necrolysis, pp. 118–20.

Further reading

Gueudry J *et al*. *Arch Dermatol* 2009;**145**:157–62 (ocular complications).
Koh MJA and Tay YK. *Curr Opin Pediatr* 2009;**21**:505–10.

Management in Stevens–Johnson syndrome or toxic epidermal necrolysis

▶▶ What should I do?

- Withdraw all drugs started in the last 4 weeks.
- Take a skin biopsy to confirm the diagnosis. Ensure the specimen includes the epidermis, which may detach during the procedure. The histology shows full-thickness epidermal necrosis in the roof of blisters.
- Rapid information about the level of blistering to differentiate TEN from SSSS may be obtained by detaching the roof of a blister, rolling this up gently, and freezing in liquid nitrogen. Frozen sections of the epidermis can be stained immediately. A Tzank preparation (see ➔ p. 647) can also be used to differentiate TEN from SSSS (see ➔ p. 112).
- Calculate the SCORTEN (see Box 5.19).
- ❗ Involve a multidisciplinary team, and remind those caring for the patient that TEN is associated with less oedema, less fluid loss, and less vascular damage than a severe burn. Do not overload with fluids (see Box 5.18).

Box 5.18 Management—essentially supportive

- Admit to intensive care unit (ICU), high-dependency unit (HDU), or burn unit (familiar with TEN), particularly if significant skin loss.
- Relieve pain with regular paracetamol and oral opiates—avoid NSAIDs. Use a VAS to assess pain in all conscious patients daily.
- Establish peripheral venous access through non-affected skin, if possible, and change every 48h.
- Central venous lines—risk of infection but can be useful for haemodynamic monitoring.
- Nutrition: nasogastric (NG) tube (Dobhoff). Avoid total parenteral nutrition (TPN). Provide continuous enteral nutrition—deliver up to 25kcal/kg/day in early phase, and increase to 30kcal/kg/day in anabolic recovery phase.
- Proton pump inhibitor in acute phase to protect GI mucosa.
- Foley catheter: keep urine output at 40–60mL/h.
- Fluids: do not overload. Pulmonary oedema is a frequent complication because of impaired alveolar barrier.
- Nurse on an air mattress (air fluidized beds increase transepidermal loss of water and heat); use non-stick sheets and heating blankets; keep ambient room temperature at 30–32°C.
- Skin: protect with 50% white soft paraffin in liquid paraffin, e.g. Emollin spray®, and non-adherent dressings, e.g. Mepitel®/ Mepilex®, or wrap the patient with Sofsorb™ impregnated with 0.5% silver nitrate solution.
- Prevent infection: reverse barrier-nursing; culture multiple skin sites, sputum, blood, and urine. Repeat every 3 days to guide therapy in the event of sepsis. Avoid prophylactic antibiotics.
- Eyes: 2-hourly application of lubricants, e.g. hypromellose eye drops; involve ophthalmologists to advise, e.g. use of topical corticosteroids,

Box 5.18 (*Contd.*)

gently separating synechiae to prevent scarring, corneal fluorescein examination for ulceration; for severe loss of ocular surface epithelia, consider amniotic membrane transplantation.

- Clean lips and oral mucosae with antiseptic mouthwashes/oral sponge.
- Use WSP 2-hourly on lips; protect oral mucosa with mucoprotectant mouthwash three times daily (tds)—e.g. Gelclair®.
- Provide local analgesia for oral mucosa, e.g. viscous lidocaine 2%.
- Consider potent topical steroid mouthwash qds.
- For urogenital skin, use white soft paraffin 4-hourly. Soft silicone dressing (e.g. Mepitel®) prevent adhesions and can be used on eroded vulval and vaginal areas. Dilators or tampon wrapped in soft silicone dressing e.g. Mepitel® should be inserted into the vagina to prevent synechiae.
- IV Ig (1g/kg) for 4 days is controversial. May not improve survival.
- Avoid systemic or potent topical corticosteroids (efficacy unproven, increased risk of infection).
- Subcutaneous heparin for DVT prophylaxis.
- Neutropenic patients may benefit from granulocyte colony-stimulating factor (G-CSF).

Box 5.19 Severity of illness score for Stevens–Johnson syndrome/toxic epidermal necrolysis: SCORTEN

- Rash usually maximal in 4 days, and the skin may start to re-epithelialize within days, but recovery is protracted.
- Sepsis and co-morbidities contribute to high mortality.
- Seven independent factors have been recognized as predictors of mortality. Allocate 1 point for each risk factor present at admission:
  - Age >40 years.
  - Malignancy.
  - Tachycardia >120bpm.
  - Initial percentage of epidermal detachment above 10%—use the rule of nines to calculate % of BSA affected (see ➔ Box 3.5, p. 51), or estimate using the patient's palms as a measure. Each palm without fingers = 0.5% of BSA.
  - Serum urea >10mmol/L.
  - Serum glucose >14mmol/L.
  - Bicarbonate <20mmol/L.

Predicted mortality

- 0–1: 4%
- 2: 12%
- 3: 32%
- 4: 62%
- 5: 85%
- 6: 95%
- 7: 99%

SCORTEN should be calculated in all patients with SJS/TEN within 24h of admission and re-measured at day 3/4.

Box 5.20 Long-term sequelae of Stevens–Johnson syndrome/toxic epidermal necrolysis

Chronic phase can develop insidiously over weeks/months.

- Fatigue—especially in early stage after discharge.
- Eyes: photophobia, pain, dryness, corneal scarring, and visual impairment.
- Skin dyspigmentation.
- Loss of nails/dystrophic growth.
- Urogenital dryness or scarring.
- Mouth dryness, scarring, or dental caries.
- Post-traumatic stress disorder, including nightmares, anxiety, and depression.
- Lungs: bronchiectasis.
- GIT—stenosis.
- Loss of muscle mass.

Further reading

Bastuji-Garin S *et al*. *J Invest Dermatol* 2000;**115**:149–53 (SCORTEN).

Harr T and French LE. *Orphanet J Rare Dis* 2010;**5**:39. Available at: www.ojrd.com/content/5/1/39.

Huang YC *et al*. *Br J Dermatol* 2012;**167**:424–32 (IVIg in TEN).

The RegiSCAR Project. Available at: www.regiscar.org/ (severe cutaneous adverse drug reactions).

Fig. 5.4 Stevens–Johnson syndrome with crusted haemorrhagic lips and extensive mucosal ulceration in the oral cavity.

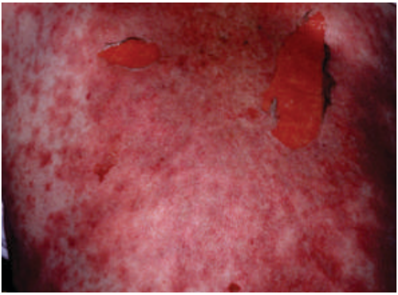

Fig. 5.5 The dusky erythema in toxic epidermal necrolysis progresses to widespread blistering, with loss of the full thickness of the epidermis. Even gentle pressure dislodges sheets of the necrotic epidermis.

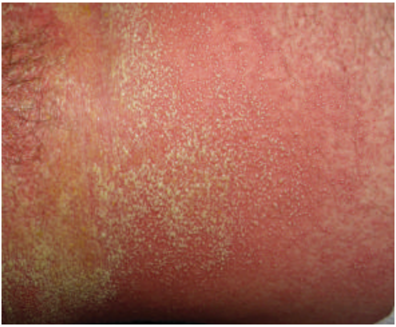

Fig. 5.6 Sheets of sterile non-follicular pustules on an erythematous background in acute generalized exanthematous pustulosis (AGEP). Generalized pustular psoriasis looks similar.

Drug rash, eosinophilia, and systemic symptoms (DRESS)

Synonym: drug-induced hypersensitivity syndrome.

DRESS is a life-threatening cutaneous drug reaction with systemic symptoms that is frequently misdiagnosed as infection. Reactivation of human herpesvirus may play a part in the pathogenesis. Diagnostic criteria are listed in Box 5.21. Commonly develops 1–2 months after the drug is started, but presentation is variable (see Boxes 5.22 and 5.23). The course may be protracted with major morbidity, including progression to liver failure. Mortality is 10%.

What should I look for?

- History of new drug started within prior 2 months (see Box 5.22).
- Flu-like symptoms, e.g. myalgia, arthralgia, headache, sore throat, abdominal pain.
- Fever.
- Sudden-onset diffuse erythematous morbilliform scaly rash with 'juicy' papules that coalesce to plaques (see Box 5.21 for clinical phenotypes). In some cases, onset subacute and protracted.
- Sometimes progresses to erythroderma.
- Pustules or vesicles or purpura.
- Facial oedema (simulates angio-oedema).
- Mild cheilitis/mucosal involvement.
- Lymphadenopathy.

What investigations should I do?

- FBC: eosinophilia common, less often lymphocytosis (sometimes atypical lymphocytes—check blood film), thrombocytopenia, neutrophilia/neutropenia.
- ESR and CRP (raised).
- Liver function tests (LFTs) (raised transaminases and γ glutamyl transferase): risk of liver failure.
- Immunoglobulins (risk of hypogammaglobulinaemia).
- Renal function tests and urinalysis: usually normal.
- Skin biopsy: lymphocytic infiltrate with few eosinophils; severe dyskeratosis may indicate greater systemic involvement.
- Blood cultures: sterile. EBV and CMV titres.

How should I manage DRESS?

- Stop suspected drugs.
- Manage temperature, fluid balance, etc., as for erythroderma (see ➔ Box 5.8, p. 109).
- Emollients, such as 50% WSP in liquid paraffin, to make the skin comfortable, e.g. Emollin® spray.
- Potent topical steroid ointment bd.
- Early treatment with oral prednisolone (0.8–1mg/kg body weight) or IV methylprednisolone recommended to prevent progression.

⚠ Box 5.21 Diagnostic criteria for DRESS

Three or more of the following:

- Acute drug rash (four recognized morphologies):
  - Urticated papular eruption.
  - EM-like (EM-like, atypical targets, often purpuric evolving into confluent areas of dusky erythema)—may be prognostic of more severe hepatic involvement.
  - Morbilliform erythema.
  - Exfoliative erythroderma.
- Fever >38°C.
- Lymphadenopathy.
- Systemic involvement variable; commonest hepatitis (↑ aspartate aminotransferase, AST), interstitial nephritis/pneumonitis, arthritis, and/or pericarditis.
- Blood anomalies: eosinophilia ± abnormal lymphocytes.

Box 5.22 Drugs associated with DRESS

Most frequent

- Anticonvulsants, dapsone, sulfonamides, including sulfasalazine.

Others

- Allopurinol.
- Antibiotics: vancomycin, minocycline, amoxicillin.
- Gold salts.
- Sorbinil.
- Calcium channel blockers.
- Ranitidine.
- Thalidomide.
- Mexiletine.
- Telaprevir (protease inhibitor to treat chronic hepatitis C).
- Imatinib.

Box 5.23 Differential diagnosis of DRESS

- Serum sickness or drug-induced vasculitis (see ➔ p. 373, p. 376).
- Infection: viral (HHV 6, CMV, EBV) or bacterial.
- Idiopathic hypereosinophilic syndrome.
- Lymphoma.
- Acute GVHD (lower T-regulatory cell levels)

Further reading

Husain Z *et al. J Am Acad Dermatol* 2013;**68**:693 and 709.
Kardaun SH *et al. Br J Dermatol* 2013;**169**:1071–80.

Generalized pustules

Introduction

Pustules are a focal accumulation of inflammatory cells and serum in skin. Pustules may indicate infection, e.g. staphylococcal folliculitis (pustules associated with hair follicles), but, in most infections, pustules are grouped, localized, and asymmetrical. For causes of widespread symmetrical pustular eruptions in sick patients, see Box 5.24.

What should I look for?

- Is the patient ill, e.g. high fever, malaise?
- What is the 1° lesion—is it really a pustule, i.e. a bump filled with creamy fluid, or is it a vesicle, i.e. a bump filled with clear serous fluid that evolves into a pustule (as in VZV infection)?
- Distribution: symmetrical (endogenous cause) or asymmetrical?
- How do pustules evolve? What is the time course?
- Are pustules associated with hair follicles, i.e. is it a folliculitis?
- Erythema, oedema, desquamation, or mucosal involvement.

Acute generalized exanthematous pustulosis (AGEP)

Over 90% of cases are drug-induced. Usually takes 1–3 weeks for rash to develop, but patients who are sensitized may develop AGEP within a few hours of exposure to the drug—for signs, see Box 5.25. May be difficult to distinguish from generalized pustular psoriasis (GPP).

What should I do?

- Reassure the patient and medical and nursing staff that this is a self-limiting condition and is not an infection.
- Culture blood and skin—these are sterile.
- Take a skin biopsy (shows subcorneal pustules).
- Stop the causative drug, most often an antibacterial.
- Prescribe an emollient, e.g. 50% WSP in liquid paraffin, a mild topical corticosteroid, and a sedating antihistamine to relieve discomfort.

Generalized pustular psoriasis

❗ Some patients have a history of psoriasis/psoriatic arthritis. Pustular psoriasis may be triggered by irritation of the skin, infections, withdrawal of systemic corticosteroids or drugs (lithium), but the trigger may not be identified. Lasts longer than AGEP (see earlier in this section and Box 5.26).

What should I do?

- Admit the patient, and manage skin failure (see ➔ Box 5.7, p. 107).
- Exclude an infection—cultures of blood and skin will be sterile. Reassure the patient and medical and nursing staff that this is not an infection.
- Check FBC, renal function, liver function, and serum calcium.
- Obtain a skin biopsy to confirm the diagnosis (subcorneal pustules).
- Avoid potential irritants. Prescribe an emollient, e.g. 50% WSP in liquid paraffin, a mild topical corticosteroid, and a sedating antihistamine to relieve discomfort.
- Continue to monitor for hypocalcaemia (rare complication).
- Consider acitretin or methotrexate—if not settling, see ➔ p. 668.

Box 5.24 Causes of widespread pustular eruptions

- Eczema herpeticum or varicella: the 1° lesion is actually a vesicle, but pustules with crusting occur as a 2° phenomenon—check for a history of atopic eczema (see ➔ p. 112 and p. 156).
- Pustular psoriasis: check for a history of psoriasis or arthritis.
- AGEP: ask about drugs started in previous 1–3 weeks.
- Acne fulminans: severe acne flare with haemorrhagic, crusted ulcers on the trunk and face and systemic symptoms (see ➔ p. 248).
- Subcorneal pustular dermatosis/IgA pemphigus (rare).

Box 5.25 Acute generalized exanthematous pustulosis—what should I look for?

(See Fig. 5.6.)
- Sudden onset of an oedematous erythema that burns or itches. Often begins on the face, axillae, and groins and spreads within hours.
- Flexural accentuation is a common feature.
- Soon followed by the appearance of hundreds of pinhead-sized, mostly non-follicular, pustules.
- Confluent pustules may produce superficial erosions.
- Some patients have marked facial swelling; few develop purpura, on legs. May also develop vesicles, bullae, and EM-like lesions.
- Over 20% have mild mucosal involvement (usually oral).
- Fever >38°C and neutrophilia (blood culture and swabs sterile).
- Systemic involvement unusual; liver, kidney, bone marrow, and lung involvement (associated with elevated CRP levels).
- Spontaneous resolution of pustules in <15 days (more rapid than in pustular psoriasis), followed by desquamation.

! Box 5.26 Generalized pustular psoriasis—what should I look for?

- Evidence of psoriasis, e.g. nail pitting, psoriatic arthritis, but many patients report no prior history of psoriasis.
- Fever and neutrophilia (swabs and blood cultures sterile).
- Scaly, erythematous, tender skin.
- Sheets of sterile non-follicular pustules, particularly in flexures and genital skin, arising on a background of tender erythema.
- Pustules may coalesce into lakes of pus.
- Pustules dry and skin desquamates, leaving a shiny erythematous surface on which waves of pustules may continue to appear.
- Normal mucosal surfaces.
- The process may continue for weeks (unlike AGEP).
- Complications are those of skin failure (see ➔ p. 106).
- Hypocalcaemia is a rare complication.

Further reading

Sidoroff A *et al*. *J Cutan Pathol* 2001;**28**:113–19.

Necrotizing fasciitis

What is necrotizing fasciitis?

Necrotizing fasciitis is a ***life-threatening*** soft tissue infection, characterized by rapidly progressive necrosis that spreads from subcutaneous tissue into the deep fascia. High mortality (25%) 2° to organ failure and to streptococcal TSS (see ➔ Box 5.2, p. 101 and Box 5.27).

Consider the diagnosis in any sick patient not responding to treatment for 'cellulitis' with pain that appears out of proportion to the signs. The depth and extent of necrosis may be much greater than the appearance of the skin suggests, and, in early stages, the condition is often misdiagnosed. Rarely can present at multiple sites.

What should I look for?

- History of recent trauma, e.g. insect bite, skin biopsy, chronic leg ulcer, surgery, IV drug abuse. Not all patients have an obvious portal of entry. For predisposing factors, see Box 5.28.
- Ill patient with high temperature, tachycardia, low blood pressure.
- Pain that is out of proportion to the signs in the skin.
- Altered level of consciousness—signs of shock are associated with 50% mortality.
- Rapidly spreading, poorly demarcated purplish erythema—commonly extremities > perineum, genitalia (Fournier gangrene—polymicrobial infection) > trunk.
- Oedema extending beyond erythema and tender induration/blisters.
- Malodorous serosanguineous exudate ('dishwater' pus).
- Crepitation in soft tissues (gas from aerobic and anaerobic bacteria).

In the later stages, you may see:

- Black necrotic plaques.
- Painless ulcers.
- Vascular occlusion and gangrene.

►► What should I do?

- Request an urgent surgical opinion—immediate deep surgical debridement of all necrotic tissue is absolutely essential. The diagnosis is clinical—do not wait for results of investigations (see Box 5.29).
- Start a combination of IV broad-spectrum antibiotics immediately, and contact a microbiologist to discuss treatment (see Box 5.30).
- Gram stain and culture any blister fluid/fluid draining from the wound.
- Check FBC, electrolytes, and renal function.
- Arrange to take deep tissue at the time of surgery for histological examination and culture for aerobic and anaerobic bacteria.
- Magnetic resonance imaging (MRI) or computed tomography (CT) to identify subcutaneous air and define the extent of involvement.
- Admit to the intensive care unit (ICU) for resuscitation and monitoring.
- Consider IV Ig for streptococcal TSS.

Box 5.27 Classification of necrotizing fasciitis

- Type I: polymicrobial. Two or more pathogens. Obligate and facultative anaerobes. Trunk and perineum, older patients with co-morbidities, e.g. diabetes.
- Type II: group A β-haemolytic streptococci (50% occur in young healthy people) ± staphylococcal species (*Staphylococcus aureus*, including meticillin-resistant strains). Aggressive course. Associated with trauma, surgery, or IV drug use. Risk of TSS.
- Type III: *Clostridium* species (*C. perfringens* commonest—gas gangrene). Associated with trauma, e.g. surgery. Also seen with Gram-negative marine organisms, e.g. *Vibrio vulnificus* (commonest in Asia)—rapid progression.
- Type IV: fungal, e.g. *Candida* species, *Zygomycetes*. Limbs, trunk, perineum. Associated with immunosuppression.

Box 5.28 What factors predispose to necrotizing fasciitis?

- Diabetes, AIDS or other immune deficiency.
- Chronic alcohol or drug abuse, malnutrition.
- Surgery and other trauma.
- Obesity.
- Malignancy.
- Peripheral vascular disease.

Box 5.29 The finger test

The finger test may help to confirm the diagnosis. Can be performed at the bedside under local anaesthesia or during surgery.
- Make a 2cm vertical incision into the skin to the deep fascia.
- Push an index finger (or probe) gently into normal-appearing tissue at the junction of the subcutaneous tissue and fascia.
- If subcutaneous tissue easily dissected off underlying fascia, finger test is positive—confirming the diagnosis of necrotizing fasciitis. Necrotic tissue or 'dishwater' pus found between fascial planes.

Box 5.30 Choice of antibiotics in necrotizing fasciitis

- Initially, prescribe broad-spectrum antibiotic cover to treat Gram-positive and Gram-negative bacteria and anaerobes, e.g. vancomycin and meropenem.
- For those with penicillin allergy, use vancomycin plus clindamycin plus an aminoglycoside (or quinolone).
- Consult a microbiologist for advice on further treatment when the results of culture are known.

Further reading

Hakkarainen TW *et al*. *Curr Probl Surg* 2014;**51**:344–62.

Chapter 12

Pustular rashes

Contents

Acne: presentation

Acne is a disfiguring disorder of the pilosebaceous unit, usually starting between the ages of 12 and 14, that is experienced to some extent by almost all teenagers. Acne may persist into adult life, and we seem to be seeing more acne in older women. Pathogenesis is multifactorial (see Box 12.1).

What should I ask?

- When did the acne start? How bad is it today?
- What impact is the acne having on the patient's life? (see ➔ p. 32.) Acne may have a major psychosocial impact, eroding self-confidence and causing depression or even suicide.
- What treatment has the patient used and for how long? Most acne treatments take up to 4 months to have a maximal effect.
- How are topical preparations, such as benzoyl peroxide or topical retinoids, being applied—all over the face or trunk (as they should be) or just on 'spots'? Any adverse effects such as irritation of the skin?
- Oral antibiotics—most patients will need prolonged treatment for several years. What does the patient mean by 'taking a course'?
- What else is being applied to the skin—greasy cosmetics or pomades?
- What is the patient's occupation? Is he or she in a hot or humid environment, working with tars or oils, or wearing a cap that occludes the forehead?
- Is trauma irritating hair follicles (acne mechanica)—friction under chin straps, helmets, or shoulder pads (American footballers), or repeated rubbing of the face with exfoliative creams?
- Is the patient taking drugs that might exacerbate acne? (see ➔ Drugs and skin diseases, p. 384 and 388.)
- Refractory severe acne (see Box 12.4).

What should I look for?

(See Figs. 12.1 and 12.2.)

- Acne affects the face and/or upper trunk. If acne is in an unusual or asymmetrical distribution, could this be related to an external factor?
- More than one type of lesion: pustules, papules, comedones (blackhead = open comedone; whitehead = closed comedone), nodules, cysts, and scars (deep 'ice-pick' or flat).
- Record the types of lesions that are present, so you can monitor the outcome of treatment.
- Do comedones predominate? Suggests exposure to tar or chlorinated hydrocarbons. Could this be pomade acne on the forehead?
- Signs of virilization such as hirsutism. Check for cliteromegaly if you suspect virilization (see ➔ pp. 474–7).
- HAIR-AN syndrome (see ➔ p. 476).
- Even mild acne may cause acne keloids or, in dark skin, long-lasting hyperpigmentation. Prolonged courses of minocycline may cause hyperpigmentation in scars.
- See Box 12.2 for clinical variants.

Box 12.1 Pathogenesis of acne

The pathogenesis is not fully understood, but a number of factors play a part in triggering chronic inflammation around pilosebaceous units:

- Sebum excretion is increased (but may remain high when acne has cleared).
- Androgens stimulate sebum secretion by the sebaceous glands.
- Occlusion of follicular ducts by hyperkeratinization (may be exacerbated by application of some greasy topical preparations or tar).
- Bacterial colonization of follicular ducts by *Propionibacterium acnes*, a skin commensal. *P. acnes* breaks down triglycerides, releasing free fatty acids that trigger dermal inflammation. *P. acnes* may also activate Toll-like receptors on keratinocytes which induce inflammation.
- Increased fibroblast growth factor receptor 2 (FGFR2) signalling (mutations in FGFR2 leading to increased signalling cause acne in Apert syndrome; see Box 12.4). Androgens upregulate signalling.
- Genes: about 50% of patients have a family history of acne.
- Diet: a link between acne and dairy products has been suggested but needs more study.

Box 12.2 Variants of acne

- Acne 'excoriée de la jeune fille': young girls with mild acne, but obsessional picking scars the skin. Difficult to manage. Even mild disease may need systemic treatment (see ➔ p. 566).
- Neonatal and infantile acne: usually face. May persist until age 3 or 4. Consider inhaled corticosteroids as a cause (see ➔ p. 244).
- Acne conglobata: severe acne with grouped comedones, deep nodules, and abscesses with sinuses oozing pus (see ➔ p. 250).
- Acne fulminans: nodulocystic acne, fever, and joint pains (see ➔ p. 248).
- Synovitis, acne, pustulosis, hyperostosis, osteitis (SAPHO) syndrome (see ➔ p. 248).
- PAPA syndrome: rare AD auto-inflammatory disease (see ➔ pp. 234–5 and pp. 248–9).
- PASH syndrome: rare auto-inflammatory syndrome (pyoderma gangrenosum, acne, hidradenitis suppurativa).
- Comedo-like acneiform lesions (chloracne) associated with exposure to dioxin.

Further reading

Bhate K and Williams H. *Clin Exp Dermatol* 2014;**39**:273–8.
Lwin SM *et al. Clin Exp Dermatol* 2014;**39**:162–7.
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Acne caused by drugs

Drugs that cause or exacerbate acne

(see ➔ p. 388.)

A number of drugs may induce acne-like eruptions or exacerbate pre-existing acne (acne medicamentosa). Consider this possibility in patients with atypical presentations.

What should I look for?

- Acneiform rash (erythematous follicular pustules and papules).
- Unusual distribution, predominantly on the trunk and upper outer arms.
- Absence of comedones (whiteheads, blackheads).

Which drugs may be implicated?

- Anti-epileptics (phenytoin, carbamazepine, gabapentin).
- Anabolic steroids—danazole, testosterone. Ask about these drugs in athletes or body builders with acne not responding to treatment.
- Antitubercular drugs (isoniazid).
- Ciclosporin.
- Corticosteroids, including corticosteroid inhalers or inhalation through a face mask, e.g. in young children.
- Epidermal growth factor receptor inhibitors (used for the treatment of some cancers).
- Growth hormone.
- Halogenated compounds (iodides, radio-opaque contrast materials, bromides in sedatives, analgesics).
- Lithium.
- Progestogens such as medroxyprogesterone.
- Vitamin B_{12} (cyanocobalamin).

Further reading

Dessinioti C *et al*. *Clin Dermatol* 2014;**32**:24–34.

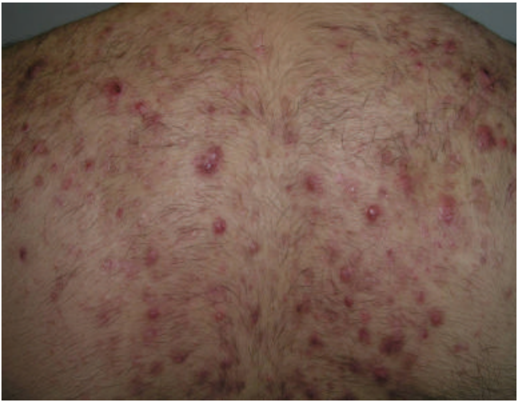

Fig. 12.1 Acne with papules, pustules, and nodules. Reproduced with permission from Warrell D *et al.* (eds) *Oxford Textbook of Medicine*, 5th edn, Layton A. Sebaceous and sweat gland disorders 2010. Oxford: Oxford University Press.

Fig. 12.2 Severe cystic acne for which isotretinoin is indicated.

Fig. 12.3 Rosacea: erythema, papules, and pustules, but no comedones. A similar pustular rash may be induced by topical or oral corticosteroids.

Acne vulgaris: management

What should I do?

- Provide written information about acne—reassure that the skin is not dirty and that you can help.
- Explain that all treatments take 3–4 months to have a maximal effect and that acne may get a little worse before it gets better. Treatment may have to be continued for months or even years.
- If acne is not responding to treatment (and the patient is following your advice), have you made the right diagnosis? Is there an underlying systemic disease? (See Boxes 12.3 and 12.4.)

Mild acne

Start one or more topical preparations (combination more effective than monotherapy). Explain treatment should be applied to all the skin, not just 'spots'. Continue treatment for at least 4 months and usually longer. Topical preparations (cream, gel, or lotion) may include:

- Antibiotics, e.g. erythromycin or clindamycin lotion.
- Benzoyl peroxide 2.5% (as effective and less irritant than higher percentage, bleaches clothing; can combine with a topical retinoid).
- Topical retinoid (avoid in pregnancy), e.g. tretinoin, isotretinoin, or 0.1% adapalene (a retinoid-like drug). All are irritants and may cause post-inflammatory hyperpigmentation in dark skin. Increase contact time gradually, washing off after 30–60min at first.

Moderate acne

- Oral antibiotics for a 4-month course initially, e.g. oxytetracycline 500mg bd, lymecycline 408mg/day, or erythromycin 500mg bd, combined with topical benzoyl peroxide to reduce the incidence of bacterial resistance to antibiotics. Do not combine with a topical antibiotic. Avoid tetracyclines in children aged <12 years.
- ♀: oral contraceptive with anti-androgen activity such as co-cyprindiol (takes 6 months to have maximal effect). Avoid contraceptive pills containing norethisterone (androgenic properties).

Severe acne or acne with a major psychosocial impact

(See Fig. 12.2.)

- Refer to a specialist. The oral retinoid isotretinoin may be the best option (0.5–1.0mg/kg/day; cumulative dose 120–150mg/kg). Isotretinoin is teratogenic. Other adverse effects include mucosal dryness and, very rarely, depression. Contraception essential in women of childbearing age during treatment and for 1 month after stopping isotretinoin (see ➔ Chapter 34, p. 669).
- Cysts: intralesional triamcinolone (0.1mL of 2.5–10mg/mL solution).

Scars

- Shallow scars become less obvious with time. Refer for cosmetic camouflage (see Changing Faces, available at: https://www.changingfaces.org.uk/Home).
- Surgical approaches (chemical peels, lasers, dermabrasion) may improve the appearance of deep scars but are not available through the National Health Service (NHS) and are not indicated until acne has been controlled.

Box 12.3 What is this pustular rash?

Papulopustular facial eruption, but no comedones

- Is this rosacea (see Fig 12.3)? Usually older patients (see ➔ pp. 254–5).
- Is the patient being treated with a systemic corticosteroid or another drug that may cause an acneiform rash? (see ➔ p. 244.)
- Is this perioral dermatitis (usually 2° to application of a topical corticosteroid)? (see ➔ pp. 256–7.)
- Could this be a bacterial folliculitis or fungal folliculitis, demodicosis, or pseudofolliculitis? (see ➔ pp. 134–5 and p. 178.)

Superficial small follicular pustules on back

- Is this pityrosporum folliculitis? (see ➔ p. 219.)
- ⚠ Could this be Behçet disease? (Rare; see ➔ pp. 284–5.)

Box 12.4 Why is severe acne not improving?

In patients with severe acne that is refractory to treatment, take the history again. Has anything been applied to the skin, or is the patient taking drugs such as anabolic steroids? Also consider:

- Polycystic ovary syndrome (obese ♀, oligomenorrhoea and/or infertility, hirsutism) (see ➔ pp. 476–7).
- Congenital adrenal hyperplasia (rare and caused by 21-hydroxylase deficiency) (see ➔ p. 474).
- An androgen-secreting tumour (adrenals, ovaries, or testes).
- XYY syndrome (tall ♂).
- Apert syndrome (low IQ, acrocephalosyndactyly) (see Box 12.1).

Further reading

Admani S and Barrio VR. *Dermatol Ther* 2013;**26**:462–6 (acne management).
Archer C *et al*. *Clin Exp Dermatol* 2012;**37 Suppl 1**:1–6 (diagnosis and management of acne).
Dessinioti C *et al*. *Clin Dermatol* 2014;**32**:24–34 (reviews acneiform rashes).
Strauss JS *et al*. *J Am Acad Dermatol* 2007;**56**:651–63 (guidelines).

Acne with arthritis

⚠ Acne fulminans

This emergency affects adolescent boys who have previously had mild to moderate acne (see Boxes 12.5 and 12.6). Haemorrhagic, crusted, ulcerative skin lesions erupt on the back, chest, arms, and face. Systemic symptoms include fever, malaise, weight loss, myalgia, and arthralgia. Patients may have hepatosplenomegaly or erythema nodosum.

Radiographs show lytic bone lesions, most often affecting the sternum and clavicle, but osteolytic lesions have been seen in the hips, ankles, and humerus. Osteolytic lesions are sterile. Sternoclavicular hyperostosis and sacroiliitis have also been described. Periosteal formation of new bone, sclerosis, and thickening occur late. Bony changes are transient, and the prognosis is good. Findings may be consistent with SAPHO syndrome.

Acne conglobata

Seronegative spondyloarthropathy and SAPHO syndrome have also been described in patients with acne conglobata (see ➔ p. 250).

What is SAPHO syndrome?

SAPHO syndrome, an auto-inflammatory bone disease, may account for 4% of all patients with seronegative spondyloarthropathies. Inflammatory bowel disease is associated with SAPHO.

The syndrome is characterized by:

- Intermittent predominantly axial **S**ynovitis (sternum, clavicles, ribs, spine, pelvis).
- **A**septic **O**steitis and **H**yperostosis. Bony pain is common.
- CT scans show lytic bone lesions—these are sterile.
- Chronic (or relapsing) sterile pustular **A**cneiform eruptions or palmoplantar **P**ustulosis (for other skin diseases, see Box 12.7).
- Leucocytosis and anaemia.

Treatment is unsatisfactory. Tetracycline antibiotics or isotretinoin may control acne. Topical steroids and acitretin may control palmoplantar pustulosis. NSAIDs, prednisolone, and methotrexate have been used for synovitis. Infliximab (anti-TNF) and bisphosphonates (suppress osteoclasts) have also been recommended.

⚠ PAPA syndrome

Very rare AD auto-inflammatory syndrome caused by mutations in *PSTPIP1/CD2BP1* gene encoding a protein that interacts with pyrin. PAPA, like FMF, is associated with decreased apoptosis and high levels of IL-1B (see ➔ pp. 234–6). Patients have severe destructive sterile arthritis (often precipitated by minor trauma), as well as scarring cystic acne, sterile abscesses (often at injection sites), and PG-like ulcers (see ➔ p. 310). Treatments include corticosteroids, etanercept, infliximab, and anakinra.

Also see PASH (Pyoderma gangrenosum, Acne, Suppurative Hidradenitis)—another very rare auto-inflammatory syndrome (see ➔ p. 234).

⚠ Box 12.5 What is acne fulminans?

Acne fulminans is a severe variant of cystic acne, characterized by:
- Sudden onset of inflamed crusted haemorrhagic plaques and ulcers on the face, trunk, and arms (usually in an adolescent boy who has had mild to moderate acne).
- Systemic symptoms that may include fever, malaise, weight loss, myalgia, synovitis, and arthralgia.

Box 12.6 Acne fulminans: what should I do?

►► Seek help from a dermatologist—acne fulminans is an emergency that needs immediate treatment.

Investigation
- Take skin swabs to exclude 2° staphylococcal infection.
- Check FBC (leucocytosis, anaemia), ESR (raised), and CRP (raised).
- Bone scans in areas of pain show increased uptake of radionuclide (not always required).

Immediate treatment
- Patients may need to be admitted, if very uncomfortable.
- Apply a potent topical corticosteroid bd to exuberant granulating ulcers. Antiseptic washes prevent 2° infection.
- Prescribe NSAID for arthralgia and myalgia.
- Start oral prednisolone 0.5–1.0mg/kg/day immediately. Continue for 4–8 weeks to control inflammation and prevent relapse.

Long-term treatment
- After 2–6 weeks of oral steroids, when the ulcerated lesions are beginning to settle, start isotretinoin, initially in a low dose (0.25–0.5mg/kg/day) to prevent a flare.
- After 4–6 weeks, provided acne has not deteriorated, gradually increase the dose of isotretinoin to 1.0mg/kg/day. Continue treatment for about 6 months.
- Tail off prednisolone, once the patient is established on treatment with isotretinoin.

Box 12.7 Skin diseases associated with SAPHO syndrome

- Acne fulminans.
- Acne conglobata and hidradenitis suppurativa.
- Palmoplantar pustulosis, GPP, plaque psoriasis.
- Other neutrophilic dermatoses, including PG, Sweet syndrome, and subcorneal pustular dermatosis.

Further reading

Hayem G. *Joint Bone Spine* 2007;**74**:123–6.

Follicular occlusion diseases: signs

Acne conglobata, hidradenitis suppurativa, and dissecting cellulitis of the scalp are diseases of adults in which follicular occlusion seems to play a part in pathogenesis. The conditions may coexist. Pilonidal cyst is another member of the 'folliculitis occlusion tetrad'. Seronegative spondyloarthropathy has been described in association with both acne conglobata and hidradenitis suppurativa, as has PG (see ➔ p. 310). ⚠ Rarely, SCC may complicate chronic disease.

Acne conglobata

What should I look for?

(See also Fig. 12.4.)

- Open comedones (blackheads) clustered into groups on the face and trunk. Papules, pustules, and nodules typical of acne.
- Fluctuant abscesses ('cysts') draining pus on the face, chest, and upper back, but also sometimes affecting the buttocks, groins, and axillae when signs overlap with those of hidradenitis suppurativa.
- Interconnecting sinus tracts linking adjacent abscesses.
- Large bowl-like scars, hypertrophic scars, or keloids.

Hidradenitis suppurativa

Hidradenitis suppurativa (acne inversa) is a chronic debilitating inflammation of the flexures, often misdiagnosed as staphylococcal furunculosis. Pathogenesis is uncertain, but follicular plugging may cause dilatation and rupture of follicles with 2° inflammation of apocrine glands. Hidradenitis is associated with obesity, polycystic ovary syndrome, and insulin resistance. Also associated with IBD, particularly Crohn disease. Cutaneous Crohn disease may simulate hidradenitis (see ➔ p. 508). Rarely associated with PG and acne (PASH syndrome) (see ➔ p. 234).

What should I look for?

(See also Figs. 12.5 and 12.6.)

- Recurrent 'boils' in axillae, breasts, groins, vulva, or perianal skin. Women tend to have involvement of the front of the body (groin, breasts), and men the back of the body (buttocks, perianal).
- Signs of insulin resistance: skin tags, acanthosis nigricans around the neck or flexures, obesity (flexural friction and sweating probably exacerbate follicular occlusion and the tendency to hidradenitis).
- Grouped open comedones (blackheads) in affected skin suggest hidradenitis—a useful sign in patients with recurrent flexural 'boils'.
- Tender nodules, abscesses, scars, and sinuses in flexures.

Dissecting cellulitis of the scalp

This rare condition, also known as 'perifolliculitis capitis abscedens et suffodiens', is commoner in black-skinned patients.

What should I look for?

- A history of recurrent and chronic painful abscesses in the scalp.
- Tender subcutaneous fluctuant scalp nodules, some of which drain pus, and intercommunicating sinuses.
- Patchy scarring with loss of hair (scarring alopecia).

Fig. 12.4 Grouped comedones in acne conglobata.

Fig. 12.5 Hidradenitis suppurativa with tender subcutaneous nodules, scars, and sinuses in the groin.

Fig. 12.6 Axillary hidradenitis suppurativa with scarring and deformity.

Follicular occlusion diseases: management

Management is difficult and at best control will be limited. Cultures of skin swabs may be negative or show a mixed growth, but *Staphylococcus aureus* may play some part in pathogenesis.

Acne conglobata

- Minimize exacerbating factors, e.g. drugs that may cause acne.
- The response to standard acne treatments with topical and oral antibiotics and topical retinoids is usually disappointing.
- Isotretinoin is worth trying, but disease can be very resistant.
- Intralesional steroids may reduce inflammation in large cysts.
- Cysts may be drained surgically.

Hidradenitis suppurativa

- Avoid smoking (some evidence nicotine exacerbates disease).
- Weight loss if obese (bariatric surgery was helpful in one case).
- Loose-fitting cotton clothing to reduce friction and sweating.
- Minimize follicular irritation by avoiding shaving and depilation.
- Antibacterial soaps and antiseptics (no evidence for efficacy).
- Topical 1% clindamycin lotion bd may control mild disease.
- Prolonged treatment (at least 6 months) with an oral tetracycline or erythromycin (as for acne) can be effective. Patients who do not respond may be treated with rifampicin 300mg bd in combination with clindamycin 300mg bd for 10 weeks.
- Anecdotal reports of therapies that may be helpful include antiandrogens (cyproterone acetate, finasteride), retinoids, and dapsone, as well as anti-TNF drugs, e.g. infliximab and etanercept.
- Severe disease requires a multidisciplinary approach—surgery or carbon dioxide laser surgery to remove scarred sinus tracts or marsupialize recurrent lesions. Leaving wounds to heal by 2° intention has been advocated, but grafts or flaps may be needed.
- Recurrence is common (less often in axillary disease), particularly after incision and drainage of abscesses, which is not recommended.

Dissecting cellulitis of the scalp

- Prolonged treatment with tetracycline antibiotics, as for acne, may be helpful. Combinations of rifampicin and clindamycin can be tried, as in hidradenitis.
- Isotretinoin has been tried (0.5–1.0mg/kg), with limited success, sometimes in combination with dapsone (50–100mg/day).

Further reading

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Gener G *et al*. *Dermatology* 2009;**219**:148–54 (hidradenitis treatment).

Rosacea

Rosacea is common in the third and fourth decades. Pathogenesis is uncertain, but genetic factors play a part. Rosacea has a particularly high prevalence in Ireland (Celts with fair skin).

What should I ask?

- Patients will often have had a long-standing tendency to flush or blush easily, now the face is permanently red (also see ➔ p. 520).
- What makes the redness worse? The skin is easily irritated by topical products. Alcohol, spicy foods, sudden temperature change, and hot drinks can all precipitate reddening. Sunlight may also exacerbate rosacea. The role of demodex is uncertain (see ➔ p. 178).
- What does the patient notice? Crops of red bumps (papules), 'yellow-heads' (pustules)? Is the skin rough or scaly? (Scale is seen more often in seborrhoeic dermatitis than rosacea, but the conditions may overlap; see Fig. 12.3).
- What medicaments is the patient using and for how long? Topical corticosteroids can cause a rosacea-like rash (perioral dermatitis; see ➔ p. 256). Oral corticosteroids cause a pustular rash and redness. Peripheral vasodilators may exacerbate redness (see ➔ p. 389).
- Patients may have noticed enlargement of the nose.
- Eye symptoms are common, e.g. burning, itching, redness, grittiness.

What should I look for?

(See Fig. 12.3.)

Signs vary. Rosacea may be predominantly erythematotelangiectatic (red), papulopustular, phymatous, or granulomatous. Look for:

- A rash affecting the central face, usually in a symmetrical pattern, but, on occasion, rosacea can be strikingly asymmetrical.
- Erythema and telangiectasia (see Fig. 12.3).
- Dome-shaped erythematous papules and pustules without comedones (see Fig. 12.3). Comedones suggest acne (see ➔ pp. 242–3), not rosacea. Facial angiofibromas in tuberous sclerosis may be misdiagnosed as rosacea (see ➔ pp. 556–7).
- Brownish yellow papules (granulomatous rosacea?) (see Box 12.9).
- Greasy skin, sometimes with diffuse fine scale. Prominent scaling suggests dermatitis (seborrhoeic or contact) or, if in a limited area, fungal infection. (Also see ➔ Demodicosis, p. 178.)
- Facial swelling (lymphoedema). Persistent erythema and solid oedema of upper 2/3 of the face (Morbihan disease) is a form of rosacea.
- A bulbous nose (rhinophyma) because of hypertrophic sebaceous glands and an overgrowth of soft tissues.
- Ocular rosacea—conjunctivitis, blepharitis, chalazion, hordeolum.
- Sparing of the trunk—unlike acne, rosacea does not affect the back or chest.
- Cutaneous B-cell lymphoma may simulate rosacea with firm erythematous papules on the forehead, cheeks, chin (rare).

Management

See Box 12.8.

Box 12.8 Rosacea: what should I do?

- Soap substitutes minimize irritation of the skin.
- A hat and sunblock creams will prevent photoexacerbation.
- Brimonidine tartrate gel ×1/day reduces redness (irritates; risk of rebound erythema and contact dermatitis).
- Mild papulopustular disease: 0.75% metronidazole cream or 15% azelaic acid gel applied bd for 3–4 months. Ivermectin 1% cream ×1/day may be more effective than metronidazole.
- Topical 1% pimecrolimus or 0.1% tacrolimus may have a role in papulopustular disease.
- An oral antibiotic (oxytetracycline 500mg bd, lymecycline 408mg/day, or erythromycin 500mg bd) taken for 4 months will control papulopustular disease but has little impact on redness. The dose can be reduced, once the disease is under control.
- Oral isotretinoin may reduce sebaceous gland overgrowth. In Morbihan variant, >6 months' treatment may reduce erythema/swelling.
- Rhinophyma can be pared down electrosurgically.
- Cosmetic camouflage or laser surgery may reduce redness.

Box 12.9 What is granulomatous rosacea?

Granulomatous rosacea is characterized by:
- Features of rosacea. Brownish yellow or red papules predominate.
- Granulomatous perifollicular inflammation without necrosis.
- Response to oral antibiotics, e.g. minocycline 100mg/day, lymecycline 408mg/day taken for 4–6 months.
- Isotretinoin is an alternative treatment.

Acne agminata (lupus miliaris disseminatus faciei) is probably a variant of granulomatous rosacea:
- Patients present with clusters of reddish brown or yellow papules, 1–3mm in diameter, distributed symmetrically around the eyes, including the eyelids, and sometimes elsewhere on the central face. Some patients have nodules in the axillae. Lesions last months, may become pustular and crusted, and eventually heal with pitted scarring.
- The histology shows superficial granulomatous inflammation with caseation necrosis, possibly a reaction to damaged pilosebaceous units. Acne agminata has no relationship to tuberculosis.
- The disease tends to regress spontaneously in 1–2 years.
- Tetracycline antibiotics, in acne doses, are usually effective in about 6 months.
- Isotretinoin and dapsone have also been effective.

Further reading

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Perioral dermatitis (periorificial dermatitis)

Perioral dermatitis is an inflammatory condition that occurs in children and adults, usually young and middle-aged women. The cause is unknown, but exogenous factors, such as potent topical corticosteroids (see ➔ p. 657), inhaled corticosteroids, irritant cosmetics, fluorinated toothpastes, and occlusive emollients, have been implicated.

The name is misleading. Perioral skin is affected more often than other sites, but the condition may affect periocular and perinasal skin. It has also been reported on the trunk. Nor is perioral dermatitis a typical dermatitis; it is more akin to rosacea. Dermatitis is treated with a topical corticosteroid; yet topical corticosteroids are one of the causes of perioral dermatitis.

What should I look for?

- A history of a rash that burns, rather than itches.
- Explore what has been applied to the skin. Many (but not all) patients with perioral dermatitis will have been applying a potent topical corticosteroid to their face prior to the onset of the rash. The corticosteroid relieves burning and causes transient vasoconstriction, which reduces redness, so the patient continues to apply the cream in the mistaken belief that the corticosteroid is helping.
- Clusters of tiny (<2mm in diameter) erythematous papules and vesicopustules (the tiny vesicles rapidly evolve into pustules) around the mouth (see Fig. 12.7).
- The background skin is erythematous and may be scaly.
- The rash tends to spare the vermillion border of the lips.
- Periocular (usually below the eye) and perinasal lesions are probably commoner in children than adults.

What should I do?

- Try to identify the underlying cause—most often applications of a potent topical corticosteroid to the face—and advise avoidance of any potential skin irritant.
- Convince the patient that the topical corticosteroid (or other preparation) is the cause of the problem—this may be difficult. (Try to avoid blaming the doctor who prescribed the strong corticosteroid—it is quite possible that the patient has not followed the doctor's instructions or has obtained the corticosteroid from a well-meaning friend or relation.)
- Warn the patient that the skin is going to look worse, before it gets better, and to resist the temptation to restart the topical corticosteroid (the redness and irritation will be more obvious when the corticosteroid is withdrawn).
- Prescribe topical or oral therapy (see Box 12.10).
- Reassure the patient—the prognosis is good, and the rash generally clears in 8–12 weeks.

Box 12.10 Treatment of perioral dermatitis

- Withdraw potential irritants and potent corticosteroid preparations.
- Topical 1% pimecrolimus or 0.1% tacrolimus ointment may be effective in mild cases. Apply bd for 4 weeks.
- An oral tetracycline (e.g. oxytetracycline 250–500mg bd) will be required for 6–8 weeks in more severe cases.
- Avoid tetracyclines in children <12 years (they stain teeth), and use erythromycin instead, but this is not as effective as tetracycline.
- Prescribe a weak topical corticosteroid (1% hydrocortisone ointment), in combination with miconazole for any underlying eczematous skin problem (possibly seborrhoeic dermatitis), if required.

Fig. 12.7 Perioral dermatitis caused by very potent topical corticosteroid.

Further reading

Dessinioti C *et al*. *Clin Dermatol* 2014;**32**:24–34.

Chapter 15

Leg ulcers and lymphoedema

Contents

Introduction

You are likely to come across a patient with a chronic leg ulcer at some stage in most branches of adult medicine. One to 2% of people will develop a leg ulcer during their life, and chronic ulcers cause morbidity and disability, particularly in older people.

Traditionally, nurses care for leg ulcers, and doctors are asked to see the patient when complications ensue or the ulcer is not healing as expected. Many doctors feel poorly trained to deal with such patients, particularly because they have no experience of what is a 'normal' leg ulcer.

Learn from the expertise of the wound care nurses, and take an interest in patients' leg ulcers. Ulcers may be associated with systemic disease, as well as vascular problems, infections, and malignancy—you may be surprised what you find underneath the dressings!

The history

Take a full dermatological history (see ➔ p. 18, pp. 20–1), asking questions that will help you to pinpoint the likely cause of the ulcer, as well as looking for risk factors and factors that may delay healing (see Boxes 15.1 and 15.2).

What should I ask?

- When the ulcer developed and how it progressed. What preceded ulceration, e.g. trauma, pressure, tender pustule (may indicate PG), or blister?
- Explore concerns (patients may fear that they may lose the leg).
- Pain: character, how bad, what makes it worse or better? (see ➔ p. 304.)
- Does the patient sleep in a bed or in a chair at night?
- How much elevation is tolerated?
- Impact of ulcer on quality of life—use the DLQI (see ➔ p. 32).
- Mobility: how much exercise?
- Weight: obesity is common in venous disease.
- Problems with skin around the ulcer: redness, pain, itch?
- Swelling of limb: does it improve after elevation, e.g. overnight?
- Present management:
  - Dressings and bandages: how often is the ulcer dressed?
  - Any compression? How much elevation?
  - Pain relief: how much, how often is it taken?
- Any previous episodes of ulceration?
- Any history of cellulitis?
- Previous treatments for ulcer:
  - Dressings or bandages?
  - Topical treatments?
  - Surgical interventions, e.g. angioplasty, skin graft?
- Past medical history and functional inquiry, including:
  - Skin cancers and sun exposure? (see ➔ pp. 26–7.)
  - Hypertension, claudication, angina, myocardial infarction?
  - Cerebrovascular accident?
  - Diabetes, peripheral neuropathy?
  - Obesity?
  - Hypercholesterolaemia?
  - DVT or pulmonary emboli?
  - Varicose veins: varicosities in pregnancy, varicose vein surgery?
  - Major orthopaedic intervention, long bone fracture?
  - RA?
- Family history of varicose veins, leg ulcers, DVT, or pulmonary emboli?
- Social history: carers, home environment; job—prolonged standing?
- Drug history, e.g. nicorandil, prednisolone, hydroxycarbamide?
- Allergies: 'reactions' to dressings or bandages? Patch tested?
- Smoker?

Box 15.1 Common causes of leg or foot ulcers

(See Figs. 15.1, 15.2, 15.3, and 15.4,.)

- *Venous disease* (70%): venous hypertension, valvular incompetence, and venous reflux lead to inflammation and ulcers. Risk factors: genetic, ♀ sex, age, prolonged standing, obesity. Past medical history of pregnancy, DVT, trauma, operations to hips or knees.
- *Arterial disease* (5–10%): intermittent claudication, rest pain, night pain, pain worse when the leg is elevated. Smokers.
- *Mixed AV* (15%): venous and arterial disease.
- *Neuropathic*: sites of pressure—often asymptomatic. Blood on the floor may be the first indication of trouble to the patient. Commonest on the feet of diabetic patients when also associated with small-vessel disease (see ➜ pp. 550–1).

Box 15.2 Other causes of leg ulcers

- *Malignancy*: BCC commonest. Misdiagnosed as an 'ulcer' (see ➜ p. 350). Other malignancies that may cause chronic ulcers include SCC, metastases, and diffuse large B-cell lymphoma (leg type).
- *PG*: rapidly growing, very painful ulcer. Often preceded by a tender pustule. Associated with RA, IBD, and myeloproliferative disorders; 50% of patients have no underlying disease (see ➜ Fig. 15.5, and p. 310).
- *Vasculitic*: associated with connective tissue diseases, particularly RA, when immobility and coexisting AV disease often contribute to ulceration (see ➜ pp. 394–5).
- *Occlusive vasculopathy*, e.g. livedoid 'vasculitis' with atrophie blanche, sickle-cell disease, thalassaemia, thrombocythaemia, hypertensive ulcer (see Box 15.4), calciphylaxis, cryoglobulinaemia, antiphospholipid syndrome (see ➜ p. 444 and pp. 452–3).
- *Inherited disorders of coagulation*: mutations in protein C, protein S, antithrombin III, fibrinogen, or factor V genes.
- *Necrobiosis lipoidica*: associated with insulin-dependent diabetes.
- *Infection*: deep fungal, treponemal.
- *Drug-related*: hydroxycarbamide, nicorandil.
- *Trauma*: including IV drug abuse.
- *Dermatitis artefacta*: deliberate self-harm (see ➜ pp. 566–7).

Further reading

Bergan JJ *et al*. *N Engl J Med* 2006;**355**:488–98 (chronic venous disease).
White-Chu EF and Conner-Kerr TA. *J Multidiscip Healthc* 2014;**7**:111–17.

Preparing to examine the ulcer

The leg ulcer is likely to be hidden away under dressings and bandages. Many doctors are reluctant to remove bandages, perhaps for fear of what they might find underneath, but more often because they are pressed for time or feel ill-prepared to re-dress the leg, a task that may not be as easy as it sounds without the help of a skilled nurse. But it is important to inspect the limb beneath the bandages—try not to be put off.

Explain to the patient what you wish to do and why. Ask if any preparation is required, before the dressing is removed. Is analgesia required?

Try to find out what dressings or bandages are being used.

Discuss your plans with the nurses; arrange a suitable time; help the nurses to set up a dressing trolley … , and take the plunge.

What do I need?

- Plastic gloves (do not need to be sterile) and a plastic apron.
- A trolley with:
  - Gauze swabs, scissors, plastic forceps, and a sinus probe (if the ulcer might be deeply penetrating).
  - A bag for disposing of contaminated dressings.
- Water for washing the leg and the ulcer—a bucket may be useful.
- Cling film to cover the ulcer—exposing the ulcer to air may be painful.
- A sphygmomanometer, stethoscope, and Doppler ultrasound probe for checking the ABPI (see Box 15.6).
- A tendon hammer, tuning fork (128Hz), and disposable neurological pin if you suspect a neuropathic ulcer. (Ideally a Semmes–Weinstein 10g monofilament for evaluation of sensation if you know how to use it.)
- A tape measure.
- An acetate sheet (or the transparent cover of a dressing pack) and a pen for tracing the ulcer.
- A digital camera.
- Liquid paraffin 50% in WSP or a bland ointment, such as Hydromol®, to moisturize the skin.
- Dressings and bandages (see ➔ p. 306), including a viscose stockinette bandage, long enough to run from the toe to just below the knee to hold the dressing in place.
- Surgical adhesive tape for securing the end of the bandage—avoid safety pins.

Fig. 15.1 Venous ulcer in the gaiter area in an obese woman.

Fig. 15.2 Ulceration in a patient with a combination of venous disease and lymphoedema (thick swollen folds of skin around the ankle, hyperkeratosis, and swelling of the dorsum of the foot, as well as the leg).

Fig. 15.3 Atrophie blanche (shiny white stellate scar with peripheral telangiectasia) in venous disease. Atrophie blanche may also be seen in occlusive vasculopathies such as livedoid vasculopathy or antiphospholipid syndrome (see ➔ pp. 314–15 and p. 452).

Fig. 15.4 Deeply penetrating neuropathic ulcer with surrounding callus and underlying osteomyelitis.

Fig. 15.5 Typical pyoderma gangrenosum with a purplish undermined border.

The examination

What should I do?

- Lay the patient comfortably on an examination couch.
- Adjust the lighting, so you can see the legs and ulcer.
- Protect the bedding beneath the legs with incontinence pads.
- Carry out a full physical examination that includes palpation of the abdomen to look for masses that might be causing external venous compression. Are the legs oedematous, or is there lymphoedema? (see ➔ pp. 316–17.)
- Check the blood pressure (see Box 15.3).
- Remove bandages to expose the leg. Gently peel off dressings, so that the ulcer is exposed. You may need to soak the dressing with normal saline prior to removal if it has stuck to the wound.
- Note the colour and quantity of the exudate on the dressing. Heavy exudate suggests heavy bacterial colonization. Green dressings indicate the presence of *Pseudomonas*.
- Remove loose crust, scab, or adherent dressing, and gently wash the ulcer with water, so you can see the base and wound edges.
- If the ulcer is painful when exposed to air, cover temporarily with cling film or a moist gauze swab.
- Assess the ulcer—site, wound edge, and wound bed (slough, granulation tissue, necrotic base) (see Box 15.4).
- Measure the ulcer (maximum diameters or trace onto an acetate sheet).
- Probe to assess the amount of undermining and depth of the ulcer.
- Record the appearance of the ulcer with photographs, if possible.
- Examine the surrounding skin, looking for signs that may be linked to the cause of the ulcer (see Boxes 15.3 and 15.4).
- Look for cellulitis—heat, swelling, tender erythema (this may be difficult to differentiate from lipodermatosclerosis (see ➔ p. 308).
- Assess the peripheral circulation, including the ABPI (see Box 15.6).
- Assess joint mobility at ankles and feet, and look for joint deformity.

Box 15.3 Hypertensive ulcer (Martorell ulcer)

- The existence of this entity is controversial, and these ulcers must be rare.
- Painful ulceration may be 2° to a thrombo-occlusive vasculopathy of cutaneous arterioles in long-standing hypertension.
- Patients present with an extremely painful superficial ulcer with an erythematous or purpuric rim, usually on the anterolateral aspect of the shin. Ulcers start as purpuric lesions which soon become necrotic.
- Management involves control of hypertension, pain relief, and compression bandaging.

Box 15.4 Signs in different types of ulcer

Venous (see Figs. 15.1 and 15.2)

- Gaiter area, just above the medial malleolus or (less often) lateral malleolus. Sloping, poorly defined wound edge. Shallow, but large, ulcer that may extend circumferentially around the limb. Pitting oedema.
- Varicose veins (assess standing): gently press any swelling to confirm it is a varicosity. The vein will empty and refill when you let go. Varicosities of the short saphenous vein are seen posterolaterally below the knee; varicosities of the long saphenous vein run more medially along the whole length of the limb.
- Venous flare (dilated superficial venules at the ankle), brownish pigmentation (haemosiderin and melanin), eczema in the gaiter area. Signs may be localized to the skin over varicosities—stand the patient up to see varicosities.
- Scar at sites of previous ulcer, atrophie blanche (see Fig. 15.3).
- Lipodermatosclerosis: erythematous, tight, indurated skin above the medial malleolus. May simulate cellulitis (see Fig. 15.6).

Arterial

- Peripheral on borders or sides of feet or initiated by trauma.
- Round ulcer, sharply demarcated with a punched-out appearance. Tendon may be exposed at the base.
- Cool, shiny, pale, or dusky skin with loss of hair. Peripheral gangrene. Reduced peripheral pulses, delayed capillary refill (for assessment of circulation, see ➔ p. 300).

Mixed AV Signs of arterial and venous disease.

Neuropathic (see Fig. 15.4)

- Pressure sites on the foot, most often under the second metatarsal head. Surrounded by callus. Limited joint mobility or deformity. (Charcot joints caused by impaired joint position sense and sensation).
- Check footwear for source of pressure or trauma. Osteomyelitis complicates deep neuropathic ulcers.
- Dry skin may indicate autonomic dysfunction (see ➔ p. 550).
- Assess vibration sense (128Hz tuning fork to the apex of the great toe), pinprick (just proximal to the great toenail), temperature (on the dorsum of the foot with tuning fork placed in iced or warm water), sensation with a 10g monofilament, and Achilles tendon reflex.

PG: bluish, undermined wound edge, base with slough (see Box 15.11 and Fig. 15.5).

Vasculitis: purpuric wound edges. Palpable purpura, nodules, livedo (see ➔ p. 440, pp. 440–7).

Occlusive vasculopathy: livedo, atrophie blanche (see ➔ pp. 312–5, pp. 444–5, p. 452, and Box 15.3).

Malignancy: non-healing ulcer. BCCs or SCCs may be misdiagnosed as 'leg ulcers'. Rarely, SCC arises within a chronic ulcer. Consider taking a biopsy in any chronic non-healing leg ulcer.

Assessing peripheral circulation

What should I do?

- Compare the signs in the legs.
- Check peripheral pulses (dorsalis pedis, posterior tibial, popliteal, femoral).
- Listen for bruits in the femoral arteries.
- Peripheral perfusion: push on the nail bed, until it turns white. Let go, and colour should return (capillary refill) in <2s—not a very reliable test.
- Buerger test: soles should be pink with the patient supine. Elevate both legs to 45° for 1–2min. Compare the soles. Marked pallor suggests ischaemia. Sit the patient up, and lower the legs over the side of the bed. Colour will return to the legs, but an ischaemic limb will become blue and then very red. Post-hypoxic vasodilatation causes hyperaemia.
- Assess the peripheral arterial circulation by measuring the ABPI using a handheld Doppler ultrasound probe (see Boxes 15.5 and 15.6).
- Pulse oximetry has been recommended as an alternative to the Doppler ABPI for assessing peripheral circulation, particularly if the leg is very oedematous, but not all vascular surgeons agree that relying on pulse oximetry is safe practice. More work is needed in this area.
- Consider referral to vascular surgeons for lower limb venous duplex scans to detect venous reflux in superficial and deep venous systems.
- Refer to vascular surgeons for arterial colour duplex scan if the ABPI is <0.8. ▶▶ Refer urgently if the ABPI is 0.5 or less.

Box 15.5 How to measure the ABPI

- Seek for permission for what you are going to do—'I would like to compare the blood pressures in your arm and in your leg. We will both be able to hear your pulse, because I am going to use this Doppler machine.' Warn that the cuff may feel uncomfortable for a short time when it is tightened on the leg.
- The patient should lie relatively flat for 20–30min, before commencing measurements. Ensure the patient is comfortable in that position.
- Apply a sphygmomanometer cuff to the arm.
- Inflate the cuff, while feeling the radial pulse to get a rough idea of the systolic pressure.
- Apply ultrasound gel to the skin over the brachial pulse.
- Angle the Doppler probe at 45–60°, and find the brachial pulse (adjust the position and angle of the probe, until the sound is maximal).
- Inflate the cuff, until the sound disappears.
- Slowly release the pressure, and record the pressure at which the sound of the brachial pulse reappears.
- Repeat in the other arm, and take the higher of the two readings.
- Place cling film around the leg, covering the surface of the ulcer.
- Place the cuff around the leg just above the medial malleolus.
- Palpate the dorsalis pedis pulse between the first and second metatarsals. Apply ultrasound gel to the skin over the pulse, and find the dorsalis pedis pulse using the probe. Angle the probe at 45–60° to the foot.
- Inflate the cuff, until the sound disappears; deflate slowly, and record the pressure at which the sound reappears.
- If possible, repeat at the posterior tibial pulse, and take the higher of the two readings—but some patients do find the tight cuff uncomfortable.
- Repeat in the other leg.
- Calculate the ABPI in each leg: ABPI = highest ankle pressure/ highest brachial pressure.

Box 15.6 Interpretation of the ABPI

- ABPI: 0.9–1.0 = safe to apply high compression.
- ABPI: 0.8 = excludes significant arterial disease. Probably safe to apply compression—start slowly.
- ABPI: 0.6–0.8 = moderate arterial disease. Very gentle compression may be tolerated, if required, to control venous incompetence and oedema, but bandages should only be applied by an expert, and the patient should be monitored. Pulse oximetry may be used to check the immediate impact of compression.
- ABPI: <0.5 = severe arterial disease. Do not use compression—refer for an urgent vascular opinion.

⚠ *Warning*: the arteries are calcified and cannot be compressed in some older patients and in some with diabetes. In this situation, the ABPI is falsely high. If the results do not fit with your clinical findings, review the patient and the test.

Investigation of leg ulcers

Investigations should be directed towards determining the underlying cause of ulceration and excluding factors that may delay healing (see Box 15.7).

Blood tests

- FBC, iron studies, ESR, CRP.
- Serum albumin (low in malnutrition) and vitamin C, if indicated (may be deficient in older patients with a poor diet).
- Random glucose.
- Clotting screen, including lupus anticoagulant, anticardiolipin antibody, factor V Leiden, protein C, and protein S levels if multiple DVTs or a family history of DVTs and leg ulcers.
- If signs suggest vasculitis—ANA, RF, hepatitis serology, and complement 4 (low C4 with raised RF may indicate type II or III mixed cryoglobulinaemia; see ➔ p. 444). Check ANCA if indications of a systemic vasculitis.
- Cryoglobulins (keep the sample at 37° in a Thermos™ flask—it may be easiest to send the patient to the laboratory)—check if you suspect ulceration is 2° to type I cryoglobulinaemia associated with a paraprotein, e.g. myeloma, lymphoma (see ➔ p. 444).

Microbiology

- Surface ulcer swabs for microbiological culture are of limited value—ulcers are always colonized with some bacteria. Only take swabs if you suspect heavy colonization (increasing pain, malodour, and large amounts of exudate) or if you detect signs of cellulitis—warmth, extending tender erythema, oedema (see ➔ p. 138).

Ulcer biopsy

- Histology of biopsies from the edge of most ulcers, including PG and many vasculitic ulcers, is not diagnostic.
- Biopsies may be indicated to exclude malignancy or for culture.
- To exclude fungal or treponemal infection, take a deep biopsy that extends into the base of the ulcer. Send for histological examination as well as for culture.

Radiology

- Radiograph, MRI, or CT may be used in chronic deep penetrating ulcers to exclude osteomyelitis, e.g. neuropathic foot ulcers in patients with diabetes.

Patch tests

- Allergic contact dermatitis can delay wound healing. Patients with leg ulcers may be sensitized to topical medicaments, including topical corticosteroids, neomycin, and Balsam of Peru (a fragrance in many creams), as well as constituents of dressings or bandages.
- All patients with chronic varicose eczema or chronic leg ulcers should be patch tested. Repeat tests every 2–3 years if the patient has persistent eczematous skin changes or non-healing ulcers.

Box 15.7 Factors that may delay healing

- *Oedema*: gravitational, congestive cardiac failure.
- *Immobility*: contributes to poor calf muscle pump and oedema.
- *Anaemia or malnutrition*: vitamin C deficiency may be a particular problem in older patients.
- *Corticosteroids* (impact of other immunosuppressive drugs is less certain).
- *Repetitive trauma*, including poor dressing technique.
- *Heavy colonization with bacteria* (suggested by increasing malodour, increasing pain, and heavy exudate). Paradoxically some bacteria may promote healing, but the role played by bacteria is not fully understood.
- *Allergic contact dermatitis*: patch test every 2–3 years.

Pain and leg ulcers

Severe pain is common in patients with arterial ulcers, but do not underestimate the pain suffered by patients with venous ulcers. The pathogenesis of pain is complex. Local dehydration caused by dressings, cutaneous ischaemia, and bacterial infection may play a part. Chronic pain demoralizes patients, limiting physical activity, but older patients may be reluctant to report pain (see Box 15.8).

What should I ask?

- Where is the pain? Ask the patient to point. Ensure the pain is caused by the ulcer—arthritis of the back, hip, or knee may cause leg pain.
- What is the pain like? (See Box 15.8.)
- How bad is the pain? Obtain a measure of the overall intensity of the pain by asking the patient to grade the pain (mild, moderate, or severe) or rate the pain on a scale of 0 (no pain) to 10 (most severe).
- When did the pain start, and how long does the pain last?
- Has the pain changed? Worsening may indicate ischaemia or cellulitis.
- What causes the pain? Typically, arterial ulcers cause severe pain at night, but many patients with venous ulcers also have pain that interferes with sleeping. Standing and walking cause pain in venous, as well as arterial, ulcers. Dressing changes may be very painful.
- What relieves the pain—elevation? Compression?
- What is the impact of the pain? Is the patient depressed or socially isolated?
- What analgesia is being taken and how often?

What should I do?

- Ensure the skin is not inflamed and there is no evidence of infection.
- Review dressings. Most 'non'-adherent dressings are drying, adhere to wounds (despite the name), and cause pain; hydrogels and hydrocolloids relieve pain.
- Review the dressing technique—use warm water for washing the ulcer; leave the ulcer uncovered for the minimum amount of time.
- Ensure that bandages are not compromising the vascular supply. Check with a pulse oximeter, if any doubt (see ➔ p. 300).
- Prescribe analgesia to be taken prior to dressing changes.
- A bed-cradle may help to relieve night pain, as may elevating the head of the bed by 12–15cm in patients with arterial disease.
- Elevation and compression bandages may alleviate much of the pain associated with venous ulcers, but not all patients tolerate this treatment, and some have neuropathic pain that does not respond to compression.
- Advise patients to avoid standing for prolonged periods.
- Refer for a vascular opinion if the Doppler ABPI is reduced to below 0.8.
- Prescribe analgesics, according to a pain ladder (see ➔ pp. 304–5).
- Reassure the patient that analgesia taken for pain is rarely addictive, and ensure the patient understands the importance of taking analgesia regularly to prevent the onset of pain.

Pain ladder

Nociceptive pain

Mild pain
Paracetamol 0.5–1g 4-hourly (max. = 4g/day).

Box 15.8 Descriptors from the short-form McGill Pain Questionnaire (SF-MPQ)

Pain is a complex sensation difficult to describe. Physical, psychological, and social factors influence the perception of pain. Patients may have a mix of nociceptive and neuropathic pain. SF-MPQ has 15 descriptors (11 sensory; 4 affective). Patients rate the intensity of each descriptor. Nociceptive pain associated with venous and arterial ulcers is usually throbbing or aching. Shooting, stabbing, or sharp pain suggests a neuropathic component. Burning may indicate damage to the skin.

- *Sensory descriptors*: throbbing, shooting, stabbing, sharp, cramping, gnawing, hot/burning, aching, heavy, tender, splitting.
- *Affective descriptors*: tiring/exhausting, sickening, fearful, punishing/cruel.

Moderate pain

- Try increasing strengths of codeine with paracetamol.
- Co-codamol weak (8mg of codeine phosphate plus 500mg of paracetamol) 1–2 tablets 4-hourly (max. = 8 tablets/day).
- Co-codamol medium (15mg of codeine phosphate plus 500mg of paracetamol) 1–2 tablets 4-hourly (max. = 8 tablets/day).
- Co-codamol strong (30mg of codeine phosphate plus 500mg of paracetamol) 1–2 tablets 4-hourly (max. = 8 tablets/day).
- If not controlled, try dihydrocodeine, 30mg every 4–6h (max. = 240mg/day). Stimulant laxatives may be required with these drugs.

Severe pain

- Day pain: short-acting morphine, 10mg 4-hourly (5mg in the elderly), and increase the dose by 50% incrementally.
- Night pain: twice 4-hourly dose, i.e. 10 or 20mg nocte. Incident or breakthrough pain 5–10mg of morphine PRN, i.e. the 4-hourly dose.
- An antiemetic may be used in conjunction with these initially, but, if still nauseated after 3 days, the dose should be reduced.
- Once pain control is stable, change from short-acting morphine to slow-release. Divide the total dose into two portions for 12-hourly administration. Ensure that the patient has an 'escape' dose of short-acting morphine for breakthrough pain. The escape dose should be 1/6 of the daily morphine dose in mg.

⚠ *Avoid NSAIDs in older patients (often used with paracetamol). Adverse effects include GI bleeding, duodenal or gastric ulceration, and renal failure.*

Neuropathic pain

Difficult to alleviate. Antidepressants, e.g. amitriptyline 25–50mg/day, may help. Increase the dose gradually. Anti-epileptic drugs, e.g. gabapentin, are also effective in some cases.

Management of ulcers

General advice

Correct factors such as oedema, iron or vitamin C deficiency, and pain (see ➔ p. 304). Wash ulcers and surrounding skin with tap water. Pick off loose scale, and moisturize the skin with 50% WSP in liquid paraffin or Hydromol® ointment. (Ointments are less likely to sensitize than creams.) Clean and dry the skin between the toes, and trim or file nails (a chiropodist may help).

Wound care

Reduce bacterial contamination with antiseptic soaks or wet dressings (see ➔ p. 662). Moist wounds heal more quickly than those exposed to air, but too much moisture irritates the skin and macerates wound edges which become friable and break down. Protect the edges with zinc oxide paste or Cavilon® (expensive).

Choice of dressing depends on the type and state of the ulcer (see Box 15.9).

Venous ulcers

- Graduated compression bandaging promotes healing. Bandages should extend from the base of toes to below the knee. Experts apply bandages with an even tension, so that the pressure steadily reduces from the ankle to the calf (see Box 15.10). ⚠ *Check the ABPI before compressing the limb* (see Box 15.6).
- Avoid prolonged standing (stand on tiptoes at intervals to activate the leg muscle pump) or sitting with the feet dependent (elevate the legs with the feet at heart level, and dorsiflex the feet regularly to activate the muscle pump).
- Encourage mobility and weight loss.
- Aspirin and pentoxifylline have been advocated to promote healing of venous ulcers, but their role is unproven.
- Skin grafts (pinch, punch, or split-thickness) may hasten re-epithelialization of ulcers with granulating bases, once oedema is controlled. Refer to a vascular service if the ulcer is not healing. Also refer patients with healed venous ulcers to a vascular service. Vascular surgery to correct venous hypertension may prevent recurrence.
- Once the ulcer has healed, ensure that the patient has compression hosiery (ideally class II). The patient may need help to put on stockings.

Arterial ulcers

- Keep the limb warm. Advise patients to stop smoking and control other risk factors for arterial disease such as hypertension or diabetes.
- Encourage regular graded exercise to promote circulation.
- Protect bony prominences and pressure areas (heels) with padding.
- Vascular surgeons may be able to improve the blood supply with balloon angioplasty or a femoropopliteal bypass graft.
- Skin grafting may have a role, once underlying vascular problems have been corrected. Consider involving a plastic surgeon.

Further reading

Fonder MA *et al*. *J Am Acad Dermatol* 2008;**58**:185–206.

Box 15.9 Primary dressings and topical treatments

Note: there are many equivalent products. Determine what is available locally, before you choose dressings or bandages.

Pain Non-adherent dressings: Atrauman®/Mepitel®/Mepilex® (also useful as blister dressings). Hydrating dressings (also encourage granulation tissue): hydrogels or hydrocolloids. May overhydrate. Reduce bacterial colonization (see below).

Topical steroid (e.g. Trimovate® cream) or silver sulfadiazine cream for short periods may relieve pain (not evidence-based).

Bacterial colonization Antibacterial wash, e.g. Prontosan®. Dressings containing iodine (may be painful). A microorganism-binding dressing, e.g. Cutimed® Sorbact. Honey (may be painful) or silver sulfadiazine.

Moderate exudate Alginates or cellulose dressings e.g. Aquacel® Ag. Alginates may also be used for wound packing.

Heavy exudate Cellulose dressings, e.g. Aquacel® Ag, and absorbent pad, e.g. Sorbion® sana, Keramax®.

Odour Reduce bacterial load with topical metronidazole (eliminates anaerobes) or dressings containing silver. Absorb odour with Clinisorb® (cheap) or Carboflex® (expensive).

Adherent slough/necrotic tissue Larvae (maggots) remove necrotic tissue but do not damage healthy tissue. In the UK, sterile maggots may be ordered from the Biosurgical Research Unit, Princes of Wales Hospital, Bridgend, Wales CF31 1RQ. Surgical debridement (avoid in arterial ulcers) to cut away necrotic tissue. Autolysis promoted by softening tissues with hydrating dressings, e.g. hydrogels, hydrocolloids.

Change dressings if wound leaks, smells, or is painful. Dressings should always be changed within 7 days.

Venous eczema Moderately potent topical corticosteroid ointment under compression. Either use elasticated tubular bandage (see Box 15.10)—easy to remove if topical treatment required—or apply a paste bandage under compression (see ➔ Box 34.4, p. 661).

Box 15.10 Compression bandages for venous ulcers

Note: there are many equivalent products. Determine what is available locally, before you choose dressings or bandages.

Apply bandages evenly from the toe to below the knee, without ridges or folds. Applied correctly, the pressure steadily reduces from about 40mmHg at the ankle to 15–20mmHg at the calf. Do not tape anything directly to the skin.

Mobile patients Tubifast® (elasticated viscose stockinette) and Soffban® with Actico® short stretch (inelastic) bandages or long stretch (elastic) bandages or compression using a multicomponent system containing an elastic bandage.

Immobile patients Long stretch (elastic) bandages, e.g. Tensopress®, Setopress®, or short stretch (see above). Shaped elasticated tubular bandage provides some compression but is not available in the community. Some graduated compression can be provided by placing a layer of unshaped elasticated tubular bandage size D from the toe to mid calf ,with a 2nd layer size E from the ankle to the knee. Always use Tubifast® under elasticated tubular bandage. Hosiery (two-layer) designed for patients with ulcers made by Activa and Medi.

Lipodermatosclerosis (sclerosing panniculitis)

- Lipodermatosclerosis is a complication of venous stasis that is seen more often in women than in men. Chronic inflammation causes fibrosis with progressive induration.
- ⚠ The diagnosis is clinical, but frequently pain and erythema in acute lipodermatosclerosis are misdiagnosed and mistreated as cellulitis (see ➔ p. 138).

What should I ask?

- Look for a history of long-standing painful 'cellulitis' that has not responded to repeated courses of antibiotics.
- Ask about factors suggesting venous insufficiency, e.g. chronic swelling, previous DVT, varicose veins, orthopaedic procedures (see ➔ pp. 294–5).
- What factors improve or exacerbate the pain? Pain in lipodermatosclerosis is worse on standing and may be relieved by elevation.

What should I look for?

- Acute lipodermatosclerosis: a well-defined, very tender, indurated, erythematous plaque in the gaiter area, usually on the medial side of the leg, with little or no swelling (see Fig.15.6).
- Chronic lipodermatosclerosis: a well-defined, slightly tender, indurated 'woody' plaque. Fibrosis tethers subcutaneous tissues (sclerosing panniculitis). Eventually, the leg adopts the shape of an inverted champagne bottle.
- Signs of venous insufficiency, including venous eczema, venous flare, pigmentation, and ulceration in the gaiter area (see Box 15.5).
- An absence of signs suggesting cellulitis: spreading erythema (the erythema is localized in lipodermatosclerosis), acute swelling (the skin is usually tight in lipodermatosclerosis), lymphangitis, systemic flu-like symptoms.

What should I do?

- Explain that lipodermatosclerosis is caused by venous disease and is not an infection. Stop antibiotics.
- Provide pain relief (see ➔ p. 304).
- Control venous hypertension with compression (first exclude peripheral arterial disease). A pressure of 30–40mmHg at the ankle may not be tolerated in acute lipodermatosclerosis. Start with less pressure, and try to increase compression gradually.
- Advise about weight loss (if indicated), elevation, and exercise, as for patients with venous leg ulcers (see ➔ p. 306).
- The role of systemic treatments, such as prednisolone (to reduce acute inflammation) or fibrinolytic agents, is unproven.
- Refer to a vascular service. Vascular surgery may be indicated to prevent venous reflux and reduce inflammation.

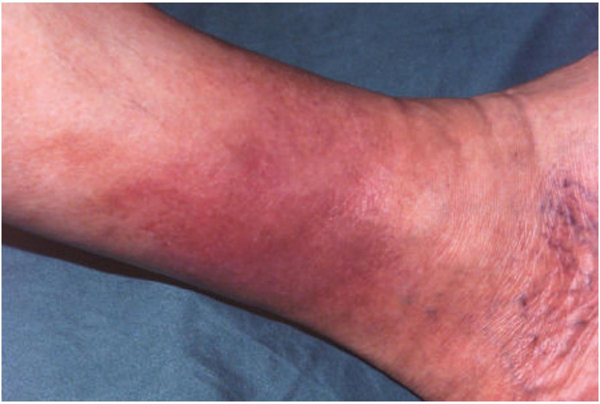

Fig. 15.6 Tender, indurated lipodermatosclerosis simulating cellulitis.

Pyoderma gangrenosum

❶ Pyoderma gangrenosum (PG) is a neutrophilic dermatosis characterized by acutely painful, rapidly enlarging deep ulcers (see Box 15.11). PG is neither an infection nor gangrenous, but is probably an aberrant immune response. Clearance of inflammatory cells may be impaired by abnormalities in, for example, neutrophil chemokines, integrins, or T-cells.

Ulcers can occur at any site, including the genitalia, but PG usually presents on the leg. It is seen most often in middle-aged adults but can affect any age. About 50% of patients have an underlying disease (see Box 15.12). Rarely, PG is triggered by surgery, often breast surgery. Diagnosis is clinical and may be difficult (see Box 15.12).

What should I ask?

(see ➔ pp. 294–5.)

- How did the ulcer start? (Was minor trauma involved?)
- Is the ulcer painful? How quickly is it enlarging?
- Has the patient another reason to have a leg ulcer (venous disease, arterial disease, infection), a past history of PG, or a systemic disease that might predispose to PG? (See Box 15.11.)

What should I do?

- Look for signs suggesting PG (see Box 15.12 and Fig. 15.5).
- Rule out common causes of ulceration, but venous or arterial disease may coexist with PG (see Box 15.12).
- FBC, ESR, CRP, protein electrophoresis, and Ig (20% have a benign monoclonal gammopathy), RF, antiphospholipid antibodies.
- Take a biopsy from the ulcer edge for histology (may be suggestive but is not diagnostic); culture (bacteria, fungi such as sporotrichosis) (see ➔ Box 33.2, p. 645).
- Prescribe analgesia.
- Dressings, such as hydrocolloids or alginates, may relieve pain.
- Try gentle compression to control oedema (if tolerated).
- Pustules, bullae, or small ulcers—a very potent topical corticosteroid ointment or 0.1% tacrolimus ointment may control pain, inflammation, and ulceration, but watch for 2° infection, e.g. *Pseudomonas aeruginosa*.
- ▶▶ Large ulcers—prompt treatment with prednisolone 40–60mg/day or pulse IV methylprednisolone is essential to minimize scarring. Steroid-sparing agents that may be tried include minocycline, ciclosporin, and dapsone.
- Anti-TNF agents may be effective, particularly in PG with IBD or RA.
- ⚠ Avoid surgery in active disease, as trauma will exacerbate ulceration. Skin grafting may be considered for large ulcers, but only when inflammation is controlled.

Further reading

Patel F *et al*. *Acta Derm Venereol* 2015;**95**:525–31.

Box 15.11 Systemic diseases and pyoderma gangrenosum

Fifty percent of patients with PG have an underlying condition:

- IBD (65%): PG may be peristomal (see ➔ Box 24.2, p. 509).
- RA and other inflammatory arthropathies, seropositive and seronegative (16%).
- Haematological disorders (12%), e.g. acute myeloid leukaemia, myelodysplastic syndromes—PG is often bullous (signs overlap with Sweet syndrome; see ➔ p. 518).
- Auto-inflammatory diseases, e.g. PAPA syndrome (pyogenic arthritis, PG, and acne), PASH (PG, acne, and suppurative hidradenitis)—vanishingly rare! (see ➔ p. 234 and pp. 248–9.)

Box 15.12 Diagnosis of pyoderma gangrenosum

Major features

(See Fig. 15.5.)

- Very painful necrotic cutaneous ulcer (pain is out of proportion to the appearance of the ulcer) with a purulent discharge.
- Rapid progression of the ulcer—may enlarge by 2cm/day.
- Irregular, violaceous, and undermined border—you may be able to insert a probe several mm under the edge.
- Rapid response to corticosteroids—sometimes one has to resort to a clinical trial if the diagnosis is not clear-cut.
- Post-surgical PG: presents in the immediate post-operative period with fever and wound dehiscence, progressing to painful ulcers with typical borders. Often misdiagnosed as wound infection.

Other suggestive features

- The ulcer was preceded by a sterile pustule or, less often, an erythematous nodule or a blister.
- History that minor trauma initiates ulcers, i.e. pathergy.
- Base with much slough or haemorrhagic base.
- Cribriform scarring (uneven sieve-like with perforations).
- Histology: follicular or perifollicular inflammation, intradermal abscesses, and/or sterile dermal neutrophilia without vasculitis. Findings are not specific.

Differential diagnosis

- Infection: deep fungal, atypical mycobacterial.
- Venous or arterial insufficiency: vasculitis or occlusive vasculopathy (see ➔ pp. 312–13, and p. 452).
- Granulomatosis with polyangiitis: ulcers affecting the limbs, perineum, or face may resemble PG (see ➔ pp. 446–7).
- Cutaneous malignancy.
- Bite of brown recluse spider.
- Dermatitis artefacta: well-defined, angular ulcers without undermined edges (see ➔ pp. 566–7).
- Drug-induced ulcers: nicorandil, hydroxyurea.

Calcific uraemic arteriolopathy

Synonym: calciphylaxis.

This life-threatening condition occurs most often in patients with chronic kidney disease on dialysis. Calcium deposited in the media of small cutaneous arteries leads to thrombosis and ischaemic necrosis with refractory ulcers often involving the lower extremity, abdomen, or buttocks. Deposition of metals (iron, aluminium) may play a part in the pathogenesis of endothelial injury and vascular calcification.

2° infection may lead to fatal sepsis, and mortality is high (60–80%). Incidence is increasing, and calciphylaxis may be seen in other situations (see Box 15.13) when it may be confused with cutaneous vasculitis or PG. Rarely, patients may have systemic problems such as proximal myopathy, aortic valve involvement, or pulmonary calciphylaxis.

What should I look for?

- A history of underlying risk factors (see Box 15.13).
- Sudden onset of painful mottled erythema on the thigh or leg, evolving to indurated plaques that eventually become necrotic and ulcerate. Initially may be misdiagnosed as cellulitis (see Fig. 15.7).
- Livedo reticularis: mottled, reddish purple, reticulated discoloration, reflecting a sluggish vascular flow in the superficial dermis (see ➔ pp. 314–15 and p. 452).
- Ischaemic necrosis with black eschars (see Fig. 15.7).
- Other cutaneous signs may include:
  - Pathergy: minor trauma triggers ulceration.
  - Recurrent digital ischaemia with gangrene, necrosis of the penis.
  - Subcutaneous nodules on the breasts and abdomen.

What should I do?

- Check for cutaneous infection—take skin swabs (but surface swabs from ulcers are generally not very informative).
- Exclude peripheral vascular disease by clinical examination.
- Check renal function, serum calcium, serum phosphate, parathyroid hormone (PTH) level, and alkaline phosphatase.
- Check fasting glucose.
- Exclude other causes of an occlusive vasculopathy—check FBC and coagulation profile, including lupus anticoagulant, anticardiolipin antibody, protein C, and protein S (see ➔ p. 452).
- Exclude cutaneous vasculitis—check complements (C3, C4), cryoglobulins, hepatitis B and C serologies.
- Consider taking a deep elliptical skin biopsy to demonstrate calcification in small arteries, vascular thrombosis, and intimal hyperplasia (endarteritis obliterans) … but the biopsy may not heal.
- Radiographs may show vessel calcification, including a net-like pattern of calcification extending beyond visible skin lesions.
- Management is difficult and requires a multidisciplinary approach (see Box 15.14) (no randomized controlled trials of therapy).

Box 15.13 Risk factors for calciphylaxis

Major risk factors

- End-stage renal disease.
- Elevated calcium phosphate product.
- Hyperparathyroidism—1° and 2°.

Non-uraemic risk factors

- ♀ gender.
- Obesity (rapid weight loss may induce calciphylaxis).
- Diabetes.
- Hypoalbuminaemia.
- Autoimmune diseases such as SLE, RA, and antiphospholipid syndrome.
- Warfarin.
- Protein C and protein S deficiency.
- Malignancy.
- Alcoholic liver disease.

Box 15.14 Management of calciphylaxis

- Pain relief and supportive care.
- Optimize wound dressings, e.g. hydrocolloid dressings.
- Avoid compression which may exacerbate cutaneous ischaemia and pain.
- Treat infections promptly with oral antibiotics, e.g. flucloxacillin.
- IV or intralesional sodium thiosulfate may promote healing.
- Attempt to normalize calcium and phosphate levels—discuss management with the renal physicians. The evidence base for management is weak, but strategies may include:
  - Discontinuing oral calcium supplements.
  - Lowering the calcium concentration in the dialysate.
  - More frequent dialysis.
  - Bisphosphonates such as IV pamidronate and oral etidronate.
  - Cinacalcet to downregulate PTH levels and correct elevated calcium phosphate products.
  - Parathyroidectomy.
- Control blood glucose.
- Hyperbaric oxygen has been suggested.
- ⚠ Avoid aggressive debridement—may trigger further ulceration.
- ⚠ Avoid immunosuppressive drugs—increase the risk of infection.
- The role of anticoagulation is controversial, but consider stopping warfarin, if relevant, and use another anticoagulant.

Further reading

Strazzula L *et al.* *JAMA Dermatol* 2013;**149**:946–9 (intralesional sodium thiosulfate).
Vedvyas C *et al.* *J Am Acad Dermatol* 2012;**67**:e253–60.

Livedoid vasculopathy

Synonyms: segmental hyalinizing vasculitis, livedo reticularis with summer/winter ulceration, livedoid vasculitis.

This painful thrombo-occlusive vasculopathy affects young to middle-aged women. Hyaline thrombi occlude small vessels in the upper and mid dermis. There is no vasculitis. The cause is unknown, but fibrinolytic and coagulation systems are usually normal. Ulceration may be chronic and recurrent. Pain reduces quality of life.

What should I look for?

- Focal purpuric lesions on the leg, usually around the ankle.
- Small, excruciatingly painful ulcers surrounded by a purpuric rim.
- Porcelain-white stellate scars with a rim of telangiectasia (atrophie blanche) (see Fig. 15.8).
- Livedo reticularis: mottled, reddish purple, reticulated discoloration, reflecting a sluggish vascular flow in the superficial dermis (see ➔ p. 452, and Fig. 20.4, p. 453).
- Signs of venous disease.
- Net-like hyperpigmentation on the leg at the site of previous livedo (differentiate from erythema ab igne, a reticulated brown staining that can develop if the skin is exposed chronically to heat from a radiator, open fire, or hot-water bottle (see ➔ Fig. 20.5, p. 454).

What should I do?

- Investigations (results usually unremarkable):
  - FBC to exclude thrombocythaemia.
  - Clotting screen, including antiphospholipid antibodies, protein S and C.
  - Protein electrophoresis to exclude paraproteinaemia.
  - ANA to exclude SLE (if indicated by the presentation).
  - Take a deep biopsy, including the edge of the ulcer, to demonstrate hyaline thrombi.
- Control pain (see ➔ p. 304).
- Venous disease: control oedema with compression, if possible.
- Try antiplatelet therapy, antithrombotic regimens, or fibrinolytic agents, but treatment is difficult, and there is no consensus about the best approach.
- Treatment with immunosuppression is ineffective.

Fig. 15.7 Calciphylaxis: mottled erythema (livedo reticularis) and painful ulceration in a patient with diabetes.

Fig. 15.8 Livedoid vasculopathy: atrophie blanche with painful ulceration around the ankles.

Further reading

Alavi A *et al.* *J Am Acad Dermatol* 2013;**69**:1033–42.

Lymphoedema

Lymphoedema causes unilateral (occasionally bilateral) chronic swelling, most often of the leg, that is associated with recurrent cellulitis and may occasionally be associated with leg ulcers. The lymphatic system is involved in immune surveillance, and all forms of lymphoedema predispose to recurrent cellulitis/erysipelas.

1° lymphoedema is caused by an intrinsic fault (structural or functional) in lymph drainage. Causal genes include *VEGFR3* (Milroy disease) and *FOXC2* (lymphoedema–distichiasis syndrome). Lymphoedema may not present until adolescence or adulthood.

2° lymphoedema is caused by lymphatic obstruction or damage to the lymphatic system, e.g. radiotherapy, lymph node dissection. Filaria parasites invade lymphatics, causing lymphoedema (40 million people affected—a huge disease burden). Rarely, pretibial myxoedema simulates lymphoedema (see ➔ Fig. 22.4, p. 473).

What should I ask?

- Where is the swelling? When did it commence, and how did it progress?
- Does the swelling change with the position of the limb, e.g. overnight? Unlike venous swelling, lymphoedema does not reduce with elevation.
- Any episodes of cellulitis? Flu-like symptoms with fevers, rigors, or vomiting precede blotchy redness. How often are these episodes?
- Any pain? (Usually not a problem in lymphoedema.)
- Any family history of lymphoedema (found in 20% of cases of lymphoedema; suggests 1° lymphoedema)?
- Past history of malignancy or surgery, e.g. lymph node dissection.
- Drug history—sirolimus is a rare cause of 2° lymphoedema.

What should I look for?

- Persistent swelling affecting the foot, as well as the leg (see Fig.15.2).
- Pitting oedema at the onset; non-pitting 'woody' swelling in chronic lymphoedema because of fibrosis.
- A warm leg and foot with thickening, fissuring, and hyperkeratosis, and papillomatous wart-like growths resembling 'elephant skin'.
- Bleb-like vesicles you can compress with your finger (lymphoceles).
- Stemmer's sign: it is not possible to pinch up a loose fold of skin on the dorsum of the foot at the base of the second toe, because the skin is thickened and fibrotic. Absence does not exclude lymphoedema.
- Ascites or pleural effusions in some 1° lymphoedemas.
- Chronic lymphoedema in obese euthyroid patients may be associated with a nodular dermal mucinosis (obesity-associated lymphoedematous mucinosis).
- Bluish nodules or plaques suggesting angiosarcoma or KS (rare) (see ➔ pp. 362–3).

- Differentiate lymphoedema of the leg from lipoedema (see Box 15.15 and Table 15.1).
- Although lymphoedema may appear to be unilateral, lymphoscintigraphy studies have shown that the contralateral limb is often also abnormal. Examine both legs.

What should I do?

- Advise on skin care to reduce the likelihood of infection. The patient should wash the skin daily, ensure that folds of skin and the skin between the toes are cleaned and dried, and use an emollient to prevent fissuring. Cuts or abrasions should be kept scrupulously clean.
- Ensure the patient can trim his/her toenails, or refer to a chiropodist.
- If cellulitis occurs more than ×2/year, prescribe prophylactic phenoxymethylpenicillin 500mg daily (clindamycin if allergic to penicillin) for at least 2 years to prevent further cellulitis and lymphatic damage with worsening swelling.
- Refer to a lymphoedema clinic for manual lymphatic drainage, advice about exercise, and compression bandaging.

Box 15.15 Swelling of the leg: differential diagnosis

Unilateral

- DVT.
- Ruptured Baker's cyst.
- Cellulitis.
- Chronic venous insufficiency.
- Lymphoedema, e.g. pelvic tumour or lymphatic damage.
- Immobility.

Bilateral

- Congestive cardiac failure, renal failure, or nephrotic syndrome.
- Liver failure and hypoalbuminaemia.
- Malnutrition.
- Drugs, NSAIDs, calcium channel blockers.
- Lipoedema (see ➔ p. 318).
- Immobility, e.g. sitting in a wheelchair.

Further reading

Mortimer PS and Rockson SG. *J Clin Invest* 2014;**124**:915–21.

Lipoedema

What is lipoedema?

- Lipoedema, also known as painful fat syndrome, affects women and starts around puberty.
- Lipoedema may be familial (40% of cases).
- Nodules containing 'wet' fat are deposited in subcutaneous tissues on the buttocks, hips, thighs, and legs. The feet are not affected.
- Treatment is unsatisfactory. Dieting does not alter lipoedema—the trunk may become slim, but the legs remain large.
- Exercise and surgical support stockings may help.
- For differentiating lipoedema from lymphedema, see Table 15.1.

Table 15.1 Lipoedema or lymphoedema?

Lipoedema	Lymphoedema
Swelling is bilateral.	Swelling is usually unilateral.
The legs are painful, and the skin is tender.	Lymphoedema is usually painless.
The fat on the ankles creates a ring of fatty tissue that overhangs normal-looking feet. Fat is deposited on the buttocks and lower limbs.	The foot is affected, as well as the leg.
Swelling increases during the day and reduces during the night when the leg is elevated.	Position does not influence the amount of swelling.
Swelling does not pit.	Swelling pits in early lymphoedema but is non-pitting in late disease.
The skin tends to be thin. Easy bruising and subcutaneous bleeding are common.	The skin is thickened, fissured, and hyperkeratotic.
Stemmer's sign is absent.	Stemmer's sign is usually present.

Chapter 16

Sun and skin

Contents

Introduction

Life on earth depends on the sun's warmth and light. Our spirits rise in the sun, and sunny days evoke memories of long holidays (as well as revision for exams). But the sun has the potential to harm—there can be 'too much of a good thing'. Sunburn, skin cancer, and most photosensitivity disorders are caused by UVR.

How does ultraviolet radiation affect the skin?

The sun emits a broad and continuous spectrum of electromagnetic energy (the solar spectrum) (see Fig. 16.1). UVR is divided into short (UVC), medium (UVB), and long (UVA, 'black light') wavelength emissions. Ninety-eight percent of the UVR that reaches the earth's surface is UVA. All UVC and a little UVB are absorbed in the atmosphere by the ozone layer. UVB is absorbed by nucleic acids and proteins in the epidermis and stimulates the synthesis of vitamin D_3 (cholecalciferol). Some UVA is absorbed in the epidermis, but UVA penetrates into the dermis.

The damage caused by UVR depends on the duration of exposure, as well as the skin phototype (see Table 16.1).

- UVB causes immediate sunburn with painful erythema and, if severe, blistering. The UVB reaction peaks at 16–24h, then the skin desquamates over 2–3 days but becomes tanned and may develop solar lentigines (sunburn freckles).
- UVR, particularly UVB, damages DNA, and cumulative damage causes skin cancer (see ➔ p. 346).
- Chronic UV exposure produces many of the signs we associate with ageing (photodamage)—wrinkling, blotchy pigmentation, telangiectasia, a sallow appearance—as well as skin cancer (see ➔ Box 3.1, p. 43).
- UVR also causes local and systemic immunosuppression, which not only leads to problems, such as recurrent herpes simplex infection, but also contributes to the pathogenesis of skin cancer.

How is the skin protected from sun damage?

Nucleotide excision repair enzymes repair photodamaged DNA, but the capacity to repair declines with age (or may be abnormal in some rare photogenodermatoses; see ➔ pp. 336–7). Epidermal melanin provides some UV protection by absorbing visible light, as well as UVR (see ➔ pp. 12–3). Melanin also quenches oxygen free radicals.

Darkly pigmented skin has the same number of melanocytes as lightly pigmented skin, but more and slightly differently packaged melanin (melanosomes) within keratinocytes provides greater resistance to solar damage. However, even dark skin burns in the sun, if the dose of UVR is high enough.

UVA causes immediate darkening of the skin (oxidation of pre-existing melanin) that lasts a few hours and is not photoprotective. UVB is the main stimulus for the delayed tanning detectable 3–5 days after exposure (more melanocytes, synthesis of new melanin, redistribution of melanosomes). This provides some photoprotection and fades slowly.

Sun exposure also induces the injured epidermis to proliferate. Thickening of the horny layer (stratum corneum) supplies a little more protection from photodamage.

The solar spectrum

The solar spectrum ranges from cosmic rays, γ rays, and X-rays, through UVR (100–400nm), visible light (400–800nm), and infrared radiation (800–17 000nm), to radio waves (see Fig. 16.1). UVB and UVA contribute to tanning; UVB causes sunburn.

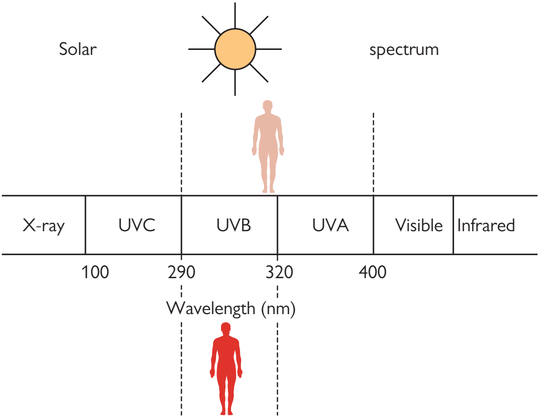

Fig. 16.1 Solar spectrum. UVA and visible light, unlike UVB, penetrate glass.

Skin phototype and photosensitivity

Skin phototype

Pale skins that burn easily and tan poorly, if at all, are most at risk of chronic photodamage and skin cancer, but it may be difficult to predict the phototype simply from the appearance of the skin. It is better to ask patients how quickly they burn and how easily they tan.

The Fitzpatrick classification (see Table 16.1) reflects the amount of melanin in the skin.

Table 16.1 Fitzpatrick classification

Skin type	Response of skin to sun	Appearance
I	Always burns, never tans	White or pale skin, many freckles, blond or red hair, blue or green eyes
II	Burns easily, tans poorly	Pale skin, possibly some freckles, blond hair, blue or green eyes
III	May burn, tans lightly	Darker white skin, dark hair, brown eyes
IV	Burns minimally, tans easily	Olive skin, brown or black hair, brown eyes—Mediterranean
V	Rarely burns, always tans deeply	Brown skin—Asian, Latin American, Middle Eastern
VI	Never burns, tans darkly	Black African

What is photosensitivity?

- Photosensitivity is defined as an abnormal cutaneous response to light/sunlight. It can be caused by a deficiency of photoimmunosuppression or as a result of an interaction between photosensitizing substances and sunlight or filtered or artificial light.
- People who are photosensitive burn easily in sunlight and may complain of reddening, swelling, blistering, itching, and/or painful skin after minimal sun exposure (see ➔ Box 4.6, p. 73).
  ❗ *Photosensitivity may lead to early skin cancers* (see ➔ p. 346).
- Photosensitivity may be 1° (i.e. idiopathic) but, more often, is acquired.

What should I do?

See Box 16.1.

Box 16.1 Photosensitivity: what should I do?

- Take a full medical history (see ➔ p. 28), and examine the patient (see ➔ Box 4.6, p. 73). Some skin diseases may be made worse by sunlight (photoaggravated) (see Box 16.2).
- Investigation of photosensitivity may include:
  - ANA, ENA (Ro antibodies are linked to photosensitivity).
  - Porphyrins in blood, urine, and stool (ensure the sample is not exposed to light by wrapping the specimen pot in tinfoil).
  - Skin biopsy for histology/fibroblasts for DNA repair disorders.
  - Patch testing and photopatch testing.
  - Phototesting to define the minimal erythema dose (MED) and action spectrum responsible. Skin can be exposed to light that simulates sunlight and also to specific wavebands by using a monochromator to fractionate the light.
  - Metabolic studies (aminoaciduria, hair analysis).
- Advise the patient how to protect the skin from UV light (see ➔ pp. 338–9).
- Manage the underlying problem.

Further reading

Hussein MR. *J Cutan Pathol* 2005;**32**:191–205.
Yashar SS and Lim HW. *Dermatol Ther* 2003;**16**:1–7.

Causes of photosensitivity

Photosensitivity in adults

Common causes

- Photoaggravated skin disease, e.g. cutaneous LE (see Box 16.2).
- Drug-induced photoallergy or phototoxicity (see ➔ p. 326 and Fig 16.2).
- Contact photosensitivity:
  - Soaps, tars, perfumes, drugs such as topical NSAIDs.
  - Plants: phytophotodermatitis (plants containing psoralen) (see Box 16.4).
- Polymorphic light eruption (PLE) (see ➔ pp. 328–9).

Uncommon or rare causes

- Cutaneous porphyrias, e.g. PCT (see ➔ p. 332).
- Pseudoporphyria (see ➔ p. 334).
- Idiopathic photodermatoses (see Box 16.5).
- DNA repair deficiencies (rare) (see ➔ pp. 336–7).
  - XP, trichothiodystrophy, Cockayne syndrome, Bloom syndrome, Rothmund–Thomson syndrome.
- Other very rare photogenodermatoses:
  - Kindler–Weary syndrome, Smith–Lemli–Opitz syndrome.

Photosensitivity in childhood

- Consider phytophotodermatitis, drug-induced photosensitivity, and PLE before porphyria or a rare genodermatosis.
- Remember that atopic eczema may be photoaggravated.

Box 16.2 Skin diseases that may be photoaggravated

Features of an underlying disease, but more severe on sun-exposed skin.

- Rosacea.
- Dermatitis—seborrhoeic, atopic.
- Psoriasis and pityriasis rubra pilaris (more often helped by sunlight).
- LP, EM.
- LE, especially subacute cutaneous LE.
- Dermatomyositis.
- Darier disease.
- Autoimmune blistering diseases—pemphigus, BP.
- Pellagra (niacin deficiency).

Fig. 16.2 Photosensitive patient with erythema of exposed skin, but with an abrupt cut-off where the skin is covered by clothing.

Drug-induced photosensitivity

Drugs are the commonest cause of photosensitivity (see Box 16.3). Photosensitivity is caused by phototoxicity more often than photoallergy, but the distinction is not always clear. Reactions may also be bullous or lichenoid. Other potential presentations of a drug-induced photosensitivity include pseudoporphyria (see ➔ p. 334) and LE (see ➔ p. 402).

What is phototoxicity?

Phototoxicity causes an exaggerated sunburn response with pain, erythema, oedema, and sometimes blistering of exposed skin. The action spectrum for phototoxins may be UVB, UVA, or visible solar radiation. The response depends on the amount of UVR (the patient may have no problems on a slightly cloudy day) and the dose of the drug but can occur on first exposure to the drug. The response is usually immediate but can be delayed. Reactions to amiodarone and chlorpromazine produce pigmentary changes. Chronic phototoxicity accelerates photoageing, e.g. patients taking voriconazole for antifungal prophylaxis. Examine the skin regularly.

Phytophotodermatitis is a phototoxic eruption caused by contact with plant juices containing light-sensitizing furocoumarin chemicals, e.g. psoralens (see Box 16.4).

What is photoallergy?

Photoallergic reactions are idiosyncratic delayed hypersensitivity reactions, similar to an allergic contact dermatitis, that do not develop, until the patient has been sensitized by prior exposure to the drug. The action spectrum for most photoallergens is in the UVA range (light coming through window glass causes problems). The response is not dose-dependent, and the reaction may spread to affect covered skin. Frequently, topical medicaments are implicated. Photopatch testing is positive.

What should I do?

- Withdraw the drug, if possible. Phototoxic reactions settle within a week. The signs of photoallergy (erythema) will also disappear within a week, but photosensitivity may persist for months after drug withdrawal (sometimes as long as 6 months).
- A potent topical steroid applied daily will help to settle the inflammation. Systemic steroids may be required in severe reactions.
- Sun protection with clothing, a hat, and a high-factor broad-spectrum UVA/UVB sunscreen is essential (see ➔ pp. 338–9).
- Photopatch testing may be helpful in suspected topical agent photoallergy, but less useful for systemic drug photosensitivity.
- ⚠ Chronic phototoxicity increases the risk of skin cancer.

Box 16.3 Drug-induced photosensitivity

- Phototoxicity: amiodarone, doxycycline, thiazides, phenothiazines, NSAIDs, quinolones, quinine, sulfonamides, retinoids, psoralens (in plants), coal tar, voriconazole, BRAF inhibitors.
- Photoallergy: topical antimicrobials, fragrances, sunscreens, angiotensin-converting agent inhibitors, non-steroidal anti-inflammatory agents, phenothiazines, sulfonamides, thiazides.
- Pseudoporphyria (a form of phototoxicity with blistering and skin fragility; see ➔ p. 334): nalidixic acid, furosemide, ciprofloxacin, bumetanide, NSAIDs.
- Lichenoid reactions resembling LP (see ➔ p. 224) (delayed in onset): hydrochlorothiazide, NSAIDs, quinidine-derived drugs.
- Photo-onycholysis (painful): tetracyclines, taxanes.

Box 16.4 What is phytophotodermatitis?

- Phototoxic reactions to light-sensitizing furocoumarin chemicals in plants can occur in anyone and are not an allergy.
- Causes of phytophotodermatitis include members of the *Apiaceae* family (previously *Umbelliferae*), e.g. Queen Anne's lace, giant hogweed, parsnip, celery; the *Rutaceae* family, e.g. garden rue (*Ruta graveolens*); and citrus fruit oils, e.g. lime juice.
- The patient, often a gardener wielding a strimmer or a child, is outside on a sunny day. Plant juice is deposited in a streaky linear fashion on exposed skin that has brushed against the plant or, with strimmers, the juice is sprayed over exposed skin.
- The psoralens in the plant juice cause a painful burning (rather than itching) erythema, often with blistering, about 24h after exposure to long-wave UVR (UVA).
- The reaction peaks at 48–72h and gradually settles over a few days.
- Linear streaks of hyperpigmentation persist at the site of inflammation and may take months to fade. These may be all you see when the patient presents (see ➔ Fig. 31.11, p. 615).
- The diagnosis depends upon the history and the bizarre linear pattern suggesting an external injury. Sometimes child abuse is suspected (see ➔ pp. 614–15).

Polymorphic light eruption

Polymorphic light eruption (PMLE, PLE) is the commonest of the idiopathic photodermatoses (see Box 16.5). It is sometimes known as 'sun allergy' or prickly heat. PLE affects about 15% of the young adult ♀ Caucasian population in northern Europe and may present in young adults or childhood. Genetic factors probably play some part in the pathogenesis. The abbreviation is confusing—PLE is not a form of cutaneous lupus erythematosus (LE).

UVR (mostly UVA; sometimes UVA and UVB) provokes a T-cell response. The mechanism is unclear. It may be caused by failure of normal photoimmunosuppression. Phototesting can be normal, but provocation testing (UVA or solar-simulated light) usually provokes a rash.

What should I look for?

- A story of an itching or burning rash that erupts within 2h of exposure to UVR but, in some patients, may not develop for 24–48h, or even longer. (The history is usually diagnostic.)
- A rash that fades within 1–6 days, leaving normal skin (no scars).
- Erythematous papules, plaques, vesicles, or, less often, haemorrhagic blisters. The rash is polymorphic in its presentation in populations, but individuals usually have one form of the disease.
- The problem usually presents in spring, but some patients only develop the rash when they go on holiday to a sunny climate on 'holiday-exposed sites'. Sunlight reflected from snow may provoke PLE.
- The rash may be triggered by as little as 10min of direct sunlight.
- Sunlight penetrating window glass (UVA) may cause the rash, as can unshielded fluorescent lights and UVA from sunbeds.
- Chronically exposed skin on the face and hands is usually spared.
- The rash usually erupts less frequently, as the summer progresses ('hardening'), provided exposure to sunlight continues.

What should I do?

- Exclude subacute cutaneous LE (see ➔ p. 402), solar urticaria and oral contraceptive (oestrogen)-induced photosensitivity (rare).
- Management options include:
  - Potent topical corticosteroids for acute eruptions.
  - Gradual sunlight exposure to induce tolerance (may be difficult to achieve) and continued sun exposure to maintain tolerance.
  - Protective clothing: broad-brimmed hats, long sleeves (see ➔ p. 338).
  - High-factor UVA/UVB sunscreens may prevent PLE. (UVB sunscreens may worsen PLE by allowing UVA through, but diminishing the protective immunosuppression produced by UVB).
  - Oral prednisolone 15–20mg/day for 2–3 days may be required to control an acute eruption, e.g. while on holiday.
  - Desensitization with UVB each spring. Treat ×2–3/week for 3–5 weeks. Patients must continue exposure to sunlight to maintain tolerance. Repeat annually if tolerance is not maintained.
  - Hydroxychloroquine can be helpful in some cases.

Box 16.5 The idiopathic photodermatoses

The pathogenesis of the idiopathic photodermatoses is incompletely understood. All are uncommon or rare, apart from PLE.

Common

- PLE: usually presents under age 30. Onset within hours of sun exposure, not minutes. Symmetrical rash (usually papulonodular or eczematous) that affects skin covered in winter. (Differential diagnosis: subacute cutaneous LE; see ➔ p. 402.)

Uncommon

- Juvenile spring eruption: mainly boys in spring/summer. Itching, uncomfortable papules and vesicles on exposed helices of the ears. Usually the rest of the skin is not affected. Generally, the rash spontaneously resolves within 1–2 weeks but can recur.
- Solar urticaria: may arise at any age. Itchy urticarial wheals appear within a few minutes of exposure to sunlight and last for <24h. (Differential diagnosis: drug-induced photosensitivity, erythropoietic protoporphyria.) Action spectra: UVA, UVB, and visible, including fluorescent, lights. Antihistamines and desensitization may be helpful.
- Chronic actinic dermatitis: most often elderly men (90%) with multiple contact allergies. Action spectra: UVB, UVA ± visible. Compositae (plants) commonest contact allergy. Potent immunosuppressants, such as azathioprine, are often required.

Rare

- Actinic prurigo: presents in childhood, commonest in American Indians/Mexicans. In the UK, strongly associated with HLA-DR4, in particular HLA-DRB1 0407. Itching papules evolve into chronic nodules and plaques. Rash is maximal in spring and summer. All exposed skin is affected but may involve covered sites (e.g. buttocks), so diagnosis often missed. May simulate atopic eczema. Solar cheilitis of the lower lip and conjunctivitis are common. Leaves pitted or linear scars. Usually resolves in early adulthood.
- Hydroa vacciniforme: presents in childhood, usually boys. Rash within minutes of exposure. Tender, itching, erythematous papules evolve into haemorrhagic vesicles that crust and heal, leaving chickenpox-like scars. Associated with EBV infection (Japanese). Usually spontaneous improvement by adulthood.
- Brachioradial pruritus: stinging or burning pain in the skin of forearms. Prevalence unknown. No rash. May be a form of light-induced chronic neuropathic damage (see ➔ Box 27.1, p. 548).

Further reading

Bylaite M *et al. Br J Dermatol* 2009;**161 Suppl 3**:61–8.
Murphy GM. *J Photochem Photobiol B: Biol* 2001;**64**:93–8.

Porphyria

What is porphyria?

The porphyrias are a group of disorders caused by genetic or acquired partial deficiencies in one of the enzymes in the metabolic pathway for haem. Haem precursors accumulate in urine, faeces, plasma, and/or erythrocytes. Disease may be acute neurovisceral, non-acute cutaneous, or mixed (see Table 16.2).

Acute neurovisceral attacks are associated with a rise in 5-aminolevulinate and porphobilinogen. Blisters/skin fragility caused by accumulation of water-soluble porphyrins (uroporphyrin and coproporphyrin) in PCT, variegate porphyria, and hereditary coproporphyria. Painful photosensitivity is caused by raised protoporphyrin in EPP.

Although mutations reduce the activity of the enzymes by 50%, most (80%) of those who inherit an AD porphyria remain asymptomatic. Other factors that increase the demand for haem and/or decrease enzyme activity are required for the diseases to be expressed.

Acute porphyria: what should I look for?

- Episodic neurovisceral attacks that start between age 15 and 35.
- Commoner in ♀ (less likely after menopause).
- Severe pain (usually abdominal), nausea, vomiting, constipation.
- Hypertension, tachycardia, low sodium.
- Neurological/psychiatric signs: psychological upset, convulsions, muscle weakness.
- Trigger: e.g. drugs (oestrogens, progesterones, barbiturates, sulfonamides, tetracyclines, etc.), alcohol, infection, calorie restriction.
- Cutaneous involvement (none in acute intermittent porphyria but may be seen in some acute porphyrias, e.g. variegate porphyria). Skin fragility, blisters, milia, erosions, and/or scars on exposed sites such as dorsum of hands or face (like PCT; see ➔ p. 332).

Acute porphyria: what should I do?

- Remove precipitants, e.g. drugs, including recreational drugs.
- Treat symptoms—fluids (monitor sodium), analgesia, antiemetics.
- Collect a random fresh sample of urine during the attack to measure porphyrins. Protect the fresh specimen from light (see Box 16.4). Analyse faecal and plasma porphyrins to determine the type of acute porphyria. Urinary, faecal, and plasma porphyrin concentrations may return to normal during remission in all the AD acute porphyrias.
- Consider early administration of IV haem arginate to suppress the production of haem precursors.
- Screen family members.

Table 16.2 Classification of the porphyrias

Porphyria and inheritance	Enzyme deficiency	Clinical	Comment
Porphyria cutanea tarda (PCT) Sporadic (80%) = type I; AD (20%) = type 2	Uroporphyrinogen decarboxylase	Non-acute. ***Skin only***. Blisters, skin fragility, erosions over photoexposed sites, pigmentation. Increased risk of hepatocellular carcinoma	***Common type*** Acquired or inherited
Erythropoietic protoporphyria (EPP) AD	Ferrochelatase	Non-acute. ***Skin only***. Burning, erythema/oedema of light-exposed skin. Baby cries on exposure to sun. Subtle scars on nose. Risk of cholestasis and liver failure	***Commonest AD type in children***
Congenital erythropoietic porphyria (Gunther disease). May have led to folklore of werewolves emerging at night to avoid sunlight AR	Uroporphyrinogen III	Non-acute. Severe photosensitivity, mutilating scarring, hypertrichosis, red teeth, haemolytic anaemia. Dark urine—stains napkin purplish red	Very rare. Presents in infancy.
Acute intermittent porphyria AD	Hydroxymethylbilane synthase	Acute neurovisceral attacks. ***Skin unaffected***. Risk of liver cancer/renal failure	***Commonest acute porphyria***
Variegate porphyria AD; founder mutation explains prevalence in South Africa/Chile	Protoporphyrinogen oxidase	Acute neurovisceral attacks (20%) or blisters like PCT (60%) or both (20%)	Common in South Africa/Chile
Hereditary coproporphyria AD	Coproporphyrinogen oxidase	Acute neurovisceral attacks. Blisters like PCT (30%)	Very rare

Very rare variants have not been included, e.g. ALA dehydratase deficiency porphyria, homozygous variants of AD porphyrias.

Further reading

British Porphyria Association. Available at: http://www.porphyria.org.uk/.

Welsh Medicines Information Centre and Cardiff Porphyria Service. *Drugs that are considered to be SAFE for use in the acute porphyrias*. Available at: http://www.wmic.wales.nhs.uk/pdfs/porphyria/2015%20Porphyria%20safe%20list.pdf.

Porphyria cutanea tarda

PCT, the commonest porphyria, is the only type that may be either sporadic (80% of cases) or familial (AD). Reduced activity of uroporphyrinogen decarboxylase results in overproduction of photoactive porphyrins. Uroporphyrinogen decarboxylase is inactivated by an iron-dependent process that is not fully understood, but a variety of factors can initiate disease (see Box 16.6).

Water-soluble porphyrins cause skin fragility and blistering on exposed skin. Uroporphyrin also stimulates fibroblasts to produce collagen in the skin, and this may explain some of the cutaneous features.

What should I look for?

PCT only affects the skin and is commonest in men. The presentation is subacute, and the relationship to sun exposure may be missed.

- Skin fragility: minor knocks produce erosions on the dorsum of the hands.
- Itching or burning may precede blisters on sun-exposed skin.
- Haemorrhagic vesicles, bullae, and crusted erosions, most often affecting exposed skin on the dorsum of the hands, face, and upper chest (see Fig. 16.3).
- Superficial scars or milia (firm, white, pinhead-sized papules—sequelae of subepidermal blisters) (see ➔ Epidermolysis bullosa acquisita, p. 272).
- Dystrophic calcification.
- Hypertrichosis usually starts on the temples and affects the cheeks and/or forehead.
- Blisters beneath the nail plate; painful discoloration (yellow, blue, or haemorrhagic) of the nail; loss of the lunula, onycholysis, or dystrophy.
- Diffuse or reticulated hyperpigmentation of sun-exposed skin.
- Waxy, yellowish thickening of sun-exposed skin ('sclerodermoid').
- Normal teeth (unlike congenital erythropoietic porphyria).
- Normal mucosa (involved in some autoimmune blistering diseases).

What should I do?

- Confirm the diagnosis, and exclude pseudoporphyria (see ➔ p. 334) by measuring porphyrins in fresh samples of urine and stool (see ➔ pp. 330–1). Fresh urine containing excess uroporphyrins is pink and fluoresces bright coral pink under UVA light (Wood light).
- Biopsy a fresh blister for histology: subepidermal cell-poor blister, periodic acid–Schiff (PAS)-positive glycoproteins at the BMZ and around blood vessels, and, for direct IMF, Igs deposited at the BMZ and around blood vessels.
- Exclude risk factors, including hepatitis C (see Box 16.6).
- Withdraw precipitating factors, including alcohol and oestrogens.
- Advise strict sun protection (see ➔ pp. 338–9).
- Regular venesection (400–500mL every 2 weeks for 3–6 months) depletes iron stores. Alternatively, desferrioxamine can be helpful.
- Consider prescribing low-dose hydroxychloroquine (100–125mg ×2/week)—higher doses may cause hepatitis.
- Monitor for the development of hepatocellular carcinoma.

Box 16.6 Risk factors for adult porphyria cutanea tarda

Factors that reduce the activity of uroporphyrinogen decarboxylase in the liver can initiate disease in genetically susceptible individuals.

- Alcohol.
- Oral oestrogens: oral contraception and hormone replacement therapy.
- Oral iron.
- Mutations in the hereditary haemochromatosis (*HFE*) gene.
- Hepatitis C virus infection.
- HIV infection.
- Polychlorinated hydrocarbons.
- Chronic haemodialysis (azotaemia reduces the activity of uroporphyrinogen decarboxylase; porphyrins are inadequately cleared by haemodialysis). Check plasma and faecal porphyrins, as urinary testing is difficult to interpret if the patient is on dialysis. If porphyrins are normal, consider pseudoporphyria (see ➔ p. 334). Venesection is usually contraindicated in renal failure, and hydroxychloroquine is ineffective. Human erythropoietin reduces iron stores and may be an effective treatment.

Fig. 16.3 Porphyria cutanea tarda: fragile skin on the back of the hand with crusted erosions and scars.

What is pseudoporphyria?

Pseudoporphyria is an acquired photosensitive blistering disease, with clinical and immunohistological features similar to PCT. The pathogenesis is uncertain, but porphyrins are normal.

Patients have PCT-like blistering, erosions, milia, and scars on sun-exposed skin (see Fig. 16.3), but, unlike PCT, patients rarely have hyperpigmentation, hypertrichosis, sclerodermoid changes, or dystrophic calcification.

Causes include:

- Chronic renal failure, haemodialysis, peritoneal dialysis (the antioxidant acetylcysteine may be an effective treatment) (see Box 16.6).
- UVA tanning beds, PUVA, excess sun exposure (generally women).
- A wide range of drugs, including:
  - NSAIDs (most frequent cause): naproxen and others (generally in women).
  - Antibiotics: nalidixic acid, tetracycline, ampicillin–sulbactam, cefepime, fluoroquinolones.
  - Antifungals: voriconazole.
  - Diuretics: furosemide, bumetanide.
  - Antiarrhythmics: amiodarone.
  - Sulfones: dapsone.
  - Vitamins: brewer's yeast, pyridoxine.
  - Retinoids: isotretinoin, etretinate.
  - Muscle relaxants: carisoprodol.
  - Anti-androgens: flutamide.

Further reading

Green JJ and Manders SM. *J Am Acad Dermatol* 2001;**44**:100–8.

Erythropoietic protoporphyria

EPP, caused by reduced activity of ferrochelatase, is a rare form of porphyria, presenting in childhood. Intense pain may cause infants to cry bitterly within minutes of sun exposure, but, as signs are subtle and the skin can look almost normal between episodes, the diagnosis is often missed. In 2% of cases, EPP is complicated by progressive liver failure.

What should I look for?

- Acute episodes of cutaneous photosensitivity that started in infancy. Window glass is not protective (sensitivity to UVA and visible light). Light reflected off snow, sand, or water may cause symptoms.
- Symptoms start in spring but reduce in autumn and winter.
- Burning, tingling, stinging, pain, and/or itching in light-exposed skin occurs within minutes of sun exposure. Children may try to relieve symptoms by plunging the hands into cold water.
- Erythema, urticaria, and/or swelling within minutes of sun exposure.
- Occasionally purpuric lesions, following sun exposure.
- Waxy skin thickening over the knuckles and on the nose.
- Subtle elliptical scars on the nose, cheeks, and/or dorsum of the hands.

What should I do?

- Check porphyrins: protoporphyrin is raised in erythrocytes and stool; urine porphyrins are not increased.
- Check FBC: some patients are anaemic.
- Advise on photoprotection (lifestyle, broad-spectrum sunscreens, clothing, UV-blocking films for window glass in home and car) (see ➔ pp. 338–9).
- Oral β-carotene taken from February until October may reduce photosensitivity, although efficacy variable. Patients may not tolerate the unsightly orange pigmentation.
- UVB phototherapy may improve tolerance to sunlight (probably the most effective treatment).
- Monitor for cholestasis and progressive liver damage.

⚠ Xeroderma pigmentosum

This rare AR photodermatosis affects all races and has an equal sex incidence (2.3 million cases/million live births in Western Europe). Patients develop premature photodamage, skin cancers, and ocular abnormalities. Thirty percent of patients develop progressive neurological disorders. The disease is devastating, and cutaneous malignancies cause early death (see Box 16.7). Normally, photodamaged DNA is excised and repaired with newly synthesized DNA (excision repair), but XP cells are unable to repair damaged DNA (see Box 16.7). The defect affects all cell types. Patients have an increased risk of cutaneous and ocular malignancy, as well as malignancy in other tissues, e.g. oral cavity, breast, lung, liver, gastric, renal, and brain.

Skin is normal at birth, but XP presents within the first few years and has a major impact on the lifestyle of the child and their family. Clinical features vary, but fair-skinned children are the most severely affected.

What should I look for?

- Easy sunburn that usually presents within the first 2 years of life.
- UVB-induced erythema that is delayed in onset and peaks in 2–3 days, rather than the usual 24h.
- A history of blistering or burning after very little sun exposure.
- Photophobia in young children.
- Dryness of sun-exposed skin (xeroderma).
- Irregular freckling appears in the first year of life on exposed skin.
- Scarring of exposed skin.
- Signs of premature photodamage—mottled hyperpigmented patches, hypopigmented macules, telangiectasia.
- Hyperkeratotic lower lip (solar cheilitis).
- Eyelid freckling, loss of lashes, ectropion, conjunctival telangiectasia, dry eyes, and corneal damage, leading to visual impairment.
- Solar keratoses (premalignant change).
- Skin cancers develop in all (median onset age 8 years)—BCC, SCC, melanoma.
- Parents with a history of skin cancers, but no other signs of XP.

Management

- The diagnosis is confirmed by skin biopsy and demonstration of the DNA repair defect in cultured fibroblasts.
- Care should be provided by a multidisciplinary team that may include paediatricians, dermatologists, ophthalmologists, neurologists, cancer specialist nurses, and plastic surgeons.
- ⚠ *It is crucial to protect the skin from photodamage* to slow the onset of life-threatening cutaneous malignancies (see ➔ pp. 338–9).
- Carers and/or patients should examine the skin regularly for signs of skin cancer. Cancers are treated surgically.
- Oral retinoids may reduce the incidence of skin cancer.
- Patients should be protected from cigarette smoke.
- Encourage patients to join XP Patient Support Group (see Box 16.8).

Box 16.7 What is xeroderma pigmentosum?

- XP is an AR disorder caused by defects in one of seven nuclear excision repair (*NER*) genes. Some XP genes are also involved in other processes, e.g. transcription and recombination.
- XP patients are photosensitive and prone to premature skin cancers, as well as internal malignancies, but patients are a clinically heterogeneous group.
- Patients were separated into different groups when it was found that the defect in excision repair in a cultured cell line from one XP patient could be corrected by fusion with a cell line from a different, unrelated XP patient. Patients whose cell lines failed to complement each other in culture were grouped together.
- Seven complementation groups have been identified (XP-A to XP-G).
- Wide variability in clinical features both between and within XP complementation groups, in part explained by the precise pathogenic mutation.
- XP-C is the commonest complementation group worldwide.
- Twenty percent of patients (mostly XP-A and XP-D) have progressive neurological abnormalities, e.g. hyporeflexia, intellectual disability, seizures, deafness, ataxia, quadriparesis. Accumulated DNA damage may cause degeneration of neural tissue, which cannot be replaced.
- Thirty percent of XP patients belong to a variant group (XP-V) with a less severe phenotype. XP-V patients have normal nuclear excision repair, but abnormal post-replicational DNA repair caused by defective DNA polymerase.
- Investigation into the genetic basis of XP has provided insights into DNA repair and the pathogenesis of cutaneous malignancy.

Box 16.8 XP Support Group

The XP Support Group is a UK charitable trust (available at: http://xpsupportgroup.org.uk/). The group supports patients and their families, keeps members up-to-date with the latest research, provides information on XP, and offers practical advice on a wide range of topics, including:

- How to deal with schools and education authorities.
- Where to obtain UV protective clothing and UV protective clear window film.
- Grants for UV protective products.
- Annual Owl Patrol night-time camps for children and their families.

Further reading

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DiGiovanna JJ and Kraemer KH. *J Invest Dermatol* 2012;**132**(**3 Pt 2**):785–96.
Fassihi H. *Br J Dermatol* 2015;**172**:859–60

Photoprotection

Patients with photosensitivity disorders must protect their skin from the solar radiation that is responsible—UVA, UVB, visible light, or a combination. Severe photosensitivity can have a catastrophic impact on lifestyle—outdoor activities may be impossible. Sunglasses, clothing, hats, and shade (indoor activities when the sun is strongest between 11.00 a.m. and 3.00 p.m.) provide more effective photoprotection than sunscreens.

Clothing, hats, sunglasses, and windows

- Clothing and hats should have a tight weave, sufficiently thick to block transmission of visible light (check the transmission of visible light by trying to look through the item held up to the sun).
- Clothing should be loose-fitting (photoprotective fabrics are available). Patients should wear hats with broad brims (baseball caps do not protect the ears or neck), as well as a flap of cloth to cover the neck, shirts with long sleeves, trousers—not shorts, and shoes, rather than sandals. Some patients may need gloves for driving.
- UV-blocking films can be obtained for window glass, car windows, and fluorescent lights, if required.
- Sunglasses (wrap-around) to protect against UV radiation and blue light.

Sunscreens

(See Boxes 16.9 and 16.10.)

- Sunscreens may be physical or chemical.
- Physical sunscreens (sunblocks): opaque creams containing particles (titanium dioxide, zinc oxide) reflect and scatter UVR (UVB and UVA) and visible light. Physical sunscreens are less cosmetically acceptable than chemical sunscreens, but more effective and safer (see Box 16.10, ➔ Box 17.3, p. 347, and Box 17.4, p. 347).
- Chemical sunscreens absorb UVR of certain wavelengths, either UVB or UVA. UVB chemical sunscreens protect against UVB-induced sunburn by blocking UVB. They contain cinnamates and other agents. UVA chemical sunscreens provide limited protection and contain chemicals such as oxybenzone. New dual UVB/UVA filters and photon absorbers are being developed.
- Chemical sunscreens may cause contact dermatitis (irritant or allergic).
- Patients using a UVB-absorbing sunscreen that prevents immediate erythema or sunburn should be warned not to stay out longer in the sun, to prevent exposure to excessive UVA.
- Lips should be protected, as well as the skin.
- The potential of topical or oral antioxidants to reduce photodamage is being investigated.
- Patients avoiding UVB may become vitamin D-deficient (see Box 16.11).

Box 16.9 How much UV protection is provided by the sunscreen?

- The sun protection factor (SPF) indicates the UVB photoprotection provided by a sunscreen. An SPF of 10 means that it takes ten times longer for the skin to go red than unprotected skin exposed to the same amount of UVB. An SPF of 20–30 is adequate for most patients.
- The star rating (maximum 5) indicates the percentage of UVA absorbed in comparison to UVB. The UVA protection should be at least 1/3 of the labelled SPF.

Box 16.10 Why is the sunscreen ineffective?

- The sunscreen is not blocking the appropriate wavelengths. Broad-spectrum combination sunscreens with an SPF of at least 15 and a high UVA protection are the most effective.
- The sunscreen is not applied sufficiently thickly (at least six teaspoons of lotion are needed to cover the body of an average adult). Opaque sunblock creams do not have to be applied very thickly to be effective.
- The sunscreen is not applied evenly to the skin or to all exposed skin, including lips, sides of the neck, ears, and temples.
- The sunscreen is not reapplied every 2–3h or straight after swimming or activities causing sweating.
- Patient allergic to the sunscreen (refer for photopatch testing).
- Recommend physical barriers—sunglasses, clothing, hats, and shade—rather than encouraging patients to rely on sunscreens.

Box 16.11 Advice on optimal vitamin D synthesis

- UVB triggers conversion of 7-dehydrocholesterol in the skin to pre-vitamin D. Adequate levels of vitamin D are maintained if 15% of the body (face, arms, legs) receives about 1/3 of the minimal erythematogenic dose of UVB on most days of the week. In the UK, this equates to 15min of sun exposure on the arms, legs, and/or face ×3/week from May to September (during the rest of the year, sunlight at this latitude is insufficient for vitamin D synthesis).
- Patients who are very photosensitive and cannot tolerate any UVB radiation should take vitamin D supplements.
- Vitamin D supplements may also be needed by individuals living in countries with low levels of UV for much of the year.
- Vitamin D deficiency is common in the elderly, people with darker skin, or those who cover the skin when outdoors.

Further reading

González S *et al*. *Clin Dermatol* 2008;**26**:614–26.
Hollick F. *Adv Exp Med Biol* 2014;**810**:1–16.

Chapter 17

Tumours

Contents

Common benign non-melanocytic tumours

Seborrhoeic wart

Also known as seborrhoeic keratosis or basal cell papilloma.

Benign epidermal tumours of unknown cause. Common in the elderly. They are commonest on the trunk but also affect the face and limbs. Patients may have one, several, or hundreds. Treatment is only required if the seborrhoeic wart is traumatized, when it may become inflamed or bleed, sometimes simulating a melanoma.

What should I look for?

- A well-defined oval, pigmented tumour, around 0.5–2cm in diameter, that appears stuck onto the surface of the skin.
- Rough surface that may feel slightly greasy. Dark pits studded over the surface—easier to see with magnification using dermoscopy.
- Colour can range from a pale pink-brown to dark brown or black.
- Seborrhoeic warts may be virtually flat or raised by several mm.

Chondrodermatitis nodularis helicis chronica

Nodules caused by prolonged pressure on the skin of the ear, but sun damage or cold may also play a part. The history is usually characteristic and differentiates this nodule from a skin cancer. The affected ear is generally the one on which the patient sleeps, and pain prevents from lying on that side. Apparently, nuns who wore wimples were affected by chondrodermatitis! Treatments include relieving pressure, topical or intralesional corticosteroids, cryotherapy and, surgery.

What should I look for?

- A tender nodule on the helix or, in women, the antihelix of the ear.
- A keratotic plug or a small ulcer in the centre of the nodule.

Dermatofibroma

Also known as fibrous histiocytoma or sclerosing haemangioma.

The localized proliferation of fibroblasts is probably triggered by minor trauma such as an insect bite. These benign growths are common in young women. If you can make a confident diagnosis, no treatment is required—excision will merely leave another scar. Multiple dermatofibromas have been reported in conditions such as HIV infection, SLE, and other diseases in which the immune state is altered.

What should I look for?

- A brownish red dermal nodule, 0.5–1cm in diameter, usually on the arm, shoulder, thigh, or leg.
- The overlying epidermis is usually smooth.
- The lesion is firm—this suggests the diagnosis.
- A central scar seen under dermoscopy with pseudo-pigmentation peripherally.
- Pinch the skin gently on each side of the tumour, and it will sink down into the dermis—the 'buttonhole sign' or 'dimple sign'.

Pyogenic granuloma

Typically, this vascular tumour presents at the site of a penetrating injury, e.g. rose thorn, usually on the fingers, lips, or face (see Fig. 17.1).

⚠ An amelanotic malignant melanoma or SCC may look like a pyogenic granuloma (see Fig. 17.11)—always ask the patient if anything preceded the development of the tumour. Was there a 'mole' at the site, or did it arise on normal skin? Always send excised lesions for histological examination.

What should I look for?

- A history of rapid growth: pyogenic granulomas grow on previously normal skin over a period of weeks to a size of 0.5–4cm.
- Fleshy vascular tumour composed of friable granulation tissue that bleeds readily on contact.
- Exclude any underlying pigmented lesion: inspect any 'pyogenic granuloma' at an atypical site particularly carefully, using a dermatoscope, if possible. Melanin can be mistaken for old haemorrhage.

Other common benign tumours or inflammatory nodules

Epidermal

- Viral infections, e.g. viral wart (see ➲ pp. 160–1) or molluscum contagiosum (see ➲ p. 158).
- Nodular prurigo (caused by chronic rubbing) (see ➲ Fig. 10.6, p. 223).
- Epidermal naevus (warty lesion, present since infancy) (see ➲ p. 560).

Dermal

- Epidermoid cyst or 'sebaceous' cyst: although not sebaceous at all, but arises from the upper part of the hair follicle. Has a punctum. Often becomes inflamed and ruptures.
- Pilar cyst: arises from the lower part of the follicle and usually found in the scalp—no punctum. Spontaneous rupture uncommon.
- Keloid: dense fibrous scar tissue spreading beyond the original wound.
- Spider naevus; Campbell de Morgan spot (cherry angioma).
- Infantile haemangioma (strawberry naevus) (see ➲ pp. 596–7).
- Neurofibroma (see ➲ Fig. 27.1, p. 555 and pp. 552–4).
- Xanthelasma or xanthoma (see ➲ p. 470).

Fig. 17.1 Pyogenic granuloma—a friable vascular papule.

Common benign melanocytic tumours

Melanocytic naevus ('mole' or 'naevus')

(See Fig. 17.2.)

Melanocytic naevi present at birth, in childhood, or in young adults.

- Acquired melanocytic naevi start as flat, evenly pigmented (junctional) naevi in which melanocytes collect in small groups (nests) along the basal epidermal layer (see Figs. 17.3 and 17.4). As melanocytes migrate down into the dermis, flat moles evolve into raised, evenly pigmented, dome-shaped papules (compound melanocytic naevi, Fig. 17.4), sometimes hairy.
- Scalp naevi in children may be an early sign of being 'moley'. Patients with many moles tend to have a 'signature naevus', i.e. all their moles follow a pigmentation pattern that is characteristic for that patient. This is more easily visualized using dermoscopy.
- Over time, the epidermal component is lost, and moles change into flesh-coloured or pale brown papules (intradermal melanocytic naevi), before disappearing in old age. A new pigmented growth in an elderly patient is much more likely to be a seborrhoeic wart, solar lentigo (see next section), or melanoma than a melanocytic naevus.
- Congenital melanocytic naevi, usually >1cm in diameter (sometimes large and disfiguring = bathing trunk naevus) present at birth/early neonatal period. Large congenital melanocytic naevi >20cm and the presence of multiple satellite naevi are associated with an increased risk of malignant change and should be monitored.
- Halo naevi: a white ring develops around a benign melanocytic naevus which gradually disappears, leaving a depigmented macule that usually eventually repigments. Halo naevi are common in adolescence and of no significance. Halo naevi in adults (age 40–50 years) may indicate melanoma elsewhere—check the skin, eyes, and mucosal surfaces (see Fig. 17.5).

Lentigo (plural = lentigines)

- Simple lentigines are small, round, flat, evenly pigmented lesions that persist in winter, unlike freckles. The basal layer of the epidermis has increased numbers of individual melanocytes.
- Numerous simple lentigines are found in some genetic disorders, e.g. Multiple lentignes syndrome, Carney complex, Peutz–Jeghers syndrome (see ➔ p. 613).
- Solar (senile) lentigines, 3–12mm in diameter, are flat, brown marks present on sun-damaged skin in older patients.

Mongolian spot

- Slate-coloured macular areas of pigmentation are present in newborns—usually on the buttocks or sacrum where pigment may simulate a bruise and even be misdiagnosed as a sign of non-accidental injury (see ➔ Fig. 31.10, p. 615).
- Mongolian spots are commonest in black or Asian infants and gradually disappear with age.
- Melanocytes are deep within the dermis, and therefore pigment appears bluish, rather than brown.

Blue naevi

- Benign acquired small slate-blue to blue-black macules or papules that present most often on the dorsum of the hand or on the scalp.
- Melanocytes are present in the dermis, and tumours appear blue, rather than brown.

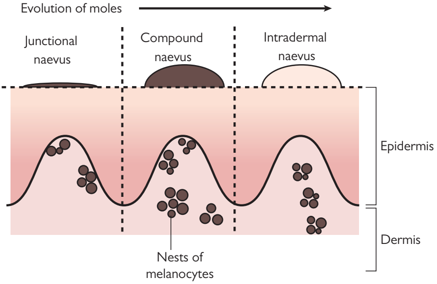

Fig. 17.2 Evolution of melanocytic naevi ('moles').

Fig. 17.3 Junctional melanocytic naevus. Flat and evenly pigmented. Reproduced with permission from Lewis-Jones, Sue (ed). Paediatric Dermatology, Oxford University Press 2010.

Fig. 17.4 Compound melanocytic naevus, with a central raised component and a flat rim of pigmentation. Reproduced with permission from Lewis-Jones, Sue (ed). Paediatric Dermatology, Oxford University Press 2010.

Fig. 17.5 Halo naevus. This mole has developed a rim of depigmentation. The mole will slowly disappear. Reproduced with permission from Lewis-Jones, Sue (ed). Paediatric Dermatology, Oxford University Press 2010.

Avoiding skin cancer

Excess UVR is the main cause of skin cancer (see Boxes 17.1 and 17.2). Encourage people to enjoy outdoor activities (some UVB maintains the levels of vitamin D), but those who have signs of sun damage, such as solar keratoses or skin cancers, or are taking immunosuppressive drugs should limit sun exposure (be sun-smart) and avoid sunburn. Spring and winter sunshine in the UK has insufficient UVB to promote much synthesis of vitamin D, but adequate levels of vitamin D will be maintained for most people through everyday exposure to the sun in the summer months, e.g. if the arms, legs, and/or face receive about 15min of sun exposure three times a week.

For more about the sun and skin, including information about vitamin D, photoprotection, and sunscreens, see ➔ Chapter 16, pp. 338–9.

A tan signals that the skin has been trying to protect itself from damage, while sunburn is foolish, unnecessary, and painful. Practical sun safety tips have been provided in the UK Shunburn campaign and Australian 'Slip–Slop–Slap–Seek–Slide' (see Boxes 17.3 and 17.4).

For signs of chronic sun damage (photoageing), see ➔ Box 3.1, p. 43.

Box 17.1 Ultraviolet radiation and skin cancer

- UVB is more photocarcinogenic than UVA, but both contribute to the pathogenesis of skin cancer by damaging DNA.
- Photodamaged DNA is usually repaired by nucleotide excision repair enzymes. Repair declines with age. Cumulative mutations lead to skin cancer by affecting the function of genes that regulate cell proliferation (oncogenes, e.g. *RAS*; tumour suppressor genes, e.g. *p53*) (see ➔ pp. 336–7).
- UV-induced immunosuppression may predispose to skin cancer by interfering with the recognition and elimination of abnormal cells.
- Fair skin types I and II with red or blond hair, and severe sunburn in childhood or adolescence, increase the risk of developing skin cancer.
- Chronic cumulative UVR, including PUVA (>200 treatments) and sunbeds, also predispose to skin cancer.
- Chronic photosensitivity (e.g. drug-induced) with accelerated photoageing increases the risk of skin cancer. Interactions between UVR and immunosuppressant drugs, e.g. azathioprine, photosensitize the skin to UVA, resulting in carcinogenesis.

Further reading

British Association of Dermatologists. *Sun awareness leaflets and posters*. Available at: http://www.bad.org.uk/for-the-public/sun-awareness-campaign/sun-awareness-leaflets-and-posters.

NHS UK. *Consensus vitamin D position statement*. Available at: http://www.nhs.uk/livewell/summerhealth/documents/concensus_statement%20_vitd_dec_2010.pdf.

UK SunSmart campaigns. Available at: http://www.sunsmart.org.uk/.

van Lümig PP *et al. J Eur Acad Dermatol Venereol* 2015;**29**:752–60.

Box 17.2 Other risk factors for skin cancer

- BCC: previous BCC, X-irradiation, immunosuppression (organ transplant, see ➔ pp. 500–1), HIV, chronic lymphocytic leukaemia, see ➔ Box 26.9, p. 541), chronic arsenic exposure, genetic conditions (Gorlin syndrome, see ➔ p. 542; albinism; XP).
- SCC: Bowen disease or solar keratosis, previous SCC, pipe smoker (lip), X-irradiation, thermal burns, immunosuppression (organ/bone marrow transplants, see ➔ pp. 500–1), HIV, chronic lymphocytic leukaemia, HPV infection, chronic wounds (sinuses, scars, or ulcers), tar, chronic arsenic exposure, *MC1R* gene variants (impaired UV protection responses), genetic conditions (albinism, XP), drugs (e.g. BRAF inhibitors), voriconazole.
- Melanoma: previous melanoma, >50 melanocytic naevi or many atypical (dysplastic) melanocytic naevi; giant 'bathing trunk' >20cm in diameter congenital melanocytic naevus (melanoma may develop early in childhood in large congenital lesions), *MC1R* gene variants (impaired UV protection responses), previous BCC/SCC, immunosuppression, family history of melanoma (mutations in *CDKN2A* and *CDK4* genes), genetic conditions (albinism, XP), drugs (e.g. BRAF inhibitors), other diseases (breast cancer, Parkinson disease, IBD).

(For XP, see ➔ pp. 336–7.)

Box 17.3 Slip–Slop–Slap–Seek–Slide (2007)

- Slip on a shirt: cover exposed skin with loose-fitting cotton clothing that has a tight weave.
- Slop on sunscreen cream (SPF30 or more) liberally and frequently to the face and other exposed skin, but creams are no substitute for hats, clothing, and a sensible lifestyle.
- Slap on a hat with a broad brim and tight weave (check how much light comes through the hat by holding it up to the sun). The neck should be protected, as well as the face and ears. Baseball caps do not provide adequate sun protection.
- Seek shade.
- Slide on wrap-around sunglasses to prevent sun damage.

Box 17.4 Shunburn Campaign 2014

- Cover up the skin—long-sleeved shirt, collar, long shorts/sarong.
- Slap on the sunscreens—use at least SPF30 generously.
- Wear a hat or cap; slip on the shades.
- Chill out in the shade. Reach for shade between 11 a.m. and 3 p.m.

Teenage Cancer Trust. *Shunburn: stay safe in the sun*. Available at: https://www.teenagecancertrust.org/what-we-do/education/shunburn/.

Premalignant epidermal lesions

Solar (actinic) keratosis

Solar keratoses are premalignant lesions (the epidermal basal cell layer is dysplastic) that indicate chronic sun damage and are a marker of the risk of developing skin cancer (see Fig. 17.6). About 20% regress spontaneously. Progression of single lesions to SCC is uncommon (<1 in 1000 per annum). Areas of coalescing solar keratoses on sun-damaged skin, i.e. 'actinic field change', have a higher risk of progression to SCC, particularly in the context of immunosuppression.

What should I look for?

- Asymptomatic or slightly itchy, brownish red, discrete scaly/keratotic patches on sun-exposed skin: bald scalp, face, helix of the ear, dorsum of the hands, and forearms (see Fig. 17.6).
- Some may be raised and warty: hypertrophic solar keratosis.
- Pain, induration, inflammation, or ulceration (suggests progression to SCC). Signs of chronic sun damage, including skin tumours: BCC, SCC, malignant melanoma.

What should I do?

- Advise patients to monitor their skin for development of skin cancers and to protect their skin from the sun (see ➔ p. 346).
- Treatment, e.g. cryosurgery, is only required if keratoses are troublesome. Topical treatments are effective for confluent areas of field change, e.g. topical diclofenac, fluorouracil, imiquimod, ingenol mebutate, or photodynamic therapy.

Bowen disease (SCC *in situ*)

Characterized by full-thickness epidermal dysplasia. Peak incidence in seventh decade. Only about 3–5% of plaques progress to SCC—but the risk of metastasis from SCC is increased. Aetiologies: chronic sun exposure (including sunbeds), radiation, oncogenic HPV (HPV-16 associated with anogenital, palmoplantar, and periungual SCC *in situ*), immunosuppression, and arsenic (lesions in sun-protected areas). Plaques are asymptomatic and enlarge slowly over years. Often misdiagnosed as 'discoid eczema' or 'ringworm' but fails to respond to treatment for these conditions.

What should I look for?

- A well-defined erythematous scaly or warty plaque (see Fig. 17.7).
- A flat edge: stretch the skin gently, and inspect the edge closely. A raised thready border suggests a superficial BCC, rather than Bowen disease.
- Dermoscopy shows glomerular vessels (focal nests of tortuous vessels resembling vessels in the renal glomerulus) with scaling.
- Nodularity or ulceration: this may indicate progression to SCC.

What should I do?

- Take a skin scrape to exclude a fungal infection, if indicated.
- Take a skin biopsy to confirm the diagnosis.

Treatment options include surgical removal (shave excision), topical fluorouracil, topical imiquimod, or photodynamic therapy. Patients should protect their skin from the sun (see ➔ p. 346).

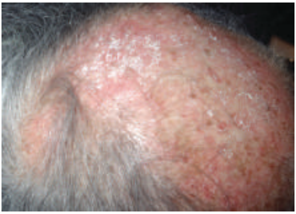

Fig. 17.6 Actinic field change: numerous solar keratoses arising in sun-damaged skin, with background solar lentigines (pigmented macules) and erythema.

Fig. 17.7 Bowen disease: an asymptomatic scaly plaque on the leg. The differential includes tinea, eczema, and a superficial basal cell carcinoma.

Further reading

de Berker D *et al*. *Br J Dermatol* 2007;**56**:222–30. Available at: http://www.bad.org.uk/library-media/documents/Actinic_keratoses_guidelines_2007.pdf.

Morton CA *et al*. *Br J Dermatol* 2014;**170**:245–60. Available at: http://www.bad.org.uk/library-media/documents/SCC_in_situ_guidelines_2014.pdf.

Keratinocyte skin cancers (1)

Basal cell carcinoma (rodent ulcer, BCC)

BCC is the commonest skin cancer. BCCs grow slowly, virtually never metastasize, and are locally invasive. Some variants can be locally destructive. BCCs present predominantly in middle-aged Caucasians. (Also see ➔ Gorlin syndrome, p. 542.)

What should I look for?

- *Nodulocystic*: dome-shaped, pearly papule with telangiectasia coursing over the surface, more easily identified with dermoscopy, described as 'arborizing' vessels which are in focus when magnified. (Common on the face and neck.) Stretch the skin gently to accentuate the translucent appearance and borders. Older tumours may present as ulcerated nodules (rodent ulcers) with smooth rolled pearly edges and telangiectasia across the surface (see Fig. 17.8). Remove the loose crust, so you can see the granulating fleshy base.
- *Superficial*: one or more scaly erythematous plaques (most often on the trunk) with well-defined raised pearly edges. Stretch the skin gently to make the pearly border of superficial BCC more obvious, and differentiate from Bowen disease or fungal infection.
- *Pigmented*: any BCC may contain pigment flecks (seen as ovoid globules under dermoscopy), but heavily pigmented tumours can simulate melanoma. A pearly appearance provides clue to the correct diagnosis.
- *Morphoeic/infiltrative*: waxy, indurated plaque that may resemble a scar. The border may be difficult to define, even when the skin is stretched.

Squamous cell carcinoma

Cutaneous SCCs are the second commonest skin cancer (25% of all keratinocyte cancers). They usually present in elderly patients on sun-exposed sites and are three times commoner in men. Prognosis depends on the potential for metastasis, which is influenced by the anatomical site, pathological features of the tumour (rate of growth, depth, degree of differentiation), and host immune response (see Boxes 17.5 and 17.7).

What should I look for?

- A keratotic nodule or an ulcerated nodule with a granulating base and a rolled undermined border (see Fig. 17.9).
- Induration or ulceration of a solar keratosis or Bowen disease.
- Regional lymphadenopathy.

Keratoacanthoma

Perhaps best regarded as a well-differentiated SCC. These grow rapidly (<6 weeks), persist for 2–3 months, and then involute over 4–6 months, leaving a depressed scar. They are usually removed surgically for histological examination.

What should I look for?

- A history of rapid growth over <6 weeks.
- A pinkish symmetrical cup-shaped nodule, 1–2cm in diameter, on sun-damaged hair-bearing skin, usually the face or neck.
- Smooth rolled edges.
- A central keratin plug.

Fig. 17.8 Nodulocystic basal cell carcinoma with a pearly appearance.

Fig. 17.9 Squamous cell carcinoma arising on severely sun-damaged skin, which is demonstrated by hypo- and hyperpigmentation.

Box 17.5 Prognostic 'high-risk' factors for squamous cell carcinoma

- High-risk anatomical sites:
  - Ear > nose = cutaneous lip = eyelid = scalp.
- Clinical diameter >20mm.
- Tumour depth >4mm (very high risk if >6mm).
- Histological features:
  - Perineural invasion.
  - Lymphovascular invasion.
  - Poorly differentiated.
  - Desmoplastic, adenosquamous, spindle cell, acantholytic, follicular subtypes.
- Squamous cell carcinoma arising within a site of:
  - Skin trauma, e.g. burns, scar tissue, or a radiotherapy field.
  - Pre-existing skin disease, e.g. venous leg ulceration or Bowen disease.
- Immunosuppression (see Box 17.7).
  - Increases the risk of multiple SCCs.
  - Increased the risk of metastases.

Keratinocyte skin cancers (2)

Treatment options for keratinocyte cancers

See Box 17.6.

Risk factors for skin cancer

see ➔ p. 346, and Box 17.7.

Box 17.6 Treatment options for keratinocyte cancers

Basal cell carcinoma

Choice of treatment depends on the age of the patient and the history, as well as the type and site of tumour, but options may include:

- Surgical excision (the gold standard); curettage and cautery.
- Mohs micrographic surgery: treatment of choice for locally recurrent skin cancers, cancers in sites where it is important to preserve tissue, e.g. adjacent to the eye/T-zone of the face, or cancers with ill-defined margins, e.g. morphoeic/infiltrative BCC.
- Topical imiquimod or topical fluorouracil: for superficial BCCs.
- Cryosurgery.
- Photodynamic therapy (PDT) (superficial BCC). Daylight PDT also demonstrated to be efficacious.
- Hedgehog signalling pathway inhibitors (Smoothened receptor inhibitors—vismodegib) (see ➔ p. 385).
  - Used to treat recurrent, locally invasive, or metastatic BCC.
  - Used to reduce the development of BCCs in Gorlin syndrome.
- Radiotherapy: second-line treatment for tumours not amenable to surgical intervention or as adjuvant therapy for keratinocyte tumours with perineural invasion.

Squamous cell carcinoma

- Surgical excision margins should aim to achieve complete histological clearance. For low-risk tumours, a clinical peripheral margin of 4mm is advised, and, in some situations, curettage and cautery is appropriate. For high-risk tumours (see Box 17.5), a peripheral margin of 6mm is advised.
- Mohs micrographic surgery can be considered in selected patients with high-risk tumours or for tumours which might be at a critical anatomical site (as described for BCC).
- 1° radiotherapy can be used for tumours which are surgically challenging. Adjuvant radiotherapy should be considered for high-risk tumours or those with close or involved margins.

Box 17.7 Immunosuppression and skin cancer

- Immunosuppressed individuals are at significantly increased risk of developing skin cancers.
- This group includes, but is not limited to, solid organ transplant recipients, patients with haematological malignancies, e.g. chronic lymphocytic leukaemia and non-Hodgkin lymphoma, bone marrow transplant recipients, HIV-positive/AIDS patients, and individuals treated with prolonged immunosuppressant therapy, in particular azathioprine (which is a direct UV carcinogen), for any inflammatory condition, e.g. IBD.
- Skin cancer is the most frequent malignancy in organ transplant recipients and contributes to significant morbidity.
  - Non-melanoma skin cancers (SCC > BCC and other rare cancers, e.g. Merkel cell carcinoma, sebaceous tumours) are the commonest. Tumours tend to occur earlier, are multiple, behave more aggressively, and appear to have increased malignant potential. Early identification is essential, and these patients are best managed in a dedicated transplant dermatology clinic.
  - Melanoma is also increased in frequency, but prognosis only appears to be worse for thicker tumours.

Further reading

Cancer Research UK: *Skin cancer (non melanoma)*. Available at: http://www.cancerresearchuk.org/about-cancer/type/skin-cancer/.

Motley R et al. *Multi-professional guidelines for the management of the patient with primary cutaneous squamous cell carcinoma*. 2009. Available at: http://www.bad.org.uk/library-media/documents/SCC_2009.pdf.

Scottish Intercollegiate Guidelines Network. *SIGN 140: Management of primary cutaneous squamous cell carcinoma*. 2014. Available at: http://www.sign.ac.uk/pdf/QRG140.pdf.

Telfer NR *et al*. *Br J Dermatol* 2008;**159**:35–48. Available at: http://www.bad.org.uk/library-media/documents/BCC_2008.pdf.

Malignant melanoma

Malignant melanoma accounts for 5% of skin cancers, but incidence in the UK has quadrupled since the 1970s (>13 300 new cases in 2013). Melanoma is commoner in men than women, and almost 1/3 occur in people aged <50 (melanoma is the commonest cancer in young adults aged 15–34 years old). Superficial spreading melanoma (SSM) is the commonest type in Caucasians (see Box 17.8).

Most melanomas arise *de novo*; only 30–50% arise in pre-existing melanocytic naevi (usually SSM subtype). For risk factors, see ➔ pp. 346–7. Melanoma accounts for 75% of deaths associated with skin cancer.

What should I look for?

- Moles that are irregular in colour, outline, or shape (see Figs. 17.10, 17.11, and 17.12,).
- Use the ABCDE criteria to identify suspicious pigmented lesions:
  - *A*symmetry in outline.
  - *B*order irregularity or blurring, sometimes with notching.
  - *C*olour variation (>3 colours) with shades of black, brown, and pink.
  - *D*iameter >6mm (cannot be covered by the end of a pencil).
  - *E*volution: tumour that is changing in size, elevation, and/or colour.
- Moles that stand out or look different from the others, i.e. 'ugly duckling sign'. Better than ABCDE criteria for identifying nodular melanoma.
- New or changing longitudinal pigmented streaks in nails, especially if associated with damage to the nail.
- Pigment extending from the nail bed onto the skin of the nail fold (Hutchinson sign), suggesting melanoma, not subungual haematoma.
- Moles that are symptomatic (itching, swollen, tender), but, more often than not, these turn out to be irritated benign melanocytic naevi, rather than melanoma.
- Regional lymphadenopathy.

What should I do?

- Obtain a history of the tumour from the patient (see ➔ p. 26).
- Document risk factors (see Boxes 17.1 and 17.2).
- Record the findings, including the site and size of the lesion.
- Photography is helpful.
- Dermoscopy is a useful tool to monitor multiple atypical naevi.
- Refer any patient in whom you are concerned about a pigmented lesion for an urgent opinion. Prognosis is determined primarily by the Breslow thickness, and mortality increases significantly with increasing tumour thickness (see Tables 17.1 and 17.2).
- Dermoscopy, performed by an expert, may help with diagnosis (see ➔ p. 640).
- Excision is the treatment of choice for suspicious pigmented lesions.

Box 17.8 Types of cutaneous melanoma

- SSM (80–85%) is the commonest melanoma in Caucasians. Presents as a slowly enlarging, slightly raised, pigmented plaque with irregularity in colour and border. Eventually, a nodule appears within the plaque, indicating deep invasion. Most often on the trunk of men or legs of women (see Fig. 17.10).
- Nodular melanoma (10–15%) grows rapidly, invading deeply from the outset. If amelanotic (often some pigment can be seen using dermoscopy), may be misdiagnosed as a vascular tumour, e.g. pyogenic granuloma or SCC (see Fig. 17.11).
- Lentigo maligna melanoma (5%) arises within a slow-growing melanoma *in situ* known as lentigo maligna (Hutchinson's freckle). This type of melanoma is associated with chronic sun damage. It usually presents on the head or neck of an elderly patient (see Fig. 17.12).
- Acral lentiginous melanoma (2–8%): palms, soles, beneath the nail. Commonest in Chinese and Japanese. Commonest subtype in Fitzpatrick skin types IV/V.

Table 17.1 Prognosis of malignant melanoma

American Joint Committee on Cancer (AJCC) stage	5-year survival (overall >90%)
Stage 0 (melanoma *in situ*)	100%
Stage I (Breslow thickness <1mm)	92–97%
Stage II	53–81%
Stage III	40–78%
Stage IV (metastatic disease)	15–20%

Further reading

Cancer Research UK. Melanoma skin cancer. Available at: http://www.cancerresearchuk.org/about-cancer/type/melanoma/.

Marsden JR *et al. Br J Dermatol* 2010;**163**:238–56. Available at: http://www.bad.org.uk/library-media/documents/Melanoma_2010.pdf.

Management of melanoma

Management of cutaneous melanoma

- Surgery is the mainstay of treatment for 1° melanoma. Excision margin is determined according to the Breslow thickness (see Table 17.2).
- Sentinel lymph node biopsy can be used as a staging procedure but has no proven therapeutic value.
- Routine imaging is not warranted for melanoma. Extent of metastatic spread is assessed in patients with American Joint Committee on Cancer (AJCC) stage III or above, using whole-body CT or positron emission tomography (PET)-CT scans.
- Better understanding of molecular signalling and immunological responses in melanoma has led to targeted therapies and immunotherapies. Selection for targeted therapies is determined by molecular testing, e.g. BRAF V600E mutations, c-KIT mutations.

Table 17.2 Breslow thickness

Breslow thickness (mm)	Recommended lateral surgical excision margins (cm)
In situ	0.5
<1	1
1.01–2	1–2
2.01–4	2–3
>4	3

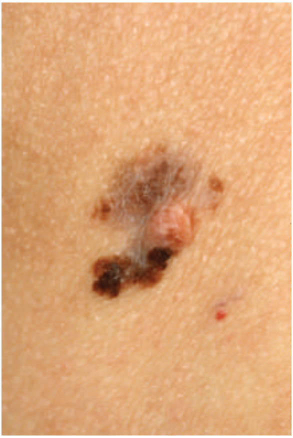

Fig. 17.10 Superficial spreading malignant melanoma with asymmetry in colour and outline. This melanoma did arise in a melanocytic naevus.

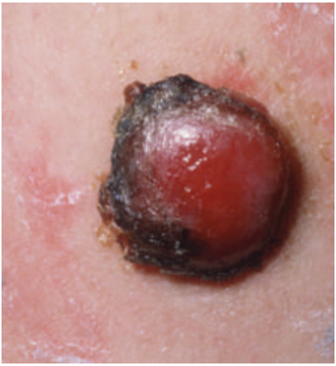

Fig. 17.11 Amelanotic nodular melanoma with a rim of pigmentation simulating haemorrhage. The tumour was misdiagnosed as a pyogenic granuloma.

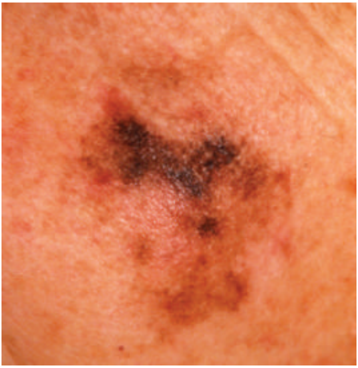

Fig. 17.12 Slow-growing lentigo maligna (melanoma *in situ*) on the cheek of an 80-year-old woman.

Cutaneous T-cell lymphoma

Clinical features, in combination with histological, immunophenotypic, and molecular studies, have helped to delineate subtypes and provide prognostic criteria in this heterogeneous group that includes:

- MF and Sézary syndrome.
- CD30-positive CTCLs, e.g. lymphomatoid papulosis—good prognosis. Anti-CD30 monoclonal antibodies (brentuximab) now available as a treatment option.
- CD30-negative CTCLs: poor prognosis.
- Adult T-cell leukaemia/lymphoma (ATLL).

Mycosis fungoides

MF, the commonest CTCL, is a malignancy of effector memory T-cells that responds well to skin-directed therapies. The aetiology is uncertain but may involve chronic antigenic stimulation (possibly viral) and mutations in oncogenes or DNA repair genes.

Disease is indolent, and the diagnosis often delayed (median age 55–60 years).

Prognosis is worse in patients aged >60 years and in late-stage disease.

MF progresses through three stages: patches, plaques, and tumours—which may overlap. MF variants include folliculotrophic, pagetoid reticulosis, and granulomatous slack skin. Early-stage disease (excellent prognosis) is managed conservatively. Options include emollients, potent topical corticosteroids, UVB, PUVA, topical nitrogen mustard, and bexarotene (retinoid that activates retinoid X receptors, causing T-cell apoptosis). Radiotherapy is effective for localized thick plaques or tumours. MF is relatively chemoresistant, and treatment of advanced disease is unsatisfactory. Sepsis is a common cause of death.

What should I look for?

- Symptoms: variable itch, ranging from minimal to intense.
- Early MF: one or more persistent and/or enlarging well-defined erythematous patches (flat) of variable size and shape with fine scale (compare with thick scale of psoriasis), often on covered sites—buttocks and breasts (see Fig. 17.13). Sparing of the elbows, knees, scalp, and nails in early MF (often involved in psoriasis). May be hypo- or hyperpigmented.
- Poikiloderma: mottled pigmentation, telangiectasia, and atrophy. Skin wrinkles like tissue paper, if pinched—not seen in psoriasis or eczema, unless ultrapotent topical steroids have been used for prolonged periods.
- First presentation of MF can be an erythrodermic emergency.
- Late MF: reddish brown, sharply demarcated, infiltrated plaques (elevated) that coalesce into annular or serpiginous shapes (see Fig. 17.13).
- Hair-bearing skin: follicular plugging and alopecia (hair is retained in psoriasis and eczema).
- Ulcerating mushroom-shaped tumours (late disease).
- Lymphadenopathy (may be 2° to inflammation in the skin and not malignant), enlarged liver, and/or spleen (late disease).

What should I do?

- Document the % of BSA involved and types of skin lesions.
- Exclude tinea (ringworm) by taking skin scrapes for mycology.

- Take two large elliptical skin biopsies for histology, immunophenotyping, and molecular studies, looking for T-cell receptor gene rearrangement. Repeated biopsies may be required, before the diagnosis is established.
- Check FBC, LFT, lactate dehydrogenase (LDH), and renal function.

Sézary syndrome

This form of CTCL accounts for about 5% of new cases. It is caused by malignant proliferation of central memory T-cells. Disease is characterized by erythroderma (see ➔ p. 108 and Fig. 5.2), lymphadenopathy, and malignant circulating $CD4^+$ T-cells (Sézary cells) forming >5% of the total lymphocyte count. The CD4:CD8 ratio is high (≥10). As disease progresses, the malignant clone of cells expands, and the normal CD4 (T-helper) and CD8 (T-suppressor) populations decrease.

Prognosis is poor (median survival 32 months from diagnosis), and patients often die from infection as a result of a failing immune system. Treatment is unsatisfactory; PUVA is poorly tolerated. Options include extracorporeal photophoresis, multi-agent chemotherapy, immunotherapy, and oral bexarotene, but prospective clinical trials are needed.

What should I look for?

- Generalized itching, often severe.
- Systemic symptoms such as fever, weight loss, and/or malaise.
- Erythroderma (generalized redness), with or without scale.
- Thickened erythematous skin on the face, with ectropion, thickening of skin on the palms and soles (keratoderma), hair loss, and nail dystrophy.
- Widespread lymphadenopathy.

What should I do?

- Carry out a full clinical examination.
- Check FBC, blood film, and count of Sézary cells (T-cells with hyperconvoluted cerebriform nuclei). Some circulating Sézary cells may be found in benign inflammatory skin conditions.
- Take an elliptical skin biopsy for histology.
- Look for T-cell monoclonal expansion in peripheral blood and skin by immunophenotyping and with molecular studies of T-cell receptor.
- Check LFT, LDH (for disease monitoring), and renal function.
- Check human T-lymphotropic virus (HTLV)-1 serology.
- Biopsy an enlarged lymph node (removal preferable to needle biopsy).
- If indicated, stage with chest X-ray (CXR), CT scan (chest, abdomen, pelvis) ± PET-CT scan, and bone marrow aspirate.

Adult T-cell leukaemia/lymphoma

This $CD4^+$ lymphoproliferative disorder, associated with infection with HTLV-1, is commonest in Japan and the Caribbean. Prognosis is poor, and opportunistic infection common.

What should I look for?

- Skin infiltration by ATLL cells leads to patches, plaques, papules, or tumours. Rarely erythroderma or purpuric rashes.
- Non-specific cutaneous inflammation (dermatitis) and infections, e.g. dermatophyte (tinea), *Candida*.
- Lymphadenopathy, hepatosplenomegaly.
- Raised LDH and hypercalcaemia.

Other malignancies and skin

Merkel cell carcinoma

This rare aggressive neuroendocrine skin cancer is associated with pathogenic factors such as UV exposure, immunosuppression, and/or Merkel cell polyomavirus. It presents as a rapidly growing bluish red or skin-coloured nodule, usually on a sun-exposed site, and spreads quickly to the lymph nodes, lungs, liver, or bone.

Dermatofibrosarcoma protuberans

A slow-growing, locally recurrent cutaneous soft tissue sarcoma caused by a chromosomal translocation t(17;22)(q22;q13) which results in a *COL1A1–PDGFB* fusion gene. Presents as an asymptomatic 'scar' or a violaceous, reddish brown, or skin-coloured nodule or plaque, which feels rubbery or firm. Has an infiltrative pattern, often with considerable subclinical and asymmetrical extension. Treatment: surgical excision (1–3cm peripheral margins) or Mohs micrographic surgery. Rarely metastasizes (<5%).

Paget disease

Paget disease is caused by direct extension of a mammary intraductal adenocarcinoma into the skin of the nipple or areola.

What should I look for?

- A weeping, erythematous, crusted eruption affecting one nipple.
- Is there a past history of atopic eczema? This is common on the nipple and sometimes is unilateral.
- Check the other nipple. Bilateral changes are much more likely to be eczema (atopic or contact) than cancer.
- Check for an associated breast lump.

What should I do?

- Biopsy any 'eczematous' nipple rash that persists despite potent topical corticosteroid ointments applied bd for 2 weeks.

Metastatic disease

Skin metastasis may be the first sign of an internal malignancy. The skin may be infiltrated by direct invasion of tumour cells or by spread from lymphatics or blood vessels. The tumour may also be implanted into surgical scars. Cancers most often associated with cutaneous metastasis are:

- Breast: skin of the chest, scalp, (rarely eyelid).
- Stomach, colon: skin of the abdominal wall, especially periumbilical.
- Lung: skin of the chest, scalp.
- Genitourinary system (uterus, ovary, kidney, bladder): skin of the scalp, lower abdomen, external genitalia.

What should I look for?

- Firm intradermal or subcutaneous nodules of varying colour.
- Scalp metastasis may cause a focal scarring alopecia.
- Cutaneous lymphatic invasion initially causes thickening and fibrosis and later firm pink papules or nodules.

- Carcinoma erysipeloides—most often associated with metastatic breast cancer. Lymphatic invasion produces a tender erythematous oedematous plaque, often initially misdiagnosed and treated as an infection (erysipelas or cellulitis), but antibiotics are ineffective.
- Extramammary Paget disease (EMPD)—a slow-growing, itchy, erythematous plaque in anogenital skin, especially the vulva. Biopsy shows mucin-containing tumour cells within the epidermis. EMPD is usually associated with an underlying adnexal carcinoma, often of apocrine origin, which may be difficult to localize, but 10–15% of patients have cancer of the rectum, prostate, bladder, cervix, or urethra.

Leukaemia cutis

Leukaemia cutis is rare but is seen most often with myeloid leukaemias, e.g. acute monocytic or myelomonocytic leukaemias. Cutaneous deposits may precede the transformation of myelodysplastic syndrome to leukaemia. It usually indicates a poor prognosis.

What should I look for?

- Firm purplish plaques or nodules.
- Infiltrates in scars or at injection sites.

Primary cutaneous B-cell lymphomas

Most B-cell lymphomas affecting the skin are systemic.

- Marginal zone lymphoma (mucosa-associated lymphoid tissue, MALT).
  - Dermal tumours in ♂ affecting the trunk and upper limbs.
  - Indolent and prone to recur.
  - Test for *Borrelia burgdorferi*.
  - Radiotherapy is the treatment of choice.
- Follicular lymphomas:
  - Multiple primaries, prominent plaques affect scalp and upper trunk.
  - Treat with radiotherapy or rituximab.
  - 95–100% 5-year survival.
- Diffuse large B-cell lymphoma:
  - Leg-type seen in older ♀.
  - Large, rapidly growing, bluish red tumours.
  - 50% 5-year survival.
  - Treatment is with radiotherapy/CHOP–rituximab.

Kaposi sarcoma (KS)

KS is a neoplasm caused by proliferation of the lymphatic endothelium, associated with HHV-8 infection of endothelial cells. May behave as a benign reactive process or pursue an aggressive life-threatening course (see Box 17.9). 2° skin infections and malignancies (lymphoma) may complicate KS. Pathogenesis is multifactorial. Chronic lymphoedema (see ➔ pp. 316–17) may predispose to KS, and HIV may exacerbate the pathogenicity of HHV-8. Circulating HHV-8-infected endothelial precursor cells localize to sites, such as the leg, where unregulated spindle cell proliferation is driven by interactions between activated $CD8^+$ T-cells, HHV-8-induced intracellular signalling pathways, products of oncogenes (viral and host), and inflammatory cytokines, including HHV-8-encoded IL-6 and, in AIDS patients, the product of the HIV-1 *tat* gene.

If immunodeficiency is corrected, e.g. by withdrawing or reducing immunosuppression, KS may regress. KS lesions also shrink in HIV-positive patients treated with highly active antiretroviral therapy (HAART). Localized disease may be controlled with intralesional vinblastine, topical alitretinoin (9-cis-retinoic acid) gel, topical imiquimod, radiotherapy, laser surgery, or cryosurgery. Systemic options for widespread disease include IFN-α and multi-agent chemotherapy, e.g. anthracyclines (doxorubicin) or paclitaxel. The role of angiogenesis inhibitors and other drugs needs further investigation.

What should I look for?

(See Box 17.10.)

- KS has a predilection for the face, ears, lower limb, including the soles of feet, genitalia, and oral mucosa.
- Bruise-like purple or brownish red macules and patches which may be subtle in early disease. Consider the possibility of KS in any immunosuppressed patient with atypical 'bruises'.
- Blue, purple, red, brown, or brownish red elliptical vascular papules lying in parallel to the natural lines of cleavage in the skin (Langer lines). These may involve surgical scars.
- Dark blue or purple vascular nodules or indurated hyperkeratotic plaques that may ulcerate (see Fig. 17.14).
- Oral mucosal lesions (vascular patches, papules, or plaques) on the hard palate or, less often, gums, sometimes with overlying candidiasis. Trauma causes bleeding.
- Lymphadenopathy.
- Lymphoedema affecting the face, genitalia, and/or leg, caused by vascular obstruction, lymphadenopathy, and local cytokines (commoner in AIDS-related KS). Swelling out of proportion to cutaneous signs.

What should I do?

- Take a skin biopsy to confirm the diagnosis.
- Exclude immunosuppression, e.g. HIV infection.
- Investigate the GIT and/or respiratory tract, if indicated.

Box 17.9 Kaposi sarcoma: clinicopathological subtypes

- Classic: men, aged 40–70 years old, Mediterranean/central to eastern European (Ashkenazi) Jewish heritage. Affects the legs and feet. Indolent and pursues a chronic course. Countries bordering the Mediterranean basin have higher HHV-8 infection rates, compared with the rest of Europe.
- Endemic (African: sub-Saharan Africa): affects younger population than classic KS. ♂ predominance less marked in childhood. Not linked to AIDS. More aggressive than classic KS. Extensive skin infiltration, especially on the lower limbs. High rates of HHV-8 infection in the general population in this part of Africa.
- Iatrogenic: particularly in transplant recipients taking calcineurin inhibitors, e.g. ciclosporin, which inhibit T-cell function. Patients either have pre-existing HHV-8 infection or, less often, acquire HHV-8 when transplanted with an HHV-8-infected organ. KS has been described in patients taking other immunosuppressive drugs and may regress if immunosuppression is withdrawn.
- AIDS-related: predominantly in homosexual/bisexual men. The trunk, arms, head, and neck involved more frequently than in classic KS. Pursues an aggressive course, involving the mucosa, lymph nodes, and viscera (GIT, respiratory tract). Poor prognosis—death from opportunistic infection, GI haemorrhage, cardiac tamponade, or pulmonary obstruction. Treatment: HAART and liposomal anthracycline chemotherapy. Occasionally, IRIS seen after treatment with HAART can trigger a flare of KS.

Box 17.10 Kaposi sarcoma: differential diagnosis

- Melanocytic tumours: melanocytic naevi, melanoma—1° or metastatic (see ➔ pp. 354–5).
- Pyogenic granuloma, haemangiomas, or other vascular tumour.
- Pseudo-KS: vascular nodules seen in venous stasis or in association with AV malformations. Histology resembles KS.
- Metastatic renal cell carcinoma (also tends to be vascular) or another metastasis.
- Bacillary angiomatosis (see ➔ p. 142). Unlike KS, vascular lesions are uncommon on the soles or in the mouth.
- Reactive angioendotheliomatosis: Rare. Occurs distal to iatrogenic AV fistulae on upper limbs used for haemodialysis (see ➔ Box 23.1, p. 493), on the lower limbs of patients with severe peripheral vascular disease, and in systemic diseases.

Further reading

Bhutani M *et al*. *Semin Oncol* 2015;**42**:223–46.

Fig. 17.13 Mycosis fungoides: fine scaly erythematous patch and a plaque with a serpiginous outline.

Fig. 17.14 Dark purple vascular nodule in AIDS-related Kaposi sarcoma.

Chapter 18

Cutaneous reactions to drugs

Contents

Introduction

Cutaneous adverse drug reactions (see Box 18.1) are common (2–3% of hospitalized patients), the signs extraordinarily diverse, time courses variable, and, worryingly, some are life-threatening. Alternative therapies, including vitamin or herbal supplements, may also cause problems. Reactions may be pharmacological or idiosyncratic (see Box 18.2). Predisposition is probably multifactorial, involving genetic and environmental factors (see Box 18.3).

Advances in pharmacogenetics may help to predict which patients are likely to benefit from, or react adversely to, certain drugs. For example, *HLA-B*1502* allele in south-eastern Asian patients has been correlated with carbamazepine-induced SJS–TEN; *HLA-A*3101* allele is associated with carbamazepine-induced reactions in Europeans, and genotyping for *HLA-B*5701*, which is linked to abacavir hypersensitivity, is used to screen HIV-positive patients prior to starting treatment.

Cutaneous adverse reactions may not be recognized, if the reaction simulates a condition such as eczema or if the medication has been taken for months without problems. The exanthem of a viral infection looks like a morbilliform drug reaction, but exanthems are more likely in children than in adults. To compound diagnostic difficulties, reactions may not settle as soon as the drug is withdrawn—some persist for months—and generally there is no definitive test to confirm the diagnosis of 'drug reaction'.

These life-threatening or severe cutaneous adverse drug reactions are discussed in more detail elsewhere:

- Anaphylaxis and angio-oedema (see ➔ pp. 104–5).
- Erythroderma (see ➔ p. 108).
- Stevens Johnson syndrome (see ➔ pp. 116–17).
- Toxic epidermal necrolysis (see ➔ pp. 116–17).
- DRESS (see ➔ pp. 122–3).
- AGEP (see ➔ p. 124).
- Anticoagulant-induced purpura fulminans and skin necrosis (see ➔ pp. 102–3).
- Vasculitis (see ➔ pp. 436–9).

Box 18.1 What is an adverse drug reaction?

'An appreciably harmful or unpleasant reaction, resulting from an intervention related to the use of a medicinal product, which predicts hazard from future administration and warrants prevention or specific treatment, or alteration of the dosage regimen or withdrawal of the product'.

Edwards IR and Aronson JK. *Lancet* 2000;**356**:1255–9.

Box 18.2 Types of reaction

- Type A (80% of reactions)—pharmacological: augmentation of the known pharmacological actions of the drug. Predictable, dose-dependent. Reversed by reducing the dose or withdrawing the drug.
- Type B (20% of reactions)—idiosyncratic: not predictable and may be life-threatening. Mechanisms include pharmaceutical variation in drug formulation, receptor abnormalities, abnormalities in drug metabolism, and immune-mediated.

Box 18.3 Factors that predispose to drug reactions

- Adults, rather than children.
- ♀ gender.
- Polypharmacy: 50% chance of an adverse interaction if five or more drugs are taken concurrently.
- Abnormal drug metabolism, e.g. reduced in renal failure or liver disease; toxic metabolites produced by metabolizing enzymes.
- Genetic polymorphisms, e.g. receptor abnormalities or enzyme deficiencies may cause pharmacological or idiosyncratic reactions.
- Previous hypersensitivity to a related drug.
- Host disease, e.g. SLE, herpesvirus infection, HIV infection. (In HIV infection, the incidence of cutaneous adverse drug reactions, most often morbilliform or urticarial rashes, increases as immune function deteriorates. Trimethoprim–sulfonamides and aminopenicillins cause most reactions.)

Further reading

Kaniwa N and Saito Y. *Ther Adv Drug Saf* 2013;**4**:246–53.
Karlin E and Phillips E. *Curr Allergy Asthma Rep* 2014;**14**:418.
McCormack M *et al. N Engl J Med* 2011;**364**:1134–43.
National Institute for Health and Care Excellence (2014). *Drug allergy: diagnosis and management*. Available at: www.nice.org.uk/cg183.
Wu K and Reynolds NJ. *Br J Dermatol* 2012;**166**:7–11.

Mechanisms of idiosyncratic cutaneous adverse drug reactions

For mechanisms of cutaneous drug reactions, see Table 18.1.

Table 18.1 Mechanisms of cutaneous adverse drug reactions

Type of reaction	Mechanism	Clinical signs
Type I immediate	IgE-mediated involving release of histamine from mast cells	Urticaria, angio-oedema, anaphylaxis. Minutes to hours after exposure to drug
Type II antibody-mediated	Cytotoxic IgG or IgM antibodies	ANCA-mediated vasculitis, drug-induced thrombocytopenic purpura. Variable time of onset
Type III immune complex-mediated	Drug-antibody immune complexes (IgM, IgG and IgA)	Vasculitis, serum sickness 1–3 weeks after exposure to drug
Type IV delayed-type	T-cell mediated	
Type IVa	Th1 cells release IFN-γ	Allergic contact dermatitis 2–7 days after drug is in contact with skin
Type IVb	Th2 cells release IL-5, IL-4, IL-13, and eotaxin (recruits eosinophils)	DRESS
Type IVc	Cytotoxic $CD4^+$ or $CD8^+$ T-cells, IFN-γ, and TNF induce keratinocyte lysis via perforin/ granzyme B, Fas/FasL, and granulysin	SJS, TEN, bullous drug reactions (Fig. 18.1)
Type IVd	Th17 cells release IL-17 and IL-22. Stimulated keratinocytes release IL-8 recruiting polymorphonuclear cells	AGEP
Pseudoallergic	Direct mast cell activation with histamine release or inhibition of kinin metabolism	Anaphylactoid reactions with angio-oedema
Autoimmune	ANAs	E.g. hydralazine-induced LE

Further reading

Harp J *et al*. *Semin Cutan Med Surg* 2014;**33**:17–27 (review).

Fig. 18.1 Bullous fixed drug eruption. The blister recurs in the same place each time the patient takes the drug (type IV reaction).

Fig. 18.2 Purplish lichen planus-like eruption caused by ramipril (type IV reaction).

Clinical approach

In any patient with a rash, ask yourself if the problem might be a manifestation of an adverse drug reaction and if this could be serious. Exclude other causes, such as viral infection, but retain a high index of suspicion.

What should I ask?

- What medicaments is the patient taking or applying to the skin (including herbal or traditional remedies, recreational drugs including kava, long-term medications such as the contraceptive pill, contrast medium, and depot injections)?
- Has the patient been exposed to the medication before?
- When medications were started (dates), and what is the temporal relation to the onset of the rash? You may have to ask the GP. The interval between the start of therapy and onset of reaction is rarely <1 week, unless the patient has had the drug before, or >1 month.
- Is the reaction less severe if the drug dose is reduced, or does it settle if the drug is withdrawn? Pharmacological adverse reactions, unlike idiosyncratic reactions, are usually dose-dependent.
- Any predisposing factors? (See Box 18.3.)
- What are the symptoms—malaise, itch, mucosal swelling, cutaneous pain or burning (may indicate TEN), arthralgia, or wheeze?

What should I do?

- Full examination, including temperature and blood pressure.
- Document the cutaneous signs—erythema, scale, macules, papules, urticarial wheals (no scale), pustules, vesicles or bullae, purpura—and decide on the predominant pattern of reaction.
- Examine mucosae (mouth, conjunctiva, and genitalia).
- Assess the likelihood of this being a serious reaction (see Box 18.4).
- Consider how the adverse reaction is likely to evolve. Your next steps will be determined by the severity (see Box 18.5).
- Document the likelihood of an adverse drug reaction (certain, likely, possible, or unlikely) and which drug(s) is/are implicated.
- Contact pharmacy if you are not sure about the likelihood of a particular drug causing an adverse reaction.
- Report any possible reaction to the regulatory agency—in the UK, return a completed yellow card to the Medicines and Healthcare Products Regulatory Agency (MHRA). Local, national, and international registries also exist for monitoring cutaneous drug reactions—consult the local dermatology department.
- Ensure that the medical records state that the drug should be avoided in future (it may be possible to continue essential treatment, despite a morbilliform reaction), and notify the GP.
- Explain the likely cause to the patient.
- Arrange further investigations when resolved (see Box 18.6).

Further reading

Barbaud A *et al. Br J Dermatol* 2013;**168**:555–62 (utility of drug patch tests).
Polak ME *et al. Br J Dermatol* 2012;**168**:539–49 (potential of *in vitro* diagnostic assays).

⚠ Box 18.4 Indications of severe cutaneous adverse drug reactions

(see ➔ p. 114.)

- General:
  - Fever >40°C, hypotension, lymphadenopathy.
  - Arthralgia, arthritis, dyspnoea, wheeze.
- Mucocutaneous:
  - Generalized scaly erythema (erythroderma).
  - Swollen face, swelling of tongue, urticaria (anaphylaxis).
  - Skin pain or burning, erosions, shearing stress detaches epidermis from dermis, bullae, mucosal erosions (SJS, TEN) (see ➔ pp. 116–17, and Fig. 5.5, p. 121).
  - Purpura (vasculitis or anticoagulant-induced).
- Laboratory results:
  - Eosinophilia >1000mm^3, lymphocytosis with abnormal lymphocytes, abnormal LFTs.

Box 18.5 Principles of management

- Withdraw non-essential medications, or reduce the dose (may help if the reaction is pharmacological). Drug withdrawal is not diagnostic immediately. Urticarial reactions settle in a few days, morbilliform rashes in 7–10 days. Other reactions may persist for >8 weeks.
- Manage life-threatening adverse reactions promptly, e.g. anaphylaxis, erythroderma, TEN, vasculitis (see ➔ p. 108, pp. 118–20).
- Exclude systemic involvement: FBC (may have eosinophilia or lymphocytosis), renal function, liver function, complement levels.
- Exclude infection with serological tests, if indicated, e.g. CMV, EBV, parvovirus B19 in patients with morbilliform rash; hepatitis B and C in urticaria; herpes simplex or mycoplasma in EM.
- Consider a skin biopsy (unhelpful in simple morbilliform reactions). The histological findings may not be conclusive.
- Emollients and soap substitutes relieve dryness (see ➔ p. 655).
- Moderately potent or potent topical corticosteroid ointments bd and sedating antihistamines reduce itch in morbilliform, eczematous, or lichenoid reactions (topical steroids are unhelpful in urticaria). (See Fig. 18.2.)
- Non-sedating antihistamines in urticaria (mediated by histamine).

Box 18.6 Further investigation of adverse drug reactions

- Once the reaction has resolved, patch testing with standardized dilutions of the suspected drug is helpful in allergic contact dermatitis and may sometimes be helpful in fixed drug eruptions, AGEP (see ➔ p. 124), and hypersensitivity syndrome (DRESS; see ➔ pp. 122–3).
- Unfortunately, skin testing (patch, prick, or intradermal) is rarely helpful in other forms of drug reaction.
- Oral re-challenge is usually inadvisable (except in fixed drug eruption).

Generally smooth erythematous drug reactions

Morbilliform (exanthematous)

- The commonest pattern of cutaneous drug reaction.
- Presents within 1–3 weeks of drug exposure.
- A symmetrical morbilliform rash (erythematous macules and papules of 2–10mm in diameter with a tendency to confluence) starts on the trunk. The rash spreads to the arms and legs and may become confluent.
- The rash may be itchy (not painful).
- The patient may have a slight (not high) temperature.
- Mucosae are normal.
- Settles with desquamation within 7–10 days of withdrawing the drug. Thick sheets of scale may detach from the palms or soles—do not confuse with the full-thickness loss of epidermis seen in TEN (see ➲ p. 116).
- Erythroderma is a rare complication (see ➲ p. 108).
- If the causative drug is essential, it may be possible to treat through a morbilliform reaction, but the patient may become erythrodermic.
- ⚠ Morbilliform reactions may resemble urticaria, but lesions are fixed, or (most often in children) a viral exanthem (take a thorough history).
- Morbilliform reactions may also precede a serious drug reaction such as TEN, hypersensitivity syndrome (DRESS), or serum sickness. Monitor for signs that suggest a serious reaction, e.g. facial swelling, mucosal involvement (see ➲ pp. 116–17 and Box 18.4).
- Common causes: ampicillin (particularly if the patient has glandular fever), sulfonamides, allopurinol, captopril, barbiturates, thiazides.

Urticaria and/or angio-oedema; anaphylactoid reactions

- IgE-mediated reactions may present within minutes of exposure, if previously sensitized. ACE inhibitors are the commonest cause of admission with angio-oedema.
- Urticaria: itchy erythematous wheals (no scale) that move around. Wheals fade within 24h to leave normal skin.
- Subcutaneous oedema (angio-oedema) affecting the lips, periorbital skin, tongue (when patient may have difficulty swallowing or breathing), external genitalia—an emergency (see ➲ p. 104).
- Common causes—penicillins, captopril, cephalosporins, thiazides, phenytoin, NSAIDs, ACE inhibitors (cause angio-oedema without urticaria and may not present until >4 weeks after the drug is started), aspirin, radiocontrast agents, opiates, quinine.
- Anaphylactoid reactions (not IgE-mediated) may present several weeks after the medication is started. Histamine is released directly from mast cells and basophils. Present as ***red man syndrome*** with flushing, erythema, and itching of the face and upper trunk, and sometimes angio-oedema, hypotension, dyspnoea, and chest pain. Caused by drugs such as vancomycin, ciprofloxacin, amphotericin, rifampicin, and teicoplanin.

⚠ Drug rash, eosinophilia, systemic symptoms (DRESS)

(see ➲ pp. 122–3.)

- DRESS is a life-threatening cutaneous drug reaction with systemic symptoms (high fever) that is frequently misdiagnosed as infection.
- DRESS may be associated with reactivation of HHV-6.
- The morbilliform rash is associated with facial oedema, simulating angio-oedema.
- Some patients progress to liver failure.
- Causes include allopurinol, anticonvulsants, minocycline, dapsone, sulfonamides, and other antibiotics.

Acute generalized exanthematous pustulosis (AGEP)

(see ➲ p. 124 and Fig. 5.6, p. 121.)

- Patients suddenly develop an uncomfortable oedematous erythema that burns or itches, and then becomes studded with tiny pustules.
- Caused by a wide range of drugs, including antibiotics such as penicillins, erythromycin, and tetracyclines.

Serum sickness

- Presents within 1–3 weeks of starting the medication.
- Erythema on the sides of fingers, toes, and hands progresses to a widespread morbilliform rash. Some patients also have urticaria.
- Malaise, fever, arthralgia, and arthritis are common.
- Causes include serum preparations and vaccines.

Interstitial granulomatous dermatitis

- Described in association with a range of drugs, including calcium channel blockers, β-blockers, lipid-lowering agents, ACE inhibitors, antihistamines, anticonvulsants, antidepressants, and TNF-α inhibitors. Also seen in association with connective tissue diseases, including RA, and lymphoproliferative disorders (see ➲ pp. 394).
- Presents months to years after initiation of treatment.
- Annular purplish red (violaceous) plaques on the arms, medial thighs, and flexures. May be mildly itchy.
- Histologically diffuse interstitial granulomatous infiltrate in the mid and deep dermis with lymphocytes, histiocytes, mucin deposition, and variable collagen necrosis.

Erythema nodosum

- Causes: oral contraceptive pill, sulfonamides, gold (but more often triggered by an infection (TB) or underlying systemic disease such as sarcoidosis or IBD) (see ➲ pp. 458–9).
- Tender erythematous subcutaneous nodules on the shins and sometimes forearms.
- Resolves in weeks, without loss of fat or scarring.
- NSAIDs relieve discomfort. Gentle compression with stockings that control swelling may speed resolution.

Also consider allergic contact dermatitis and SDRIFE

see ➲ p. 374.

More scaly erythematous drug reactions and pruritus

Allergic contact dermatitis (eczema)

- A localized allergic contact dermatitis is caused by topical preparations to which the patient has been sensitized by previous contact (see ➲ p. 214). The reaction appears within 48h of exposure.
- Skin is itchy, oedematous, erythematous, and scaly (may not be scaly if acute). It may blister if the reaction is severe (see ➲ Fig. 10.1, p. 216).
- Eczema may generalize if contact persists or if the patient takes a cross-reacting oral preparation.
- Consider allergic contact dermatitis in chronic leg ulcers, hand dermatitis, facial dermatitis, otitis externa, or pruritus ani. Causes include topical neomycin, benzocaine, incipients in topical medicaments, and topical hydrocortisone.
- Allergy to the constituents of subcutaneous heparin presents with well-demarcated infiltrated eczematous plaques at injection sites.
- Treat eczema with emollients and topical corticosteroid ointments.
- Investigate possible allergic contact dermatitis by patch testing.

SDRIFE: symmetrical drug-related intertriginous and flexural exanthema

- A distinctive erythematous flexural reaction to systemic drugs and iodinated contrast medium. Commonest cause: aminopenicillins.
- Latency hours to days. Occurs without prior exposure to the drug.
- Itchy, erythematous papules coalesce to produce a symmetrical, well-demarcated erythema on buttocks (baboon's bottom) and/or V-shaped erythema of the lower abdomen, groins, and thighs. Involves at least one other skinfold. May become bullous.
- No systemic symptoms or signs.
- Encompasses 'baboon syndrome', a distinct form of systemic contact dermatitis (sensitization by skin contact, and subsequently patient takes the agent by mouth).

Drug-induced photosensitivity

(see ➲ p. 326.)

- Drugs are the commonest cause of photosensitivity, more often phototoxicity than photoallergy.
- Look for an eczematous rash on exposed sites, sometimes with pigmentation (see ➲ Box 4.6, p. 73).
- Drugs may also cause photosensitivity by triggering PCT, pseudoporphyria (see ➲ p. 332, p. 334), or subacute cutaneous LE (SCLE) (see ➲ p. 402).

Lichenoid (lichen planus-like) reaction

- Very itchy, flat-topped, purplish red papules that may coalesce into erythematous plaques and become generalized. Often also an eczematous scaly component (see Fig. 18.2).
- Examine the buccal mucosa for reticulated white areas and erosions—found more often in LP than lichenoid drug reactions (see ➔ p. 224).
- Resolution is slow (months) after the drug is stopped.
- The papules fade to leave macular hyperpigmentation that persists for months, particularly in dark skin. Pigmentation tends to be more marked in lichenoid drug reactions than in LP.
- Very potent topical corticosteroid ointments may be required to relieve itch.
- Common causes: gold, antimalarials, penicillamine, β-blockers, thiazides, NSAIDs. Also reported with TNF-α antagonists.

⚠ Erythroderma (exfoliative dermatitis)

- Generalized scaly erythema—an emergency (see ➔ p. 108).
- Common causes: sulfonamides, carbamazepine, antimalarials, phenytoin, and gold.

Chronic eczematous eruptions

- Common in the elderly, but it may be difficult to identify the drug responsible.
- Calcium channel blockers (commonest cause).

Pruritus

- Itch without a 1° rash may be caused by a number of drugs, including statins (also dryness of the skin), ACE inhibitors, opiates, barbiturates, antidepressants, and oral retinoids (also dryness of the skin).
- Hydroxyethyl starch products (colloid volume expanders—in 2013, withdrawn in the UK because of risk of renal damage). Cause discrete attacks of severe intractable pruritus lasting for minutes up to an hour. Patients may have several episodes a day. Itch may persist for years.
- Signs are those of scratching sometimes superimposed on dry skin.
- Exclude other causes of itching, including urticaria (the rash is transient) (see ➔ p. 228), scabies (see ➔ p. 172), and renal, hepatic, thyroid, haematological, or neurological disease, including iron deficiency.
- Menthol 1% in aqueous cream is a soothing antipruritic. Sedating antihistamines may be helpful (ineffective in hydroxyethyl starch itch).
- Also see ➔ Eczematous reactions, p. 388.

Purpuric or pigmented drug reactions

Vasculitis (palpable purpura)

- Usually 1–3 weeks after starting drug, but interval may be longer.
- Palpable purpura (small-vessel cutaneous vasculitis) initially on legs but may become widespread. Purpuric papules may evolve into haemorrhagic blisters or purpuric plaques (see ➔ p. 440).
- Other signs may include urticaria, ulcers, and nodules.
- Vasculitis may affect the kidney, liver, GIT, and/or nervous system.
- Consider other causes of a small-vessel cutaneous vasculitis, including infection (see ➔ Box 20.3, p. 441).
- Common causes: penicillins and other antibiotics, allopurinol, phenytoin, thiazides, and thiouracils.
- Cocaine, adulterated by levamisole, may induce retiform purpura on the body and tender purpura of the ears, nose, cheeks, lips, and hard palate, sometimes with necrosis (vasculitis and/or occlusive vasculopathy).

Macular purpura: may be pigmented

- Aspirin, anticoagulants, calcium channel blockers, ACE inhibitors, analgesics.

Anticoagulant-induced purpura fulminans and skin necrosis

- Rare, but life-threatening, complication of treatment with warfarin.
- Risk greatest in obese ♀ patients with heterozygous deficiency of protein C or protein S. Warfarin inhibits protein C and protein S, inducing a hypercoagulable state.
- Three to 5 days after starting warfarin, painful, erythematous, indurated plaques develop on fatty areas—breasts, hips, and buttocks. The large, irregularly outlined plaques become haemorrhagic, bullous, and eventually necrotic. Biopsy shows microthrombi in capillaries, venules, and veins.
- ▶▶ Treatment: stop warfarin; administer vitamin K and/or fresh frozen plasma to restore levels of protein C/S. Use another anticoagulant, e.g. heparin.
- Heparin necrosis is rare. Presents 5–14 days after starting heparin. Erythema at injection sites (rarely distant sites) progresses to painful necrosis. Platelet aggregation is 2° to antibodies to heparin–platelet factor 4 complex. Platelets may fall. Stop heparin. Use non-heparin anticoagulant.

Cutaneous hyperpigmentation

- Mechanisms of hyperpigmentation include increased melanin synthesis and deposition of drug or drug metabolites in the skin.
- Photosensitivity may contribute to the colour change.
- Ask if the colour change was preceded by erythema or itching to differentiate drug-induced pigmentation from post-inflammatory hyperpigmentation, e.g. after dermatitis, lichenoid rashes, or fixed drug eruptions (see ➔ p. 378), particularly in individuals with dark skin.
- What is the colour and distribution of the change, e.g. is it predominantly in scars or on light-exposed skin?

- Increased melanin (brown pigmentation) may be caused by drugs such as cytotoxics, hydroxycarbamide (hydroxyurea), pegylated IFN (in chronic hepatitis C infection), ACTH, oral contraceptives (melasma on the face), and some antiretroviral agents (zidovudine and lamivudine).
- Flagellate brown hyperpigmentation is caused by bleomycin (see ➔ p. 383).
- Bluish grey pigmentation (may be worse on light-exposed skin) is caused by drugs such as minocycline (often in scars or on the shins) (see Fig. 18.3), phenothiazines, antimalarials, ezogabine (anti-epileptic), vandetanib, gold, and amiodarone (facial pigmentation with photosensitivity).
- Orange pigmentation is caused by the antimalarial mepacrine.
- Pink discoloration is caused by clofazimine.
- Facial hyperpigmentation caused by skin-lightening creams containing hydroquinone (acquired ochronosis).
- Resolution of pigmentation may be very slow (months or years) or pigment may persist, despite drug withdrawal. Photoprotection is often an important part of management.

Oral mucosal pigmentation

- Changes are often seen along the gingival margins but may develop on the lip, tongue, or palate.
- May be caused by drugs such as oral contraceptives (general darkening), minocycline (grey-blue), antimalarial drugs (grey-blue), phenothiazines (grey-blue), and ezogabine (blue-grey).
- Pigmented macules or patches may develop on the tongue in heroin addicts who inhale smoke.
- A black hairy tongue may be caused by drugs such as oral antibiotics (cephalosporins, chloramphenicol, clarithromycin, penicillins, sulfonamides), as well as corticosteroids and antidepressants.

Fig. 18.3 Greyish pigmentation after prolonged treatment with minocycline.

Further reading

Magliocca KP *et al*. *J Oral Maxillofac Surg* 2013;**71**:487–92 (cocaine abuse).
Thornsberry LA *et al*. *J Am Acad Dermatol* 2013;**69**:450–62 (hypercoagulable states).

Drug-induced blisters or mucocutaneous ulcers

Blisters may be widespread or localized. Oral manifestations may be erythematous, vesicular, erosive, or ulcerative and can involve the tongue, as well as the buccal mucosa or gums. Blistering eruptions, such as TEN or SJS, may cause severe oral ulceration (see ➔ p. 116).

⚠ Any persistent oral ulcer should be biopsied to exclude malignancy.

Fixed drug eruption

- One or more well-defined circular, erythematous plaques that usually blister and then resolve over 7–10 days, leaving a hyperpigmented macule. Itching is uncommon (see Fig. 18.1).
- May also affect the oral mucosa, and, in severe cases, mucosal lesions may ulcerate.
- Each time the patient is exposed to the drug, the problem recurs within 24h in exactly the same place or places. Recurrent herpes simplex infection is the main differential diagnosis, but this does not usually leave macular pigmentation.
- Take a skin biopsy during the reaction to confirm the diagnosis.
- Causes: tetracycline, sulfonamides, allopurinol, aspirin (ask the patient what analgesic is taken for menstrual pain or headaches), phenolphthalein. Also consider herbal medicines and other over-the-counter preparations that are ingested sporadically.
- Oral challenge may help to confirm the diagnosis.

Aphthous ulcers

(see ➔ pp. 282–3.)

- Aphthous ulcers are caused by drugs such as azathioprine, captopril, ciclosporin, fluoxetine, sertraline, and sulfonamides.
- NSAIDs and nicorandil cause oral ulcers that may simulate giant aphthae, but the ulcers do not have the erythematous halo or yellow base of an aphthous ulcer.

Erythema multiforme

(see ➔ p. 111.)

- EM is an acute, self-limited illness. Infection (most often herpes simplex) is a much more likely trigger than a drug.

⚠ Stevens–Johnson syndrome and toxic epidermal necrolysis

(see ➔ pp. 116–17.) Life-threatening cutaneous reactions, with overlapping features, caused by drugs such as sulfonamides, aminopenicillins, anti-epileptics, barbiturates, NSAIDs, and allopurinol. Patients may have underlying diseases, particularly AIDS. Course protracted—at least 3 weeks.

- Tender EM-like rash, with severe mucosal involvement (oral, ocular, genital), blistering, and erosions that presents 2–3 weeks after drug is started.
- Epidermal involvement is more severe and widespread in TEN (>30% BSA) than in SJS (<10% BSA).

Coma-induced blisters

- Seen in association with decreased level of consciousness caused by overdose of agents such as barbiturates, benzodiazepines, heroin, methadone, imipramine, and alcohol. (Also reported in some neurological and metabolic causes of reduced levels of consciousness.)
- Blisters, preceded by erythematous plaques, are usually few in number, appear within 24h, and resolve in 10–14 days.
- Blisters are predominantly over bony prominences (pressure may cause hypoxic damage), but some may also be found elsewhere.
- Histologically: subepidermal bullae with sweat gland necrosis.

Chemotherapy agents

- Mucosal ulcers are common.

Contact stomatitis

- The patient may notice an oral burning sensation and xerostomia.
- Reactions develop after days or years of exposure.
- Patch testing will confirm the diagnosis.
- Caused by agents such as topical anaesthetics and antiseptic mouthwashes.

Cutaneous ulcers

- Nicorandil causes large, deep, painful persistent ulcers on perianal skin and other sites. Ulcers may resemble pyoderma gangrenosum (PG) or a cutaneous malignancy (see ➔ p. 310).
- Cocaine abuse may cause PG.

Other drug-induced blistering reactions

- Pseudoporphyria (see ➔ p. 334).
- Pemphigus, most often foliaceus-type with anti-dg1 autoantibodies, usually caused by thiol-containing drugs, e.g. *D*-penicillamine, captopril (see ➔ Box 13.4, p. 269).
- Paraneoplastic pemphigus caused by fludarabine (see ➔ Box 13.4, p. 269).
- Pompholyx caused by IV Ig. May evolve into a generalized eczematous rash.
- Linear IgA bullous dermatosis induced by vancomycin (most often) and other drugs, e.g. penicillin, ceftriaxone, metronidazole, moxifloxacin, diclofenac, phenytoin, amiodarone. Tends to be more severe than the spontaneous form and may mimic TEN.
- Iododerma (see ➔ p. 388).

Further reading

Chanal J *et al. Br J Dermatol* 2013;**169**:1041–8 (drug-induced linear IgA).

Drug-induced abnormalities of hair and nails

Diffuse non-scarring alopecia

- Exclude other causes, including iron deficiency, thyroid disease, and telogen effluvium after a severe illness.
- Diffuse alopecia caused by drugs such as cytotoxics, anticoagulants (heparin and warfarin), antithyroid drugs, and oral contraceptives.
- Temporary interruption of cell division may cause focal narrowing of hair shafts (Pohl–Pincus marks). If severe, hair shafts may fracture (correlates with Beau lines in nails—see later in this section).

Scarring alopecia

- Rare after chemotherapy with taxanes.

Increased hair growth

- Hirsutism = increased hair growth in ♀ in an androgen-dependent pattern (normal ♂ sexual pattern)—face, lips, chest, arms, thighs. Causes: oral contraceptive pill, other androgenic drugs.
- Hypertrichosis = increased hair growth at all body sites. Causes include: ciclosporin, corticosteroids, acetazolamide, phenytoin, IFN, minoxidil, and cetuximab.

Nail abnormalities

Common causes: tetracyclines, antimalarials, retinoids, antiretroviral agents, and chemotherapy agents. Drug-induced nail changes usually involve many or all of the nails and resolve when the drug is withdrawn. Drugs may affect the nail matrix, nail bed, periungual tissues, or blood vessels. Several mechanisms may be involved.

- Nail matrix damage:
  - Beau lines = transverse grooves in the nail plate caused by a temporary interruption of cell division in the nail matrix (common). If severe, nails may be shed (onychomadesis). (Correlates with Pohl–Pincus marks in hair shafts—see earlier in this section.)
  - Nail fragility, altered rate of nail growth, leuconychia.
  - Pigmentation: diffuse or in bands (longitudinal or transverse).
- Nail bed damage:
  - Separation of the nail plate from the nail bed = onycholysis (common) (see Fig. 18.4).
  - Photo-onycholysis (painful, may be pigmented).
  - Splinter haemorrhages.
  - Subungual hyperkeratosis or thickening of the nail bed.

- Proximal nail fold damage:
  - Periungual pyogenic granulomas (common).
  - Acute paronychia, sometimes with subungual abscess (unusual) (see Fig. 18.5).
- Nail blood flow alterations:
  - Ischaemic changes; subungual haemorrhages (purpura).
  - Nail atrophy.

Fig. 18.4 Painful onycholysis caused by chemotherapy.

Fig. 18.5 Painful paronychia caused by tretinoin.

Reactions to chemotherapy agents: 1

Chemotherapy-induced hand–foot skin reaction

Synonyms: acral erythema or palmar–plantar erythrodysaesthesia.

- Occurs in association with systemic drugs such as cyclophosphamide, cytarabine, capecitabine, doxorubicin hydrochloride, fluorouracil, BRAF inhibitors (vemurafenib, dabrafenib), imatinib, and sunitinib. Eccrine glands involved in the pathology. Severity is dose-related. Occurs on skin exposed to friction or pressure.
- Causes paraesthesiae, tingling, burning, or pain predominantly of the palms and soles. Reduced tolerance to contact with hot objects.
- Symmetrical erythematous, oedematous papules develop on palms and soles. Skin may fissure, blister, and ulcerate. Sides of fingers and periungual skin may be erythematous.
- Erythema may extend to the dorsum of the hands and feet.
- Hyperkeratosis develops late (seen early with sorafenib). Tends to localize to pressure areas or areas of friction.
- Tenderness and pain may lead to withdrawal of treatment.
- Management:
  - Minimize pressure and friction: soft, loose-fitting shoes; cotton socks; padded gloves.
  - Pare hyperkeratotic areas gently—regular chiropody.
  - Keratolytic moisturizers with 10–25% urea or salicylic acid, topical anaesthetic creams (EMLA®), topical corticosteroids.
  - NSAIDs, pregabalin 50mg tds.

Neutrophilic eccrine hidradenitis

- Mainly in patients receiving chemotherapy for malignancy, most often haematological such as acute myelogenous leukaemia.
- Delay of about 10 days between start of chemotherapy and onset.
- Rash may be asymptomatic or painful.
- Presents with one or more erythematous, oedematous papules, nodules, or plaques. Sometimes pustular or purpuric centres to plaques.
- Lesions either grouped or disseminated widely in an asymmetrical distribution.
- Distribution is mainly proximal, affecting the upper trunk, upper limbs, and face, particularly periorbital skin, or distal involving the extremities.
- Usually spares groins and axillae.
- Mucosae are not affected.
- May show pathergy, i.e. lesions at sites of IV injections.
- Fever is common but may be linked to underlying neutropenia, rather than the rash.
- Skin biopsy shows neutrophilic infiltrate around degenerative eccrine glands.
- Lesions resolve without scarring in a few days or weeks. Some, but not all, patients relapse when re-exposed to the same drug regimen.

- Topical treatments are ineffective. The role of oral corticosteroids is unproven. NSAIDs may relieve pain and fever.
- Differential diagnosis includes cutaneous infection (culture tissue), leukaemia cutis, EM (symmetrical, target lesions, mucosal ulcers), and Sweet syndrome (see ➔ p. 111 and p. 518).

Eccrine squamous metaplasia

- Oedematous, erythematous plaques or macules that may become confluent. Sometimes vesicles. Typically affects the face, axillae, groins, and legs. May also have erythema and oedema on the palms and legs.
- Itchy or painful.
- Possibly a non-specific reaction to damaged eccrine ductal epithelium.
- Triggered by many chemotherapy drugs. Also seen with excessive sweating and in some skin diseases, e.g. infection (herpesvirus, CMV), inflammatory skin diseases, and skin tumours.
- Sometimes considered to be in the spectrum of neutrophilic eccrine hidradenitis.

Flagellate dermatitis

- Mainly seen in association with bleomycin.
- Appears 1 day to several months after administration of the drug.
- Intensely itchy, linear, erythematous, urticated plaques may affect the arms, back, and scalp, mimicking self-inflicted injury produced by a whip.
- Lesions fade, leaving post-inflammatory linear (flagellate) pigmentation.
- Treat with sedating antihistamines and potent topical corticosteroids.

Radiation recall dermatitis

(see ➔ Box 26.1, p. 531.)

- Well-defined cutaneous reaction at a previously irradiated site. Resembles an acute radiation reaction.
- Triggered most often by IV chemotherapy.

Nail changes are common

(see ➔ pp. 380–1 and Fig. 18.4.)

Reactions to chemotherapy agents: 2

BRAF inhibitors

Agents, e.g. vemurafenib, dabrafenib. Cutaneous toxicity, common, and dose-dependent. Regular skin monitoring recommended. Cutaneous toxicities may include:

- Itch and dry skin.
- Grover disease (itchy erythematous papules, acantholytic histology).
- Rashes, with or without itch—maculopapular, papulopustular, folliculocentric, seborrhoeic dermatitis-like, milia, keratosis pilaris-like.
- Hyperkeratotic hand–foot skin reaction (see ➔ p. 382).
- Photosensitivity (UVA-dependent) may cause painful blisters.
- Squamoproliferative tumours, e.g. papillomas, viral warts, seborrhoeic warts, warty dyskeratomas, palmar/plantar hyperkeratosis, solar keratoses, keratoacanthomas, SCCs. More at onset of treatment, then the rate of development of tumours levels off.
- Melanocytic proliferation, e.g. atypical melanocytic proliferation, second 1° melanomas, activated naevi, and lentigines. Monitor melanocytic lesions with a dermatoscope.
- Painful lobular panniculitis.
- Hair loss (non-scarring).
- Firm purplish papules with granulomatous histology.
- Pyogenic granuloma (vemurafenib).
- Radiation recall dermatitis (vemurafenib).

Epidermal growth factor receptor inhibitors

Agents, e.g. cetuximab, panitumumab, erlotinib, gefitinib. EGFR is expressed by basal keratinocytes and sebocytes, as well as the outer root sheath of hair follicles. Cutaneous adverse effects may include:

- Initial oedema, redness, and a burning sensation on the face and upper trunk. May be mildly itchy.
- Sterile itchy (unlike acne) papulopustular follicular eruption on the face and trunk 7–10 days after starting treatment (no comedones) (see Fig. 18.6). May spare skin previously treated by radiotherapy. Gradually crusts and resolves. Severity dose-related. Presence of rash may correlate with more effective treatment and increased survival time. Swab to exclude 2° staphylococcal infection. Treatment options: antihistamines (for itch), topical clindamycin, topical calcineurin inhibitors, oral tetracyclines, low-dose isotretinoin (20mg daily).
- Painful paronychia, periungual abscess, pyogenic granuloma of the nail fold.
- Brittle, curly scalp hair, increased facial hair, long eyelashes, scarring alopecia with pustules, i.e. follicultis decalvans (erlotinib).
- Mucosal ulcers, dry mouth.

Taxanes

Agents: docetaxel, paclitaxel. Cutaneous adverse effects include:

- Hand–foot skin reaction (common) (see ➜ p. 382).
- Radiation recall dermatitis (see ➜ Box 26.1, p. 531).
- Alopecia—usually regrows. Rarely scarring and permanent.
- Nail abnormalities:
  - Subungual haemorrhage leading to onycholysis (see Fig. 18.4).
  - Orange-brown discoloration.
  - Beau lines, transverse loss of the nail plate.
  - Acute paronychia and subungual abscess (see Fig. 18.5).
- AGEP.
- SCLE, photosensitivity, scleroderma-like changes of the legs, and erysipelas-like erythema.

Multikinase inhibitors

Agents, e.g. sorafenib, sunitinib, and vandetanib. Inhibit a variety of tyrosine kinase receptors, e.g. vasoactive endothelial growth factor receptor, platelet-derived growth factor receptor. Adverse effects may include:

- Hyperkeratotic hand–foot skin reaction (see ➜ p. 382).
- Morbilliform rashes, dry skin, itch.
- Comedonal acneiform eruptions, folliculitis.
- Alopecia, subungual splinter hemorrhages.
- Depigmentation hair/skin.
- Photosensitivity.
- AGEP, TEN.
- Sweet syndrome (nilotinib).
- Papillomas, BCCs, and SCCs.

Inhibitors of mitogen-activated protein kinase (MAPK) kinase (MEK inhibitors)

Agents, e.g. selumetinib, cobimetinib, trametinib. Adverse effects include:

- Morbilliform eruption.
- Papulopustular eruption (similar to EGFR inhibitors). 2° staphylococcal infection is common.
- Dry skin and itch.
- Paronychia, mild hair loss.
- Dusky erythema with urticated or targetoid patches.

Smoothened (Smo) receptor inhibitors

Agents: vismodegib (blocks hedgehog signalling pathway). Adverse effects include:

- Alopecia, taste disturbance.
- Possibly keratoacanthoma/eruptive SCCs (sun-damaged skin).

Mammalian target of rapamycin (mTOR) inhibitors

Agents: rapamycin, temsirolimus, everolimus. Mucocutaneous adverse effects are common and may include:

- Stomatitis—dose-related. Responds to potent topical corticosteroids.
- Morbilliform, eczematous, or acneiform eruptions on the trunk, spreading to the extremities, neck, face, and scalp.
- Paronychia, pyogenic granuloma-like lesions of the nail fold.
- Alopecia, itch, dry skin, vasculitis, poor wound healing.

Cytotoxic T-lymphocyte-associated protein 4 (CTLA-4) inhibitors

Agents: ipilimumab. Stimulates the immune system. Induces autoimmune adverse effects:

- Itch, morbilliform rash, vitiligo-like hypopigmentation.

Purine and pyrimidine analogues

Agents: fludarabine, cladribine, capecitabine, tegafur, gemcitabine. Cutaneous adverse effects include:

- Hand–foot skin reaction (see ➔ p. 382).
- Acral hyperpigmentation most marked in skin creases.
- Autoimmune phenomena, including paraneoplastic pemphigus (linked to fludarabine).
- Inflammation of actinic keratoses (caused by capecitabine which is a prodrug of 5-fluorouracil).

Hydroxycarbamide (hydroxyurea)

Cytostatic agent used in myeloproliferative disorders. Cutaneous adverse effects include:

- Diffuse hyperpigmentation and brown nails.
- Dermatomyositis-like eruption on the dorsum of hands (ANA-negative) (see ➔ Box 19.14, p. 411).
- Acral erythema and palmoplantar keratoderma (PPK).
- An atrophic poikilodermatous (see ➔ p. 410) appearance on the legs.
- One or more persistent, painful, well-defined shallow leg ulcers, usually over the malleoli, but may affect the calf, dorsal foot, or toes. Develop after an average of 5 years of treatment. Surrounding skin may be purpuric and atrophic. Heal when hydroxycarbamide is stopped.
- Oral mucosal ulcers.

Tretinoin

Acid form of vitamin A. Used in acute promyelocytic leukaemia. Cutaneous adverse effects include:

- Pruritus, erythema, and dry skin; dry mucosae and cheilitis.
- Alopecia, periungual pyogenic granulomas, acute paronychia (see Fig. 18.5).

Fig. 18.6 Acne-like eruption caused by erlotinib (EGFR inhibitor).

Further reading

Belum VR *et al*. *Curr Oncol Rep* 2013;**15**:249–59.
Macdonald JB *et al*. *J Am Acad Dermatol* 2015;**72**:203–18.
Macdonald JB *et al*. *J Am Acad Dermatol* 2015;**72**:221–36.

Drugs and skin diseases

Acne or acne-like pustular eruptions

(see ➔ p. 244.)

- Causes include topical and oral corticosteroids; androgens, including anabolic steroids, phenytoin, phenobarbital, lithium, bromides, isoniazid, rifampicin, and EGFR inhibitors (see ➔ p. 384).
- A papulopustular rash develops within weeks of starting the drug. Comedones are unlikely, unless the reaction is caused by exogenous androgens. May take months to resolve, despite treatment for acne.
- Iododerma (rare, but commoner in renal failure when clearance of iodine in contrast medium is impaired). Acneiform papules/pustules, haemorrhagic bullae/pustules, plaques, or nodules.

Dermatomyositis-like

(see ➔ pp. 410–11.)

- Most often hydroxycarbamide (hydroxyurea) (see ➔ p. 386).
- Also reported with penicillamine, statins, IFN-β, anti-TNF agents, ipilimumab, and others.

Eczema

(see ➔ pp. 212–13.)

- Calcium channel blockers have been linked to generalized chronic eczematous eruptions in the elderly.
- IV Ig may precipitate pompholyx (vesicles on palms and soles) and eczema.

Elastosis perforans serpiginosa

(see ➔ Box 19.28, p. 431.)

- Penicillamine.

Erythema gyratum repens

- Azathioprine, pegylated IFN-α.

Halo naevi

- Multiple halo naevi (halo of hypopigmentation around regressing melanocytic naevi) may be induced by immune-modifying drugs, including infliximab, IFN-β1a, adalimumab, tocilizumab.

Ichthyosis

Kava dermopathy is a common reaction to heavy use of kava, a psychoactive drink prepared from *Piper methysticum*. Used recreationally throughout the Pacific and increasingly elsewhere. May be added to herbal remedies. Dermopathy is characterized by:

- Scaly skin (ichthyosiform dermatitis) that starts on the head, face, and neck, progresses to the body and feet, and becomes generalized.
- Sometimes facial oedema, alopecia, and hyperpigmentation of exposed skin.
- Resolves slowly when kava withdrawn.

Lichen planus-like

(see ➜ p. 224.)

- Widespread lichenoid reactions may appear weeks or months after exposure to the drug (see ➜ Fig. 18.2, p. 369, and p. 375).
- Discrete papules, resembling LP or plane warts, have been caused by palifermin, a recombinant human keratinocyte growth factor.

Neutrophilic dermatoses

- Pyoderma gangrenosum-like (see ➜ p. 310). Nicorandil induces painful ulcers, as may cocaine.
- Azathioprine hypersensitivity syndrome resembles Sweet syndrome, with fever, neutrophilia, and papules/plaques with dense infiltrate of polymorphonuclear cells. Signs mainly on extremities, unlike Sweet syndrome (see ➜ p. 518).
- Granulocyte colony-stimulating factor (G-CSF) is the commonest cause of drug-induced Sweet syndrome.

Psoriasis and palmoplantar pustulosis

(see ➜ p. 196, p. 198.)

- Lithium exacerbates psoriasis more often than inducing disease.
- Antimalarials and IFNs may exacerbate psoriasis.
- β-adrenergic receptor-blocking agents may induce or exacerbate psoriasis.
- Systemic corticosteroids: an exacerbation of psoriasis, which may be pustular, can be triggered by withdrawal of systemic corticosteroids prescribed for an unrelated condition.
- NSAIDs, angiotensin receptor blockers, and ACE inhibitors possibly exacerbate psoriasis.
- Alcohol: psoriatic patients with a high alcohol intake tend to have more severe disease.
- TNF-α antagonists have been reported to induce plaque, guttate, erythrodermic, and pustular psoriasis, although these drugs are used to treat psoriasis. They also cause palmoplantar pustulosis.
- Ustekinumab (IL-12/23 inhibitor) may trigger pustular psoriasis.

Rosacea

(see ➜ pp. 254–5.)

- Exacerbated by peripheral vasodilators, including glyceryl trinitrate, isosorbide, and alcohol (no increased risk found with calcium channel blockers in a study published in 2014).

Pigmented purpuric dermatoses

- A variety of drugs, including diuretics, NSAIDs, sedatives, antibiotics, cardiovascular drugs, isotretinoin, and vitamins.

Pityriasis rosea-like

(see ➔ p. 151 and Fig. 18.7.)

- A mildly itchy rash, mainly on the trunk and proximal upper limbs. Papules and plaques are aligned in a 'Christmas tree' pattern with long axes parallel to the ribs (parallel to Langer's skin lines). Drugs that have been implicated include:
- Gold compounds, imatinib mesylate, barbiturates, *D*-penicillamine, captopril, terbinafine, isotretinoin, bismuth compounds, omeprazole, arsenicals.

Pemphigus

(see ➔ pp. 268–9.)

Subacute cutaneous lupus erythematosus

(see ➔ p. 402.)

SCLE appears weeks to several years after starting the drug and resolves 1–24 weeks after stopping the drug. Anti-Ro antibodies may persist when the drug is withdrawn. Consider in older patients presenting with SCLE for the first time. A wide variety of drugs have been implicated, including:

- Diuretics (hydrochlorothiazide and spironolactone), calcium channel blockers, β-blockers, ACE inhibitors, proton pump inhibitors, antifungals (terbinafine and griseofulvin), chemotherapeutic agents, immunomodulators, biologics, NSAIDs, hormone-altering drugs, statins, and antihistamines

Porphyria cutanea tarda

(see ➔ p. 332.)

- In genetically susceptible individuals, some drugs exacerbate PCT, including alcohol, oral oestrogens, oral contraception and hormone replacement therapy, oral iron, antimalarials, and griseofulvin.

Fig. 18.7 Pityriasis rosea-like drug reaction with annular erythematous lesions.

Further reading

Ahronowitz I and Fox L. *Semin Cutan Med Surg* 2014;**33**:49–58 (drug-induced dermatoses).
Lowe G *et al*. *Br J Dermatol* 2011;**164**:465–72 (drug-induced SCLE).
Spoendlin J *et al*. *Br J Dermatol* 2014;**171**:130–6 (antihypertensive drugs and risk of rosacea).

Chapter 31

Skin in infancy and childhood

Contents

Skin of the neonate: the first 28 days

The skin of a full-term infant

- Fine desquamation at 24–48h of age. Increased desquamation in post-mature babies. Persists up to 1 week. WSP is a safe emollient.
- Increased heat loss with reduced vasoconstriction.
- Permeable, particularly if occluded or in high temperatures or humidity. Topical agents are easily absorbed and may be toxic.
- Wash with unperfumed, neutral-pH cleanser. Rinse and dry completely. Aqueous chlorhexidine is a safe antimicrobial.
- Minimal eccrine sweating, more on the forehead than trunk and limbs. Emotional sweating (palms, soles) in response to pain or hunger.

The skin of a premature infant

- Thin, very fragile skin prone to mechanical injury (e.g. from tapes or electrodes) may leave scars, apparent after a few months. Protect injuries with occlusive dressings. Avoid adhesive tapes.
- Very permeable. Large transepidermal water loss (20–50% of body weight in 24h). Loss is increased by phototherapy and heat.
- Immature thermal homeostasis—loss of body heat. No sweating.

Some transient neonatal skin changes

- Vascular changes:
  - Cutis marmorata. A physiological net-like (reticulate) pattern of purplish red mottling that disappears on warming.
  - Capillary ectasia (salmon patch) on the nape (stork bite), eyelids (angels' kisses), and/or glabella. Most resolve in months or years. May persist on the nape of the neck.
  - Petechiae on upper body caused by pressure during delivery.
  - Harlequin colour change (rare). Reddening of dependent side of body, with sharp midline demarcation and blanching of the upper side. Episodes last seconds to minutes. Seen most often in premature infants lying on one side. Episodes cease after 3 weeks.
- Sebaceous hyperplasia: creamy papules on nose. Last a few weeks.
- Milia: 1–2mm firm, yellowish papules on face. Last a few months.
- Pigmented macules:
  - Mongolian spots (dermal melanosis). Blue-grey macules on the sacrum or low back. Most often in black or Asian neonates. May be multiple and can be confused with bruises. Disappear by age 4 (see Fig. 31.10).
- Blisters and pustules (see ➔ p. 592).
- Perianal dermatitis:
  - Erythematous macules, 2–3mm, that may be eroded or bleed. Occurs between days 4 and 7. Commoner in premature infants. Associated with feeds of formula milk (see ➔ p. 604).

⚠ More worrying signs

- Perinatal injury (see Box 31.1).
- Subcutaneous nodules or plaques (see Box 31.2).
- Purpuric macules or nodules (see Box 31.3).

Box 31.1 Cutaneous signs of perinatal injury

- Caput succedaneum: swelling over presenting part of scalp caused by pressure on the head during labour. Vacuum extraction—haematoma and oedema. Forceps—facial erythema or abrasions.
- Needle marks: hypopigmented speckles. Heel pricks may leave calcified nodules. Scalp electrodes may leave cuts, superficial ulcers, or, uncommonly, abscesses.
- Ring of alopecia 'halo scalp ring' caused by pressure during labour. Usually over the vertex and associated with caput succedaneum. Sometimes necrosis. If severe, may leave permanent scarring alopecia.
- Amniocentesis scars: more apparent after a few weeks; 1–5mm cutaneous dimple-like scars. Most often on extremities.

Box 31.2 Subcutaneous nodules or plaques in neonates

Neonate appears well

Subcutaneous fat necrosis. One or more circumscribed, firm, mobile subcutaneous nodules in first few months of life. Most often on the back or buttocks. Sign of preceding fetal stress, hypothermia, or difficult labour with ischaemic fat injury. May calcify or ulcerate. Usually resolves within 6 months. Prognosis good, but monitor for hypercalcaemia.

Sick neonate

⚠ Sclerema neonatorum. Generalized yellowish hardening of the skin that occurs in very sick preterm neonates. Treat the underlying condition, and, if the infant survives, hardening resolves.

⚠ Box 31.3 Blueberry muffin baby

Bluish purple, non-blanching (purpuric) macules, papules, and/or nodules are produced by extramedullary erythropoiesis in the dermis. Usually involves the head, neck, and trunk. Causes include:

- Congenital infections, e.g. toxoplasmosis, other, rubella, cytomegalovirus (commonest), herpes simplex ('TORCH').
- Haematological disease, e.g. haemolytic disease of the newborn, hereditary spherocytosis, twin–twin transfusion syndrome.
- Malignancy/proliferative diseases, e.g. leukaemia, neuroblastoma, congenital rhabdomyosarcoma, Langerhans cell histiocytosis.

Investigate to determine the cause (including skin biopsy) (see ➔Box 33.4, p. 649), and manage the underlying condition.

Pustules and blisters

Blisters are discussed in detail in ➔ Chapter 13, pp. 259–75, but here we consider some problems specific to infants and children. Blisters may evolve into pustules. Consider infection, infestation, dermatitis, or trauma, before much less common or very rare blistering diseases.

What should I ask?

- Is the child sick—any fever, lethargy, loss of appetite?
  - Could blisters/pustules be caused by a systemic disease or drug?
- When did the problem start—neonatal or childhood?
- Is the problem localized or generalized?
- What triggers blisters or pustules?
  - Are sun-exposed sites affected more than covered sites?
  - Is the skin fragile? Do blisters occur at sites of friction, e.g. feet? Or in heat? (Sweating may exacerbate EB; see ➔ pp. 624–5).
  - Are mucosal surfaces affected?
- Any post-inflammatory pigment, scarring, or milia?
- Is there a family history of skin problems, e.g. maternal autoimmune blistering disease (IgG autoantibodies cross the placenta and may cause transient blisters in the neonate)?

Neonatal vesicles, pustules, or papulopustules

See Box 31.4.

Vesicles, pustules, or papulopustules in childhood

- Infections and infestations:
  - Scabies: vesicopustules on palms and soles in infants usually at least 4 weeks of age. Is the mother itchy? (see ➔ p. 172).
  - Virus: HSV, including eczema herpeticum (see ➔ p. 152), VZV, CMV, EBV, Coxsackie virus (hand–foot–mouth disease).
  - Bacterial: *Staphylococcus aureus* (bullous impetigo, SSSS; see ➔ pp. 136–7), *Streptococcus, E. coli, Haemophilus influenzae, Klebsiella pneumoniae*.
- Contact dermatitis, including phytophotodermatitis (see ➔ Box 16.4, p. 327).
- Trauma, including burns (see ➔ p. 110).
- Erythema multiforme (see ➔ p. 111).
- Drug-related: phototoxic or photoallergic (see ➔ p. 326), SJS or TEN (see ➔ pp. 116–20).
- Mastocytosis (see Box 31.5).
- Rare: EB (see ➔ pp. 624–5, and p. 593), photosensitivity (see ➔ Box 16.5, p. 329), acrodermatitis enteropathica (zinc deficiency—unusual before age 4 weeks; see ➔ p. 513), autoimmune (see ➔ pp. 260–75).

What should I do?

- Identify the 1° lesions: vesicle or pustule. Remember these may evolve into erosions or ulcers.
- Exclude infection: culture fluid or pus for bacteria, *Candida*, and herpesvirus.
- A Tzank smear may help you to make the diagnosis (see ➔ p. 647).

Box 31.4 Vesicular and pustular eruptions in neonates

Swabs sterile

- Erythema toxicum neonatorum: onset age 1–3 days. Mainly trunk. One to 2mm yellowish papules or pustules with an erythematous halo (like flea bites). Fluctuates. Resolves in 2–3 days. Eosinophils in smear (very common).
- Transient pustular melanosis: usually in black neonates. Present at birth. Flaccid, 1–2mm pustules on the chin, neck, and trunk evolve to pigmented macules. Persists a few months. Neutrophils in smear.
- Miliaria: clear, fragile vesicles, papules, or pustules caused by occlusion of sweat ducts. Induced by heat and high humidity.
- Sucking blisters or trauma, including burns.
- Aplasia cutis may look bullous (see Box 31.6).
- Acropustulosis of infancy: crops of very itchy vesicles and pustules on hands and feet. Usually black neonates (neutrophils in Tzank smear). Lasts 1–2 years (uncommon). Exclude scabies (see ➔ p. 172).
- Benign cephalic pustulosis (neonatal 'acne'): pustules on the face and scalp. No comedones. Onset age 2 weeks. Persists about 2 months. May be reaction to *Pityrosporum* yeasts (see ➔ p. 602).
- Rare: maternal immunobullous disease (IgG autoantibodies cross the placenta), EB, incontinentia pigmenti (eosinophils in Tzank smear; look for linear pattern; examine the mother) (see ➔ pp. 636–7).

⚠ *Infection: neonate may not appear unwell*

- Bacterial infection, e.g. staphylococcal/streptococcal impetigo (common), SSSS (see ➔ pp. 136–7), listeriosis (rare cause of pustules and petechiae).
- Cutaneous candidiasis (may be congenital).
- HSV or VZV: look for scars (multinucleated giant keratinocytes in Tzank smear) (see ➔ p. 647).

Box 31.5 What is mastocytosis?

Uncommon conditions in which mast cells accumulate in skin and other organs. Linked to mutations in *KIT* gene. Systemic mast cell disease (raised serum tryptase) is extremely rare in children.

Presentations in childhood:

- Mastocytoma: localized reddish yellow macule, plaque, or nodule. Develops in infancy. Swells and may blister when rubbed (Darier sign). Blistering less frequent with age. Nodules resolve slowly.
- Urticaria pigmentosa: freckle-like macules. Urticate (become erythematous and raised), if rubbed. Improves by adolescence.
- Diffuse cutaneous mastocytosis: very rare. Mast cells disseminated widely in skin. Skin may be erythematous, leathery, or normal. Generalized blistering in infants aged <3, may mimic SSSS. Systemic symptoms include itch, flushing, hypotension, syncope, diarrhoea, and dyspnoea. Tends to improve with age.

Congenital cutaneous lesions

⚠ Midline congenital skin lesions on the spine may be a marker for occult spinal dysraphism (incomplete fusion of midline mesenchymal, bony, or neural structures). Cord tethering is a common complication. Ask about a family history of neural tube defects. Midline lesions on the face or scalp may have intracranial connections. Consider imaging, if clinically suspicious.

Back or perianal skin: what should I look for?

Occult spinal dysraphism is more likely in the presence of two or more cutaneous signs. Look for neurological symptoms or signs (may be subtle or none in neonate) and urogenital or anorectal abnormalities. Suggestive signs include:

- Lumbosacral markers, e.g. deep or large midline dimples >2.5cm from the anus or dermal sinus tract (may leak cerebrospinal fluid), localized tuft or patch of hypertrichosis (faun's tail), aplasia cutis, skin tags (check the gluteal cleft), lipoma, telangiectatic patch (PWS) or haemangioma near the midline.
- Asymmetrical curved gluteal cleft.
- Palpable vertebral defects.

Face or scalp: what should I look for?

- Midline dermoid cyst (firm subcutaneous nodule): those on the nose or occipital scalp are most likely to have intracranial extension. Punctum with hairs increases the chance of intracranial connection.
- Midline nasal mass or pit ± overlying skin discoloration may have an intracranial connection. Hair may protrude from the pit.
- Nasal glioma: firm, non-compressible blue or skin-coloured nodule on the side of the nose. May widen the nasal bone with hypertelorism. May have intracranial connection.
- Cephalocoeles: midline, 1–4cm, bluish, compressible, subcutaneous nodule on the nose (encephalocoele) or scalp (most often occipital) or circular area of alopecia, with a translucent membrane (may look bullous). Indications of possible intracranial involvement:
  - Enlarges with crying. May be pulsatile.
  - Transilluminates.
  - Scalp—overlying PWS (see Fig. 31.1).
  - Scalp—hair collar sign: a rim of dark coarse hair around the bald patch. This marker for ectopic neural tissue is also seen in membranous aplasia cutis congenita (see Fig. 31.1). If hair collar present, image to look for skull defect and to exclude cephalocoele. If no connection, may either be heterotopic brain tissue (meningothelial and/or glial tissue in subcutaneous tissue or dermis) or membranous aplasia cutis congenita (see Box 31.6).
- Sebaceous naevus: yellowish plaque with loss of hair. Most often oval. If linear (following Blaschko lines), may be associated with neurological problems (see ➔ p. 560 and Fig. 31.2).

Box 31.6 What is aplasia cutis congenita?

In this rare condition, localized areas of skin are absent at birth. Can occur anywhere. Aetiology is uncertain but may relate to incomplete closure of embryonic fusion lines.

- Aplasia cutis congenita has been classified into nine groups. All, but group 1 (commonest type), have associated abnormalities.
- Most cases are sporadic, rarely inherited in an AD or AR pattern.
- Usually presents as a 1–2cm circular or oval bald patch near the vertex of the scalp. Rarely affects the face (focal facial dermal hypoplasia) and other areas.
- Appearance varies—most often a bald patch appears to be covered by a shiny translucent membrane (membranous aplasia cutis), but the skin may be eroded, deeply ulcerated, or scarred.
- Membranous aplasia cutis is usually surrounded by a hair collar (see ➲ pp. 594–5 and Fig. 31.1).
- On scalp, may be associated with an underlying skull defect.
- May be >1 lesion.
- Extensive aplasia cutis of the trunk is associated with fetus papyraceus (defects may be caused by placental infarcts after death of a twin fetus). Also malformations in other organs.
- Differential diagnosis: obstetric trauma (e.g. forceps injury), HSV or VZV infection, EB (rare).

Fig. 31.1 Membranous aplasia cutis congenita with dark hair collar and surrounding capillary malformation.

Fig. 31.2 Linear sebaceous naevus—yellowish smooth plaque and absence of overlying hair.

Further reading

Baselga E *et al*. *Pediatr Dermatol* 2005;**22**:13–17.
Bellet JS. *Semin Perinatol* 2013;**37**:20–5.
Frieden IJ. *J Am Acad Dermatol* 1986;**14**:646–60.
Guggisberg D *et al*. *Arch Dermatol* 2004;**140**:1109–15.

Infantile haemangioma

Synonyms: strawberry haemangioma, capillary haemangioma.

Infantile haemangiomas are common benign proliferative tumours, both disfiguring and alarming. To differentiate from other vascular tumours and malformations (see ➔ p. 598, pp. 600–1), endothelial cells in infantile haemangiomas are glucose transporter 1 (GLUT1)-positive (biopsy rarely required).

What should I look for?

- Seen most often in girls and preterm babies (may be multiple).
- Fifty percent occur on the head and neck.
- Infantile haemangiomas are not apparent at birth, but parents may notice a pale, telangiectatic or erythematous macule (may resemble a PWS).
- Bright red, dome-shaped nodules or plaques evolve in the first month of life. Some have a deeper bluish component (see Fig. 31.3).
- Growth phase: rapid for 3–4 months; slow for 6–18 months. Deep components grow for longer.
- Involution. Colour slowly fades; nodules soften and shrink; 50–60% involute completely by age 5 years, 70–75% by age 7 years, and 90% by age 9 years. Residual changes may include pallor and lax skin.
- Deeper components may persist.

⚠ What to do and when to worry

- Most do not require treatment, regressing spontaneously, without significant sequelae. Discuss the likely outcome with the parents. Photographs may be useful to document evolution and resolution.
- A minority of enlarging infantile haemangiomas need early treatment to prevent disfigurement or organ compromise (see Box 31.7).
    - Periocular: may obstruct vision and lead to permanent amblyopia. Refer for urgent visual assessment. MRI if displacement of the globe or thickening of the lid.
    - Mandibular or neck area: internal infantile haemangiomas may cause airway obstruction. If noisy breathing, stridor, or difficulty breathing when feeding, refer for urgent ENT assessment.
    - Lips: infantile haemangiomas may interfere with the ability to suck.
- Ulceration in the growth phase (see Box 31.8).
- Plaque-like facial haemangioma may be associated with PHACE (Posterior fossa malformations, Haemangioma, Arterial abnormalities, Coarctation of the aorta, and Eye abnormalities). Called PHACES if also Sternal clefting and/or Supraumbilical raphe.
- Plaque-like lumbosacral haemangioma: look for tags or dimples in the genital and sacral areas. May be associated with spinal dysraphism and/or multiple congenital anomalies (see ➔ p. 594).
- Large plaque-like haemangiomas are more likely to be associated with a systemic problem. Investigate with imaging, e.g. magnetic resonance angiography (MRA), MRI.
- Five or more cutaneous haemangiomas: rarely associated with disseminated haemangiomatosis (liver, heart, and other internal organs). Most not life-threatening. Age >6 months: unlikely to be problems.
- Bleeding disorder: review the diagnosis. (See Box 31.9.)

Box 31.7 Management of infantile haemangiomas

An enlarging infantile haemangioma may lead to irreversible disfigurement (central face, nasal tip, lips) or compromise organ function. β-blockers are the treatment of choice, ideally from age 4–6 weeks.

- Propranolol: gradually increase from 1mg/kg/day to 2mg/kg/day in three equal doses. Continue for 12–18 months. Well tolerated. Side effects include hypoglycaemia (feed regularly) and sleep disturbance. Risk of rebound growth of infantile haemangioma on stopping treatment.
- Use with caution in PHACE: risk of stroke if arterial anomalies.
- Topical timolol maleate 0.5% solution may have a role.
- Other options:
  - Topical potent corticosteroid or imiquimod for superficial periorbital infantile haemangiomas.
  - Intralesional corticosteroids in localized infantile haemangiomas—nasal tip, lip.

Box 31.8 Ulcerated infantile haemangiomas

- Large infantile haemangiomas growing rapidly may ulcerate, particularly infantile haemangiomas affecting anogenital skin, the neck, or the lower lip. Appearance of a grey-white colour on the surface of infantile haemangioma may herald ulceration.
- Ulceration is painful, particularly in the anogenital region.
- Ulcers usually heal in 2–3 weeks, but deep ulcers cause scarring.
- Bleeding is usually minimal.
- Monitor for infection (uncommon)—spreading erythema, warmth, and pain—and swab for bacterial culture.
- Treatment options:
  - Wound care: soak off crusts; apply white soft paraffin to protect skin.
  - Antibiotic ointments. Occlusive dressings.
  - Topical lidocaine has been advocated for pain relief (pea-sized amount, no more than qds).
  - Paracetamol for pain control.
  - Oral propranolol is now the treatment of choice (see Box 31.7).
  - Topical potent or very potent corticosteroid ointment may relieve pain and speed involution, as well as re-epithelialization.
  - Pulsed dye laser may relieve pain and speed healing.

Further reading

Luu M and Frieden IJ. *Br J Dermatol* 2013;**169**:20–30 (reviews infantile haemangioma and propranolol).

Solman L *et al*. *Arch Dis Child* 2014;**99**:1132–6 (propranolol protocol).

Other vascular tumours

Infantile haemangioma, the commonest infantile vascular tumour, should be differentiated from other vascular tumours, mostly much less common, including:

- Pyogenic granuloma: very common in childhood, but rare in infants <3 months of age. Friable vascular papule, often on the head or neck. Bleeds readily (see ➔ p. 343).
- Eruptive pseudoangiomatosis: sudden eruption of vascular papules. Usually facial. Age infancy to 6 years. Possibly viral aetiology. Resolves in 1–2 weeks.
- Glomangiomas: many bluish nodules (some may be congenital). Numbers increase slowly. Familial, AD inheritance.
- Congenital haemangiomas (CHs): rare. Resemble infantile haemangiomas, but fully developed at birth. May regress (RICH) by 12–14 months or no involution (NICH). Endothelial cells GLUT1-negative, unlike infantile haemangioma.
- Verrucous haemangioma: congenital linear vascular plaque with overlying hyperkeratosis. Mainly limbs. Wilms tumour antigen (WT1) and GLUT1-positive. Has some features of a vascular malformation (see ➔ pp. 600–1).
- Tufted angioma and kaposiform haemangioendothelioma: rare. Enlarging purplish red patches, indurated plaques, and/or nodules. Partial lymphatic endothelial immunophenotype. Associated with Kasabach–Merritt syndrome (see Box 31.9).

Box 31.9 What is Kasabach–Merritt syndrome?

The rare association of a vascular tumour with thrombocytopenia (platelet trapping in tumour) and sometimes microangiopathic haemolytic anaemia and a 2° consumptive coagulopathy.

- Occurs with some rare enlarging vascular tumours of infancy/childhood, e.g. tufted angioma and kaposiform haemangioendothelioma (not seen in infantile haemangioma).
- Podoplanin-positive endothelium in tumours binds CLEC-2 receptor on platelets, activates platelets and initiates clotting in tumour.
- ⚠ Look for a purplish, tender, rapidly enlarging deep dermal nodule or plaque surrounded by petechiae or bruising and prolonged bleeding at other sites.
- May be life-threatening. Check FBC with film (low platelets, fragmented RBC) and fibrinogen level.

Further reading

Colmenero I and Hoeger PH. *Br J Dermatol* 2014;**171**:474–84.
Hoeger PH and Colmenero I. *Br J Dermatol* 2014;**171**:466–73.

Fig. 31.3 A large infantile haemangioma with both superficial and deep components. The superficial component will resolve, but some of the deeper component may persist.

Vascular malformations

Malformations may be predominantly capillary, venous, lymphatic, arterial, or AV, or any combination of these. Distribution may be localized or diffuse. Malformations may be present at birth or only become apparent in childhood. Unlike infantile haemangiomas, most malformations do not involute but persist and may slowly worsen with age. Many capillary malformations occur in CM–AV malformation syndrome (see Box 31.10).

What should I look for?

- Port wine stain (PWS): malformation with dilated capillaries. Somatic mosaic activating mutation in GNAQ in affected tissue. May be associated with complex vascular syndromes:
  - Limb PWS may be associated with Klippel–Trenaunay (varicose veins, capillary, or mixed capillary–lymphatic malformation) or Parkes–Weber (multiple AV shunts) syndromes. Look for varicosities, haemangiomas, lymphangiomas, and hypertrophy, with gradual asymmetrical lengthening of the affected limb.
  - ! Facial PWS involving the forehead and upper eyelid may be linked to Sturge–Weber syndrome (see ➔ p. 562 and Fig. 31.4). Seizures are a frequent complication. Early and regular ophthalmological review required to detect glaucoma.
- Cutis marmorata telangiectatica congenita: very rare malformation with dilated capillaries and veins in the dermis and subcutis.
  - A network of purplish red bands with some telangiectasia. Persists when skin is warm.
  - Involved skin may be atrophic and may ulcerate.
  - Girth of the involved limb may be reduced—less fat, muscles, and/or bone.
  - Rarely associated with other congenital problems.
  - Fades partially over years.
- Venous—slow flow: elevated D-dimer levels.
  - Subtle bluish compressible vessels—no thrill or bruit.
  - May swell when dependent.
  - Can be associated with chronic localized or diffuse intravascular coagulopathy. ! In patients with large venous malformations, minor surgery may trigger DIC.
- AV—fast flow: normal D-dimer levels.
  - Most often head and neck.
  - May be subtle initially, like a PWS.
  - Look for local warmth, subtle thickening. Rarely a thrill or bruit.
  - Potentially life-threatening haemorrhage.
- Lymphatic: normal D-dimer levels.
  - Cystic hygroma: mass, usually in the neck or axilla that presents in the neonate. Collections of dilated lymphatics in the dermis, with superficial vesicles that intermittently leak clear fluid.
  - Localized group of vesicles: lymphangioma circumscriptum. These appear during childhood. Commonest on the proximal limbs and limb girdle.

Fig. 31.4 The distribution of facial port wine stain probably follows the embryonic vasculature of the face, rather than the innervation of the trigeminal nerve. Facial port wine stain involving the forehead may be associated with cerebral and ocular vascular malformations (Sturge–Weber syndrome). Complications include seizures, stroke, and glaucoma (see ➔ p. 562).

Box 31.10 Capillary malformation–arteriovenous malformation syndrome

AD disorder, usually with loss-of-function *RASA1* mutations, but clinically and genetically variable. Features include:

- Numerous CMs—congenital and acquired (more with age). Multifocal and randomly distributed; 1–3cm in diameter, and oval or circular. Often surrounded by a pale halo.
- Less often large or irregular CMs.
- Partial or total absence of vellus hair over CMs.
- Punctate red spots surrounded by pale halos on upper extremities.
- Grouped telangiectasias on the neck/upper trunk.
- Naevus anaemicus has been noted in some cases.
- AV malformations and/or AV fistulae in one-third. Involve the skin, muscle, bone, spine, and brain. Risk of cerebral haemorrhage and sensorimotor deficits. MRI may be indicated to exclude intracranial and spinal AV malformations, if the diagnosis is suspected.

Further reading

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Orme CM *et al. Paediatr Dermatol* 2013;**30**:409–15.

Red scaly skin in neonates and infants

Atopic eczema (dermatitis), the commonest cause of redness and itching in children, does not usually present until around the age of 3 months (see ➔ p. 606). Neonates with red scaly skin are much more likely to have infantile seborrhoeic dermatitis, which presents in the first month and usually clears after 3 or 4 months.

⚠ Severe generalized erythema and scale (neonatal erythroderma) associated with failure to thrive is rare but suggests an underlying problem (see Boxes 31.11 and 31.12). In some forms of ichthyosis, the neonate is born encased in a tight, smooth, shiny collodion membrane (collodion baby). The tight skin causes dysmorphic features, including ectropion (lower eyelid turned outwards), eclabium (lip turned outwards), malformed ears, and flexion contractures. In both erythroderma and collodion babies, the barrier function of the skin is abnormal, leading to fluid loss, infection, poor thermoregulation, and absorption of topical agents. Use bland emollients, and control humidity (see ➔ pp. 626–7).

Infantile seborrhoeic dermatitis

What should I look for?

- A well infant, not particularly itchy (atopic eczema is very itchy).
- A greasy, erythematous, scaly, orange-red papular rash involving the face, scalp ('cradle cap'), and flexures. Look in skinfolds, particularly around the neck and in the napkin area (usually spared in atopic eczema) (see Fig. 31.6). Irritant napkin dermatitis involves convex surfaces and spares the skinfolds (see Box 31.12 and Fig. 31.5). Rash may be secondarily infected by *Candida*. Some infants develop tiny sterile pustules (see neonatal acne in Box 31.4).

What should I do?

- Explain that the condition should clear by about 4 months of age.
- Bath with emollients to reduce scale and crust. Avoid soaps which may irritate the skin.
- A mild topical corticosteroid combined with an antifungal bd will reduce redness. Avoid more potent corticosteroids.
- Remove scalp scale with olive oil or an emollient such as Hydromol® ointment.

Langerhans cell histiocytosis

This rare disease may be misdiagnosed as infantile seborrhoeic dermatitis. ⚠ Purpura and ulceration should alert you to 'something different'. A skin biopsy will confirm the diagnosis. Look for:

- Erythematous pink or brown, scaly, crusted papules on the trunk, in the napkin area, and in flexures that may become confluent.
- Purpuric papules that may ulcerate and weep.
- Scaling and crusting on the scalp. The scalp bleeds, when scale is removed (a useful sign).
- Fever (sometimes).
- Isolated skin involvement may regress spontaneously, but refer for assessment and treatment.

Box 31.11 Scaly skin—differential diagnosis

Common

- Atopic eczema (see ➔ pp. 606–7).
- Seborrhoeic dermatitis (see ➔ p. 602).
- Scabies (see ➔ p. 172).
- Keratosis pilaris: scaly follicular papules on lateral aspects of arms and thighs, and sometimes cheeks. Not itchy. Skin feels rough.
- Psoriasis (much less common than eczema in childhood) (see ➔ pp. 189–208). Usually a family history of psoriasis.
- Ichthyosis vulgaris: fine pale scale, increased skin markings on palms. Presents after age 3 months. Mutation in filaggrin gene.

Rare

- Langerhans cell histiocytosis (see ➔ p. 602). Refer to a specialist for assessment and treatment.
- Netherton syndrome: erythroderma (diffusely red and scaly), failure to thrive, hypernatraemic dehydration, and sparse hair. Mutations in *SPINK5* gene that encodes the lymphoepithelial Kazal-type 5 serine protease inhibitor (LEKTI) lead to uncontrolled proteolytic degradation of the corneodesmosomes and a thin, highly permeable stratum corneum. May find characteristic hair shaft abnormality—trichorrhexis invaginata (bamboo hair) on the scalp or eyebrow, and later a characteristic rash—ichthyosis linearis circumflexa. Also raised IgE, atopy, and food allergies. Skin very permeable—risk of infection and absorption of topical agents.
  - ⚠ Avoid topical corticosteroids or topical calcineurin inhibitors.
- Nutritional deficiency, including:
  - Zinc deficiency (see ➔ p. 513): well-demarcated periorificial and perineal glazed erythema, paronychia, dermatitis on the fingers and toes.
  - Cystic fibrosis: periorificial and perineal rash like in zinc deficiency (see ➔ Box 24.6, p. 517)
  - Biotin deficiency: eczematous or psoriasiform rash around the eyes, face, and perineum. Also vomiting, seizures, developmental delay, hypotonia, ataxia.
- Immunodeficiency (see ➔ p. 612).

⚠ Box 31.12 Features that should make you re-evaluate the diagnosis of atopic eczema or seborrhoeic dermatitis

- Congenital onset.
- Erythroderma, large scaling plaques, or indurated skin.
- Purpura or ulceration.
- Failure to thrive, diarrhoea.
- Repeated infections.
- Poor response to emollients and mild/moderately potent topical corticosteroids.

Napkin rashes

Atopic eczema, the commonest form of dermatitis in infancy, tends to spare the moist skin in the napkin area. Instead, most napkin rashes are caused by urine and/or faeces irritating macerated skin (irritant contact dermatitis). Prevalence has fallen, because highly absorbent nappies wick moisture away from the skin.

What should I look for?

- Irritant dermatitis occurs between the ages of 3 weeks and 2 years. The rash involves the convex surfaces (e.g. buttocks) in contact with the napkin and spares skinfolds. The erythematous skin may have a rather glazed shiny appearance (see Fig. 31.5). Severe chronic irritant dermatitis may be associated with well-demarcated erosions (Jacquet dermatitis) or extensive papules.
- *Candida* napkin dermatitis, in infants aged 6–24 months, starts in skinfolds but may extend to involve the entire perineum. Look for satellite papules (less often pustules) and oral candidiasis.
- Infantile seborrhoeic dermatitis is an asymptomatic rash that usually presents around age 4–6 weeks. Look for well-demarcated erythematous patches, without much scale, in skinfolds. Inspect the scalp, face, and axillae (see ➔ p. 602 and Fig. 31.6). May be difficult to differentiate from napkin psoriasis (see Box 31.13).
- Allergy to preservatives in wet wipes may present as dermatitis.
- Erosive perianal eruption is caused by diarrhoea, often in breastfed infants aged 6 weeks to 3 months. The rash is characterized by erythema, with well-demarcated superficial erosions.
- Staphylococcal impetigo may present with flaccid bullae in the napkin area, often in the first few weeks of life.
- Group A streptococcal dermatitis (infants, >6 months) causes painful perianal erythema (may involve inguinal folds) (see ➔ p. 132).

What should I do?

- Culture skin swabs for *Candida* and bacteria.
- Minimize irritation of the skin:
  - Wash skin with plain water and a soap substitute.
  - Ensure that baby wipes are alcohol- and fragrance-free.
  - Change the napkin, as soon as it is soiled by faeces.
  - Avoid talcum powders which may irritate the skin.
- Paraffin-based ointments, such as Hydromol® or 50% WSP in liquid paraffin, hydrate the skin and act as a barrier to external irritants.
- Treat inflamed skin with twice-daily applications of an ointment containing 1% hydrocortisone, in combination with an antifungal agent such as miconazole or clotrimazole.
- Treat bacterial infection with oral antibiotics.
- Rashes unresponsive to treatment: think again (see Box 31.13).

Further reading

Stamatas GN and Tierney NK. *Pediatr Dermatol* 2014;**31**:1–7.

Fig. 31.5 Painful irritant dermatitis with a shiny glazed erythema that spares the skinfolds.

Fig. 31.6 Seborrhoeic dermatitis favours the skinfolds.

Box 31.13 Napkin rash unresponsive to standard treatment

- Find out how the skin in the napkin area is being washed, how often the napkin is being changed, what wipes are being used (could this be allergic contact dermatitis?), and what ointments are being applied.
- Take skin swabs, and culture for *Candida* and bacteria … again.
- Psoriasis responds poorly to treatments used for napkin dermatitis. The well-demarcated, but asymptomatic, erythema has minimal scale. Napkin psoriasis tends to start on the convex surfaces of the buttocks but may spread to the flexures, trunk, face, and scalp. Most infants do not have evidence of psoriasis in nails. Any family history of psoriasis? The rash is difficult to differentiate from infantile seborrhoeic dermatitis, but seborrhoeic dermatitis is usually controlled after 2–4 weeks of treatment. Aim to keep the skin comfortable using emollients and mild topical corticosteroids, rather than clear psoriasis.
- Dermatophyte infection ('ringworm') presents with well-demarcated scaly, erythematous papules and plaques, usually in toddlers, rather than infants. Sometimes there are pustules. Often involves the thighs and lower abdomen, as well as the napkin area. Has anyone in the family got tinea pedis? Take a skin scrape for mycology (see p. 164).
- Blisters or erosions? SSSS, caused by a circulating exfoliative exotoxin, presents with a tender peeling erythema, sometimes limited to the napkin area (see p. 112). Allergic contact dermatitis is uncommon in infants but may present as a vesicular eruption in infants aged >6 months. Rarely, autoimmune blistering diseases may present in the napkin area in infancy.
- Purpura or ulceration? Langerhans cell histiocytosis involves the flexures and the napkin area, but may also affect the scalp, the skin behind the ears, and the trunk (see p. 602).
- Zinc deficiency presents with psoriasiform dermatitis around the mouth, as well as in the napkin area. Look for paronychia, sparse hair, and failure to thrive, as well as recurrent *Candida* infections (see p. 513).

Atopic eczema (atopic dermatitis)

Atopic eczema is a chronic itchy skin condition that affects 20–25% of children in Western Europe. Prevalence is increasing. Atopic eczema usually starts in infancy. Most have mild disease. Disease clears in about 50% of children by puberty. Others have a chronic and recurrent lifelong illness. Pathogenesis is multifactorial (see Box 31.14). The prevalences of bronchial asthma and allergic rhinitis are increased in atopic patients, and skin-prick tests are positive to many common environmental allergens.

What should I ask?

Explore:
- Age of onset (usually age 2–3 months) and distribution of eczema.
- Response to previous and current treatments.
- Impact on the child and family/carers (especially on sleep).
- Growth, development, and progress at nursery/school.
- Dietary history, including any dietary manipulation.
- Immediate reactions (urticaria, wheeze, angio-oedema) to foods.
- Exposure to trigger factors, including heat and irritants such as wool clothing, soaps, shampoos, bubble baths, and shower gels. Allergic dermatitis is unusual in children, but irritant reactions to creams are common.
- Contact with airborne allergens such as house dust mite and pet dander (more important in older infants/children).
- Personal and family history of atopy (e.g. asthma, allergic rhinitis).

What should I do?

- Confirm the diagnosis: usually clear-cut, but remember scabies in any itchy child. For diagnostic criteria, see Box 31.15, and Figs. 31.7, 31.8, and 31.9).
- Make a global assessment: mild, moderate, or severe—based on symptoms (such as itching and sleep disturbance); the quantities or strengths of treatment, clinical signs, and the area affected.
- Record impact on quality of life of the child/family (see ➔ pp. 32–4).
- Using 'POEM' (patient-orientated eczema measure), get the patient's view (see www.nottingham.ac.uk/dermatology/POEM.htm).
- Scores, such as EASI (Eczema Area and Severity Index) or SCORAD (SCORing Atopic Dermatitis), will help to monitor response to treatment objectively.
- Crusted weeping eczema or pustules: swab for bacterial culture.
- Pain, grouped vesicles or small well-defined crusted erosions: exclude herpes simplex infection (eczema herpeticum), a medical emergency (see ➔ Box 7.2, p. 154).
- Immediate reactions to foods: refer for further investigation, if clinically indicated (see Box 31.18).
- Refer for patch testing, if you suspect allergic contact dermatitis, but irritancy is much commoner than allergy.
- Approaches to management, including top tips, are discussed on ➔ pp. 608–9 and pp. 610–11.

Box 31.14 Pathogenesis of atopic eczema

- Genetic and environmental factors interact.
- Atopic diseases are common in the family of affected individuals. Risk is more closely related to inheritance of maternal than paternal genes.
- No single atopic gene has been identified, but potential loci have been mapped to chromosomes 1q, 3q, 3p, 17q, and 20p, among others.
- Defective barrier function and innate immunity seem likely to play a role in the pathogenesis. Loss-of-function mutations in the gene for filaggrin, an epidermal barrier protein, predispose to atopic eczema, and the lipid/protein ratio is reduced in the stratum corneum. Defective stratum corneum may allow pathogens/allergens to penetrate the skin and trigger a hyperactive immune response, with activation of T-lymphocytes, dendritic cells, macrophages, keratinocytes, mast cells, and eosinophils. Pro-inflammatory cytokines and chemokines may perpetuate disease.
- Skin shows increased susceptibility to colonization by *Staphylococcus aureus*, which may also damage the skin barrier.
- *S. aureus* enterotoxins with superantigenic activity may play some part by activating T-cells and macrophages.

Box 31.15 UK criteria for diagnosis of atopic eczema

A child with an itchy skin condition in last 12 months, plus three or more of the following:

- Visible flexural dermatitis involving the skin creases such as the bends of the elbows or behind the knees (or visible dermatitis on the cheeks and/or extensor areas in children aged 18 months or younger).
- Personal history of flexural dermatitis (or dermatitis on the cheeks and/or extensor areas in children aged 18 months or younger).
- Personal history of dry skin in last 12 months.
- Personal history of asthma or allergic rhinitis (or history of atopic disease in a first-degree relative if child aged <4 years).
- Onset before age of 2 years (not used in children <4 years of age).

Further reading

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Margolis JS *et al. JAMA Dermatol* 2014;**150**:593–600.
National Institute for Health and Care Excellence (2007). *Atopic eczema in under 12s. diagnosis and management*. Available at: https://www.nice.org.uk/guidance/cg57.
Nottingham Support Group for Carers of Children with Eczema. *Information*. Available at: http://www.nottinghameczema.org.uk/nsgcce/information/index.aspx.
Werfel T. *J Invest Dermatol* 2009;**129**:1878–91.

Managing atopic eczema: top tips

Living with and managing atopic eczema is challenging for child and family. The psychosocial burden can be profound. Aim to address concerns and to educate (see Box 31.16), so that the child /family have the tools to manage this distressing chronic disease.

- Intractable itch/scratch. Relieve dryness, and control eczema. Protect skin from damage—suggest stroking or pinching skin to relieve itch; use occlusion. Therapeutic stories, in which the child is champion, may facilitate behavioural change, improve self-esteem, and help the child engage with treatment (overcoming 'Evil Eric Eczema').
- Complaints that the skin 'reacts to'/'is allergic to' emollients. Irritation of inflamed skin is much commoner than allergy. Ointments may be tolerated less well than creams. Suggest trying a creamy emollient on normal skin for a few days, before applying to eczematous skin. Explain that, if normal skin does not react, allergy is unlikely. Control acute eczema with topical corticosteroids or calcineurin inhibitors, and then apply emollients.
- Steroid phobia. Some parents refuse to use 'bad' topical steroids (a misconception that may have been reinforced by other healthcare professionals). Explain the different strengths of steroids and the proper use of topical steroids (see ➔ p. 610). State the specific quantity of topical corticosteroid you expect to be used over 1 or 2 weeks.
- Once eczema is controlled, empower the child and family by introducing proactive therapy, i.e. low-dose, intermittent applications of anti-inflammatory therapy to previously affected skin ('hot spots') (see Box 31.17).
- 'He has a food allergy'. Find out exactly what the parent means and what has been observed (see Box 31.18). Inappropriate withdrawal of foods may cause nutritional deficiencies. Most children with mild eczema do not need investigation for food allergies.
- Recurrent 2° bacterial infection (oozing golden-crusted patches), usually with *S. aureus*. Use antibacterial emollients, e.g. products containing benzalkonium chloride and/or chlorhexidine, to cleanse the skin and remove crusts. Bleach baths (1/4 cup of household bleach in a half bathtub of water) for 5–10min, 2–3×/week may reduce bacterial colonization. Treat nasal carriage of *S. aureus* (child and family) with topical mupiricin. Short courses of oral antibiotics may be required.
- Molluscum contagiosum. May be spread by scratching. Reassure the child/parent that molluscum will go, but this may take some time, particularly if topical steroids are being used. Flares of eczema around molluscum may herald regression. (Also see ➔ p. 158.)
- Attention-deficit/hyperactivity disorder (ADHD). Caring for children with eczema is exhausting. Some are just itchy and active, but others, in particular those with more severe eczema, may have ADHD. Consider referral to a child psychologist.

Box 31.16 Child/parent education

Adherence to treatment is difficult. Education is crucial, and the help of a dermatology nurse invaluable.

- Explore beliefs, fears, preferences, and hopes.
- Identify barriers to adherence, e.g. steroid phobia, time for treatment, complexity of regimen, cost, and discomfort.
- Agree educational objectives, e.g. proper use of topical steroids.
- Implement a written action plan.

Box 31.17 Proactive therapy for atopic eczema

Normal-looking non-lesional skin in patients is not normal—there is subclinical inflammation, as well as a barrier defect. Proactive therapy targets invisible inflammation in problem areas and reduces overall use of topical anti-inflammatory therapy.

- After clearance of visible eczema, continue to apply anti-inflammatory therapy, e.g. a potent topical corticosteroid ointment or 0.1% tacrolimus ointment, to previously affected skin (hot spots or problem areas) ×2/week.
- Continue daily application of emollients to all skin.

Box 31.18 Food allergy

- Consider food allergy in children with atopic eczema who have reacted previously to a food with immediate symptoms (e.g. urticaria, wheeze, angio-oedema), or in infants and young children with moderate or severe atopic eczema that has not been controlled by optimum management, particularly if associated with gut dysmotility (colic, vomiting, altered bowel habit) or failure to thrive. Refer to a paediatric allergist.
- Most children will grow out of food allergies, except allergy to nuts. Food allergy is very unlikely to play any part in the pathogenesis of atopic eczema in adults.

Further reading

Gillette C and Wallenberg A (2014). *Br J Dermatol* **170**(**Suppl S1**):19–24 (proactive therapy).

Naidoo RJ and Williams HC (2013). *Paed Dermatol* **30**:765–7 (therapeutic use of stories).

University of Nottingham, Centre of Evidence-Based Dermatology. *Psychology and eczema*. Available at: www.nottingham.ac.uk/research/groups/cebd/resources/psychology-and-eczema.aspx (story templates).

Atopic eczema: gaining control

Step 1: Listen and explain

- Explore beliefs, fears, and hopes, and answer questions (see ➲ pp. 606–7 and pp. 30–31).
- Discuss the multifactorial cause, particularly the abnormal skin barrier, and trigger factors, including soaps and bubble baths.
- Provide written information about eczema and support groups.
- Explain that treatment can control, but not cure, eczema.
- Reassure that most children improve, as they grow older.
- Explain that eczema will fluctuate and can be frustrating to treat.
- Discuss how to recognize flares of eczema and infections, including eczema herpeticum (rare).

Step 2: Emollients, antiseptics and soap substitutes

- Emollients are the cornerstone of management and should be used all over the body, even when eczema has cleared. Supply a range of unperfumed emollients. The best emollient is the one the patient likes. Creamy emollients, rather than ointments, may be preferred for daytime use. Prescribe in large quantities (>500g/week) to be used liberally and frequently (see ➲ p. 655).
- Use an emollient as a soap substitute.
- Reduce bacterial colonization with antibacterial emollients, e.g. products containing benzalkonium chloride and/or chlorhexidine or bleach baths (see ➲ p. 608).

Step 3: Topical anti-inflammatories and other treatments

- Discuss and demonstrate the use of topical corticosteroids. Explain that topical corticosteroids, unlike emollients, should only be applied to eczematous skin. (Apply 20–30min after emollient.) Explain how to step up and step down the strength of treatment. Show how to measure the application in fingertip units (FTUs) (see ➲ Box 34.3, p. 658).
- Most mild infantile atopic eczema can be controlled with mild to moderate corticosteroid ointments used once daily. Ointments are preferable to creams, as they are less likely to irritate the skin. Mild corticosteroid ointments can be used safely on the face.
- Potent topical corticosteroids or topical calcineurin inhibitors may be needed ×1/day for 7–14 days for more active disease.
- Very potent corticosteroid ointments should not be prescribed, without the guidance of a specialist.
- Once eczema is controlled, switch to proactive therapy (see Box 31.17).
- Wraps or paste bandages can help children with very itchy skin (see ➲ p. 660, and Box 34.4, p. 661).
- Treat 2° bacterial infection with antibiotics.
- Treatments, such as UVB, methotrexate, azathioprine, or ciclosporin, may be indicated in severe chronic atopic eczema.
- Prednisolone should be avoided, except for acute severe flares.

Fig. 31.7 Atopic eczema typically spares the napkin area.

Fig. 31.8 Atopic eczema with secondary bacterial infection causing crusting and weeping.

Fig. 31.9 Lichenification with increased skin markings secondary to chronic rubbing.

Step 4: General measures

- Involve a dermatology nurse to provide education and support.
- Discuss strategies to deal with problems such as sleep disturbance, irritability, or clingy behaviour.
- Eliminate irritants and other trigger factors—no pets in the bedroom!
- Use loose-fitting cotton clothing.
- Keep the bedroom cool, and avoid too much bedding.
- A sedating antihistamine may be helpful at night—but the impact may be disappointing.

Further reading

National Institute for Health and Care Excellence (2007). *Atopic eczema in under 12s: diagnosis and management*. Available at: http://www.nice.org.uk/CG57.

Cutaneous signs in rare syndromes

Immunodeficiency syndromes

⚠ Cutaneous signs may provide a valuable early clue to the diagnosis of these rare syndromes which may present insidiously. Problems suggesting immunodeficiency include recurrent, prolonged, or severe infections at more than one site, infections with unusual organisms, and infections poorly responsive to antibiotics (often requiring IV antibiotics).

Look for failure to thrive, chronic diarrhoea, and hepatosplenomegaly. Infants with severe T-cell immunodeficiency develop GVHD 2° to maternal engraftment at or around the time of delivery and may become erythrodermic. These patients also have an increased risk of malignancy.

- Omenn syndrome: AR. Severe combined immunodeficiency. Generalized eczema, failure to thrive, alopecia, eosinophilia, diarrhoea, repeated infections. Also hepatosplenomegaly and lymphadenopathy.
- Wiskott–Aldrich syndrome: X-linked disorder. Severe atopic eczema in first few months of life is associated with recurrent pyogenic infections and bleeding (petechiae, bruising), due to thrombocytopenia. May present with epistaxis and bloody diarrhoea.
- Ataxia telangiectasia: AR. Ataxia noticed in infancy. Oculocutaneous signs develop from age 3, conjunctival bulbar telangiectasias (by age 6), and cutaneous telangiectasias (ears, eyelids, malar, anterior chest, flexures). Also premature ageing of the skin (loss of fat), café au lait macules, hypertrichosis, chronic seborrhoeic dermatitis, and painful cutaneous granulomas (non-infectious). May have sinopulmonary infections, growth retardation, intellectual disability, and endocrine abnormalities.
- Hyper IgE syndrome (Job syndrome, Buckley syndrome): AD. *STAT3* mutations. Recurrent severe cutaneous and sinopulmonary infections in association with papulopustular dermatitis in first few months of life. 'Cold' abscesses with limited erythema. Also repeated fractures and retained 1° teeth. Elevated IgE levels (>2 000IU/L) and eosinophilia.
- Chronic mucocutaneous candidiasis. Most do not have positive family history. Progressive *Candida* infections of the skin, nails, and mucous membranes. May be associated with polyglandular autoimmune syndrome type 1 (autoimmune polyendocrinopathy–candidiasis–ectodermal dystrophy = APECED) (see ➔ p. 484).
- Chronic granulomatous disease: X-linked or AR. Chronic staphylococcal infections of skin, suppurative lymphadenitis, bronchopneumonia, hepatosplenomegaly, chronic diarrhoea, and malabsorption.
- Chediak–Higashi syndrome: AR. Silvery sheen to hair, partial albinism (strabismus and nystagmus), photophobia, hyperpigmented exposed skin, severe recurrent infections, and progressive neurological deterioration.

Multiple lentigines

What is a lentigo? (plural: lentigines or lentigos)

Lentigines are small (1–3mm in diameter), well-circumscribed, flat, brown to black 'freckles' that may appear on any body site. Lentigines do not fade when the skin is no longer exposed to sun, and are not necessarily triggered by sun exposure.

Some normal individuals have numerous lentigines, but, in a child with 'freckles', you should consider:

- Neurofibromatosis 1—a RASopathy: freckling in skinfolds (see ➔ p. 552). Incidence around 1 in 3000.
- Segmental neurofibromatosis 1: unilateral lentigines in the axilla or groin (see ➔ pp. 554–5). Incidence about 1 in 40 000.
- XP: premature sun damage with many lentigines mainly on sun-exposed skin. High risk of early-onset skin cancers, including malignant melanoma (see ➔ pp. 336–7). Rare—incidence around 1 in 250 000 to 1 in 500 000.
- Peutz–Jeghers syndrome: freckling on vermillion borders of the lips, oral mucosa, perioral, perianal, and periorbital skin, over joints, and on palms and soles (see ➔ p. 543). Incidence uncertain—estimates range from 1 in 25 000 to 1 in 300 000.
- Carney complex (includes NAME and LAMB syndromes): mucocutaneous pigmentation (lentigines on the lips, conjunctiva, inner or outer canthi, vaginal and penile mucosa, blue naevi; may have small café au lait spots that tend to fade with age), myxomas (cardiac, cutaneous—eyelid, external ear, nipple), breast myxomatosis or ductal adenoma, endocrine tumours (1° pigmented nodular adrenocortical disease, pituitary adenoma, testicular neoplasms, thyroid adenoma or carcinoma, ovarian cysts), psammomatous melanotic schwannoma, osteochondromyxoma. Inactivating mutation in regulatory subunit 1A of protein kinase A gene (PRKAR1A) located at 17q22-24 in 2/3 of cases (very rare indeed).
- Multiple lentigines syndrome—a RASopathy (previously known as LEOPARD syndrome). **L**entigines (hundreds of freckle-like spots all over the skin), **E**lectrocardiographic cardiac conduction defects, **O**cular telorism, **P**ulmonary stenosis, **A**bnormalities of genitalia, **R**etardation of growth, **D**eafness. Mutations in protein tyrosine phosphatase, non-receptor type 11 gene (*PTPN11*) in 90% cases (very rare indeed).

This list is not exhaustive, and other rare syndromes are associated with multiple lentigines, either in a generalized or localized distribution.

Physical and sexual abuse

Sadly some children are abused, and it is important to recognize cutaneous signs that may indicate abuse—but it is equally important to realize that some dermatoses can be confused with abuse. Record all signs, and refer urgently to a paediatrician with expertise in this area, if you suspect abuse. The welfare of the child should be your 1° concern.

What should I look for?

- Carer has delayed seeking medical help.
- Inconsistent history, incompatible with the injury or signs.
- Parental hostility or lack of interest.
- Sad, withdrawn, or frightened child.
- A child displaying frequent sexualized behaviour.
- Physical abuse—the upper head and body are frequently targeted in cases of deliberate physical injury. Look for:
  - Many bruises of different ages, often in clusters.
  - Pinch marks on the ear or petechiae on the pinna.
  - Linear slap marks, belt marks, finger impressions, or buckle marks.
  - Cigarette burns to the face or hands, or burns to the lower limbs (child held forcibly in scalding water).
- Sexual abuse:
  - Pregnancy.
  - HIV infection or another sexually transmitted disease in the absence of perinatal acquisition. Anogenital warts in children may be acquired by autoinnoculation, but the possibility of sexual abuse should be considered, particularly in children <2 years (see ➔ p. 160).
  - Signs of acute penetrating anogenital trauma, including acute hymenal injury, laceration, bruising, or perianal lacerations.
  - Genital abrasions, lacerations, bites, suction marks, or burns.

Dermatological conditions that may simulate abuse

- Mongolian blue spots (present at birth or soon after) may simulate bruises (see ➔ p. 344 and Fig. 31.10), as may the blue/black pigmentation in naevus of Ota (face) or naevus of Ito (upper trunk).
- Bullous impetigo may simulate cigarette burns (see ➔ Fig. 6.1, p. 130).
- SSSS, as the name suggests, can simulate scalds (see ➔ Fig. 6.2,p. 131).
- Phytophotodermatitis with blistering in linear streaks may simulate burns, or the residual pigmented streaks may simulate linear bruising (see ➔ Fig. 31.11, and Box 16.4, p. 327).
- Angio-oedema with swelling of the face or lips may simulate an early bruise (see ➔ p. 104).
- Cutaneous vasculitis with purpura may simulate bruises (see ➔ p. 442).
- Group A streptococcal perineal infection with itch, pain, erythema, discharge, and localized bleeding may suggest sexual abuse (see ➔ p. 132).
- Lichen sclerosus with haemorrhage may suggest sexual abuse (see ➔ p. 288).
- EB (rare) with skin fragility and blistering may suggest physical injury (see ➔ pp. 624–5).

Fig. 31.10 Mongolian blue spot.

Fig. 31.11 Phytophotodermatitis. Linear pigmentation was preceded by streaky erythema and blisters where the sun-exposed skin had been in contact with the juice of a plant that contained psoralen.

Further reading

Department for Education, Home Office, Department for Communities and Local Government, Department of Health, Foreign & Commonwealth Office, Official Solicitor and Public Trustee, and Ofsted. *Safeguarding children*. Available at: https://www.gov.uk/childrens-services/safeguarding-children.